

This dissertation explores the rise of virtual sports, using virtual cycling as its primary case study. It establishes virtual sports as fundamentally distinct and unprecedented sporting disciplines, defining them as competitive, physical-digital hybrid events that closely mimic their traditional parent sports while being co-governed by algorithmic, virtual-world laws. Through an integrative multi-theoretical approach, this dissertation analyses the adoption determinants, institutional strategies, and unique governance challenges, such as digital doping and algorithmic transparency, inherent to virtual sports. The five empirical studies included highlight that successful diffusion requires managing the complex interplay of digitisation, institutional dynamics, and innovation strategies across the micro-, meso- and macro-level of the sport sector. Ultimately, the legitimisation of virtual sports into society requires interdisciplinary collaboration and proactive co-governance by both traditional federations and technology developers.

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Real Sport Performances, Uncharted Virtual Worlds
An Institutional-Strategic Perspective on the Rise of Virtual Sports Through the Case of Virtual Cycling | Thibault Fouquaert

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**A dissertation submitted at Ghent University to
obtain the degree of Doctor of Health Sciences**

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REAL SPORT PERFORMANCES, UNCHARTED VIRTUAL WORLDS

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Thibault

English Summary

This dissertation examines innovation in the sport sector, and in particular the emergence of virtual sports disciplines and their impact for sport organisations and sport policy. We examine this through the most popular and legitimate case of virtual sports so far, namely virtual cycling. Virtual sports are characterised by digitally measured performances, and specifically performances that realistically and accurately mimic the bodily participation of their “parent” or “traditional” sports’ physicality. As we explain how virtual sports situate within the broader field of sports digitalisation, we posit that they are fundamentally distinct from any previous digitalisation such as esports, simulations in sports, or exergames. By systematically developing an understanding of virtual sports along nine dimensions of sports, we reveal their distinction based on (1) their unique physicality (i.e. *highly decompressed*) and (2) their partial use of the actual reality of real-world laws, given that performances are actively mediated and altered by virtual-world laws via algorithms, creating a *distinct physical-digital sporting reality*. This leads to an important theoretical contribution of this dissertation, namely the development of a universal definition of virtual sports as: “*Institutionalised, rule-governed contests of human physical and cognitive skills that operate within a distinct physical-digital interreality that are characterised by voluntary human participation, a niche following, and high movement decompression mimicking a traditional parent sport, made possible through virtuality yet unbound to specific configurations of remoteness, (a)synchrony or representational fidelity*”.

Moving forward with this definition, we adopt an integrative multi-theoretical approach to study the innovation adoption determinants, processes, and strategies behind virtual sports. We point to research gaps in our overarching framework of Institutional Theory (e.g., lacking understanding of virtual sports’ legitimacy), and complementary frameworks used such as Organisational Innovativeness and Diffusion of Innovation (e.g., lacking micro-meso integrative models), Disruptive Innovation Theory (e.g., lacking insights on technical versus institutional strategies), and Good Governance (e.g., lacking virtual sports specific governance frameworks). We address these gaps using insights from five empirical studies. Study 1 focusses on adoption determinants, revealing that adoption is primarily dependent on positive perceptions and attitudes of managers (i.e. micro) rather than structural capacities at the meso-level. Study 2 focusses on virtual sports as-a-club-service within institutional logics, revealing potential for both sport-for-all and result-oriented (elite) applications within clubs. Study 3 focusses on sport governing bodies’ legitimisation strategies for virtual sports, revealing that an institutional strategy (i.e. normative framing) is predominantly employed while a technical

strategy (i.e. developing features) lacks, despite their equal importance for successful diffusion. Study 4 focusses on governance issues pertaining to virtual cycling, revealing multiple issues specific to virtual sports like platform standardisation, platform fragmentation, digital doping, and a lack of development in athlete-related support and structures. Study 5 focuses on the differences between virtual cycling platforms, revealing that MyWhoosh's algorithm and Zwift's algorithm significantly differ in how physical performances are handled, which in turn opens the door towards governance issues such as algorithmic opacity and asymmetric power relations, for which dual governance and scrutability-without-disclosure models are needed.

Reflecting on our aggregated results, we discuss how successful diffusion of an innovation like virtual sports ultimately relies on managing three interdependent domains of sport digitisation, institutional dynamics, and innovation strategies across micro-, meso-, and macro-levels. We discuss how the digitisation aspects such as hardware and software technicalities affect its adoption and use for (1) individual athletes, volunteers, and decision-makers within clubs, for (2) meso-level organisations such as clubs, national sports federations, or commercial platform providers, but also for (3) the larger macro-level institutions such as international standards or sector-wide regulatory networks. With regard to institutional dynamics, we discuss (1) how lacking governance structures negatively affect individual competing athletes, (2) how governing bodies' organisational constraints make them highly dependent on commercial platforms and shape our sporting policies, and (3) how institutional change theory enables us to view virtual sports development as two different deinstitutionalisation-reinstitutionalisation trajectories: one within the sport-for-all logic and one within the result-oriented logic. Last, innovation strategies in virtual sports are discussed and demonstrate (1) how individualistic perceptions and attitudes are at the core of innovation strategies but are also shaped by both digitisation and institutionalisation, (2) which managerial and human-resources determinants are important for clubs, and (3) how sector-wide diffusion is achieved through actively constructing norm-compliant narratives around the ontology of sports: what sport *is*, what it should *be*, and how virtual disciplines support that development.

In the future, courage, a clear vision, strategic planning, dedicated resources, and interdisciplinary research will be needed to increase virtual sports' legitimacy and acceptance by society as a new sporting discipline. For sport governing bodies, a transition from passive overseers to active participants in developing the virtual sports landscape is required by incorporating technical knowledge that mitigates the technical discrepancies currently undermining the virtual disciplines' legitimacy. On top, close collaboration must be sought with platform developers to employ dual-governance and scrutability-without-disclosure models.

Dutch Summary (Nederlandstalige samenvatting)

Dit doctoraatsproefschrift onderzoekt innovatie in de sportsector, en in het bijzonder de opkomst van virtuele sportdisciplines en de impact daarvan op sportorganisaties en sportbeleid. We onderzoeken dit aan de hand van de meest populaire en legitieme casus van virtuele sporten tot nu toe, namelijk virtueel fietsen. Virtuele sporten worden gekenmerkt door het digitaliseren van fysieke prestaties en dit realistisch te vertalen naar een virtuele wereld. Het onderzoek start met een theoretische uitleg hoe virtuele sporten zich situeren binnen het bredere veld van sportdigitalisering. Hierbij stellen we vast dat virtuele sporten fundamenteel verschillen van eerdere vormen van sportdigitalisering zoals *esports*, sportsimulaties of *exergames*. Om dit verschil duidelijk te maken is er een definitie van virtuele sporten ontwikkeld aan de hand van negen universele sportdimensies. Daarbij tonen we aan dat virtuele sporten zich onderscheiden op basis van twee dimensies. Ten eerste is er hun unieke hybride digitaal-fysieke component, wat betekent dat fysieke bewegingen sterk gelijkend op de “traditionele” sport absoluut nodig zijn om de sport te beoefenen én accuraat worden vertaald naar de virtuele wereld. Ten tweede is er hun fysiek-digitale realiteit, wat betekent dat fysieke prestaties tijdens hun vertaling naar een virtuele wereld worden beïnvloed door de algoritmes die deze vertaalslag maken. De uiteindelijke prestatie in de virtuele wereld is met andere woorden een combinatie van de fysieke – echte – realiteit én van de digitale – virtuele – realiteit. Dit leidt tot een eerste belangrijke theoretische bijdrage van dit proefschrift, namelijk de ontwikkeling van een universele definitie van virtuele sporten die berust op eerdergenoemde negen universele sportdimensies: “(1) Competities waaraan (2) mensen (3) vrijwillig deelnemen om zich tegen elkaar te meten op basis van hun (4) cognitieve vaardigheden en (5) digitaal-fysieke vaardigheden die accuraat de traditionele discipline nabootsen, en waarbij de competitie gekenmerkt wordt door een (6) fysiek-digitale realiteit, (7) een niche aanhang, en (8) gereguleerde (9) institutionele context. Deze virtuele competities worden daarbij mogelijk gemaakt door virtualiteit (d.i. het digitaal meten van prestaties) maar zijn niet noodzakelijk gebonden aan tijd, ruimte, of manier van weergave”.

Voortbouwend op deze definitie hanteren we een multi-theoretische benadering om de determinanten, processen en strategieën van innovatie-adoptie achter virtuele sporten te bestuderen. We wijzen op hiaten in bestaand onderzoek binnen ons overkoepelend theoretisch kader van de institutionele theorie (bijv. een gebrek aan kennis over legitimiteit van virtuele sporten). Dit geldt ook voor aanvullende theoretische kaders zoals organisatorische innovativiteit en innovatie-diffusie theorie (bijv. het ontbreken van micro-meso integratieve

modellen), disruptieve innovatie theorie (bijv. het ontbreken van inzichten in gebruik van technische versus institutionele strategieën), en de theorie rond goed bestuur (bijv. het ontbreken van een theoretisch kader voor de regulatie van virtuele sporten). We pakken deze hiaten aan met behulp van inzichten uit vijf empirische studies. Studie 1 richt zich op adoptiedeterminanten van virtueel fietsen en laat zien dat adoptie grotendeels afhankelijk is van positieve percepties en attitudes van managers (micro) in plaats van structurele capaciteiten op organisatieniveau (meso). Studie 2 richt zich op virtueel fietsen als een clubdienst binnen institutionele logica's, en toont het potentieel aan voor zowel sport-voor-allen als resultaatgerichte (elite) toepassingen binnen clubs. Studie 3 richt zich op de legitimeringsstrategieën van internationale federaties voor virtueel fietsen. Hieruit blijkt dat overwegend een institutionele strategie (normatieve framing) wordt toegepast, terwijl een technische strategie (het ontwikkelen van technische functies) ontbreekt, ondanks hun gelijkwaardig belang in succesvolle diffusie van virtueel fietsen als legitieme discipline. Studie 4 richt zich op bestuurskwesties met betrekking tot virtueel fietsen en brengt meerdere problemen aan het licht die specifiek zijn voor de virtuele variant van fietsen, zoals platformstandaardisatie, platformfragmentatie, digitale doping, en een gebrek aan ontwikkeling in atleetgerelateerde ondersteuning en structuren. Studie 5 richt zich op de verschillen tussen de algoritmen van online virtuele fietsplatformen. Deze studie toont aan dat het algoritme van het MyWhoosh platform en het algoritme van het Zwift platform aanzienlijk verschillen in de manier waarop fysieke prestaties worden vertaald naar virtuele prestaties. Dit opent op zijn beurt de deur naar kwesties rond goed bestuur zoals gebrek aan algoritmische transparantie en asymmetrische machtsverhoudingen, waarvoor modellen van duaal-besturen (federatie en commercieel platform) en controleerbaarheid-zonder-openbaarmaking (*scrutability-without-disclosure*) nodig zijn.

Reflecterend op onze gebundelde resultaten, bespreken we hoe de succesvolle diffusie van een innovatie zoals virtuele fietsen, en in uitbreiding virtuele sporten, uiteindelijk neerkomt op het goed en strategisch beheren van drie onderling afhankelijke domeinen: sportdigitalisering, institutionele dynamiek, en innovatiestrategieën over micro-, meso- en macroniveaus heen. We bespreken hoe de digitaliseringsaspecten, zoals hardware- en softwaretechnieken, de adoptie en het gebruik ervan beïnvloeden voor (1) individuele atleten, vrijwilligers en besluitvormers binnen clubs, voor (2) organisaties op mesoniveau zoals sportclubs, nationale federaties, of aanbieders van commerciële platforms, maar ook voor (3) institutionele instellingen op macroniveau zoals internationale federaties, standaarden of sectorbrede regelgevende netwerken. Wat betreft de institutionele dynamiek bespreken we (1)

hoe ontbrekende bestuursstructuren individuele competitieve atleten negatief beïnvloeden, (2) hoe de beperkte kennis en structuur van federaties hen sterk afhankelijk maken van commerciële platformen en zo ons sportbeleid mee vormgeeft, en (3) hoe de theorie van institutionele verandering ons in staat stelt de ontwikkeling van virtuele sporten te zien als twee verschillende de-institutionalisierungs- en re-institutionaliseringstrajecten: één binnen de sport-voor-allen logica en één binnen de resultaatgerichte-topsport logica. Tot slot worden innovatiestrategieën in virtuele sporten besproken. Deze laten zien (1) hoe individualistische percepties en attitudes de kern vormen van innovatiestrategieën, maar ook worden gevormd door zowel digitalisering als institutionalisering, (2) welke determinanten op het gebied van management en human-resources belangrijk zijn voor clubs, en (3) hoe sectorbrede diffusie wordt bereikt door actief normconforme narratieven te construeren rond de ontologie van sport: wat sport is, wat het zou moeten zijn, en hoe virtuele disciplines die ontwikkeling zouden kunnen ondersteunen.

In de toekomst zullen een duidelijke visie en strategie, extra toewijding van middelen, en interdisciplinair onderzoek nodig zijn om de legitimiteit en acceptatie van virtuele sporten als een nieuwe sportdiscipline in de samenleving te verzekeren. Voor sportbonden is een overgang nodig van passieve toezichthouders naar actieve deelnemers in de ontwikkeling van het virtuele sportlandschap. Dit kan bereikt worden door technische kennis te integreren die de technische discrepanties vermindert, iets wat momenteel de legitimiteit van virtuele disciplines ondermijnt. Daarnaast is het voor federaties essentieel om nauw samen te werken met commerciële platformontwikkelaars om modellen van duaal-bestuur en controleerbaarheid-zonder-openbaarmaking toe te passen.

List of Abbreviations

AAM	Author Accepted Manuscript
ACE	Arbitration Court for Esports
AR	Augmented Reality
CAS	Court of Arbitration for Sport
CQA	Collective Qualitative Analysis
EPO	Erythropoietin
ESIC	Esports Integrity Commission
IOC	International Olympic Committee
IP	Intellectual Property
ITA	International Testing Agency
KMO	Kaiser-Meyer-Olkin (test for factor analysis suitability)
LDA	Linear Discriminant Analysis
LLM	Large Language Model
MANOVA	Multivariate Analysis of Variance
MR	Mixed Reality
RQ	Research Question
SD	Standard Deviation
SGB	Sport Governing Body
UCI	Union Cycliste Internationale
VR	Virtual Reality
VSRN	Virtual Sports Research Network
WADA	World Anti-Doping Agency
XR	Extended Reality
ZADA	Zwift Accuracy and Data Analysis

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Part I

Introductory framing of context, literature, gaps, and research questions

1 Introduction

Writing a doctoral thesis on the topic of innovation in sport management seems intriguing yet forgiving at first due to the widely applicable nature of innovation, and as such, what that dissertation can therefore be about. However, this exact freedom makes it above all a challenging task in which careful consideration of numerous different perspectives, both theoretical and practical, is due. Innovations naturally encounter resistance when introduced within traditional sporting norms, particularly when new or distinct hurdles remain unmanaged and unregulated. Therefore, the purpose of this thesis is to develop insights into *how* and *when* organisational innovation takes place, *how and why* organisational innovation introduces barriers surrounding institutionalisation and legitimacy, and *what* value organisational innovation can create in sport organisations, for its members, and the broader field of sport policy. This is done through studying the specific case of virtual cycling, a virtual sporting discipline within cycling, as an emerging technological innovation that many sports organisations have begun to adopt. Furthermore, this research aims to draw implications that are not only relevant for sport management, but also for closely related fields such as sports digitalisation, sports engineering, sports marketing, sociology of sports, or sports law.

Technological innovation continues to shape the development of sport, which can be situated within the broader trends of sport digitalisation and digitisation (Goebeler et al., 2024; Miah, 2017; Westmattelmann et al., 2025). These developments introduce multifaceted opportunities and barriers that require appropriate management by the sport organisations faced with them. As many practices in sport (e.g. policy implementation, professionalisation, technology) tend to change frequently (Skille, 2008; Slack & Hinings, 1992; Tjønndal, 2021), managing innovation might seem evident for sport organisations. Yet, embracing technological innovation (like virtual cycling) in a way that activates its full potential (e.g. leveraging it for service or social innovation) has proven to be a difficult endeavour for many sport organisations (Corthouts et al., 2020, 2022; Delshab et al., 2022; Fouquaert, Tjønndal, De Clerck, et al., 2025; Fouquaert, Tjønndal, & Willem, 2025; Gerke & Dickson, 2025; Hoeber & Hoeber, 2012; Mazurova et al., 2022; Svensson et al., 2020; Winand & Hoeber, 2017). These sport organisations are often faced with governance issues, the risk of missed opportunities, or strategic dilemmas during their innovation process. As such, new research-based insights are needed to inform academics, policymakers, and practitioners regarding both successful and unsuccessful approaches to consider, implement, and make innovations socially accepted

within sport organisations and beyond. For instance, several studies have pointed out that new practices in sport are primarily used for gains in sport performance, undermining their potential and social acceptance as a tool to also address social goals (De Bock et al., 2022; Loland, 2002; Stenling & Fahlén, 2009). Consequently, understanding how to manage innovations in a way that safeguards organisational interests, athlete interests, and broader societal values remains a critical challenge to leverage the full potential of innovation in a sporting context.

To explore these competing dynamics, this thesis relies on the specific example of virtual cycling to examine the development of virtual sporting disciplines as organisational innovation in sport. Virtual cycling was chosen because it is the most advanced, legitimised, and institutionalised virtual sport discipline at the time of writing (i.e. 2022-2026). For instance, Zwift (i.e. an online platform for consumers to participate in virtual cycling from home using peripheral equipment) reported over a million paid subscriber worldwide in 2024 (Maker, 2024). In support of its accessibility, the financial investment for individuals to participate in virtual cycling also significantly decreased over the years as prices of smart trainers (compatible with almost any bike) now start at €250 (e.g. Decathlon D100) and average €500, while new platforms like MyWhoosh are free to use (versus Zwift's €20/month) (Maker, 2025). Other examples include the Union Cycliste Internationale (UCI) hosting a Cycling Esports World Championship since 2020, and a total prize pool of over US\$3.75 million being paid out to athletes competing in virtual cycling races in 2024 (Schwenker, 2025). This illustrates how virtual cycling has matured and gained acceptance since its debut in 2014. Meanwhile, virtual disciplines of other sports (e.g. running, triathlon, golf)¹ are currently being launched and have much to learn from the development of virtual cycling. This underpins the importance and timeliness of studying virtual cycling and virtual sports in general.

Building on this timeliness, the research presented in this dissertation is needed because virtual sports' unprecedented features (see section 9.5) introduce novel integrity threats such as digital doping, weight manipulation, and algorithmic discrepancies that challenge traditional governance structures and approaches (Dyer, 2022; Dyer & Taylor, 2023; Fincoeur & Bongiovanni, 2021; Richardson, 2025; Westmattelmann et al., 2022). In addition, the sporting landscape stands at a critical juncture where virtual sports are transitioning from a niche training tool into legitimate sporting disciplines with their own socially embedded structure (Berger et al., 2025; Reed et al., 2023). Combining these insights, it becomes clear that the process of

¹ See Thibault's blogpost "Virtual Sports: an overview of what exists and what is possible" for illustrative examples at <https://www.virtualsports.be/2025/03/24/virtual-sports-an-overview-of-what-exists-and-what-is-possible/>

virtual sports gaining awareness, legitimacy, and social acceptance within sport organisations is one of many unresolved hurdles, issues, and underutilised opportunities. This means that if we fail to develop a deeper understanding of the phenomenon of virtual sports and its transitioning process now, we risk overlooking the threats to virtual sports' integrity, inclusivity, and sustainability in the future. Beyond risk mitigation, a deeper and more contextualised understanding of virtual sport disciplines is essential for engaging leisure sport participants for whom being physically active indoors using a virtual platform might be the only alternative to being not physically active at all under certain circumstances (Devine et al., 2024; Evsikov et al., 2024; Reed et al., 2023). In that sense, it is also important to understand how virtual sports can complement Sports-for-All policies and how they can be leveraged to turn individual sport participation (e.g. "light" sport settings unbounded to formal structures and rules like clubs or schedules) into socially embedded (e.g. club-organised) sport participation given the benefits of the latter (Bentvelzen et al., 2022; Borgers et al., 2018; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021). Altogether, immediate research on virtual sports ensures the strategic and sustainable long-term development of these novel technologies, which is something that remains underdeveloped (Edwards & Hettinga, 2023; Reed et al., 2023).

This doctoral dissertation consists of six separate parts. Part I provides an introduction to the context, theory, gaps in the literature, and research questions. To begin, section 1 explains the purpose of and rationale behind the research in this dissertation. Next, section 2 provides context to introduce virtual sports and situate it within the broader field of sport digitisation and digitalisation. Section 3 focuses in on our theoretical approach towards studying virtual sports, outlining the adopted overarching perspective, the theories and frameworks used, and the existing research gaps in the literature related to virtual sports. Section 4 presents our research aims and research questions (RQs) addressing the previously outlined gaps. Part II delineates our philosophy of science in section 6, while section 7 clarifies the empirical and methodological considerations by reflecting on our methods, data collection, and case selection. At this point, we arrive at the parts that provide answers to our research questions. This begins with Part III, that presents a theoretical *pre-study* into the ontology of virtual sports (i.e. what makes an activity a "virtual" sport) in section 9. The literature review and self-developed definition of virtual sports presented in this pre-study were written in function of assembling our earlier completed studies. While a clear definition of virtual sports is ultimately required to bridge our empirical studies in terms of their object of study, the result of this pre-study also adds a distinct theoretical contribution to sport management and adjacent fields of study as an

additional output of this dissertation. We continue by presenting our five empirical studies in Part IV, listed from section 12 to 16 (cf. overview in Table 4, section 11). Next, Part V delineates our general discussion, in which both theoretical and empirical insights from our pre-study and empirical studies are linked back to theory, the literature, and the research gaps identified in section 3. We start by summarising our empirical results in relation to their RQ in section 17, which then introduces the guiding structure for the upcoming discussion sections. This guiding discussion structure outlines how sections 18, 19, and 20 cover three domains (digitisation, institutional dynamics, and innovation strategies) across micro-, meso-, and macro-levels. This structure serves to reflect upon and contextualise the answers to our research aims and research questions. Lastly, we close off in Part VI by discussing the ongoing societal outreach project initiated during this doctoral process, which is the Virtual Sports Research Network².

2 Research Context of Sports Digitalisation: a Layered Approach

In this section, we examine the intertwining of sports and digitalisation. In particular, sport digitisation and digitalisation get discussed through three distinct but interconnected layers as a way to situate virtual sports within the broader and ever-evolving sporting landscape. This section, as part of the general introduction (Part I), does however only introduce virtual sports to clarify the theories by which it will be studied, alongside the research gaps in the literature, and the research questions that address those gaps. For a more extensive literature review on virtual sports, we refer to Part III, where we develop a conceptual definition of virtual sports as a way to clarify the object of study of this dissertation.

To begin with, a widely recognised distinction is made between sport digitisation and sport digitalisation. Where sport digitisation concerns the “what” of technological conversion of sporting activities (e.g. biometrics), sport digitalisation concerns the “why” and “how” of the technological transformation of sport activities (e.g. strategy, governance, experiences). This distinction is often central to studies situated in sport management, sociology of sport, and sport ethics (Miah, 2017). In this sense, digitisation can occur without digitalisation. For example, a club may buy software (digitisation) yet fail to beneficially integrate it into their operations (digitalisation).

The literature on sport digitalisation covers various aspects of integrating digital technology in sport related to competition, governance, management, ethics, consumption, and

² See www.virtualsports.be

more (Miah, 2017; Westmattelmann et al., 2025). However, sport digitalisation is not to be confused with “technology-driven sports”. These are sports that by their nature require the adoption of “new” equipment (e.g. bicycles, automobiles) to perform the activity itself (e.g. cycling, motorsports) (Thomas & Tierney, 2021). Sport digitalisation may thus be thought of as the use of sports technology in a broader sense. It involves modernising of many aspects across any sport by augmenting or mediating it with digital technologies, rather than technology being fundamental to the sport’s existence (though, that still depends on the reader’s definition of “technology”). The scope of this thesis concerns both sport digitisation and sport digitalisation. We develop an understanding of *what* virtual sports entail technically and conceptually, but also *when*, *how*, and *why* issues and value creation could take place in sport organisations adopting virtual sports like virtual cycling. Some of our empirical work presented (e.g. study 5) analyses the conversion of physical to digital performances (digitisation), which is more closely related to sport digitisation. Nonetheless, we seek to interpret these results within the broader perspective of sport digitalisation, for example by focussing on governance issues related to such digitisation outcomes.

Another central aspect of both sport digitisation and digitalisation is the role of the physical artifact involved. Goebeler et al. (2024) define physical artifacts in this context as objects characterised by their manifestation *in* and *of* our environment, or in other words, characterised by their physical nature and their purposeful being in space. Artifacts in our context are technical systems (e.g. devices) and have a relationship with the sporting activity that interacts with the artifacts’ matter, form, or function. These properties can be designed, conditioned or used in such a way that allow certain actions while disallowing others, which provides a grounding for researchers to make statements about their role with respect to digitalisation. Explicitating the role of the physical artifact does not only provide insights into *what* sport digitalisation entails, but also allows to further analyse and theorise about the *when*, *how*, and *why*.

Over the last few decades, different layers of sport digitalisation have unfolded that add up and blur into each other. As illustrated in Figure 1, each layer is marked by a period of rapid and transformative growth, though development is not halted when a new layer emerges.

Figure 1: Three layers of sport digitalisation. Own conceptualisation based on literature. Layer 1 involves the use of technology as a peripheral supplementary tool for performance enhancement, monitoring, and officiating. Layer 2 shifts toward blending physical and digital environments by using Extended Reality (XR) to interactively augment the sporting experience. Layer 3 represents the deeper digital mediation of sports performances as either mere simulations of reality or as new virtual distinct sporting realities. Source: own conception.

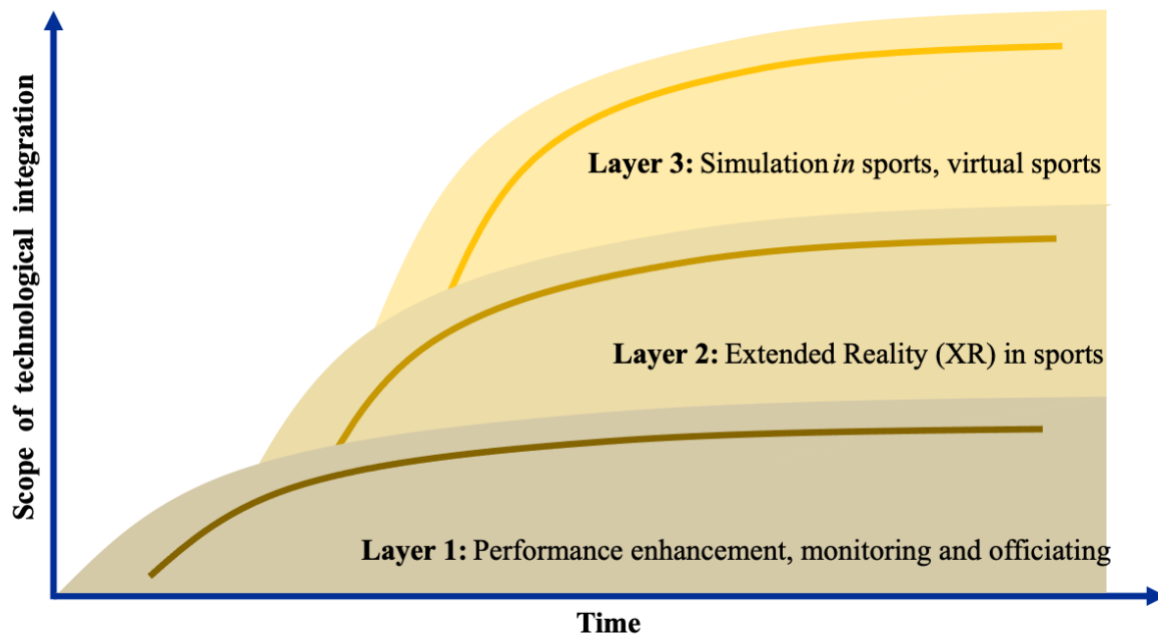


Figure 1 is not intended to be exhaustive in terms of digitalisation areas, but captures the literature of Fouché (2017) and Miah (2017) relevant to situate virtual sports. Therefore, the conceptual boundaries of Figure 1 are such that the layers' temporal nature (cf. X-axis) do not imply a strict interdependency but rather represent the order of their natural emergence. In addition, Figure 1 shows how each layer saturates and matures over time (e.g. similar to adoption rates in diffusion theory S-curves). This means that each layer's development becomes marginal as technology matures or regulatory refinement limits it (e.g. sport governing bodies allowing new technology, or purposefully limiting it) (Fouché, 2017; Synrem, 2019). For instance, during this continued development, rules are put in place with the idea of protecting the “purity” of the sport from “unnatural” technological intrusion (Magdalinski, 2009).

2.1 *Sport Digitalisation Layer One: Enhancing and Monitoring*

The first layer primarily involves the digitisation of performance enhancement (e.g. engineering of hydrodynamic swimwear), performance monitoring (e.g. power meters or continuous glucose monitors in cycling), and officiating (e.g. VAR in football and Hawk-Eye in tennis)

(Fouché, 2017). These systems are mainly aimed at increasing performances and reducing errors in the human judgement of performances (Fouché, 2017). The first layer of digitalisation may also overlap with the initial phase of Lupton's (2016) notion of "datafication" of the human body, where digitisation turns into digitalisation as captured data is applied in an often prescriptive rather than descriptive way, for instance using AI-powered system in artistic gymnastics judging (Mazurova et al., 2022). Examples include selecting athletes, creating a culture of "dataveillance", or transferring trust from human judgement to algorithmic certainty (Lupton, 2016). While technology in this first layer might indeed be assumed to be objective and mitigate subjective human judgement, research has shown that it reproduce biases and generate ambiguous outcomes (Mazurova et al., 2022; Skirbekk, Moen, et al., 2025; Skirbekk, Røsten, et al., 2025). This offers "the illusory power of automatically enforcing what they prescribe" (Lupton, 2016, p. 56). Overall, the physical artifacts utilised in the first layer can mainly be thought of as changing little about how the nature of sport is performed or experienced because the role of the artifact is to function as as a peripheral support tool. Performance-enhancing, monitoring, or officiating technologies do not change how performances are executed or delivered by athletes, but they do impact the performance measures (e.g. speed), approaches to the game (e.g. tactics or pacing), or outcomes (e.g. penalties or timing). Consequently, the *purpose* of technological integration in layer one is to supplement sports in a broad context through technological capturing and monitoring: indirectly increasing performances, adapting training approaches, optimising equipment designs, assisting officiating practices, and so forth.

2.2 *Sport Digitalisation Layer Two: Extended Reality in Sport*

The second layer of sport digitalisation marks a shift from technological integration supplementing sport practices and performances in the first layer to a different scope of technological integration, that is, blending the physical and the digital. This layer is largely characterised by Extended Reality (XR), which represents the collective term used to create digitally interactive environments through Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) (Cranmer et al., 2021; Miah et al., 2020). While AR overlays graphics onto the real world, such as information for fans via in-stadium apps (Uhlendorf & Uhrich, 2024) or via broadcasts (Goebert et al., 2022), VR features computer-generated simulations of sports via headsets (Miah et al., 2020). MR encompasses both AR and VR to blend virtual and physical objects in real-time. For example, it can project altered dimensions onto a table tennis

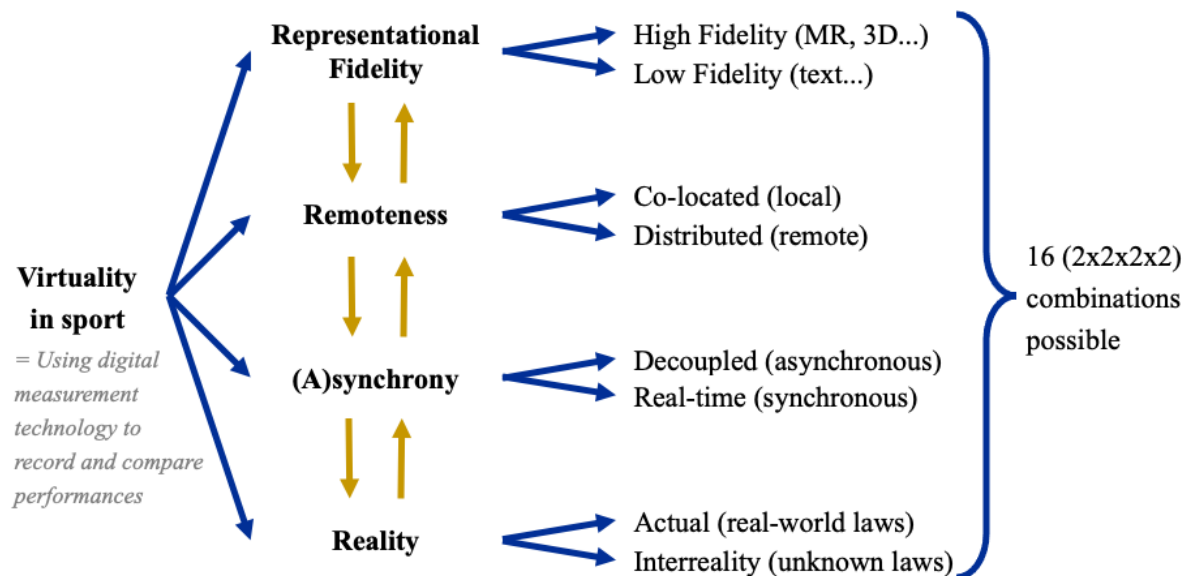
surface to balance competition between differing skill levels (Cranmer et al., 2021). Furthermore, this second layer also includes “digitally augmented sports” (Goebeler et al., 2021, p. 5846) that introduce mechanisms where digital feedback directly influences physical output. An example is Formula E’s “FanBoost”, which allows spectators to vote for an in-race speed advantage for certain athletes. In these hybrid contexts, the physical artifact functions as a “nexus for digital innovation” according to Goebeler et al. (2024), where physical and digital components mutually constitute the sporting experience. While earlier work argued that these interventions change little about the ontology of sport, as “the competition can still carry on without the digital components” (Goebeler et al., 2021, p. 5846), these boundaries are becoming increasingly overlapping. This is pointed out by Miah’s (2017) concept of “Sport 2.0”, which argues for moving beyond viewing these technologies as mere supplements to sports. Miah (2017) suggests that the depth of this integration demands abandoning the binary distinction between “physical” and “digital” because digital technologies become constitutive of sport itself. This fundamentally alters the ontology of how sport is played, viewed, and understood in the contemporary sporting landscape. Consequently, the *purpose* of technological integration in layer two is to blend digital artifacts in the physical environment, moving beyond the supplementation of sport practices and performances in layer one. Within broader management and consumer experience literature, the deep integration of physical and digital realms is often referred to as the concept of “phygital” (Batat, 2024). The implications of these phygital spaces for defining the reality of modern sporting experiences will be further elaborated below.

2.3 Sport Digitalisation Layer Three: Virtualising Sports

The third layer of sports digitalisation extends beyond interactively augmenting sports with physical-digital blended environments (i.e. layer two) to virtualising sports in hybrid environments that deeply merge physical and virtual domains (Cranmer et al., 2021; Goebeler et al., 2021; Miah et al., 2020). This layer revolves around the concept of “virtuality” of sport, which Parry and Giesbrecht (2023) conceptualise as: the application of physical artifacts (e.g. digital measurement technology) to record and compare performances. While the technological integration in layers one and two also relies on measuring sport performances, its *purpose* in layer three is not supplementary to sports and environment-blending. Rather, it aims to allow the mediation of sport through technical artifacts. This, in turn, also leverages the possibilities in terms of space and time boundaries (Goebeler, 2024). Figure 2 illustrates the “spatial and temporal decoupling” (Goebeler, 2024, p. 52) virtuality can bring to sports (i.e. not to be

confused with institutional decoupling, cf. Section 3.2.1), which we respectively label as changes in sports' remoteness (e.g., from locally co-located to remotely distributed) and their (a)synchrony (e.g., from real-time synchronous to decoupled asynchronous) as sports become increasingly digitised (Parry & Giesbrecht, 2023). However, Parry and Giesbrecht (2023) argue that regardless of "virtuality" happening under the form of tracking timing results, motion capture and body tracking, or capturing power output, any sort of digital measurement is merely a way to mediate and facilitate competition. This can then be held remotely (distributed instead of co-located) and asynchronous (decoupling from time instead of real-time) without fundamentally altering the nature of the sport. This suggests that a visual representation is not required and definitely not the same as virtuality, as virtuality may exist without a visual representation of sport. In other words, it exists regardless of representational fidelity (i.e. the degree of detail and quality by which the performance is visually represented; see Figure 2). For instance, Parry and Giesbrecht (2023) refute to label digitally measured and mediated sports as "virtual sports" regardless of their visual fidelity, remoteness, and (a)synchrony. They argue against the labels of "virtual" in sports "since measurement-virtuality does not make them any the less real" (p. 220) and "do not fundamentally change what is required for a [sporting] competition to occur, 'virtually' or otherwise" (Parry & Giesbrecht, 2023, p. 219). To them, virtual sports should just be called "sports". Even if these newly labelled "virtual sports" change in representational fidelity (e.g. performances shown through platforms with visual 3D simulations), change in remoteness (e.g. distributed from-home competitions), or change in (a)synchrony (e.g. technology allowing athletes to compete in real-time instead of serial competitions), the physical exertion is real (i.e. not to be labelled "virtual" according to Parry and Giesbrecht) and remains the same. Ultimately, this means that physical artifacts such as measurement technology still simply enables comparison. Parry and Giesbrecht (2023) therefore argue that virtuality functions merely as a contest mediator, regardless of how representational fidelity, remoteness and (a)synchrony are implemented, or in their words, "virtuality is merely a reference to the digital measurements necessary for comparison of performances, that are represented on a screen." (p. 220).

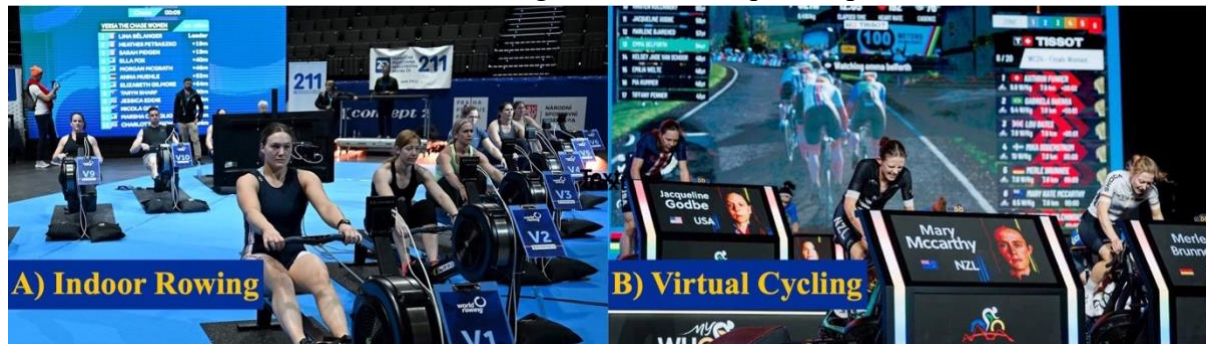
Figure 2: Overview of virtuality in relation to representation, space, time, and reality, resulting in sixteen distinct configurations of how virtuality can be implemented. Source: own conception.



To better capture this understanding of virtuality, we will illustrate with some examples how sports relying on virtuality (i.e. digital measurement of performance) can be, but not need to be, visually simulated, remote, or asynchronous. For instance, the Cycling Esports World Championship is an event of sports virtuality involving digital measurement. Yet, it is a local co-located (i.e. physically held live on a location), synchronous event where athletes compete in real-time against each other via a locally connected virtual platform visually representing their performances in a high-fidelity 3D simulated virtual world (see Figure 3 B for an illustration) (Reed et al., 2023; Tjønndal et al., 2025). Another example is indoor rowing, which has traditionally allowed for both asynchronous formats (e.g., time trials where athletes compete at different times) and synchronous formats (e.g., side-by-side races in the same venue, see Figure 3 A for an illustration). During COVID-19, these competitions shifted to remote setups where competition was held physically apart, sometimes synchronous with real-time online events, and sometimes asynchronous where virtuality merely enabled comparison over time. Organisers labelled these events “virtual” rather than “indoor”, even though the core physical activity (i.e. “indoor rowing”) remained the same. Only its remoteness (local to remote) differed and its (a)synchrony and representational fidelity remained the same (e.g. low-fidelity textual ranking overview for indoor rowing as can be seen on Figure 3A). These two examples make clear that representational fidelity, remoteness or (a)synchrony are intertwined and may leverage each other (i.e. yellow arrows in Figure 2), but are not mutually dependent. For the aspect of representational fidelity, it is important to note that virtuality can exist

independently of any medium (e.g., AR vs. VR vs screen) and fidelity (e.g., ranking table vs. 3D animations, see Figure 3). As such, virtual reality or VR (explained earlier) in sports and virtuality in sports are thus not to be confused; VR is only one way to implement the representational fidelity of virtuality (see Figure 2).

Figure 3: Illustration of two different configurations of sports virtuality within the third layer of sport digitalisation: Virtual Rowing (A) and Virtual Cycling (B). Source: World Rowing Federation and UCI Yearbook 2024. Images used with requested permission.



To conclude, virtuality in sport is not a recent phenomenon. Using digital measurement technology to record and compare performance has been done for a long time, for example, in downhill mountain biking or skiing, where the performance is translated to a (digitally) timed result. Consequently, recent advancements in representational fidelity, remoteness, or (a)synchrony do not change the sport’s nature and physicality. Therefore, it could be “question[ed] whether they should be called ‘virtual’ sports at all” (Parry & Giesbrecht, 2023, p. 220). However, the statement in which Parry and Giesbrecht (2023) question virtual sports as a distinct concept, requires nuance because they did not consider the *reality* in which sports take place (see Figure 2). This dissertation presents several arguments, elaborated upon in section 9.5 and section 16 (study 5), on how recent advancements of virtuality in sports may introduce a new sporting reality that is incomparable to the actual reality (e.g. real-world laws of nature, governing laws) and *does* change the sport’s nature and physicality, justifying the use of “virtual sports” in certain cases. For now, we conclude that the “reality” feature of virtuality in sports allows us to distinguish two types of virtuality configurations within the third layer of sport digitalisation:

- (1) **Simulations *in* sport** (Parry & Giesbrecht, 2023, p. 216), involving actual bodily participation that mimics the physicality and performances of an existing ‘parent’ sport that is measured, though remains unaltered, and gets translated to a digital environment using the actual reality of existing and known real-world laws. Here, the physical artifact

merely allows to capture and represent the physical performance, but does not interact with it.

While it is debatable whether measures can truly remain “unaltered” when being translated, we understand “unaltered” as not actively or purposefully altered. We do acknowledge some unavoidable alteration (e.g. systematic error of measurement) that is assumed to be equal for all measurements. To give an example of simulations *in* sports, we can build on our earlier example of indoor rowing and refine it to show how it can take certain configurations of (1) reality, (2) representational fidelity, (3) remoteness, and (4) (a)synchrony (cf. 16 possible combinations in Figure 2). In the case of indoor rowing as a simulation *in* sport, the (1) reality will thus be the actual reality. This means that an ergometer captures the athlete’s power output as physical performance and uses these unaltered measures to translate the performance to a digital output. Meanwhile, the (2) representational fidelity can for instance be a static textual leaderboard with power numbers. The (3) remoteness can be a co-located live event where (4) the (a)synchrony is set up so that athletes compete in real-time (e.g. Figure 3 A).

The second type of virtuality configuration within the third layer of sport digitalisation entails:

- (2) **Virtual sports** (Fouquaert, Tjønndal, & Willem, 2025; Westmattellmann et al., 2025), where performances based on bodily participation mimicking sports’ physicality are also digitally measured, though compared to simulations in sport, their translation does *only partially* use the actual reality of existing and known real-world laws. The virtual environment is thus not a mere one-on-one representation of the physical performance, but instead creates its own *distinct sporting reality* (i.e. physical-digital) that affects the nature of sport, such as virtual physics combining actual real-world laws, in-game laws, new tactics, or new physical skills that are required to master features of the game and the sport. Here, the physical artifact mediates and interacts with the performance as it both enables and bounds.

To illustrate a virtual sport, we can build on our earlier example of virtual cycling. Unlike the simulation of indoor rowing, virtual cycling can be argued to possess a distinct sporting reality. This means the (1) reality is an interreality characterised by unknown laws. While a smart trainer captures physical power output, algorithms actively alter and mediate this measure. Platforms like Zwift attempt to simulate the laws of nature (e.g. gravity, rolling resistance, and

air resistance), but they, at best, approach actual real-world laws and therefore introduce their own undisclosed ways of mediating physical performances into virtual outcomes. Building on the interreality that distinguishes virtual sports from simulations in sport, virtual cycling can take various configurations (i.e. to illustrate some of the 16 possible combinations mentioned in Figure 2). In a first example, athletes might compete in a (2) high-fidelity 3D simulated world like Zwift or MyWhoosh, during a (3) distributed remote event from their own homes, where the (4) (a)synchrony is set up as a real-time synchronous competition. Conversely, a second example could involve the exact same interreality but configured differently. It could use a (2) low-fidelity text-based interface like a textual reading on a Garmin bike computer and a textual leaderboard summary, be hosted in a (3) co-located local venue, and operate as (4) asynchronous decoupled time trials.

Collectively, our conception of the progression of sport digitalisation in layers illustrates how digitalisation increasingly shapes the ontology of sport: shifting from peripheral tools that support performances, monitoring, and decision-making to immersive technologies that fundamentally redefine the nature of the sporting reality itself. While governing bodies often attempt to preserve the ‘purity’ of sport and resist these perceived ‘unnatural’ technological intrusions (Magdalinski, 2009), their regulatory responses frequently lag behind the pace of innovation. Constraints like institutional inertia and the complexity of integrity dilemmas (Synrem, 2019; Washington & Patterson, 2011) underscore the organisational challenges inherent in managing organisational innovation. This is particularly evident in terms of the transition throughout the three layers of sport digitalisation, and in particular the most recent and third layer of sport digitalisation. In the following section, we zoom in on our theoretical approach of layer three of sport digitalisation, and in particular of virtual sports.

3 Theoretical Perspective, Utilised Frameworks, and Research Gaps

To explain how, when, why and under which conditions innovation management in sport organisations takes place, theory is needed (Cunningham, 2013). Consequently, this section takes a closer look at the theoretical approach towards virtual sports as an innovation in a sporting context, highlighting sensible perspectives and frameworks to study it.

3.1 Innovation, Schools of Thought, and Organisational Innovation

Given our focus on virtual sports as innovation management in and by sport organisations, it is sensible to zoom out and ask ourselves: (1) what innovation is, and (2) how it can be studied?

First, having a definition of innovation to start from seems indispensable, as it may above all help to understand the perspectives and best practices by which research on innovation can be approached. However, a single definition of innovation is hard to provide. Godin (2008) historically traced the idea of innovation to its modern understanding, defining it as a phenomenon representing “novelty” that includes, depending on the scientific discipline, related terms such as discovery, creativity, change, imitation or invention. A more detailed definition of innovation is offered by Crossan and Apaydin (2010) in their widely cited systematic literature review: “*Innovation is: production or adoption, assimilation, and exploitation of a value-added novelty in economic and social spheres; renewal and enlargement of products, services, and markets; development of new methods of production; and establishment of new management systems*” (p. 1155). The latter definition offers a more refined understanding of innovation compared to Godin’s (2008) conception. Second, to study innovation, there is no single unified innovation theory because innovation is a multidisciplinary field (e.g. economics, sociology, management, engineering) where each discipline applies its own lens. Even then, within each discipline, the perspective on innovation and theoretical frameworks used may vary greatly, and are not tied to a specific discipline. Hence, innovation could best be approached as distinct “schools of thought” (Baregheh et al., 2009; Damanpour, 2020, Chapter 1). The most recognised schools of thought include: the economic school (Hagedoorn, 1996; Schumpeter, 1983), the sociological school (Rogers, 2003), the system school (Lundvall, 1992), the strategic management school (Chesbrough, 2003, 2010; Christensen, 1997; Christensen et al., 2018), and the user-centric school (Von Hippel, 1988, 2006).

Innovation studies can be said to subscribe to these schools of thought. In other words, different *perspectives* (i.e. schools) are possible when studying the innovation process, its antecedents, and its outcomes within and beyond organisations. Moreover, when an innovation study positions itself in relation to these perspectives, it may rely on one or multiple theories to study a specific aspect of innovation (e.g. legitimisation, adoption) regardless of its innovation perspective. Hence, the same theory can be applied from different perspectives. While theories might lean towards one perspective, they do not, however, exclusively subscribe to one perspective or school of thought. To give an example, the legitimisation of an innovation may be studied using Disruptive Innovation Theory, which could infer implications for policy and society (e.g. sociological school) as well as internal organisational decisions (e.g. strategic management school). Alongside theory, different topics of interest, aims, and research

questions can be approached from any innovation perspective or school of thought on innovation (Damanpour, 2020, Chapter 1).

Alongside innovation, *organisational innovation* can be considered a subfield of innovation studies that refers to a variety of innovations in different kinds of organisations, for instance technological, administrative, or service innovations in for-profit or non-profit organisations (Damanpour, 2020, Chapter 1). More specifically, organisational innovation concerns the study of internal and external conditions that result in organisations engaging in innovation adoption, as well as the study of the impact such adoption has on organisational performance or adjacent systems like industries, policy, or society (Damanpour, 2020, Chapter 1). This thesis predominantly employs organisational innovation, as our empirical studies will discuss different aspects of virtual sports adoption within the context of sport organisations, the sport industry, and sport policy.

With respect to the field of organisational innovation, Crossan and Apaydin's (2010) definition of innovation teaches us that innovation in organisations can be studied as both an outcome and a process, respectively denoted in the literature as studies of *innovativeness* and *innovating* (Baregheh et al., 2009; Van De Ven & Rogers, 1988; Wolfe, 1994). Studies on *innovativeness* aim to identify the determinants and context influencing several innovation outcomes (e.g., adoption attitudes, adoption rate) that are proxies for the innovativeness of an organisation (Damanpour, 2020, Chapter 1; Wolfe, 1994). Consequently, these studies are often cross-sectional and quantitative in nature and aim to describe the variance in measures of innovativeness based on one or more sets of determinants. Studies on *innovating*, on the contrary, aim to analyse the innovation process of generating, adopting, and implementing innovations, and therefore are often qualitative in nature, such as case studies (Damanpour, 2020, Chapter 1; Van de Ven et al., 2000; Wolfe, 1994).

This makes it clear that many different approaches to study innovation and organisational innovation can be undertaken. While it is not in the scope of this thesis to discuss each innovation school, we will however focus our attention on the innovation schools used in the studies within this dissertation, namely, the sociological and strategic management schools.

3.1.1 The Sociological School of Thought

The sociological school of thought offers a distinct perspective on innovation, rooting it in social interaction and imitation rather than pure invention (i.e. the more economic view by Schumpeter). This tradition can be traced back to Gabriel De Tarde's laws of imitation (1903), which state that while invention drives change, society progresses primarily through the

socially embedded phenomenon of imitation. Tarde (1903) argued that innovation acts as an instrument of social progress, emphasising social change over the technological or economic focus found in other schools of thought. This view was later developed in the literature under the field and theory of diffusion of innovation by Rogers (2003), who presented innovation as the dissemination of new products and practices in a social system according to the rate of adoption by members of that system. According to Rogers' theory, adoption is largely facilitated by perceived innovation attributes that determine the speed and extent to which something new is accepted. Applied to virtual sports diffusion, this may explain the socialised processes of virtual sports adoption and implementation, such as how athletes and managers perceive virtual disciplines as advantageous, compatible, legitimate, and so forth.

From an organisational innovation perspective, diffusion of innovation theory distinguishes more complex organisational processes consisting of multiple phases (e.g. agenda-setting to routinising) where the structure of the organisation and socialised processes like innovation champions, more so than organisational resources, are critical for an innovation to be successfully adopted (Rogers, 2003, Chapter 10). This sociological perspective is closely linked to the adaptation-progression approach to organisational innovation (Damanpour, 2020, Chapter 1). Within this approach, the organisation is viewed as a social entity that innovates to adapt to its external environment, ensuring its continued operation and progress. Innovation, therefore, becomes a mechanism for coping with environmental changes and fulfilling societal needs, rather than solely a tool for technological invention and competitive advantage (Abrahamson, 1996; Damanpour & Schneider, 2006; Van de Ven et al., 2000; Van De Ven & Rogers, 1988). The adaptation rather than invention perspective reflects virtual sports adoption within sport organisations, because clubs or governing bodies do not invent the underlying software or hardware but rather adapt to these environmental shifts by looking for ways to integrate them into their existing services.

3.1.2 The Strategic Management School of Thought

In contrast, the strategic management school, which is more linked with the Schumpeterian competition-wealth approach, views the organisation primarily as an economic entity striving for competitive advantage and wealth creation through the generation of new products and processes (Damanpour, 2020, Chapter 1). Within this school, one theory that is particularly relevant for this dissertation is the theory of disruptive innovation. This theory elucidates how established and leading innovations can be displaced by simpler rather than technically superior innovations, while targeting new markets that are often ignored by incumbents who favour

sustaining innovations (Christensen, 1997; Christensen et al., 2018). Other widely adopted theories within this school, such as open innovation theory, transcend the limitations of internal resources and state that firms must abandon closed systems and instead leverage external ideas to increase their competitive position (Chesbrough, 2003). The critical importance of business model innovation is also highlighted in this school, as new business models are required to overcome “dominant logics” or internal cognitive rigidities that resist deviating from established (but potentially superseded) value creation logics (Chesbrough, 2010). The strategic perspective also applies to virtual sports diffusion, as virtual disciplines can be considered as a new and “simpler” innovation compared to traditional sports (e.g. not as advanced structures and rules, ways to compete, or specialised equipment needed to compete at the highest level). Consequently, it may attract and target new markets (e.g. non-cyclists) or compete with existing ones (e.g. road cyclists), but also requires organisations to strategically invest in resources internally or seek them externally (e.g. publishers), or experiment with new business models (e.g. online mass participation events).

3.1.3 An Institutional–Strategic perspective

Taken together, the sociological and strategic management schools provide a complementary foundation for this dissertation. Rather than treating these schools as mutually exclusive, the studies conducted in this thesis operate at their intersection. To capture what it means for virtual sports to be discovered, evaluated, adopted, and implemented by sport organisations, we approach innovation as a dual process of institutional and strategic work (Oliver, 1997). This means shifting the analytical focus away from innovation as a technological object, and instead treating it as a socially constructed process where organisations must secure social legitimacy (i.e. the sociological aspect) and competitive advantage (i.e. the strategic aspect) (Crossan & Apaydin, 2010; Oliver, 1997).

This synthesis aligns with the commentary of scholars like Oliver (1997), who argues that sustainable organisational success is rarely achieved if an organisation only pursues economic or performance-based rationality. Instead, an organisation’s actions must also align with the cultural norms, values, and expectations of its broader environment. Applied to this dissertation, this means that sport organisations make adoption decisions not just to rationally optimise performance, but to conform to stakeholders’ expectations of digitalisation, respond to peer pressure from competing federations, or seek societal approval. Sport organisations must therefore navigate the strategic acquisition of new technology while simultaneously managing the institutional context of these decisions. They are required to actively interpret,

evaluate, imitate, and negotiate innovations like virtual sports until they become fully accepted, culturally legitimate, and routinised within the traditional sporting landscape.

Innovations like virtual sports are consequently not primarily treated as an objectively *better* technology, but as a social construction in which organisations and stakeholders interpret, evaluate, imitate, and negotiate new practices until they become accepted, legitimate, and routinised (Rogers, 2003; Wolfe, 1994). This implies that the “newness” and “value” of an innovation, as referred to in the definition of innovation by Crossan and Apaydin (2010) (cf. section 3.1), are not inherent properties but perceptions shaped by the social interactions of the adopter. In this way, organisational innovation is shaped by how decision-makers perceive innovation attributes, how organisations mobilise support (e.g., innovation champions), and how innovations align or clash with prevailing values and institutionalised norms (Van de Ven et al., 2000). The adoption of an innovation like virtual sports throughout this thesis is therefore regarded as not solely driven by its technological features, but equally (if not more) by its social framing internal and external to the organisation (Oliver, 1997). Similar to other scholars like Oliver (1997), we refer to this as the use of an *institutional–strategic perspective* throughout this dissertation.

This institutional–strategic perspective reflects the theoretical coherence throughout the five empirical studies in this thesis (Part IV), all of which prioritise the social and institutional dimensions of adoption over purely technical ones. In what follows, we highlight clubs’ perceptions, beliefs, and capacities as antecedents of virtual cycling adoption (i.e. studies on *innovativeness*), and examine how virtual cycling’s diffusion and institutionalisation depend on the construction of meaning, legitimacy, and good governance practices across clubs, federations, and platform actors (i.e. studies on *innovating*). In other words, the case of virtual cycling is studied here as a technological innovation whose organisational relevance and societal value are ultimately determined by social processes of acceptance, contestation, and institutional embedding. Meanwhile, the strategic management perspectives (e.g., disruption or organisational capacities) remain important for explaining how and why these actors engage with the innovation of virtual cycling (Christensen, 1997). The utilisation of an institutional–strategic perspective thus offers a coherent approach for this thesis as it links (1) perceptions and attitudes of individual decision-makers at the micro-level, with (2) organisational beliefs and capacity differences at the organisational meso-level, and with (3) broader dynamics of legitimacy, governance, and societal norms at the macro-level (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002).

At this point, it is crucial to clearly distinguish between the institutional-strategic approach as a higher-level paradigm and the lower-level theories used to execute it in our empirical work. Our institutional-strategic perspective, in which innovation is presented as a dual process of seeking social legitimacy and competitive advantage, acts merely as a philosophical boundary for this dissertation, rather than a specific applicable framework. However, alongside this umbrella perspective, specific testable theories are required. Consequently, institutional theory (DiMaggio, 1988; DiMaggio & Powell, 1983) is utilised as the primary analytical anchor to explain sport organisations' normative rationality when evaluating, adopting, and implementing virtual sports. To operationalise the more strategic counterpart of our perspective, our primary anchor of institutional theory is complemented by additional theories used in our empirical work. Diffusion of innovations theory (Rogers, 2003) is used to map how these socially constructed innovations are adopted over time within voluntary sports clubs. Disruptive innovation theory (Christensen, 1997) is applied to understand strategic market shifts and strategies used for the successful diffusion of virtual sports. Finally, organisational capacity frameworks (Balduck et al., 2015; Doherty & Cuskelly, 2019; Hall et al., 2003) are applied to analyse clubs' capacities in managing future virtual sport integration, while governance frameworks (Burger & Goslin, 2005; Kelly et al., 2022) are applied to analyse how federations strategically regulate new digital-led disciplines. These specific theories and frameworks, explained in sections 3.2 to 3.5, are our transition from a broad philosophy of innovation management to testable theories and frameworks.

Before delving into the aforementioned theories, we will shortly present how the institutional-strategic perspective on innovation translates to our empirical studies presented in Part IV. For a single overview of which theory is used in which study, and which research gap it addresses, we refer to Table 1 and Table 2. Study 1 demonstrates the application of combining diffusion of innovations theory with organisational capacity frameworks. By relying on the premises of these theories, it reveals that perceived innovation attributes and managerial attitudes are stronger predictors of virtual cycling adoption in voluntary sports clubs than objective capacities like financial resources. Furthermore, institutional theory elucidates the findings of study 2, where adoption is framed as a negotiation of competing institutional logics. This negotiation results in distinct club archetypes (e.g., holistic clubs vs. performance-focussed clubs) based on how virtual sports align with their beliefs about engaging and inclusive sport participation versus competitive-oriented sport participation. Study 3 integrates disruptive innovation theory with institutional theory. It reveals that the diffusion of a disruptive novelty relies on institutional rather than technical legitimisation strategies, which means that

sport governing bodies rely on a normative narrative about virtual cycling to enhance its social acceptance rather than developing technical features. Study 4 implements the institutional-strategic perspective by applying governance frameworks to examine the issues and tensions between traditional sport regulation and virtual sport regulation. It identifies how stakeholders socially construct potential issues of integrity, such as the lack of doping control and algorithmic transparency, within the newly formalised discipline of virtual cycling. Finally, study 5 deals with the technical specificities of algorithms in virtual sports viewed through a governance lens and links this to institutional implications. It highlights that regardless of technical implementation, the main challenge is found in the governance gaps resulting from these implementations, such as differences in platform algorithms shaping selection procedures or competitions in an unfair way. In this way, the studies in this dissertation all clearly posit that organisational innovation in sport is, next to technological development, fundamentally a human endeavour of adapting to, legitimising, and governing a changing reality.

3.2 *Institutional Theory in Sport*

This section outlines institutional theory as the primary anchor of the theories that underpin our work. To understand how sport organisations navigate an increasingly digitised landscape, the following subsections systematically unpack key concepts in institutional theory relevant for our work. Section 3.2.1 explores how organisations aim to sustain themselves through institutional legitimacy and isomorphism. Subsequently, section 3.2.2 examines how competing institutional logics, resulting from isomorphism, create conflicting tensions during the adoption of innovations. Finally, section 3.2.3 discusses the processes of deinstitutionalisation and reinstitutionalisation when new practices such as virtual sports begin to replace traditional sporting norms.

3.2.1 *Institutional Legitimacy, Isomorphic Pressures and Decoupling*

Institutional theory has served as a dominant framework in organisational research for decades. The work by Meyer and Rowan (1977) and DiMaggio and Powell (1983) fundamentally shifted the analytical focus away from viewing organisations as purely rational, efficiency-driven actors. Instead, institutional theory posits that organisations are deeply embedded within a complex social context. Institutions, which are not necessarily organisations or physical structures but equally taken-for-granted social norms and ideologies (Washington & Patterson, 2011), establish the “rules of the game” consisting of formal regulations, cultural norms, and shared cognitive beliefs (North, 1990). At its core, institutional theory answers why

organisations take actions that ultimately defy rational explanations (Boxenbaum & Jonsson, 2008).

Within this theoretical perspective, a primary driver of organisational behaviour is the pursuit of institutional legitimacy. Suchman (1995) defines legitimacy as a publicly shared perception that an entity's actions are desirable, proper, and appropriate within a socially constructed system of norms. To withstand external pressures, secure critical resources, and effectively compete with others, sport organisations must align their structures and practices with these external societal expectations (Suchman, 1995).

The continuous pursuit of legitimacy or social acceptance inevitably triggers a process known as institutional isomorphism. Isomorphism explains how organisations that operate within the same institutional environment gradually become structurally and behaviourally similar over time (Boxenbaum & Jonsson, 2008). In their seminal work, DiMaggio and Powell (1983) identify three distinct isomorphic pressures that drive this gradual homogeneity in structures and behaviours. First, coercive pressures arise from external power structures, such as governmental regulations or strict funding requirements imposed on sport federations. Second, mimetic pressures occur during periods of uncertainty, prompting organisations to imitate peers they perceive as highly successful. Finally, normative pressures stem from professionalisation, where increasingly developed backgrounds and professional networks create shared ideas across a specific sector. Combined, these pressures help explain “whether and to what extent organizations adopt similar structures and forms because they adapt to what they believe society expects of them” (Greenwood et al., 2017, p. 7).

In the sport sector, these isomorphic pressures are highly prevalent. The drive for legitimacy often leads sport federations and local clubs to adopt highly similar, formalised organisational forms (Washington & Patterson, 2011), making it difficult for new practices such as virtual sports to become socially accepted. While this drive for conformity ensures stability and predictability, it simultaneously creates a highly rigid environment. Sport organisations are often reluctant to deviate from established norms because doing so risks their institutional legitimacy. Consequently, such isomorphic pressures can significantly complicate and stifle innovation adoption. This means that successful adoption of an innovation requires strategies that cope with these institutional dynamics.

Sport organisations must thus engage in active, strategic legitimisation when introducing innovations. Sport management research indicates that the successful adoption of innovations relies heavily on social framing to gain social acceptance rather than on technical features (Stenling, 2014). For instance, Stenling’s (2014) study showed how non-traditional

drive-in sports in Sweden gained institutional acceptance through problem-solving narratives that strongly resonated with established sport-for-all principles. In a different study, Pizzo et al. (2022) demonstrated that traditional sport organisations successfully integrated esports by framing them as complementary activities that aligned with norms and beliefs shared among their stakeholders. Similarly, the legitimacy of novel sporting formats like “The Hundred” in cricket was secured by anchoring this new and innovative format of cricket to broader normative societal values, such as gender equity and modernised broadcasting (Read, 2023). In study 3, we zoom in on the social framing of virtual cycling as a strategy to increase its social acceptance. Conversely, innovations face the risk of rejection if their technical realities conflict with institutional norms and “rules of the game”. For instance, literature has pointed to several issues in virtual sports and esports that threaten their institutional trust and legitimacy, such as algorithmic exploitation (e.g. abusing software design to gain unintended advantages) , digital weight manipulation, and vulnerabilities around data privacy (Dyer & Richardson, 2026; Richardson, 2025; Westmattelmann et al., 2025).

As governing bodies strive to implement new practices and innovations using strategic framing to increase their legitimacy, a phenomenon known as institutional decoupling can occur (Boxenbaum & Jonsson, 2008), which is not to be confused with spatial and temporal decoupling (cf. Section 2.3). In the case of technological innovations, sport governing bodies aim to align their internal technical operations required to implement the innovation with institutional norms and expectations. However, this is often a challenge because they lack the necessary structures and organisational capacity to manage these technicalities (Boxenbaum & Jonsson, 2008; Washington & Patterson, 2011). This creates a decoupling between governing bodies’ narrative about an innovation (e.g. aimed to increase inclusivity) and their actual implementation of it (e.g., having features that enable exclusion). In studies 3 and 5, we examine how a decoupling currently exists between governing bodies’ beliefs and narratives around virtual cycling (e.g. using it to enter new markets of women or youth, or using it for professional racing), and the actual implementation of it (e.g. almost 90% of virtual cyclists are male, or the lack of technical knowledge and systems to guarantee fair professional racing).

This dynamic reveals a significant gap in the current sport management literature regarding technological innovation. Existing research in sport management still lacks a general understanding of how radically new technological innovations, like unproven virtual sporting disciplines, lacking any form of legitimacy, require specific legitimisation strategies to overcome the challenges of isomorphic pressure and institutional decoupling when sport governing bodies engage with such innovations.

3.2.2 *Competing Institutional Logics*

The concept of institutional logics provides a framework to understand how different norms and values coexist but also compete. Thornton and Ocasio (1999, p. 804) define institutional logics as “the socially constructed, historical pattern of material practices, assumptions, values, beliefs, and rules by which individuals produce and reproduce their material subsistence, organize time and space, and provide meaning to their social reality”. Three logics have been identified in literature: (1) the sport-for-all logic promoting democratic participation, (2) the result-oriented logic promoting competitive sports, and (3) the commercialisation and professionalisation logic promoting revenue-generating activities (Skille, 2011; Stenling & Fahlén, 2009; Szerovay, 2022). Therefore, despite the notion of isomorphism, different “dominant” ways for sport implementors to arrange their activities still exist (Bettis & Prahalad, 1995; Thornton et al., 2012).

Over the past two decades, sport policy has focussed heavily on social goods like health and broad inclusion, for which sport organisations (e.g. clubs, federations) are considered the implementors of such policy (De Bock et al., 2022; De Bock, Scheerder, Theeboom, De Clerck, et al., 2023; Skille & Stenling, 2018). However, in doing so, they find themselves involved in a struggle between conflicting logics they seek to implement due to the abovementioned coercive pressures or normative pressures. This fundamental tension between conflicting institutional logics (e.g., the sport-for-all logic versus result-oriented logic) has been broadly recognised in the literature (De Bock et al., 2022; Stenling, 2014). Furthermore, research moreover consistently shows that the result-oriented logic often remains central to federation and club activities, whereas sport-for-all goals receive less emphasis and remain difficult to fulfil (De Bock, Scheerder, Theeboom, De Clerck, et al., 2023; May et al., 2013; Skille, 2010). Furthermore, an overemphasis on the result-oriented logic is argued to sustain social exclusion and hinder social inclusion in and through sport (Spaaij et al., 2014).

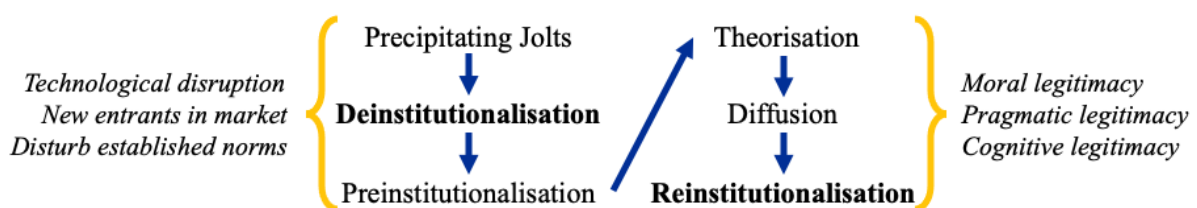
When an innovation such as a virtual sporting discipline is introduced, sport organisations consequently do not evaluate it in a vacuum, but from their dominant logic. Because platforms like virtual cycling can simultaneously be used for result-oriented goals (e.g. structured training, competition) and sport-for-all goals (e.g. gamifying sports, participation of disabled athletes) (Devine et al., 2023; Evsikov et al., 2024; Tjønndal, 2022; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021), they are situated at the intersection of these conflicting logics. However, a notable gap remains in the current sport management literature for such instances. While existing research analysed the historical tension between practices

that drive either a result-oriented or sport-for-all logic, there is limited empirical insight into how innovations that may serve both simultaneously are perceived, approached, and implemented. In addition, it remains unclear how sport organisations' beliefs and convictions about such innovations result in distinct pathways of innovation adoption, and how these align with dominant institutional logics in sport policy. In study 2, we discuss how the beliefs around virtual sports like virtual cycling can be situated within the conflicting institutional logics of sport-for-all versus the result-oriented logic, and how this affects clubs' willingness to experiment with virtual disciplines as-a-club-service.

3.2.3 *Deinstitutionalisation and Reinstitutionalisation*

Institutional change is a complex, multi-stage process where organisations adapt to new practices to maintain their legitimacy and comply with institutional pressures and logics. Greenwood et al. (2002) outline a comprehensive six-stage model to understand this change process, as illustrated in Figure 4. The first three stages on the left describe the breakdown of the status quo. Disruptive events, such as technological or regulatory changes, introduce new entrants into the market, which actively disturb established practices and norms (DiMaggio, 1988; Oliver, 1992). This triggers the gradual erosion of existing, socially constructed consensus and introduces the possibility for change (Greenwood et al., 2002). These initial three phases put pressure on established actors, who cope with these pressures by searching for new alternatives to the traditional ways of operating (Greenwood et al., 2002).

Figure 4: Stages of institutional change, with key changes summarised for stages on the left and stages on the right. Source: own illustration, adapted from (Greenwood et al., 2002).



For these new alternatives by established actors to become formalised and legitimised, they must progress through stages four (Theorisation) to six (Reinstitutionalisation), shown in Figure 4. Greenwood et al. (2002) argue that this progression depends on acquiring different forms of legitimacy, as introduced by Suchman (1995). Innovations must secure pragmatic legitimacy by demonstrating tangible, functional value to immediate stakeholders. Subsequently, they require moral legitimacy, which is achieved by aligning the new practices with broader societal values and normative expectations. Finally, the innovation must attain cognitive legitimacy,

which occurs when new practice becomes comprehensible, widely accepted, and taken for granted as the natural way of doing things (Suchman, 1995). In study 3, we discuss how each form of legitimacy helps to legitimise virtual cycling as an official sporting discipline of the UCI.

While Greenwood et al.'s (2002) entire six-stage model provides a comprehensive theoretical overview of institutional change, this dissertation will focus primarily on the two anchoring processes: deinstitutionalisation (stage 2) and reinstitutionalisation (stage 6). Focusing on these two specific stages is theoretically and empirically justified because they represent important inflection points in the process of institutional change, which is argued to occur in the two possible directions of deinstitutionalisation or reinstitutionalisation (Borgers et al., 2018; Gilmore & Sillince, 2014; Oliver, 1992). Moreover, Greenwood et al. (2002) argued that “institutional theorists have given very little attention to conceptualizing and specifying the processes that move innovations beyond preinstitutionalization toward full institutionalization” (p. 2002). Hence, our focus is on deinstitutionalisation and reinstitutionalisation.

Deinstitutionalisation is defined by Oliver (1992) as “the delegitimation of an established organizational practice or procedure as a result of organizational challenges to or the failure of organizations to reproduce previously legitimated or taken-for-granted organizational actions” (p. 564). Conversely, reinstitutionalisation represents the reinterpretation of institutional logics creating new normative structures and processes (Dacin et al., 2002), which does not necessarily replace existing structures but can also involve strategic adaptation to a changing institutional context (De Bock, Scheerder, Theeboom, Schyvinck, et al., 2023; Greenwood et al., 2002).

Within sport management literature, Borgers et al. (2018) effectively applied the concepts of deinstitutionalisation and reinstitutionalisation to analyse structural shifts in leisure-time sports participation. They conceptualised how the historical dominance of traditional organised sports (e.g. in clubs) underwent a clear deinstitutionalisation process. This erosion made way for the reinstitutionalisation of highly flexible, commercialised, and unorganised “light sports” (Borgers et al., 2018). In our general discussion, we discuss how this theoretical application mirrors the current evolution of cycling, as we notice a similar societal shift towards flexible participation in cycling through virtual cycling, both in a leisure and competitive context (Berger et al., 2025; Botwina et al., 2024; Devine et al., 2023; Rojas-Valverde et al., 2022; Westmattelmann, Grotenhermen, Sprenger, & Schewe, 2021). The rise of participation in cycling through virtual platforms forces a re-evaluation of what constitutes the cycling. This

potentially deinstitutionalises cycling and its governing bodies, as cyclists become attracted to new players in the market that offer alternatives for both leisure and competitive participation (e.g. Zwift, MyWhoosh). Therefore, deinstitutionalisation and reinstitutionalisation provide a relevant framework to explore the disturbance of established outdoor cycling and the subsequent legitimisation of virtual cycling as an official discipline by governing bodies, which is further reflected upon in our general discussion section 19.3.

3.3 *Innovation Determinants*

While institutional theory explains the societal pressures that shape sport organisations, additional theories are needed to understand the specific organisation-related mechanics of technology adoption. This section explores the field of innovation diffusion theory (i.e. the rate and patterns of innovativeness) and organisational innovativeness (i.e. what determines the innovativeness of an organisation) in order to comprehend the determinants of organisational innovation adoption.

3.3.1 *Innovation Diffusion Theory for Organisations*

Innovation diffusion theory provides a complementary lens to understand how new technological practices spread through a population. Rogers (2003) posits that the adoption of an innovation is heavily dictated by its *perceived attributes* rather than its objective technical features or primary attributes (e.g., size, cost). Key attributes repeatedly identified in and beyond sport management literature include relative advantage, compatibility, image, perceived ease of use, trialability, visibility, result demonstrability, and perceived enjoyment, to name but a few of the most frequently studied attributes (Corthouts et al., 2020; Karahanna et al., 1999; Moore & Benbasat, 1991; Rogers, 2003; Venkatesh, 2000; Venkatesh & Brown, 2001; Wolfe, 1994). These attributes are all subjective assessments based on personal characteristics as well as the organisational culture and environment one operates in, and they determine and explain to a great extent (i.e. up to 80% of explained variance) the likelihood of an organisation adopting an innovation (Frambach & Schillewaert, 2002; Wolfe, 1994).

Crossan and Apaydin (2010) advocate that these determinants, which are thus perceptions about an innovation living *in* the organisation, are essential for understanding organisational innovation adoption. A single technological tool can be perceived entirely differently by various organisations depending on their internal norms and subjective beliefs, or the aforementioned “dominant logics” (Wolfe, 1994). However, despite perceived innovation attributes being a powerful predictor of innovation adoption and frequently studied

in user-centric studies in a variety of fields (de Vries et al., 2018), a clear research gap in sport management literature remains. Perceived innovation attributes have only been sparsely studied in the sport management literature (Corthouts et al., 2020, 2022; Park et al., 2024; Williams et al., 2021), regardless of their acknowledged value, predictive power and advocated use in integrated innovation models combining “innovation-in-an-organisation”-specific determinants with other types of determinants (e.g. structural organisational characteristics) (Frambach & Schillewaert, 2002; Wolfe, 1994, p. 409). In study 1, we examine the predictive power of perceived innovation attributes compared to other innovation determinants regarding clubs’ willingness to innovate with virtual cycling as-a-club-service, effectively applying an integrated innovation model that considers virtual cycling as an “innovation-in-an-organisation” where individual perceptions live *in* the structural context of an organisation (i.e. cycling or triathlon club).

3.3.2 Organisational Innovativeness and Organisational Capacity

As we just discussed “integrated innovation models”, it should be clear that alongside perceived attributes, an organisation's capacities and its volunteers’ abilities to implement an innovation also determine the success rate of organisational innovation adoption. Based on organisational innovativeness literature (Damanpour, 1991; Wolfe, 1994), innovation determinants in the context of sport have been identified as managerial, organisational capacity, and environmental determinants (Corthouts et al., 2020; Hoeber & Hoeber, 2012; Winand et al., 2013). Managerial determinants cover an organisation’s characteristics in terms of board attitude towards innovation, knowledge management, and innovation-supportive volunteers and actions. Organisational capacity is defined by Hall et al. (2003) as the ability of a non-profit organisation to deploy various types of capital to achieve its goals, and has since been well developed within sport management literature (Balduck et al., 2015; Casey et al., 2012; Doherty & Cuskelly, 2019; Ehnold et al., 2020; Millar & Doherty, 2021). It can be generally categorised into human-related and non-human-related dimensions. Human-related capacity encompasses the number of volunteers, coaches, board members, but also their competencies and relations with external stakeholders. Non-human-related capacity encompasses the financial, infrastructural, and formalised planning initiatives and structures (Doherty & Cuskelly, 2019). Lastly, environmental determinants are external factors that impact organisations’ innovativeness, such as engagement with external stakeholders or (perceived) competition in the market (Hoeber & Hoeber, 2012; Winand et al., 2013).

While innovation determinants have proven their relevance to study innovation adoption in sport organisations, a significant gap in sport management remains related to the fact that determinants of technological service innovations are often far less resource-intensive, for instance, offering Zwift group rides or competitions as-a-club-service while relying on members' personal Zwift setup and subscription. In such cases, the financial investment would be none for the club when relying on individual members' personal investment (see Section 1). Another relatively inexpensive option to subsidise the equipment would be renting (hour- or session-based) places that offer a room with the necessary equipment for the club's members to participate in virtual cycling. For such innovations, traditional financial or human-related capacity constraints may not hinder their adoption as much as other administrative, service or technological innovations (Hoeber & Hoeber, 2012; Winand et al., 2013). Consequently, organisational innovation determinants could benefit from a reconsideration in that regard. As indicated above in section 3.3.1, study 1 examines various dimensions of organisational capacity (i.e. managerial, human-related, non-human-related, environmental) as determinants of clubs' willingness to innovate with virtual cycling as-a-club-service, alongside the innovation determinant of perceptions around virtual cycling living *in* the club.

3.4 *Disruptive Innovation Theory*

To complement our institutional perspective, disruptive innovation theory provides a framework for understanding how novel technological innovations penetrate established markets, for instance, virtual cycling entering the cycling market. Disruptive innovations are defined as products or services that initially underperform on mainstream performance criteria but introduce new functionalities to niche or non-consumers, such as enhanced accessibility, convenience, or affordability (Christensen, 1997; Christensen et al., 2015). When we compare the “performance” of a newly introduced innovation, we rely on performance evaluation criteria that are suitable to assess mainstream products or service (e.g. accuracy or ease-of-use). However, disruptive innovations introduce new functionalities that mainstream performance evaluation criteria cannot adequately assess (Christensen & Bower, 1996), which means that new evaluation criteria must also be developed. Innovations are thus compared based on both newly developed technical features and newly developed evaluation criteria. As a way to increase the social acceptance and legitimacy of new innovations apart from their technical features, new evaluation criteria can leverage their coherence with institutional and societal norms (Das & Van de Ven, 2000). For instance, study 3 applied this lens to highlight how

virtual cycling gains acceptance by developing new evaluation criteria like accessibility (indoor cycling regardless of time or weather), remoteness (no travel needed), or safe sport participation (avoiding crashes). If virtual cycling were to be assessed against cycling's mainstream evaluation criteria, it would perform poorly. This is because it would be evaluated against criteria assessing established features like platform and calendar standardisation that virtual cycling currently lacks. Meanwhile new functionalities like accessibility or remoteness would remain unassessed despite their newly added relevance. An example outside of sports can be found in disruptive products like cardboard coffins, which gained social acceptance despite being a technically inferior product by shifting market evaluation criteria away from traditional material quality and towards ecological impact and price (Ben-Slimane et al., 2020).

Furthermore, we want to highlight how disruptive innovation theory strongly complements institutional theory. While disruptive innovation theory explains the market dynamics of *why* technically inferior innovations can capture new or existing markets (i.e. through assessing evaluation criteria), institutional theory helps us to understand *how* organisations can develop diffusion strategies to build new evaluation criteria (i.e. through leveraging the social mechanisms of institutional dynamics). In study 3, we apply Das and Van de Ven's strategy model (2000) to highlight how governing bodies must balance (1) a "technical strategy" to improve existing technical features for mainstream evaluation criteria (e.g. calendars, qualification pathways), and (2) an "institutional strategy" to legitimise new functionalities (e.g. remote racing) by socially constructing the perceived value of new evaluation criteria.

This perspective has, however, to our knowledge, not yet been applied to sport innovation, leaving a research gap in the sport management literature. Current literature lacks empirical insight into how traditional sport governing bodies strategically navigate the introduction of new technological products and services, and whether they rely on a "technical strategy", an "institutional strategy", or both, in their efforts to increase the social acceptance of such innovations.

3.5 Governance of Sports and Esports

As sport organisations adopt digital innovations, they must establish new regulatory structures to maintain their institutional legitimacy. This section explores the complex governance within an increasingly digitalised sporting context. First, we examine how traditional good governance frameworks are challenged by the unique complexities of proprietary platforms and new

commercial stakeholders. Second, we delve into the specific regulatory challenges of algorithmic transparency, highlighting the difficulties of governing physical-digital realities (as explained in section 9.5.3).

3.5.1 Good Governance Frameworks

Complementary to institutional theory and our institutional-strategic perspective, governance frameworks can be used to explain the formal mechanisms used to enforce norms and maintain societal legitimacy. While institutional theory highlights the external pressure to conform to expectations, governance frameworks complementarily provide the required dimensions and tools needed to enact these norms. In traditional sports, good governance relies heavily on established principles including: accountability (liability for actions and outcomes), responsibility (fulfilling assigned duties and obligations), transparency (ensuring open and clear communication), social responsibility (acting in the broader public interest), independence (maintaining freedom from undue influence), fairness (providing impartial and equitable treatment), and discipline (adhering strictly to established rules and standards) (Thompson et al., 2023).

However, as sports increasingly digitalise (cf. layers of sport digitalisation, section 2), traditional governance frameworks face unprecedented challenges. The literature on esports highlights several governance issues that complicate traditional institutional oversight. These challenges include fragmented regulatory bodies, toxic online behaviours, and complex potential doping exploits such as body weight manipulation or mechanical doping (Dyer & Taylor, 2023; Tjønndal, 2023; Tweedie et al., 2023). In study 4 and 5, we shed light on the good governance principles in virtual cycling and the issues pertaining to these principles. We explain how virtual sports ultimately require a complex combination of traditional governance and esports-specific governance (Kelly et al., 2022). For instance, traditional sport governing bodies are suddenly faced with commercial platform monopolies and game publishers, potentially hindering governance due to commercial or legal conflicts of interest (e.g. intellectual property over the game). To date, however, sport management literature lacks an understanding of the issues in virtual sports, and is consequently unable to develop well-integrated sports and esports governance frameworks that are required to assess the effectiveness of regulation for virtual sporting disciplines such as virtual cycling, pointing to a critical research gap.

3.5.2 Algorithmic Transparency

A critical and distinct component of esports and virtual sports governance is algorithmic

transparency, which refers to the fact that the algorithms through which the sporting competitions take place are largely an undisclosed black box (Westmattmann et al., 2022). In traditional sports, the laws of nature are well known and allow for effective governance of the competitive environment. In virtual sports and esports, proprietary algorithms replace these laws of nature (see section 9.4.3 and section 9.5.3).

The translation of human physical exertion into digital sporting performance is entirely mediated by software, opening up novel governance and integrity issues that challenge traditional governance structures (Westmattmann et al., 2025). For instance, in study 5 we examine the differences in such translation between two platforms, and discuss the (unintended) impact of these differences when organising competitions across different platforms or when unexpectedly switching platforms to host competitions (as was the case for the 2024 UCI Esports World Championships, cf. study 5). In this way, athletes are affected by commercial choices (i.e. platform bidding process, cf. section 9.5.4). In addition, not all athletes have equal access to these platforms as they are financially restricted or legally restricted due to sponsorship agreements (Berger et al., 2025; Devine et al., 2024).

The new digital-physical reality with which sport governing bodies are faced reveals a considerable gap in the current sport management literature. While existing research thoroughly documents good governance principles in traditional sports (Shilbury et al., 2016; Thompson et al., 2023), there is a lack of empirical insight into how governing bodies can and should regulate physical-digital integrity. To our knowledge, literature has not yet adequately explained how governing bodies can maintain their role of regulatory oversight and retain institutional legitimacy under the changing technology, commercial interest, and stakeholders of sports. Furthermore, it remains unclear how stakeholders in sport socially construct, negotiate, and ultimately regulate these novel integrity issues of virtual sports.

3.6 An Integrative Multi-Theoretical Approach to Understand Virtual Sports

The preceding sections established the foundational theories required to understand the complex nature of sport digitisation and digitalisation, and in particular the emergence of virtual sports. Table 1 provides an integrative summary of the specific research gaps identified across institutional theory, organisational innovativeness, innovation diffusion, disruptive innovation, and governance frameworks. It highlights how the theories used throughout this dissertation are complementary to the primary institutional anchor and outlines the specific scope of the empirical studies that address them.

Table 1: Outline of the multi-theoretical perspective used in this dissertation. The complementary nature between institutional theory and other theories is indicated after the ↗ symbol. For each theory, the research gap(s) and used scope are listed.

Perspective and complementary theories	Research Gap	Scope
Institutional-Strategic Perspective	-	Dissertation
Institutional Theory See section 3.2	(1) Sport management literature lacks an understanding of virtual sports, and of their legitimisation processes and strategies. Yet these are required to overcome the isomorphic pressures and normative framing delegitimising virtual sports	Dissertation (pre-study), Study 2, 3
Organisational Innovativeness & Diffusion of Innovation See section 3.3 ↗ Understand how meso- and micro-level dynamics are socially constructed within macro-level institutional logics.	(2) Integrative innovation models are seldom used in sport management literature, despite perceived innovation attributes being powerful predictors. In addition, structural determinants need revisiting for virtual sports because their barriers and constraints are different (e.g. complexity rather than financial or infrastructural)	Study 1, 2
Disruptive Innovation Theory See section 3.4 ↗ As legitimacy and technical superiority are lacking for disruptive innovations like virtual sports, an institutional diffusion strategy that constructs new evaluation criteria based on normative societal values and logics is needed.	(3) While several technological innovations could fit the definition of disruptive innovation (e.g. virtual sports), sport management literature lacks empirical insight into how traditional sport governing bodies navigate a technical strategy, an institutional strategy, or a combination of both to increase the social acceptance and adoption of an innovation.	Study 3
Good Governance See section 3.5 ↗ Understand how societal normative expectations translate to (the lack of) governance mechanisms and tools.	(4) Sport management literature lacks an understanding of the issues in virtual sports, and is consequently unable to develop a well-integrated sports and esports governance framework that allows for the assessment of the effective regulation of virtual sports.	Study 4, 5

To bridge these theoretical gaps listed in Table 1 (i.e. based on the gaps described in depth in sections 3.2 - 3.5), the pre-study and the five empirical studies in this dissertation are operationalised in such a way that they address each gap. Each study thereby isolates a specific mechanism of the overarching institutional-strategic perspective to understand exactly how virtual sports are evaluated, adopted, and governed:

- (1) A **pre-study** (section 9) that addresses the lack of universal definition and limited understanding of virtual sports. It focuses on the ontology of virtual sports by developing a universal definition of virtual sports, strengthening our understanding of all the aspects and configurations virtual sports can entail.
- (2) **Study 1** (section 12) addresses the gap in organisational innovativeness by employing an integrative innovation model. It investigates how perceived innovation attributes and human capacities *simultaneously* drive virtual cycling adoption in voluntary sports clubs. By doing so, it captures the perceptions and managerial attitudes rooted within the institutional logics and normative values around virtual sports.
- (3) **Study 2** (section 13) addresses individual and organisational beliefs about virtual cycling in voluntary sports clubs, which can be considered the micro- and meso-level dynamics within institutional theory. It empirically evaluates how clubs construct meaning about the potential uses of virtual cycling, and how these fit within the macro-level institutional logics, such as the conflicting logics of result-oriented and sport-for-all.
- (4) **Study 3** (section 14) investigates the specific legitimisation strategies required for sport governing bodies to gain social acceptance for virtual cycling and virtual triathlon by their stakeholders and society at large. It reveals how institutional framing, rather than developments in technical superiority, is utilised to overcome the mechanisms that delegitimise virtual cycling, such as isomorphic pressures.
- (5) **Study 4** (section 15) examines the gap in good governance by developing a better understanding of the novel integrity issues present in virtual cycling, and why these are particular to virtual sports rather than traditional sports. It highlights the need for better-developed governance frameworks and mechanisms to regulate virtual sports effectively, which is important to achieve institutional legitimacy.
- (6) **Study 5** (section 16) further assesses the governance and regulatory gaps by demonstrating how significant, yet hidden differences exist between the algorithms of Zwift versus MyWhoosh. It further investigates the integrity issues this introduces in

virtual cycling and the acute need for algorithmic transparency and governance frameworks that capture this integrity-threatening dimension.

4 Aims and Research Questions

Up to now, we have introduced the concept of virtual sports and situated it within the digitalisation of sport (see section 2). We have also discussed relevant theories and, based on (the lack of) existing studies, refined the research gaps situated when applying these theories to virtual sports (see section 3). Together, these highlight the areas holding potential for closer examination.

The introduction (see section 1) already briefly touched upon the aims of this dissertation: “the purpose of this thesis is to develop insights into *how* and *when* organisational innovation takes place, *how* and *why* organisational innovation introduces barriers surrounding institutionalisation and legitimacy, and *what* value organisational innovation can create in sport organisations, for its members, and the broader field of sport policy [...] through studying the specific case of virtual cycling” (p. 2). Given our theoretical outline above, we can now refine this purpose into more specific research aims that capture which type of organisational innovation, which type of value, for whom, and so forth. Hence, we formulate the following research aims:

- (1) **Aim 1:** Understand virtual sports as an (organisational) innovation in sports and esports and its institutional development since its emergence.
- (2) **Aim 2:** Understand the micro-, meso-, and macro-level drivers and barriers of the organisational innovation adoption process of virtual sports, and in particular, virtual cycling.
- (3) **Aim 3:** Understand the governance issues that pertain to virtual sports and how these challenge virtual sports’ legitimacy.

To uncover these research aims and address the specific research gaps listed in sections 3.2 to 3.6, the research questions of this dissertation read as follows. We want to note that RQ 1 is the main RQ of the dissertation, while RQs 2-6 actually repeat the five empirical studies’ RQs (in terms of completeness, cf. summary in Table 2). Each RQ provides a useful point of reference to guide our general discussion.

- (1) **RQ 1:** How can virtual sports be defined and understood, and what implications does their emergence hold for sport management and its adjacent fields?
- (2) **RQ2:** How can the identification of distinct adopter categories among voluntary cycling and triathlon clubs and their underlying differences in virtual cycling adoption determinants inform future innovation and marketing strategies for virtual sports?
- (3) **RQ 3:** What archetypes of sports clubs emerge from their convictions and hesitations regarding virtual sports, and how do these profiles align with overarching sport-for-all and result-oriented institutional logics?
- (4) **RQ 4:** How do governing bodies in cycling and triathlon legitimise or delegitimise virtual cycling, and what mechanisms do they employ in these processes?
- (5) **RQ 5:** How do stakeholders in virtual cycling and virtual triathlon experience current governance issues in virtual cycling?
- (6) **RQ 6:** how do differences between virtual sports platforms shape virtual sports competitions and their governance?

A schematic overview of all our research questions, their related research gaps they aim to address, and the scope where they are discussed, is shown in Table 2, which should be read in conjunction with Table 1.

The pre-study in in Part III section 9 provides a theoretical development into the ontology of virtual sports based on literature. It offers a conceptual framework that defines the fundamental nature, boundaries, and core characteristics of what constitutes activities that can be considered virtual sports, resulting in a self-proposed universal definition of virtual sports. As such, our theoretical pre-study partly answers RQ 1. In our general discussion in Part V, we will elaborate our answer to RQ 1 by discussing the implications this holds for different domains in sport management and adjacent fields of study. The other RQs (2-6) are empirically answered in our five studies listed in Part IV (see a summary in Table 18) but will also be reflected upon and contextualised in our general discussion in Part V. As a sidenote, studies 1 and 2 each list multiple RQs but are condensed into one single RQ per empirical study (i.e. RQ 2 and RQ 3 respectively) to maintain a more focussed overview and reflection on the different research directions this dissertation and its empirical studies highlight.

Table 2: Overview of research questions

Aims	RQ's	Institutional Theory complemented by	Gap	scope
Aim 1	RQ 1	Sport and eSports ontology debates	Table 1 gap 1, Literature lacks non- normative definition	Dissertation (pre-study)
Aim 2	RQ 2	Organisational Innovativeness, Diffusion of Innovation	Table 1 gap 2	Study 1
Aim 1,2	RQ 3	Institutional Logics, Diffusion of Innovation	Table 1 gap 2	Study 2
Aim 1	RQ 4	Disruptive Innovation Theory	Table 1 gap 1 & 3	Study 3
Aim 3	RQ 5	Good Governance	Table 1 gap 4	Study 4
Aim 3	RQ 6	Good Governance	Table 1 gap 4	Study 5

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Part II

Philosophy of Science and Methodological Outline

6 Philosophy of Science

6.1 *Critical Realist Approach*

A research philosophy represents the system of beliefs and assumptions guiding the researcher and the development of knowledge (Saunders et al., 2015). To effectively navigate the intersection of objective technological platforms and socially constructed sporting logics, this dissertation adopts critical realism as its philosophy of science.

While the predictive quantitative designs in the first and second studies might initially suggest a post-positivist approach, post-positivism ultimately relies on a “flat” ontology (Skinner et al., 2024). It assumes that a single objective reality exists and treats subjective human beliefs merely as imperfect measurements of that reality (Okasha, 2016). However, relying on such a flat ontology restricts the analytical capacity to address the aims of this research, which are heavily connected to institutional theory. Specifically, it limits our ability to examine socially constructed concepts that do not have direct objects, such as institutional legitimacy or sport-for-all logics, as actual causal forces that shape organisational behaviour (Scambler, 2023).

Critical realism, pioneered by Bhaskar and recently applied to the sociology of sport by Scambler (2023), resolves this tension through a stratified ontology. This ontology posits that our reality is “composed of not only experience and events but also underlying ‘mechanisms’ that exist whether or not detected and that shape or facilitate events” (Scambler, 2023, p. 46). In this way, an ontology of intransitive (i.e. not having or containing a direct object) structures and mechanisms (i.e. the “real” domain) is distinguished from a transitive reality that is used to experience and observe (i.e. the “empirical domain”) and describe these experiences and observations (i.e. the “actual” domain) (Scambler, 2023). This means that while we may observe and describe the direct (i.e. transitive) platform algorithms or organisational capacities, the reality in which these are constructed into existence and are described is one of indirect mechanisms such as institutional norms, (de)legitimising narratives, institutional or commercial logics, and so on.

Epistemologically, critical realism acknowledges that while an independent world exists, our human understanding of it is inherently socially constructed and fallible (Scambler, 2023; Skinner et al., 2024). Therefore, existing frameworks and concepts, such as good

governance and institutional legitimacy, are not treated as facts but rather as actively negotiated understandings of what is referred to as “good governance” or “legitimate” by stakeholders.

Axiologically, this paradigm recognises that a researcher’s theoretical biases inevitably influence the research process, requiring a reflexive approach to data interpretation (Okasha, 2016). This is further developed in our researcher positionality in section 6.2.

This philosophical stance directly translates to the epistemological pluralism (i.e. using different epistemological methodologies to acquire a fuller description) of this dissertation by employing two quantitative, two qualitative, and one mixed-methods approach in our studies. While the quantitative studies use statistical means to uncover the observable “real” structures at the micro- and meso-levels, the qualitative investigations explore how these structures are institutionally and subjectively constructed by governing bodies at the macro-level.

6.2 *Researcher Positionality*

Building on the critical realist lens, we view knowledge as something that is underpinned by the socially constructed frames of reference of the researcher in relation to the research object. As a researcher, the author’s positionality is somewhat between an “insider” and an “outsider”, depending on the specific domain investigated in virtual sports (Dwyer & Buckle, 2009).

In some aspects, the author has experience from an insider perspective on virtual sports because he regularly participates in sports (triathlon) himself, which also conveys the personal use of different virtual cycling platforms (e.g. Zwift, TrainingPeaks Virtual). The author is also a member of a voluntary triathlon club that in fact uses Zwift (a cycling platform) as a complementary club service offered, providing an alternative to outdoor group rides when time or weather constraints do not allow for these. Being part of a sport organisation experimenting with virtual cycling adoption provides the author with direct insights and biases about the drivers and barriers of virtual sports adoption. As the author also participates competitively (though not professionally) in sports, he gets to experience at first-hand potential governance issues or algorithmic discrepancies in sport and virtual sports.

In other aspects, the author keeps an outsider perspective as he is not actively involved in any sport governing body at a national or international level. The author is also not directly involved in the decision-making processes of virtual cycling implementation within his club, nor does he actively shape the narrative about it. As the author does not compete professionally in sports himself, he is also unable to experience the variety of ways that governance issues and algorithmic differences impact elite athletes or organisations that depend on sports’ integrity.

Occupying an insider role offers methodological advantages such as immediate trust and rapid participant acceptance, yielding greater data depth (Dwyer & Buckle, 2009). However, this proximity introduces significant subjectivity and the risk of overlooking discrepant views. Conversely, an outsider perspective provides the critical distance necessary to conceptualise phenomena clearly and identify latent patterns, although participants may doubt the researcher's capacity to truly understand “their” reality (Dwyer & Buckle, 2009). Ultimately, researchers such as the author navigate a complex space between these two positionalities.

Lastly, the author’s academic background adds another layer to this positionality. As a sport management researcher, the author approaches the phenomenon primarily through organisational and institutional lenses. However, the interdisciplinary nature of virtual sports means the subject spans across various fields of study, such as sports digitalisation, sports engineering, sports marketing, the sociology of sport, and sports law. Consequently, the author operates with a limited background in specific technical or legal intricacies of algorithmic programming and data privacy frameworks. This disciplinary positioning naturally influences the research, as it will often consider technological complexities from a managerial perspective (e.g. governance challenges) rather than technological or engineering problems.

7 Empirical and Methodological Outline

Structuring our methodological and empirical approaches within the theory on organisational innovation described in section 3.1, this dissertation balances studies on “Innovativeness” (i.e. determinants and outcomes) and “Innovating” (i.e. the process) (Wolfe, 1994). Not only do we investigate multiple facets of organisational innovation, this dissertation is also underpinned by triangulation and multi-methods because we employed a variety of theories and methods such as quantitative, qualitative, and mixed-methods techniques (Skinner et al., 2024). Our overarching study design is summarised in Table 3. As already stated in the introduction, all our studies analyse virtual cycling, some of which also examines the virtual cycling discipline within virtual triathlon, as the specific case or application of “virtual sports”. We will elaborate on these design choices more in depth in section 7.1.

Table 3: Empirical and Methodological overview of studies, listing their focus within Organisational Innovation theory, data collection instruments, sample sizes and geographical location, and applied analyses.

Study	Organisational Innovation scope	Data collection	Sample	Analyses
Study 1	Innovativeness	Survey	70 clubs, Flanders	Clustering, MANOVA, Discriminant analysis
Study 2	Innovativeness	Survey	90 clubs, Flanders	Clustering, MANOVA
Study 3	Innovating	Interviews	16 interviewees, International	Collective Qualitative Analysis
Study 4	Innovating	Interviews	16 interviewees, International	Collective Qualitative Analysis
Study 5	Innovating	Data scraping ³ , interview vignette	102 race results, 13 interviewees, International	Regressions, t-test, Collective Qualitative Analysis

From a higher level (i.e. across studies), we deliberately chose a methodological process that balances studies on innovativeness and innovating. That is because they have distinct merits and are complementary in nature (Wolfe, 1994). As such, combining them provides a holistic understanding of technological diffusion within complex social systems. Consequently, this dissertation bridges the main critique on *innovativeness* studies concerning their “static orientation [and] variance research approach” (Wolfe, 1994, p. 409) with the methodological recommendation to also employ *innovating* studies that “move away from a static, overdetermined, perspective towards investigations of the nature of, and the factors which influence, innovation processes” (Wolfe, 1994, p. 409). By initially focussing on innovativeness in studies 1 and 2, the research captures the specific organisational capacities and perceived attributes driving variance in adoption. These insights are complemented by more in-depth insights on the broader process of innovating in studies 3, 4, and 5, transitioning to meso- and macro-level legitimisation strategies, issues, and dynamics (DiMaggio, 1988; Thomas & Tierney, 2021). This approach accommodates the critical realist philosophy underpinning this dissertation. It enables the identification of objective structural barriers alongside the subjective social negotiation of virtual cycling’s legitimacy and governance, thereby not limiting insights to descriptive adoption rates or prescriptive adoption determinants.

³ The scraped data used in study 5 was collected by prof. dr. Daniel Westmattmann and complemented by our own data collection (see Table 4).

From a lower level (i.e. specific to each study), we also made deliberate methodological choices so that our methods within each study are best suited to answer each study's RQs. Following our overview in Table 3, we will now shortly discuss the rationale behind each study's choice of methods. For the methodological details of sample, analysis, and procedures, we refer to the methods section within each separate study (cf. section 11 - 16).

For studies 1 and 2, our quantitative approach using self-reported survey data allowed us to differentiate adopter categories or archetypes based on their perceptions and capacities (RQ 2) or based on their beliefs about virtual cycling's potential use cases (RQ 3). This methodology is effective for exploratory classification, but also allows us to compare determinants between different types of innovations. Such an approach yields insights into the way adoption determinants differ between different innovations and organisational contexts, particularly for virtual cycling. The combined use of clustering and multivariate techniques is well-established within strategic profiling in organisational research (Hair et al., 2019), allowing future research to compare their results. Surveying key volunteers introduces inherent self-selection and social desirability biases, which must be kept in mind when interpreting results as they can often only be partially addressed methodologically within the boundaries of what is pragmatically possible (see details in studies 1 and 2). A detailed reflection on our sampling of clubs is presented in section 7.1. Lastly, we acknowledge that our study 2 could have benefitted from including empirical data on the institutional logics. This is a theoretical connection that was only made after data collection had been completed. We have addressed this in the limitations section 23.1 and future research section 22.1.

For studies 3, 4, and 5, a qualitative approach using semi-structured interviews is optimal for capturing the complex, socially constructed processes of institutional legitimisation (RQ 4) and the nuanced stakeholder experiences regarding governance issues (RQs 5 and 6). Semi-structured interviews provide the necessary flexibility to uncover deep institutional narratives and subjective interpretations that rigid quantitative models cannot easily grasp (Bryman, 2016). Furthermore, study 5 employs a mixed-methods design by integrating quantitative platform data as an interview vignette (Venkatesh et al., 2016). Utilising vignettes in qualitative research is highly effective for eliciting specific reactions to complex scenarios (e.g. how algorithms between virtual cycling platforms differ, how this affects athletes). It grounds the discussion in concrete evidence, thereby stimulating participants to reflect deeply on abstract concepts like algorithmic differences and their implications for governance (Barter & Renold, 1999). A detailed reflection on our purposive sampling of interview participants is presented towards the end of section 7.2.

7.1 *Reflection on data collection*

As mentioned above, methodological triangulation was achieved by integrating quantitative surveys, data scraping, and qualitative semi-structured interviews. Methodological diversity and source triangulation ensured a comprehensive perspective on virtual sports. Data were gathered from a wide array of stakeholders, including from voluntary sport club managers, national federations, international federations and governing bodies, elite athletes, media reporters, media content creators, and virtual cycling platform developers. Despite the focus on the single case of virtual cycling, our data collection with diverse actors allows us to cross-validate evidence between different actors and check how these align with theory-informed institutional narratives and logics, mitigating individual participant biases (Eisenhardt, 1989).

The quantitative data collection (see Table 4 in section 11) in this dissertation is geographically restricted to Flanders, which naturally limits our generalisability in terms of location. The sample sizes ranged from 70 to 90 sports clubs due to different missing values for different analyses. Specifically, Study 1 (Section 12) and Study 2 (Section 13) rely on the same initial data collection comprising both cycling and triathlon clubs but use different variables in their analysis. Because Study 1 incorporates a broader set of variables to measure adoption determinants and outcomes, more clubs were excluded during data cleaning due to missing values compared to Study 2. There are multiple reasons why we accepted these numbers as our final sample sizes. First, it was difficult to get a good overview and contact information of our population, i.e. cycling and triathlon clubs in Flanders that (1) actively offer sports services (e.g., facilitate group training sessions) to their members, (2) are accessible to the public (i.e., everyone can become a member), and (3) are members of one of the recognised Flemish cycling federations or the Flemish Triathlon Federation. Since we were unable to obtain all the (e)mail addresses of all cycling clubs in our population to contact them to participate, we resorted to a self-selection sampling strategy that maximised the chances of cycling clubs being included by spreading the survey as widely as possible. Second, the response rate remained very low despite sending multiple reminders and maintaining online and offline visibility for our survey. Third, the dropout for those who started the survey was significant, likely due to its length (i.e. approximately 15-20 minutes).

While these sample sizes may appear small compared to individual-level surveys, they can nonetheless be argued as appropriate for organisational research. In fact, previous sport management studies by other researchers experienced similar response rates in the context of popular sports such as football (having 3822 clubs), which confirms that access to club-level

data is inherently difficult and naturally limits sample sizes. Some examples are: $n = 106$ (Constandt & Willem, 2019); $n = 73$ (De Waegeneer et al., 2017); $n = 51$ (Clerck et al., 2022).

A second argument to support our sample sizes can be found in their statistical (mathematical) determination. We can calculate the minimum required sample size according to the widely applied method presented by Bartlett et al. (2001). To do so, we first need to know the size of our population of triathlon and cycling clubs in Flanders (see criteria above). For triathlon clubs, the triathlon federation provided a register of 101 clubs, which we screened and filtered for clubs that met our population criteria (by checking their website or contacting them), resulting in a population of 90 triathlon clubs. For the cycling clubs, however, the definition of “club”, as used by the Flemish cycling federations’ registers, is very broad (e.g. including many “informal” clubs with very few members or volunteers). Consequently, the officially reported number of clubs is overestimated in view of our criteria (e.g. 60% of cycling clubs have less than 30 members⁴, likely existing of an informal group of friends who registered as a “club” for administrative reasons). This was confirmed by the federations, who told us that over half of the officially registered clubs would not meet our selection criteria (e.g., actively offering public sports services). We were nonetheless able to estimate the population of cycling clubs through people who indicated *themselves* in a survey of Sport Flanders (governmental sport agency) that they “participate in road cycling in a club context”, which can be considered in line with our criteria. The estimation is as follows: 251.194 people of the Flemish population aged older than 6 are known to participate in road cycling⁴, cycling clubs in Flanders are on average sized 42,02 members⁴, while 15.3% of road cyclists indicated *themselves* that they participate “in a club context”⁵. Together, this results in an estimated cycling club population of 915 clubs ($251.194 \times 0.153 \div 42,02$) that fit our criteria, which is reasonable in light of the 285 municipalities in Flanders. Applying our total estimated population size (i.e. triathlon + cycling; $N = 1005$) to the corrected formula⁶ of Bartlett et al. (2001), this results in a determined minimum total sample size of 70 clubs, which is in line with our actual sample size (see Table 3).

⁴ Data by Sport Vlaanderen for 2023 available via <https://www.sport.vlaanderen/kennisplatform/themas/thema-sportclubs/db-sportclubs/>, tab “grootte” (size) and “aantal sporters” (amount of sport participants).

⁵ Data by Sport Vlaanderen for 2023 available on <https://www.sport.vlaanderen/kennisplatform/themas/thema-sportparticipatie/db-vlaamse-monitor-sport-en-bewegen/>, tab context.

⁶ Parameters used: $\alpha \text{ level} = 0.025$ in each tail, $\text{margin of error} = .05$, $t = 1.96$, $s = 1.11$, $d = 0.25$, $\text{Population} = 1005$, $n_0 = 75.87$

Overall, we acknowledge that our sample size and geographical diversity are limited, but they should still be considered acceptable to derive meaningful insights that are statistically valid. Consequently, we decided to proceed using this sample and its size.

7.2 *Reflection on case selection of virtual cycling*

The selection of virtual cycling as the primary focus of this dissertation functions as a specific case within the broader phenomenon of virtual sports. The seminal work of Flyvbjerg (2006) corrects several misunderstandings about the use of cases and discusses their advantages when studying a new phenomenon. Flyvbjerg (2006) also discusses the strategies for case selection, which have been used to select virtual cycling. First, the “extreme case” allows researchers to obtain unusual information, either problematic or especially good, because it “activate[s] more actors and more basic mechanisms in the situation studied” (Flyvbjerg, 2006, p. 229). Second, the “paradigmatic case” highlights the core characteristics of a new phenomenon and serves as a foundational example for a developing research domain for which “no standard exists [...] because [the paradigmatic case] sets the standard” (p. 232), meaning that it often “operates as a reference point” (p. 232) for newly developed theory or practice to learn from (Flyvbjerg, 2006). Because the UCI was among the first, if not *the* first to our knowledge, traditional governing bodies to formally recognise and engage with a virtual discipline, virtual cycling stands as the most advanced, legitimised, and institutionally embedded example of virtual sports. It may therefore provide crucial and critical insights into how traditional federations navigate the technical challenges and novel integrity issues of virtual sports (Dyer, 2022; Dyer & Richardson, 2026; Richardson, 2025; Tjønndal et al., 2025; Westmattmann et al., 2022, 2025). Studying this highly dynamic environment yields rich, transferable lessons for the future governance and adoption of other emerging virtual disciplines.

For these reasons, we believe virtual cycling is the most insightful virtual sports case to study at this time. It is both an extreme case (e.g., in terms of general apprehension, formal adoption or legitimisation) and a paradigmatic case (e.g., no standards exist for virtual sports platforms), a standpoint that is defended by Flyvbjerg himself: “various strategies of selection are not necessarily mutually exclusive. For example, a case can be simultaneously extreme, critical, and paradigmatic.” (Flyvbjerg, 2006, p. 233).

To examine this extreme and paradigmatic case, the selection of specific actors and platforms followed a deliberate purposive sampling approach in studies 3, 4 and 5 (Palinkas et al., 2015). To assure comparability and control for overarching institutional norms, pressures,

or logics, the focus remained strictly on one sport and its associated governing structures (e.g. UCI, IOC). Simultaneously, we aimed for differences by investigating various platforms and perspectives across multiple levels, from voluntary clubs to governing bodies, elite athletes, and commercial platforms (for an overview, see Table 4 Section 11). These sampling criteria provided a balance of similarities and differences. By sticking to our case of virtual cycling and purposefully not investigating other upcoming virtual sports, a bounded system was chosen wherein the heterogeneous patterns supplied enough diversity to thoroughly examine the complex phenomena, such as adoption and digital governance. This, in turn, enables transferability to other contexts (Eisenhardt & Graebner, 2007).

7.2.1 *Virtual cycling compared to other virtual sports cases*

Because virtual cycling serves as our case, it is insightful to recognise why it has achieved a level of institutional legitimacy that other emerging virtual sports currently lack, and therefore provide an explanation of why other virtual sports are less suitable cases. Virtual cycling's success can be argued to stem from meeting specific technological, commercial, and institutional preconditions that often remain developmental barriers for other virtual sports. Based on literature and our own empirical work, we identify five primary drivers that aided virtual cycling's popularity compared to other virtual sports.

- (1) **Measurement:** cycling's nature as an individual endurance sport enabled it to become a sport where performances are already continuously and accurately measured in traditional disciplines (e.g. outdoor road cycling). The measurement of relative power output (i.e. W/kg) is a reliable metric that can be easily translated to an indoor virtual setting by integrating a power meter into virtual cycling hardware (Dowd et al., 2024; Kyle, 2003). However, this is not to say that all individual endurance sports are easily measured. For instance, running still often relies on speed over power as a measurement, despite speed being more subject to environmental parameters (shoes, wind, incline, weight...).
- (2) **Hardware:** virtual cycling benefits from a large market in terms of available hardware, ranging from cheaper entry-level smart trainers to highly accurate models. Tied to cycling's measurability, smart trainers can accurately and reliably measure physical power output. This is in turn instrumental for sport governing bodies to sanction fair competition (cf. studies 4 and 5). The hardware available in other sports sometimes still struggles heavily with accurate and reliable measures (e.g. running treadmills, non-

endurance sports involving motion tracking), making hardware more expensive and performance verification more challenging.

- (3) **High Movement Decompression:** as illustrated in section 9.5.2 below, the movement of pedalling on a stationary bike closely mimics the required movement of the parent sport (e.g. road cycling). While there naturally are differences (e.g. biomechanically) between virtual cycling and road cycling, the highly accurate and sensitive movement comparison (i.e. “high movement decompression”) can be argued to support virtual cycling in comparison to other sports having lower movement decompression (e.g. virtual boxing, virtual taekwondo).
- (4) **Commercial market:** cycling possesses a relatively large global market of people participating in it compared to other sports (Hulteen et al., 2017). This incentivises publishers to build and maintain platforms, despite the high investment costs, given that they can benefit from network externalities and economies of scale (Rochet & Tirole, 2003). Moreover, the virtual cycling market for both software and hardware is highly competitive resulting in lower prices (Maker, 2023, 2025). This is for instance in contrast to the Concept2 ergometer’s dominance in the rowing market (Maker, 2020).
- (5) **Institutional Culture:** While the UCI is often criticised for stifling innovation by overregulating, they did however demonstrate innovativeness by proactively engaging with virtual cycling since the early stages (cf. study 3). While it is difficult to compare the traditional or conservative cultures of sport governing bodies, the UCI embracing virtual cycling definitely supported its legitimisation. In other sports, such as action sports, tensions towards digitalisation have been reported (Standaert & Mazurova, 2025).

Combined, these five conditions can be argued to favour cycling compared to other sports in their development of a virtual discipline. Projecting which sports will match virtual cycling’s success remains however difficult, and will highly depend on their ability to overcome these barriers. Virtual golf, for instance, is currently navigating some of these hurdles successfully through developing highly accurate hardware and proactive institutional backing (cf. Section 9.5.2).

7.3 *Ethical statement*

The general outline of this doctoral dissertation was approved by the ethical commission (code: 2021-106) and the doctoral commission of the Faculty of Medicine and Health Sciences of

Ghent University. This approval also included the first two studies (see Table 4). In addition, studies 1 and 2 were checked to be in line with, and hence implicitly accepted by, the ethical protocol of the Faculty of Political and Social Sciences at Ghent University under the specific regulation that (1) all data is collected anonymously (i.e. no personal data that can uniquely identify the respondent) and (2) participants are presented with an information letter and with an informed consent form whereby they provide permission to participate in our study according to GDPR legislation for anonymous data. Studies 3, 4 and 5 (see Table 4) were approved by the ethical committee of Nord University, SIKT – The Norwegian Agency for Shared Services in Education and Research. Overall, each study obtained ethical approval.

7.4 Statement on the Use of Artificial Intelligence

The author acknowledges the use of generative artificial intelligence (specifically, Google Gemini) during the preparation of this doctoral thesis. The technology was utilised strictly as an auxiliary tool to support the writing process. Its application was limited to language refinement, such as correcting grammar and spelling, improving sentence flow, and image generation (acknowledged in the figure caption with specific prompt used). It was also consulted to iteratively brainstorm about structuring new ideas and explore their possible connection to existing theory, concepts, or findings from the literature.

Artificial intelligence was not employed to generate empirical data, conduct data analysis, or formulate the scientific conclusions of this research. All AI-assisted suggestions were rigorously reviewed and validated against the source material. The author takes full responsibility for the originality, accuracy, and academic integrity of the final manuscript.

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Part III

An Ontological Pre-study on Virtual Sports

9 Pre-study: Towards a Definition of Virtual Sports

This pre-study presents a piece of research that is ultimately required to conceptualise virtual sports as the object within this dissertation. In addition, it bridges our empirical studies in terms of their object of study. By focussing on the ontology of virtual sports, it addresses **aim 1, RQ 1**, and research gap 1 (cf. Table 1 and section 4). As indicated in section 4, this is done through developing a conceptual framework that defines the fundamental nature, boundaries, and core characteristics of what constitutes activities that can be considered virtual sports. Naturally, this conception is not something we evidence through empirical data, but through a discussion of literature around virtual sports. The result of this section is twofold: First, we provide a self-proposed universal definition of virtual sports, allowing us to reflect on the implications of virtual sports being a distinct ontological sporting category in our general discussion (cf. Part V). Second, such definition and reflection upon it presents a distinct theoretical contribution to the field of sport management and sport digitalisation.

To begin, this pre-study delineates an array of phenomena that are (1) at the core of the third layer of sport digitalisation (i.e. simulations in sport and virtual sports), but also that are (2) closely related to it. To do so, we will discuss Loy's (1968) dimensions of sport and utilise them as a guiding framework (section 9.1) to understand the following interrelated phenomena: sports-linked technological phenomena in section 9.2 (i.e. gamification, simulation *of* sports, simulation *in* sports, interactive sports training), exergaming in section 9.3, esports in section 9.4, and virtual sports in section 9.5. While these concepts are often confused or used interchangeably, they are definitely not the same. Making a clear distinction and comparison between them is important to put forward a definition of virtual sports and set out what it actually is this doctoral thesis investigates (i.e. its scope). It also sets out how this further impacts any approaches to and findings of its empirical studies. The development of our own definition for virtual sports is necessary because current literature lacks a unified conceptualisation. Despite virtual sports' increased use in literature, few universal (i.e. not applied to the specific research case or technology) definitions exist. For example, the definition of virtual sports used by Westmattmann et al. (2022, 2025) is actually one of "Meta-sports" integrating both sports and the Metaverse. This seems valuable yet too restrictive for our studies, or other studies, on virtual sports (see section 9.5). Consequently, and following the recommendations of Wolfe (1994), we aim to present a clear and universally applicable definition of virtual sports. This is essential because the determinants of adoption differ as the

characteristics of the defined innovation differ (Wolfe, 1994). This means that failing to specify the innovation's attributes obscures the findings and limits research generalisability. For example, treating distinct phenomena like esports and interactive training under the generic label of “virtual sports” or even “technological innovation” renders context-specific theoretical and empirical approaches difficult, as well as preventing potential meaningful comparison.

9.1 *Dimensions of Sport to Understand Sport Digitalisation Phenomena*

By using the dimensions of sport as an *analytical tool*, our aim is to move away from using definitions of sport as a normative checklist to define what can and cannot be labelled “sport”. Instead, we use definitions of sport to explore how technological innovations related to sport (e.g. esports, exergaming) map against the dimensions constituting sport. This allows us to understand how these innovations are rooted in, similar to, and different from sports. Such a non-normative perspective also facilitates an understanding of what can be learned from sport to develop innovation in sport, and vice versa.

Sport activities can be defined based on their incorporation of eight key elements according to Loy (1968), Guttman (2004), Suits (2007), and Parry (2019). These elements comprise:

- (1) **Human** activities, meaning that human interferences are essential whereby any technology used is not autonomously performing the activity
- (2) **Voluntary** play so that participation is noncoercive and motivated by intrinsic or extrinsic enjoyment
- (3) **Competition** that tests participants' capabilities in contests against each other
- (4) **Physical** skills, meaning that the physical engagement of human beings is crucial to its outcomes.
- (5) **Cognitive** skills such as tactical abilities, decision-making and game strategy that complement participants' physical prowess
- (6) **Following** that is widespread, attracting audiences that extend beyond local interests or communities
- (7) **Governance** structure relying on formal established rules enforced by legitimate governing bodies such as international and national federations, which does not, however, mean a complete absence of potential governance issues or threats
- (8) **Institutionalisation** achieved through legitimacy, rules and norms securing the role of sport as a significant social practice

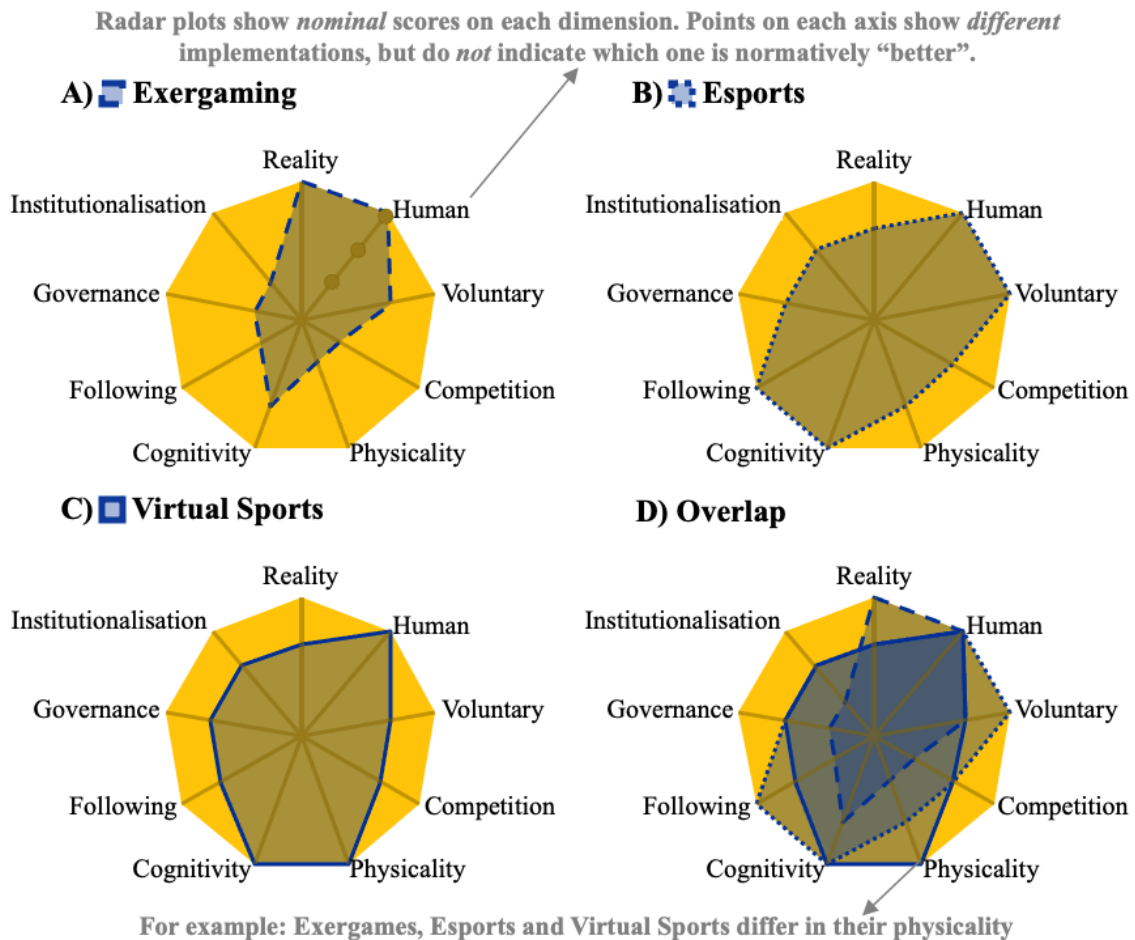
Putting these eight dimensions together, we arrive at our definition of sport as: *institutionalised rule-governed contests of human physical and cognitive skill, characterised by voluntary participation of a widespread social practice*. Interestingly, conceptualising sport through these dimensions also provides an unintended distinction between “sports” and other forms of (physical) activities, often based on the (absence of) dimensions like competition or physicality (e.g. walking, chess). While such a distinction in itself is irrelevant to this thesis, there is nonetheless merit in mapping activities against these specific elements. It allows us to better understand how, in our case, virtual sports are utilised in practice. For example, when an individual engages in virtual cycling purely for recreational or training purposes without partaking in a rule-governed contest, virtual cycling is not a sport but physical activity according to the above strict definition. Because we do not intend to diminish the relevance of such practices, it becomes immediately clear why we rely on the definition of sport as a non-normative analytical tool (i.e. use each dimension separately to guide analyses) rather than assessing virtual sports and other phenomena against the complete definition of sport. Such an approach enriches our understanding of how virtual sports can be implemented, managed, and create value beyond the rigid boundaries of earlier defined competitive “sports”, such as within sport-for-all contexts. While it should be clear already, we want to emphasise that our purpose is not to open up a normative debate over which innovative practices can be considered sport versus physical activity versus virtual sports.

In addition, our explanation of the third layer of sport digitalisation in section 2.3 introduces the notion of virtuality. This naturally leads to a distinction in which *reality* sports can take place, i.e. in actual physical-world conditions or in a distinct sporting reality creating its own conditions. Hence, we account for this by complementing our analytical tool of sport dimensions and adding a ninth dimension that comprises:

- (9) **Reality** of the specific environment and laws (i.e. nature, governing...) under which a performance occurs in the case of sports virtuality (i.e. comparing digitally measured performances)

Together, these nine dimensions will act as a guide in discussing the often blurred boundaries of technological sport innovation that have evolved substantially over recent years on these dimensions. Before continuing more in depth, we present Figure 5 as a guiding visualisation of how exergames, esports and virtual sports are similar or different in each of these nine dimensions.

Figure 5: Nine-dimensional nominal scoring of exergaming, esports and virtual sports visualised in subfigures A, B, and C respectively, to be compared by their filled area overlap in subfigure D. The way scores in subfigure A-C differ in each dimension, is explained in subfigure E. Source: own conception.



E) Nominal scoring comparison

Differences	Reality	Human	Voluntary	Competition	Physicality	Cognitivity	Following	Governance	Institutionalisation
Esports	physical -digital inter- reality	sports- alike	sports- alike	mix of compe- titive, casual, social, & fun	means, highly compres sed	sports- alike	signifi- cant	existing but rigour and authority debated	
Virtual Sports			gamifi- cation outside gaming context		means & end, decomp ressed		niche, rising		
Exer- gaming	mainly physical			social & fun	designed for training and rehabilitation		minor	not designed for	

While all concepts in this figure will be explained in depth throughout the following sections, it should for now be clear that the radar charts in subfigures A-C show how different phenomena score on different dimensions, making them visually comparable. Such comparison can also be seen in subfigure D, which overlaps the nine scores for exergaming, esports and virtual sports. It can for instance be immediately seen that each radar chart (subfigure A-C) looks quite different and few points overlap (subfigure D). This means that exergaming, esports, and virtual sports greatly differ on all dimensions. Furthermore, analysing how these dimensions are configured in virtual sports helps us understand various practical applications and their implications. For instance, virtual cycling can be a casual leisure activity or a high-level competition, or we may understand its growth based on either active participants or spectator following. Examining different dimensions provides valuable insights across entirely different fields (e.g., Sport-for-All policies, technology adoption, sport governance) regardless of whether the activity is labelled as “sports”. In what follows, we apply our systematic analysis of sports dimensions introduced in this section and summarised in Figure 5 to sports-linked technological phenomena in section 9.2 (i.e. gamification, simulation *of* sports, simulation *in* sports, interactive sports training), exergaming in section 9.3, esports in section 9.4, and virtual sports in section 9.5. Combined, these analyses form the basis on which we draw to develop our own definition of virtual sports, as introduced in section 9.6.

9.2 *Sports-linked Technological Phenomena*

To develop a better understanding and work towards a definition of virtual sports, it is useful to first distinguish between an understand multiple closely related technological phenomena: gamification, simulations in sport, simulations of sport, and interactive sports training.

9.2.1 *Gamification*

Gamification can be defined as the application of game-design elements in non-gaming contexts to enhance engagement and motivation (Tu et al., 2019). Gamification is thus a technique that, in the context of sports, is used to motivate consumers to stay active by addressing psychological needs like enjoyment or social interaction (Tu et al., 2019). Applying our framework of nine dimensions of sports to a universally applicable technique (gamification) rather than a specific activity in itself is not a meaningful analysis (e.g. how can a technique like gamification possess a certain degree of governance or competition?). It is moreover not the intended use of the guiding framework. Consequently, we will not discuss gamification against all nine dimensions. However, we can acknowledge its potential to amplify the

cognitivity and voluntary dimensions of sport, without necessarily altering other dimensions such as physicality.

9.2.2 *Simulations in sport and simulations of sport*

Parry and Giesbrecht (2023) establish a critical distinction between simulations *in* sport and simulations *of* sport. A simulation *in* sport occurs when an individual physically replicates (i.e. mimics) the bodily movements of a parent discipline. For example, virtual cycling requires the athlete to cycle on a stationary bike and smart trainer to propel their in-game avatar forward. Conversely, simulations *of* sport provide merely a visual and virtual representation of sports. In other words, simulations *of* sport are sports-themed video games, also frequently labelled sports video games or sports simulation games (Rogstad & Skauge, 2023). In these instances, gamers use a controller or keyboard to manipulate digital athletes, such as playing a virtual football match on a console (e.g. *FIFA* or *NBA 2K*).

Understanding this distinction is vital because it separates the terminology used to refer to traditional video games about sports and less traditional video games that are not played in a sedentary manner and require a different kind of motoric input (Parry & Giesbrecht, 2023). Such a distinction was, for example, not made in the communication of the International Olympic Committee's (IOC) up to 2020. They referred to "esports" to possibly refer to both categories of games (e.g. during IOC's Olympic Summit events) (S. E. Jenny et al., 2024). Failing to recognise this difference creates the risk of conflating or even disturbing stakeholders from two environments (i.e. sports and esports) that are to date still considered very distinct (Pizzo et al., 2022). From 2020 on, the IOC made the distinction between "non-physical virtual sports" and "physical virtual sports" (S. E. Jenny et al., 2024). This was seen, for instance, during the Olympic Virtual Series which featured simulations *in* sport like virtual cycling on Zwift, alongside sedentary simulations *of* sport, such as virtual baseball video games (International Olympic Committee, 2021a).

It should be noted that we have decided not to use the labels "non-physical virtual sports" and "physical virtual sports", but instead denote these as simulations *of* sport and simulations *in* sport to prevent confusion with our definition of virtual sports (see section 9.5). The ways in which they relate to the nine dimensions of sports are not discussed separately to avoid repetition. Instead, they are rather discussed in light of esports (section 9.4) and virtual sports (9.5), which is more relevant to developing our definition of virtual sports.

9.2.3 *Interactive Sports Training*

Interactive sports training concerns the use of XR technology (e.g. interactive led pods,

interactive basketball hoops, illustrated in Figure 6) to mainly promote skill acquisition (e.g. reaction time, passing accuracy) (Jensen et al., 2014; Kong & Wang, 2019). Because these systems are designed to be used in existing activities, they are considered auxiliary tools in sports. Consequently, their place in this dissertation is important to distinguish them from exergames and virtual sports. However, discussing them against all nine dimensions of sports would introduce repetition without adding much new insight. To avoid this, we refer to interactive sports training systems when analysing the dimensions of exergames and virtual sports since they may influence specific dimensions of these activities.

Such influences for instance include that they may isolate and target the physicality and cognitivity dimensions of sport by simulating in-game scenarios for athletes to practise (Kong & Wang, 2019). Furthermore, interactive sports training may integrate extrinsic elements like gamification techniques to motivate participants by making repetitive drills inherently rewarding and fun. This potentially boosts intrinsic enjoyment and hence the voluntary dimension (Grosprêtre et al., 2023; Zhang et al., 2025). In conclusion, these tools may indirectly enhance the voluntary, physical, and cognitive dimensions of sport (related) activity, though they do not facilitate a holistic rule-governed contest.

Figure 6: Illustration of Interactive Sports Training technology. Source: image generated by Gemini 3.1 Pro, prompt used: “Create an image to illustrate Interactive Sports Training. It could for example contain Blazepods, smart basketball hoop 'Huupe' and more.”



9.3 Exergaming

Exergaming, a combination of exercising and gaming, refers to video games requiring motoric movements to play — often gross motoric in nature (Mura et al., 2018). As illustrated in Figure 7, these movements are detected and captured by cameras, handheld controllers, or balance boards, among other devices (Mura et al., 2018). Exergaming can be considered as an activity in itself, as opposed to our definition of gamification and interactive sports training. When performed, it significantly influences the required physical and cognitive required capabilities of players. This explains why they are largely used in physical training or in rehabilitation settings, or even for entertainment purposes (Grosprêtre et al., 2023; Martin-Niedecken & Schättin, 2020; Mura et al., 2018).

Figure 7: Exergaming at Sport Flanders’ and Howest’s Sportscube for Active Gaming.
Source: Howest. Images used with permission of Howest.



Mapping exergaming onto the nine dimensions of sport (see Figure 5 A, D and E) reveals a configuration that can be largely understood as tailored towards health and entertainment rather than competition. This is especially evident when considering the dimensions of physicality and cognitivity related to the design purposes of exergaming (i.e. training or in rehabilitation). Regarding the human dimension, exergames explicitly require human interaction through physical movements to operate the system. Furthermore, the interactive nature of exergames enhances their voluntary dimension, as they primarily rely on gamification techniques to

achieve sustained participant motivation and engagement (Brauner et al., 2013).

The reality dimension could best be described as a “basic translation” or mediation of physical performances into the virtual environment, where they are presented in an unaltered way. For instance, players are often represented as virtual avatars to receive direct and unaltered feedback on their body movements and activities (Brauner et al., 2013). Exergames are specifically designed for players to “focus on the real environment” and their actual physical exertion, while still maintaining a playful character (Martin-Niedecken & Schättin, 2020, p. 5). In this way, exergames *mainly* take place within the actual real-world laws. Players focus on accounting for actual laws (e.g. gravity, fatigue) and adjusting their motoric input accordingly, rather than adapting their motoric input to an algorithmically altered reality and laws of the game (e.g. the “metagame” in esports, as described in section 9.4.3).

Lastly, given the design of exergames for health and entertainment purposes, they generally lack structured competition. If any competition exists, it is situated within the “Casual–Social” or “Casual–Fun” classification (Jung Hong & Wunsch, 2024, p. 167), which is explained more in-depth regarding esports in section 9.4.1. The design purposes and configuration of the abovementioned dimensions have clear implications for governance, institutionalisation, and following. These specific dimensions can all be described as relatively minor or absent by design. This understanding of exergaming provides a useful bridge to understanding the different and more complex games found in esports.

9.4 *Esports*

Esports or competitive video gaming has evolved into a global phenomenon characterised by structured tournaments, institutionalisation by governing bodies, and increasingly professional environments (Cranmer et al., 2021; Reitman et al., 2020). While conceptualisations vary across disciplines, a widely cited definition of esports is: “a form of sports where the primary aspects of the sport are facilitated by electronic systems; the input of players and teams as well as the output of the esports system are mediated by human–computer interfaces” (Hamari & Sjöblom, 2017, p. 213). This definition refers to esports being a form of sports, meaning that Hamari & Sjöblom consider esports to theoretically align with the elements constituting sports (see section 9.1). These elements include human activities, voluntary participation, competition, skills, structured governance, a wide-following, and institutional legitimacy (Borkowski et al., 2025; Hallmann & Giel, 2018; S. E. Jenny et al., 2016; Reitman et al., 2020). While debatable, esports is described in the literature to increasingly mirror the socio-economic structures of traditional

sports through leagues, sponsorships, and growing professionalisation (He et al., 2025; Reitman et al., 2020; Scholz & Nothelfer, 2022).

Still, ongoing debate exists over whether or not esports are sports (Funk et al., 2018; Naraine, 2021; Parry, 2019; Parry & Giesbrecht, 2023). In our discussion of esports within this section, we do not take such a “hard” standpoint on whether these phenomena should be labelled as sports. From a research perspective, such labelling is subordinate to the fact that much can be learned from esports to be transferred to other sporting contexts such as exergaming, virtual sports, or sports. Given that many studies in literature refer to virtual sports (e.g. virtual cycling) as “esports” rather than sports (Berger et al., 2025; Borggreffe & Hoffmann, 2024; Dyer, 2020, 2022; Dyer & Richardson, 2026; Dyer & Taylor, 2023; Hemmingsen, 2024; Parry & Giesbrecht, 2023; Postma et al., 2022; Richardson, 2025), it seems evident that we discuss the dimensions of esports in depth to uncover esports’ link with virtual sports. In what follows, we explain Figure 5 B and E, that is, how esports maps onto the nine dimensions of sports introduced earlier.

9.4.1 Human, Voluntary, Cognition and Competition

Certain dimensions of esports are generally accepted as aligning closely with traditional sports. For instance, esports involve human activity and voluntary participation (Heere, 2018; Naraine, 2021). Furthermore, esports require high levels of cognitive skills, such as tactical decision-making and game strategy, to successfully compete in. In terms of the competition dimension, esports are in the literature generally characterised as highly structured elite-level contests. Yet, it must be acknowledged that in esports as well, a comprehensive classification of players in terms of competition exists: (a) Competitive, (b) Casual, (c) Casual–Social, (d) Casual–Fun, and (e) Casual–Competitive (Jung Hong & Wunsch, 2024, p. 167). This means that much like sports, esports may also create value beyond the rigid boundary of “competition”, with activities, for instance, being rooted in fun, social, or leisure contexts. In fact, many participants engage in esports titles purely for leisure or social interaction without any intention of taking part in any sort of competition. This would move us to the realm of “recreational video gaming” rather than esports (Nothelfer et al., 2024, p. 11), similar to what physical activity is to sports (see section 9.1).

9.4.2 Physicality

Regarding the dimension of physicality, perspectives vary. Naraine (2021) contends that esports requires physical skills such as elite manual dexterity and physiological control, which resembles other modern or digitally enhanced sporting disciplines. Conversely, other scholars

highlight a lack of direct gross motoric physicality, noting that the activity is largely sedentary (Parry, 2019; Parry & Giesbrecht, 2023). Borggreffe and Hoffmann (2024) further clarify this by explaining that “communication of physical performance as an end in itself is not present in eSport” (Borggreffe & Hoffmann, 2024, p. 1201). This means that the physical performance of operating an input device like a mouse or keyboard in esports is merely a means to demonstrate a player’s capabilities. The motor activity or physicality of clicking in itself is thus unsuitable for observing the difference between winning or losing in esports and therefore does not function as an end in itself (Borggreffe & Hoffmann, 2024). In this way, we can argue that esports possesses physicality, though it makes more sense to understand it as having a “means finality” rather than as fine-motoric versus gross-motoric (Parry, 2019; Parry & Giesbrecht, 2023). This is because it is only by observing the game that spectators know what actually happens and how a player is performing. In fact, if gross-motoric movements were the criteria to assess physicality in sport, then even some Olympic sports such as shooting sports would be up for debate (Naraine, 2021). Also, other distinctions have been made such as virtual vs. real and mediated vs. unmediated. However, these are all argued to put importance to facts that are actually not important to understand physicality in esports, such as the fact that it is mediated, or where the outcome takes places (Hemmingsen, 2024). Instead, Hemmingsen (2024) proposes the concept of “Movement compression” to discuss physicality in esports, which is understood as “the degree to which the quality of the initial action translates to a difference in outcome in the activity along various dimensions.” (p. 176). That is to say that physicality in esports is “compressed” in a way that not every detail of the physicality is captured. Therefore, various slightly different actions lead to the same outcome, which goes against the traditional view in sports that movement directly affects outcomes (Hemmingsen, 2024). This is an interesting approach by Hemmingsen, as it helps us to understand that esports are physical. Even though that physicality is compressed but doesn’t make it less physical, nor does it take away the fact that physical movements still affect the outcomes of the game to a certain degree. As indicated in Figure 5 (subfigure E), movement compression should be understood on a continuum and can become “decompressed” as the required physicality changes (e.g. in virtual sports where sports movements are mimicked) or as the accuracy and sensitivity of the captured physical movements changes (Hemmingsen, 2024).

9.4.3 Reality

This nuanced understanding of as-a-means and compressed physicality directly shapes the reality dimension of esports. As illustrated in Figure 5 (E), we present the environment of

esports to operate within a physical-digital interreality. Rather than existing purely in the physical or digital realm, Ekdahl (2022) notes that players synthesise a “hybrid space” where their physical surroundings and the virtual arena are intrinsically linked. Within this hybrid space, the reality is ruled by the platform’s algorithms rather than the laws of the actual real world (e.g., laws of nature, to the extent these are represented within these games). It is precisely these algorithms that affect the movement compression discussed above. They actively translate and mediate physical inputs into digital outcomes based on programmed constraints or sometimes even commercial logics (Lv & Schotter, 2025). The notion of reality being a digital-physical interreality builds upon the concept of the “meta” or the “metagame” in literature, i.e. the optimal way to play within the laws of the virtual reality. As explained by Jenny (2024):

Collectively, esports players attempt to adapt and master the meta by utilizing the most dominant and optimal mechanics, resources, and strategy to win. However, the meta can be dynamic within a game title as the meta can shift, especially after a patch (i.e., game developer/publisher update). (p.180)

This confirms how esports’ reality is shaped by both the actual reality and the digital reality, and how both may change over time (e.g., new equipment, patches). It also highlights how both require specific adaptation, skills, and expertise, making it also connected to the dimension of cognitivity in esports.

9.4.4 Governance and Institutionalisation

The dimensions of governance and institutionalisation also present a complex configuration in esports. On the one side, esports possesses established governance structures such as governing bodies (e.g. the International Esports Federation) (Naraine, 2021). On the other side, institutional instability remains a valid concern since esports lacks a “standardised” public institutionalisation. Some governing bodies are self-proclaimed, some are unrecognised by public authorities, and above all, esports remain legally owned and governed by game publishers who sometimes fail to regulate (S. E. Jenny et al., 2024, p. 495). Esports are fundamentally built upon commercial products, meaning the intellectual property rights are owned and controlled by corporate publishers rather than independent governing bodies (Abanazir, 2019). These challenges have led to the fact that esports are faced with governance issues such as cheating, doping, or toxicity to name but a few (Felczak, 2022; Hayday et al., 2021; Piggott & Tjønndal, 2024; Richardson, 2025).

9.4.5 Following

Lastly, esports easily satisfies the dimension of having a widespread following, though it should be acknowledged that this specific following is rooted in its own community and ecosystem (He et al., 2025; Reitman et al., 2020). From the literature, we know that minimal crossover between traditional sports communities and esports communities exists (Pizzo et al., 2018, 2022), and that sports fans generally view esports brand extensions as a peripheral activity (Bertschy et al., 2019). Understanding esports' configuration on these nine dimensions is crucial for sport management, as integrating such activities in a traditional sport context can either serve as an important strategy for brand innovation or at the same time pose a potential risk of sport brand dilution (Ke & Wagner, 2022, p. 155).

9.5 Virtual sports

Virtual sports have in section 2.3 already been situated as one branch within the third layer of sport digitalisation next to simulations *in* sport (see section 9.2.2), where their difference has also been explained. In this section, we will focus on virtual sports and work toward a definition based on its configuration within the nine dimensions of sport, and their differences and similarities to esports, exergames, and other sports related phenomena discussed above. While research on virtual sports is still developing given its recent emergence, scholars have begun to explore these virtualised sporting disciplines using terms such as esports (Borggreffe & Hoffmann, 2024; Hemmingsen, 2024; Postma et al., 2022), mixed-reality sports (Westmattmann, Grotenhermen, Sprenger, Rand, et al., 2021; Westmattmann, Grotenhermen, Sprenger, & Schewe, 2021), digitally replicated sports (Goebeler et al., 2021), meta-sports (Westmattmann et al., 2025), virtual active sports (Evsikov et al., 2024), virtual leisure environment (Reed et al., 2023), or virtual sports (Bentvelzen et al., 2022; Devine et al., 2024; Fouquaert, Tjønndal, De Clerck, et al., 2025; Fouquaert, Tjønndal, & Willem, 2025; S. E. Jenny et al., 2024; Ningning & Wenguang, 2023; Parry & Giesbrecht, 2023; Tjønndal et al., 2025; Westmattmann, Grotenhermen, Sprenger, Rand, et al., 2021; Westmattmann, Grotenhermen, Stoffers, & Schewe, 2021).

The term “virtual sports” has also gained institutional traction and was picked up by the IOC. It was first put forward in their Declaration of the 9th Olympic Summit (12 December 2020) and has since been a recurring topic in the IOC's communications, including the Olympic Agenda 2020+5 (15 February 2021), the Inaugural Olympic Virtual Series (7 July 2021), the Olympic Esports Series (22-25 June 2023), the announcement of the Olympic Esports Games

(14 October 2023), and so on (S. E. Jenny et al., 2024, p. 502). It is important to note, however, that the IOC makes a distinction between “Physical Virtual Sports”, which we refer to simply as “virtual sports”, and “Non-Physical Virtual Sports” which we refer to as simulations *of* sports (see section 9.2.2) or esports with sports as content such as FIFA or NBA2K (S. E. Jenny et al., 2024)

Despite virtual sports’ increased prevalence in literature and practice, little attention has been paid to putting forward a universal definition that is not strictly tied to a specific research case or technology (e.g., Zwiift) (Bentvelzen et al., 2022; Devine et al., 2024; Fouquaert, Tjønndal, De Clerck, et al., 2025; Fouquaert, Tjønndal, & Willem, 2025; Westmattelmann et al., 2022; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021; Westmattelmann, Grotenhermen, Sprenger, & Schewe, 2021). One notable exception is the study of Westmattelmann et al. (2025) that defines virtual sports as “*replicated sports in virtual environments fulfilling both the sport and Metaverse criteria*” (p. 74), thereby relying on the definition of sports (see section 9.1) and their definition of the Metaverse as: “*a technology-mediated, immersive, and persistent virtual environment that integrates physical and virtual realities, allowing users represented by avatars to engage socially and economically with one another in real-time*” (p. 74). The latter definition is based on the Metaverse review of Ritterbusch and Teichmann (2023). While the Metaverse certainly presents an interesting conceptual lens to approach virtual sports, it remains questionable if it should be regarded as a necessary component in defining virtual sports specifically yet parsimoniously. For instance, events like the UCI Cycling Esports World Championships can take place on a local offline network without relying on the interconnected “social” features of a Metaverse, though they are an exemplary case of virtual sports.

Consequently, there is a need to establish a clear, universally applicable definition of virtual sports. To achieve this, we will systematically map virtual sports across the nine dimensions of sport introduced earlier, as visualised in Figure 5 (subfigures C, D and E). By exploring how virtual sports are configured across these dimensions compared to exergaming and esports, we can progressively construct a robust definition that accurately captures their distinctness from these other phenomena. To arrive at such a definition, the following sections will explain how esports maps onto the nine dimensions of sports as outlined in Figure 5 (subfigures C, D, and E).

9.5.1 Human, Voluntary, Cognition, and Competition

Virtual sports share several similarities with how sports, esports and exergaming are configured

in their dimensions (see Figure 5 D & E for a side-by-side comparison). First, virtual sports also rely on human interaction to be performed, though these interactions are distinct in that they require sports-mimicking movements in their very nature, as further explained in the physicality dimension. Second, virtual sports also demand a certain amount of cognitivity to be performed. Participants must, for instance, process real-time digital feedback, execute tactical pacing, and make strategic decisions within the virtual environment, mirroring the tactical demands found in both traditional sports and esports (Devine et al., 2024; Parry & Giesbrecht, 2023). Third, we regard participation of virtual sports as voluntary in nature. Similar to the motivational design of exergames, virtual sports' voluntary dimension can be amplified through gamification (e.g. rewards) that fosters sustained motivation and engagement outside of its strict gaming context (i.e. the physical context) (Evsikov et al., 2024; Mura et al., 2018; Schwarz et al., 2020; Tu et al., 2019). Fourth and last, the competition dimension in virtual sports is highly versatile. While virtual sports are fully capable of hosting elite-level competitions, the competitive landscape is not exclusively elite. Similar to the categorisation found in esports described in section 9.4.1, virtual sports accommodate a wide spectrum of users who are also seeking casual, social, or fun competitions (Jung Hong & Wünsch, 2024). In fact, individuals frequently utilise virtual sports platforms for personal health, rigorous training, or pure leisure (Bentvelzen et al., 2022; Berger et al., 2025; Devine et al., 2023; Edwards & Hettinga, 2023; Evsikov et al., 2024; Felczak & Filiciak, 2025; Fouquaert, Tjønndal, De Clerck, et al., 2025). In these instances, the user may have for example no intention of entering any sort of competition. Hence, virtual sports' competition dimension is multifaceted and mirrors the health and training applications of exergaming. This highlights its potential use outside of strictly speaking "sports" (i.e. competitive), but rather for inclusion, sport-for-development, sport-for-all etc. (Fouquaert, Tjønndal, De Clerck, et al., 2025; Reed et al., 2023; Tjønndal et al., 2025).

9.5.2 *Physicality*

The dimension of physicality allows us to clearly differentiate virtual sports from other sports-lined technological phenomena and from esports. As discussed earlier in our esports analysis of physicality (see section 9.4.2), scholars often highlight the fact that the required physicality in esports is of a different nature (e.g. fine vs. gross motoric, means vs. end finality, highly compressed), noting its largely sedentary nature. In contrast, virtual sports explicitly require substantial physical exertion that is designed to actively mimic the movements of the traditional parent sport, such as pedalling on a smart trainer (see Figure 3 B) or pulling on a rowing

ergometer (see Figure 3A).

To further understand this distinction, we can build on the concept of “movement compression” introduced in section 9.4.2 on esports’ physicality (Hemmingsen, 2024). In esports, physical input is highly compressed, meaning that complex movements in the virtual world are reduced or *compressed* to controlling a game controller or keyboard and mouse, which is of course very different in its motoric nature (e.g., moving a joystick versus swinging an actual golfing club). Virtual sports, however, reduce this compression and require inputs that are more *decompressed* in their physicality. That is because the mediating technology is designed to capture a much broader and more realistic spectrum of physical exertion, such as mimicking the movements of the parent sport as realistically as possible. Still, some amount of compression does exist, as the inputs given to advanced peripheral equipment are, at best, a very realistic approach to capture the actual real physicality of the sport and are therefore *not* the same as this actual real physicality of the sport.

In Figure 8, we illustrate and build upon the *continuum* of movement (de)compression as outlined by Hemmingsen (2024) in their exemplary case of golf. As can be seen, advanced technology significantly increases the sensitivity and accuracy of the required physical inputs to perform these activities. A fitting case of this is illustrated in Figure 8 example 3: the SoFi Center simulator features a massive screen (274 m²) where players swing actual golf clubs tracked with sub-millimetre sensitivity (e.g. 18 radar-based launch monitors and 8 optical cameras). They hit into a virtual landscape for long shots *and* a physical indoor “GreenZone” as hits come closer to the hole, ultimately combining virtual and digital worlds to achieve maximum movement decompression. We propose that for activities to be regarded as virtual sports, their physicality has to occur in the right half (i.e. centre to right) of the movement (de)compression continuum seen in the blue line on Figure 8. It could even be argued that when a virtual sport expresses enough movement decompression, mimicking the required exertion, accuracy, and sensitivity of the parent sport close enough, it is ready to be perceived as legitimate by the public, at least on the dimension of physicality. The cut-off or range where that could occur in this continuum is however undefined and remains open for debate, and classifying an activity as virtual sports would moreover also depend on its configuration of other dimensions.

Figure 8: Illustration of movement (de)compression using the example of golf. Source: own conception. Images generated by Gemini 3.1 Pro, prompts used: (1) playing EA Sports PGA Tour on PS5; (2) playing Wii Sports golf on TV; (3) golfer in the TGL SoFi simulator; and (4) golfer playing real world.



Consequently, with enough movement decompression, it could also be argued that the physicality of virtual sports is not merely used as a means to an end, as is the case in esports (see section 9.4.2). More specifically, that is because the movement activity (physical performance) in itself allows one to observe the capabilities of a player in their play. This is not the case for esports players clicking a mouse, as only observing the clicking would not render a meaningful image of that player's capabilities. In fact, spectators would have no clue what is going on when *only* observing the clicking. In some cases of virtual sports, such as in virtual cycling or in virtual golf, *only* observing the physical performance allows one to capture the outcomes of the game at least partially (e.g. winning or losing, missed shots, cadence...), meaning that the physical performance must therefore be observed in combination with its virtual representation. However, we would argue that this does not shift the finality of the physical performance to a means rather than an end in itself, as both the digital *and* the physical environment would in such a case equally demonstrate the capabilities of a player. In other virtual sports cases, such as augmented table tennis, observing *only* the physical performance would render a whole image of the players' capabilities and the outcome of a game. This renders a need to redefine the means-end-finality logic (Borggreffe & Hoffmann, 2024) for every case, as it appears not universally applicable to all virtual sports.

9.5.3 Reality

As already elaborated upon in section 2.3 and section 9.1, the dimension of reality takes a central role in defining and distinguishing virtual sports from other digitised sports phenomena. We argue that virtual sports do not exist *solely* in the actual, real-world reality such as the laws of nature or governing rules by established governing bodies. Instead, virtual sports exist within a physical-digital interreality. This physical-digital interreality is closely related to but not the same as the concept of 'phygital' sports and customer experiences (Batat, 2024). A phygital environment seamlessly blends physical and digital realms to co-create a holistic and new experience (Batat, 2024). Consequently, the phygital nature of virtual sports is relevant within adjacent management debates about commercial and experiential features (Heidemann et al., 2024). For instance, recent studies highlight how 'home phygital environments' significantly shape user retention and perceived value in online home fitness platforms by blending digital interfaces with users' physical spaces (Langaro & Werling, 2025), while other applications demonstrate the impact on blended learning experiences in sport (Ahmadi et al., 2026). Building upon this notion of phygital, the concept of virtual sports' physical-digital interreality focuses in particular on the *sporting rules, laws and performances* that become hybrid rather than the entire experience. To further unpack this physical-digital interreality, we can start by drawing some parallels to esports' reality dimensions (see Figure 5 D and E).

Esports' reality is discussed (section 9.4.3) to take place in a "hybrid space" (Ekdahl, 2022). A similar understanding can be drawn for virtual sports, where the advanced games and online platforms express a single competitive environment that intrinsically links their physical surroundings with a virtual arena (Ekdahl, 2022). Reed et al.'s (2023) underpin this notion in their multi-perspective study on Zwift (i.e. an exemplary case of virtual sports), proposing the argument that virtual sports have an "*interreality*" (p. 198), based on a micro-sociological, gender, and moral inquiry. They argue that the physical and virtual domains are bridged to create a feedback loop where an athlete's physical exertion directly shapes their virtual experience, while the virtual environment's design reciprocally impacts their physical reality. Beyond physics, this interreality introduces unique sociological dimensions, such as impression management through avatar customisation and the accumulation of symbolic capital via in-game levels and equipment, fundamentally defining the social fabric of virtual sports. In addition, the interreality of virtual sports is also shaped by specific moral and gendered architectures like the potential to express and receive care (e.g. group rides, encouragements) or the algorithmic reproduction of gender norms (e.g. visual representations of gender and

strength) (Reed et al., 2023). It is however important to acknowledge that this interreality is partly possible because of Zwift's (the platform enquired by Reed et al.) unique features and its resemblances of a Metaverse where athletes can interact online and in real-time. In this way, a Metaverse-design and unique features incomparable to real-world scenarios (e.g. PowerUps) can be argued to enrich virtual sports' distinct sporting reality, though they should not be regarded as a prerequisite for virtual sports to exist (e.g. the UCI Cycling Esports World Championships happening on a local 'offline' network without gamification features).

Next to the arguments presented by Ekdahl (2022) and Reed et al. (2023), other evidence in the literature exists that indirectly supports the notion of virtual sports' physical-digital interreality. The studies of Westmattmann et al. (2022, 2025) discuss how virtual sports platforms (e.g., Zwift) rely on algorithms that introduce variables such as 'new' physics or gamified elements, which fundamentally alter the nature of the competition compared to its real-world counterpart (e.g., road cycling). Their studies put forward empirical evidence of how measured physical performance (input) does not perfectly predict the digital competition outcome (output). In some cases, only a minimum of 77% of the virtual results are explained by the athlete's physical performance while the remaining 23% is dependent upon the use of virtual, non-real, features and mechanisms (e.g. drafting dynamics, tactics in using gamification elements). While information on the exact mechanisms and parameters that generate this discrepancy remain undisclosed, it can be conceptually explained. Platforms aim to simulate complex real-world physics using proxy variables (i.e. simpler or reduced estimations). For instance, because standard smart trainers do not account for a rider's dynamic posture or shifting crosswinds, aerodynamic drag is calculated using variables that are known or calculated in real time: rider height, weight, (mathematic) positioning to other riders, in-game equipment (and likely other parameters that remain proprietary). Consequently, an avatar adopting an aerodynamic 'tuck' position on-screen is a visual animation (i.e. increasing the representation fidelity, cf. section 2.3) triggered by in-game speed thresholds, rather than a true reflection of the athlete's physical reality. Next to aerodynamic drag, many other mechanics of real-world cycling behaviour remain only limitedly captured or even uncaptured, such as terrain, weather conditions, vehicles in race, and rolling resistance, to name but a few.

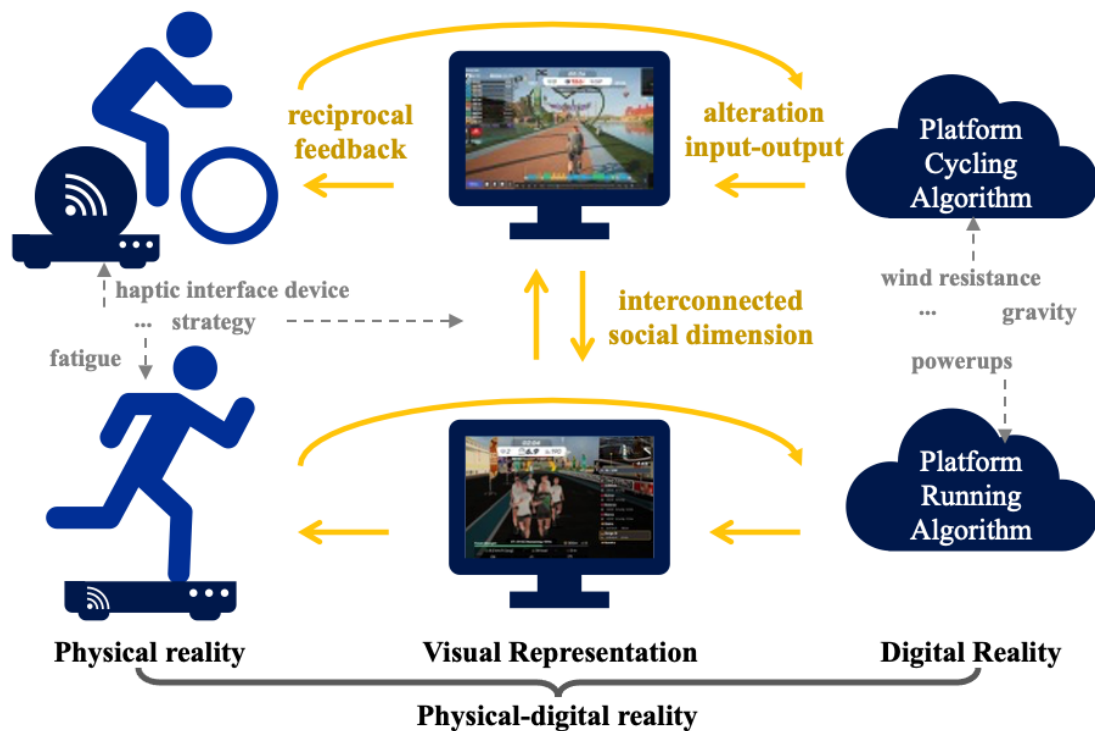
In addition, this discrepancy between physical performance and virtual performances resembles its own reality shaped by algorithms actively altering these physical performances when translating them into virtual performances. As shown in figure 8, this creates a feedback loop to which athletes ultimately have to adapt (Westmattmann et al., 2022, 2025). While these insights might be new to virtual sports in specific, adapting to the "rules of the game" as

part of the performance is not a new phenomenon. Esports literature has long described how esports players aim to master the “metagame” (S. E. Jenny, 2024, p. 178), something we covered in section 9.4.3. In this way, we argue that the reality of the virtual sport is one that requires athletes to understand and manipulate the platform’s unique physics alongside their physical exertion.

One last and more theoretical argument that indirectly supports the notion of virtual sports’ physical-digital interreality, can be found in Loland’s (2002) theoretical framework of sport technology, which distinguishes between “thin” and “thick” perspectives. Under a “thin” theory, technology serves merely as a passive tool or visual representation of reality; it does not fundamentally alter the nature of the activity, meaning “little is gained in terms of adding value to the sport” (Loland, 2002, p. 8). Simulations *in* sport, for example indoor rowing ergometer simply reproducing power metrics on a screen, struggle to find justification beyond this thin perspective. In contrast, virtual sports align strongly with Loland’s “thick” theory of sport. In this view, the technology is not passive but possesses a “fundamental constitutive function and value in sport” (Loland, 2002, p. 10). Because the virtual environment diverges from actual physical reality by introducing unique in-game features (e.g. PowerUps, platforms’ own “laws of nature” of gravity, rolling resistance, and air resistance), virtual sports introduce “technology that requires athlete efforts and skills” (Loland, 2002, p. 10) and ultimately establish a distinct physical-digital sporting reality unique to the virtual format of the sport. It is exactly this thick integration of technology, where virtual sports hardware or software shapes new skills alongside physical exertion, that helps to develop virtual sports into a discipline of its own rather than a “thin” digital reproduction and visualisation.

Critics like Parry and Giesbrecht (2023) fail to acknowledge this crucial distinction of virtual sports and simulations *in* sport (see section 9.2.2) despite the fact that it has significant implications for some of their conclusions, including: “we question whether [virtual sports] should be called ‘virtual’ sports at all, since measurement-virtuality does not make them any the less *real*” (p. 220). Yet, we argue that despite virtual sports indeed not being less *real*, they are also *not entirely* real as they are shaped through a reciprocal feedback loop between the physical (real) and the digital (virtual) reality, which we have summarised in Figure 9.

Figure 9: Schematic overview of virtual sports reality dimension. Source: own conception.



9.5.4 Governance and Institutionalisation

The dimensions of governance and institutionalisation in virtual sports have some parallels to those of esports. As indicated in Figure 5, virtual sports are increasingly experiencing formal institutionalisation and governance, though their rigour and authority are still debated upon. Somewhat similar to esports, the software platforms used in virtual sports are owned by game publishers (e.g., Zwift, MyWhoosh) (Fouquaert, Tjønndal, & Willem, 2025). Governance initially emerged ‘bottom-up’ organically through game publishers engaging in self-regulation of the sport on their platform. Yet, it appears they fail to regulate appropriately on aspects such as privacy, software-, and hardware-related manipulation, toxicity, and performance verification, to name but a few (Bjärehed & Bjärehed, 2022; Fouquaert, Tjønndal, & Willem, 2025; Richardson, 2025; Richardson et al., 2022). Virtual sports are also characterised by increased institutionalisation as recognised international sporting federations begin to acknowledge disciplines of virtual sport (e.g. virtual cycling as a formal cycling discipline) and seize partnerships with game publishers to regulate federation-sanctioned competitions (Dyer & Richardson, 2026; Tjønndal et al., 2025). For instance, the UCI officially regulates the Cycling Esports World Championships and its associated qualifying races in partnership with MyWhoosh. Similarly, World Triathlon partners with SuperTri to organise and regulate an indoor virtual triathlon World Championship (i.e. pool swimming, virtual cycling, and virtual running). Such collaborations and acknowledgements could be seen as the start for virtual

sports towards a formal degree of traditional institutional legitimacy.

Within such partnerships between publishers and traditional sport governing bodies, a new element is the bidding process sanctioned by governing bodies to decide which publisher's platform will be used as the "official" platform on which sport governing body sanctioned races will take place (Fouquaert, Tjønndal, & Willem, 2025; World Triathlon, 2024). While similar to other bidding processes in traditional sports and esports, for instance for hosting cities, it is uncommon to have a bidding process in esports over the actual games to be played. This is because the specific game itself is often a rather fixed starting point where tournaments and event branding are built around (McCauley & Baker, 2024). It appears that for virtual sports, the game itself is less of a fixation and could in that sense be understood as a means to organise virtual sports. In fact, governing bodies seek to develop standardisation across platform performance metrics and rankings, meaning that virtual sports tournaments could in the future host multiple games (e.g. both Zwift and MyWhoosh) (Fouquaert, Tjønndal, & Willem, 2025). However, and as already discussed above under the section of reality (cf. 9.5.3), allowing virtual sports tournaments to take place on multiple platforms does not appear evident because "the choice of virtual platform emerged as a significant concern among athletes [due to] differences in game physics across platforms". Hence, it would require substantial efforts in good governance (Fouquaert, Tjønndal, & Willem, 2025).

Despite this traditional oversight, the governance of virtual sports clearly presents unique and unprecedented regulatory challenges. That is, governing bodies must suddenly regulate entirely new digital frontiers alongside human physical performance. This includes overlooking hardware calibration, auditing proprietary software algorithms, preventing novel forms of digital and analogue cheating (e.g. body weight manipulation), implementing standards to mitigate differences between platforms' unique physical-digital realities, and more (Dyer & Taylor, 2023; Fouquaert, Tjønndal, & Willem, 2025; Richardson, 2025). Consequently, the governance dimension of virtual sports requires a complex blend of traditional sporting rules and digital regulations.

9.5.5 Following

The dimension of "following" reflects this institutional integration. Virtual sports have cultivated a substantial and dedicated participant and spectator base (Berger et al., 2025; Schwenker, 2025). Unlike the following dimension in esports, of which literature suggests minimal crossover exists with traditional sports fans (Bertschy et al., 2019), the virtual sports community shares cultural and demographic overlap with the traditional parent sport

(Fouquaert et al., 2026; Reed et al., 2023). Athletes and fans of traditional road cycling, for example, frequently engage with virtual cycling platforms for structured training, community group rides, or off-season competition (Evsikov et al., 2024). While we may argue this cultural fit further, little research exists about how virtual sports is perceived within the sporting niche or even society at large as a legitimate, modern extension of the traditional sporting landscape, rather than a completely isolated digital ecosystem. To better understand this argued “niche” positioning, virtual sports can be viewed through the lens of the Long Tail theory (Anderson, 2006). Because virtual sports platforms operate as digitised goods, they can aggregate geographically dispersed individuals and cater cost-effectively (i.e. online) to specialised interests rather than mass-market appeal (Elberse, 2008). Therefore, possessing a ‘niche’ follower base does not imply a small total number of participants, but instead indicates a highly segmented and stable audience that can be reached through online channels rather than physical ones. Nonetheless, as virtual sports’ global adoption continues to grow, this niche following could eventually shift towards a more mainstream one if it would ever be included in major “traditional” sports events such as the Olympic Games.

9.6 *A Universal Definition of Virtual Sports*

Through our systematic analysis across the eight dimensions of sport and a ninth dimension of reality (see section 9.1), it is evident that virtual sports occupy a highly specific configuration that separates them from sports, esports, and exergaming. Next to that, we also account for the central role of virtuality (i.e. using digital measurement to record and compare performances, see section 2.3) in virtual sports allowing for; (1) vast flexibility in remoteness, (2) realistic and immersive virtual representation of the performance, and (3) races that are online, real-time, and synchronous to asynchronous decoupled time trials. Putting these elements together allows us to propose a definition that accurately captures this complex phenomenon. We define virtual sports as:

*Institutionalised, rule-governed contests of human physical and cognitive skills
that operate within a distinct physical-digital interreality that are
characterised by voluntary human participation, a niche following, and high
movement decompression mimicking a traditional parent sport, made possible
through virtuality yet unbound to specific configurations of remoteness,
(a)synchrony or representational fidelity.*

In conclusion of this definition, Figure 5 synthesises our thorough literature review by systematically highlighting the nine constitutive dimensions of sport across sports-linked technological phenomena, exergaming, esports, and virtual sports. This analysis reveals that while exergaming and interactive training fulfil human and cognitive criteria, they lack formal governance and structured competition. Esports present a complex case, with much debate in the literature regarding several dimensions such as highly compressed physicality and institutionalisation.

This literature review and definition development demonstrates that while virtual sports align with and integrates some dimensions of sports, exergaming, and esports, they also present a new and unique configuration in other dimensions. As visualised in Figure 5, this is particularly evident within the dimensions of physicality and reality where virtual sports exhibit highly decompressed physicality and a distinct physical-digital interreality. Furthermore, they increasingly achieve institutional legitimacy through regulated events sanctioned by traditional sport governing bodies. Using these insights, we propose a universal definition of virtual sports in section 9.6. This, in turn, justifies our consideration of virtual sports as an emerging, distinct sporting discipline relative to their parent sports. Such consideration is a vital distinction in framing the subsequent empirical studies that explore the adoption and management of virtual sports as a legitimate physical-digital sporting discipline.

10 References Part III

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Part IV

Five empirical studies

11 Overview of Empirical Studies

As an introduction to our five empirical studies in Part IV (sections 12 to 16), we start by providing a schematic overview on the next page in Table 4. This table indicates the two different data collections that were performed as part of this dissertation. This is supplemented by a third dataset which was collected by Prof. Dr. Daniel Westmattmann. The latter dataset served as a complement to data collection #2 and was part of the co-authored mixed-methods study 5. Table 4 also shows how each data collection or dataset relates to the empirical studies. For each study, we also list a brief version of its main RQs, theories and methods used, and its publications status. This is followed by a full reference, which includes all co-authors and the journal in which the study was published or is currently still under peer review.

Table 4: Overview of empirical studies. “Manuscript” refers to unpublished works.

Data	Study #	RQ (in full in section 4)	Theory	Methods	Status
 Data collection #1 	Study 1	What are the adopter categories and determinants of virtual cycling adoption in clubs?	Organisational Innovativeness, triathlon and Diffusion of Innovation Theory	Survey to cycling clubs (n=70), parametric tests	Accepted for publication in International Journal of Sports Marketing & Sponsorship
		Fouquaert, T., De Clerck, T., & Willem, A. (2026). Understanding Determinants of Virtual Cycling Adoption Among Voluntary Sports Clubs. <i>International Journal of Sport Marketing and Sponsorship</i> . 10.1108/IJSMS-10-2025-0527			
  Data collection #2 	Study 2	What are clubs' beliefs about virtual cycling in relation to institutional logics?	Institutional Logics, Diffusion of Innovation Theory	Survey (n=90) to triathlon and cycling clubs, clustering	Published in Innovation: The European Journal of Social Science Research
		Fouquaert, T., Tjønndal, A., De Clerck, T., & Willem, A. (2025). Virtual cycling in sports clubs: Innovation adoption within institutional logics. <i>Innovation: The European Journal of Social Science Research</i> , 1–22. 10.1080/13511610.2025.2580521			
	Study 3	How is virtual cycling (de)legitimised by governing bodies?	Institutional & Disruptive Innovation Theory	Interviews (n=16) governing bodies & stakeholders, Collective Qualitative Analysis (CQA)	Accepted for publication in International Review for the Sociology of Sport
 Data set #3 	Study 4	Which governance issues pertain virtual cycling?	Good Governance	Interviews (n=16) governing bodies & stakeholders, CQA	Published in Performance Enhancement & Health
		Fouquaert, T., Tjønndal, A., & Willem, A. (2025). “There are a lot of vulnerabilities still left”: Governance Issues in Virtual Sports. <i>Performance Enhancement & Health</i> , 13(4), 100360. 10.1016/j.peh.2025.100360			
	Study 5	How do virtual cycling platforms shape competitions and their governance?	Good Governance	Mixed: Scraping (n=102), interviews (n=13) governing bodies & stakeholders, t-tests, CQA	Manuscript with major revisions in European Sport Management Quarterly
		Fouquaert, T., Westmattmann, D., Tjønndal, A., & Willem, A. (2025). <i>When Algorithms Define Sports: Algorithmic Differences and Governance Gaps in Virtual Cycling</i> . Manuscript submitted for publication			

12 Study 1: Understanding Determinants of Virtual Cycling Adoption Among Voluntary Sports Clubs

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Abstract

This study investigates the determinants of virtual cycling adoption by voluntary sports clubs, considering how determinants differ across and discriminate between adopter categories.

Using an integrated innovation model that bridges innovation diffusion theory and organisational innovativeness research, we surveyed Flemish (Belgium) sport clubs (n = 70). Clubs were clustered into adopter categories based on a composite innovativeness index (attitude, evaluation, adoption-intention). MANOVA and discriminant analysis assessed differences in determinants and their predictive power.

Not resources but human capabilities, such as perceptions about innovation attributes and managerial capacity, emerged as the strongest predictors of clubs' innovativeness, followed by human-related capacity (e.g. sufficient skilled volunteers). Compared to non-human-related and environmental capacity, these determinants significantly differ and discriminate between innovative, intermediate, and lagging clubs.

Findings support integrated models, highlighting the need to study both innovation-specific and organisation-specific determinants. Results suggest that adoption can be people-driven rather than resource-driven, which should receive greater attention in future sport marketing and management research.

Insights inform advice on knowledge management for federations and targeted marketing strategies for virtual cycling providers by identifying which perceptions and organisational factors to emphasise in their approach.

This study introduces club-level adoption insights of virtual cycling using an integrated approach that simultaneously considers organisation-specific and innovation-specific determinants. It also extends sport management literature by applying adopter categorisation to voluntary clubs using a composite index to measure club's innovativeness.

Keywords: Virtual cycling; Voluntary sports clubs; Innovation Diffusion; Organisational Innovativeness; Adoption Determinants; Adopter Categorisation

12.1 Introduction

The future of sports is set to increasingly take place online, opening a race to capture the next generation of sport participants or engage differently with existing audiences (Evsikov et al., 2024). Virtual sports, blending physical activity with mixed-reality environments, are redefining how people experience sport (Fouquaert, Tjønndal, De Clerck, et al., 2025). Virtual cycling is the virtual sport that has been embraced most rapidly by consumers, with platforms like Zwift or MyWhoosh surging in popularity. While virtual cycling can be viewed as a distinct sporting discipline affecting the nature and core dimensions of cycling as a sport (Fouquaert, Tjønndal, & Willem, 2025; Westmattelmann et al., 2025), this study conceptualises virtual cycling as a complementary technological service for engaging in cycling. From this perspective, virtual cycling classifies as an output-based service innovation based on Gerke and Dickson's typology (2025), which are innovations adopted by clubs to enhance value for their members such as hedonic or practical benefits of virtual cycling (e.g. safety, efficiency, motivational features, decoupling of space and time) (Fouquaert, Tjønndal, De Clerck, et al., 2025; Westmattelmann et al., 2021).

Within this context, we focus on voluntary sports clubs (here, cycling and triathlon) as the potential adopters of virtual sports (here, virtual cycling) as a club service complementary to their core outdoor disciplines. Despite the open race between clubs to cater new and existing audiences, many clubs remain hesitant about adopting virtual sports. Their reluctance can be considered a strategic blind spot, a missed opportunity and a threat to their future. If they fail to embrace virtual sporting activities, they risk becoming less relevant to the adapting needs of existing customers, to new digitally native audiences, or to miss out on new revenue streams and sponsorship models (Evsikov et al., 2024). Despite having many opportunities like redefining engagement, offering hybrid experiences, or enhancing retention, the adoption of virtual sports by clubs remains a gap in sports marketing literature.

Research on virtual sports adoption has primarily focused on virtual cycling adoption by individuals, elite sports or sport governing bodies (Evsikov et al., 2024; Fouquaert, Tjønndal, & Willem, 2025; Park et al., 2024; Tjønndal, 2022; Westmattelmann et al., 2021). Little is known about what motivates or discourages clubs from adopting virtual sports, even though these grassroots level clubs play a vital role in sport participation (Balduck et al., 2015). This gap in the literature further limits the development of targeted marketing strategies for virtual cycling providers and slows the diffusion of innovations that could enhance clubs' relevance in a rapidly evolving sports market.

To address this gap, our study investigates the determinants of virtual cycling adoption by voluntary sports clubs. Building on innovation diffusion research (Rogers, 2003) and organisational innovativeness research (Wolfe, 1994), we categorise clubs into adopter categories and examine how determinants differ across these categories. We consider both organisation-specific determinants (e.g., management, human resources) and innovation-specific determinants (i.e. perceptions about virtual cycling use in clubs like complementarity with the club), as prior research suggests both dimensions significantly shape adoption decisions (Corthouts et al., 2022; Damanpour & Schneider, 2006; Frambach & Schillewaert, 2002), yet they have rarely been considered simultaneously (Corthouts et al., 2020; Hoeber & Hoeber, 2012; Winand et al., 2013; Winand & Anagnostopoulos, 2017).

This approach offers three contributions. First, it extends sport management and marketing literature by examining virtual cycling adoption at the club level. Second, it responds to the call for integrated models by simultaneously considering organisational- and innovation-specific determinants (Frambach & Schillewaert, 2002). Third, it offers practical guidance for virtual cycling providers and marketers by highlighting which perceptions shape virtual cycling adoption to prioritise in communication strategies and how adopter segmentation can inform targeted marketing strategies.

Relying on adopter category segmentation (Corthouts et al., 2022; Rogers, 2003), we pose the following research questions (RQs): (1) To what extent do voluntary cycling and triathlon clubs adopt virtual cycling? (2) Which determinants of virtual cycling adoption differ across adopter categories, and to what extent can these determinants effectively discriminate between those categories? (3) Which implications does this hold to inform marketing strategies for virtual sports?

12.2 Defining Clubs' Innovativeness to Distinguish Adopter Categories

In order to categorise clubs based on their *innovativeness*, in our case related to virtual cycling adoption (RQ1), it is important to first define clubs' innovativeness. Based on adjacent research (Corthouts et al., 2022; Damanpour & Schneider, 2006; Frambach & Schillewaert, 2002; Wolfe, 1994), the present study defines organisational innovativeness as a composite metric of three measures: how clubs feel about virtual cycling (attitude), how they assess its value (evaluation), and whether they plan to start using it (adoption-intention). These three aspects of innovativeness are rooted in innovation diffusion and acceptance models and are each closely related to actual innovation adoption and use (Davis et al., 1989; Rogers, 2003; Venkatesh &

Brown, 2001).

These three aspects of organisational innovativeness can be integrated into an index that can be used to distinguish and categorise clubs based on their innovativeness (Corthouts et al., 2022; Martínez & Polo, 1996). Adopter categories are an insightful analytical tool to use in subsequent analyses, like assessing the differences in determinants of innovative clubs compared to less innovative clubs (RQ2). This approach has for instance been used by Corthouts et al. (2022) to study determinants of innovativeness within sport federations. The innovation diffusion theory (Rogers, 2003) distinguishes universally applied categorisation of potential adopters (e.g. clubs) based on their innovativeness. This theory proposes five normally distributed adopter categories: Innovators (2.5%), Early Adopters (13.5%), Early Majority (34%), Late Majority (34%), and Laggards (16%). A high level of heterogeneity in innovativeness is observed between categories while high homogeneity in innovativeness is observed within categories. This is due to the fact that potential adopters within each category share common characteristics (e.g. managerial capacity, human resources) and perceptions about the innovation (e.g. perceived complexity) that distinguish them from other categories (Rogers, 2003).

12.3 Organisational Innovation Adoption: Theoretical Framework

To address the inconsistencies often found in single-lens studies, this study employs an integrated innovation model (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002; Wolfe, 1994). Wolfe (1994) critiqued earlier organisational innovativeness research for not admitting the perceptually-based nature of innovation attributes making these attributes essentially organisation-contingent, whereas treating them as generic innovation-contingent attributes “confounds and dilutes research outcomes” (p. 409). Consequently, we bridge innovation diffusion theory (i.e. what is the rate and patterns of innovativeness) and organisational innovativeness theory (i.e. what determines the innovativeness of an organisation) (Wolfe, 1994). While clubs are the unit of analysis for assessing and categorising them on their innovativeness (RQ1), we follow Wolfe’s recommendation to advance to “innovation-in-an-organisation” (p. 409) as the unit of analysis when exploring how determinants vary across adopter categories (RQ2) (Wolfe, 1994). We combine organisation-specific determinants (managerial, capacity, environmental) (Damanpour & Schneider, 2006) with innovation-specific determinants living *in-an-organisation* (perceived innovation attributes) (Rogers, 2003) to capture the dynamic interaction between organisations’

innovativeness, their characteristics and the specific technology. While employing integrated innovation models (i.e. the combined use of these different streams of research and determinants) has thus been long advocated for (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002; Wolfe, 1994), its use within studies on sport organisations' innovativeness remains unused (Corthouts et al., 2020, 2022, 2023; Delshab et al., 2022; Hoeber & Hoeber, 2012; Winand et al., 2013; Winand & Anagnostopoulos, 2017; Winand & Hoeber, 2017).

Table 5: Constructs, subdimensions, and summary definition (study 1)

Construct & subdimensions	Summary definition
<i>Organisation-specific determinants</i>	
<i>Managerial determinants</i>	
Innovation champion	Having evangelists, supporters of virtual cycling
Pro-innovation characteristic	Positive board/volunteers around new changes
Subjective knowledge	Confidence to handle & use virtual cycling
<i>Human-related capacity determinants</i>	
Coaches for adults	Sufficient qualified coaches for adults
Coaches for youth	Sufficient qualified coaches for youth
Volunteer sufficiency	Having sufficient volunteers to operate
Volunteer quality	Having proficient volunteers to operate
Board	Sufficient qualified board members to operate
Inter-organisational relationships	Pursuing external collaborations and contacts
<i>Non-human-related capacity determinants</i>	
Financial	Having sufficient finances to operate
Infrastructure	Having sufficient facility access to operate
Planning and development	Having long-term policy outlines
<i>Environmental determinants</i>	
Competitive environment	Having to compete for resources and members
<i>Innovation-specific determinants</i>	
<i>Perceived innovation attributes (of virtual cycling)</i>	
Relative advantage	Superiority over earlier practices
Compatibility	Compatibility with existing operations and beliefs
Image	Raising the clubs' image and status when adopted
Perceived ease of use	Simple to understand and integrate in the club
Trialability	Can be easily and inexpensively experimented with
Visibility	Visibly in its use or integration towards others.
Result demonstrability	Direct results that are observable and measurable
Perceived enjoyment	Accompanying fun-factor for members when using

In that sense, our integrated innovation model approach is relatively novel and contributes to sport management and marketing literature. We refer to Table 5 for a guiding overview of the considered determinants and their subdimensions that will be discussed throughout the literature review that follows.

12.3.1 Organisation-Specific Determinants of Organisational Innovativeness

Organisational innovativeness research aims to uncover innovation adoption determinants that help explain the extent to which organisations are innovative (Damanpour & Schneider, 2006; Rogers, 2003). Damanpour and Schneider (2006) suggest that innovativeness within organisations is determined by three distinct dimensions: (1) managerial determinants; characteristics of the individuals who manage the organisation, (2) organisational determinants; characteristics of the organisation, and (3) environmental determinants; the contextual factors.

12.3.1.1 Managerial Determinants

Extant literature reveals that club volunteer characteristics such as attitudes, commitment, and knowledge are positively related with clubs' innovativeness (Hoeber & Hoeber, 2012; Winand & Anagnostopoulos, 2017; Winand & Hoeber, 2017). Three managerial determinants have been consistently identified. First, presence of innovation champions, i.e., individuals who possess social skills, a compelling vision towards change, and a commitment to pursue innovation integration (Hoeber & Hoeber, 2012; Rogers, 2003; Winand & Hoeber, 2017; Wolfe, 1994). Second, pro-innovation attitudes of key volunteers (e.g. board) such as perceived need for change, risk-taking, openness to change, forward-thinking, challenging the status quo, and visionary leadership (Corthouts et al., 2020, 2023; Crossan & Apaydin, 2010; Hoeber & Hoeber, 2012; Winand et al., 2013, 2016; Winand & Anagnostopoulos, 2017). These attitudes appear moreover critical during the initial adoption stages, when board approval is required (Hoeber & Hoeber, 2012). Third, volunteers' subjective knowledge or self-perceived degree of expertise in the innovation topic (Corthouts et al., 2022; Evsikov et al., 2024). Delshab et al. (2022) advanced the notion of knowledge as "an essential organisational process that leverages all value-creating activities, such as innovativeness" (p. 152), demonstrating the importance of active knowledge management like acquiring and sharing internal and external knowledge.

H1. Managerial determinants are higher for more innovative adopter categories than for less innovative adopter categories.

12.3.1.2 Human-related organisational capacity determinants

The *capacity* of non-profit organisations, i.e. their “ability to draw on or deploy a variety of types of organisational capital” (Hall et al., 2003, p. 4), is also a determinant of innovativeness (Winand & Hoeber, 2017). To maintain parsimony and increase comparability with existing studies measuring similar variables (for example, Balduck et al., 2015; Corthouts et al., 2020, 2023; Delshab et al., 2022; Doherty & Cuskelly, 2019; Winand et al., 2013, 2016; Winand & Anagnostopoulos, 2017), we split organisational capacity in human-related capacity (human resources and external relationships) and non-human-related capacity (financial resources, infrastructure, planning & development).

Human-related capacity encompasses human resources, i.e., the number, competencies, and attitudes of volunteers, board members, and coaches, and external relationships, which refer to collaborations with external partners that provide expertise unavailable within the club (e.g. for technology use) (Doherty & Cuskelly, 2019; Hall et al., 2003). Human resources are consistently found to drive indicators like clubs’ performance, perceived need for change, and innovativeness (Balduck et al., 2015; Corthouts et al., 2023; Doherty & Cuskelly, 2019; Winand & Hoeber, 2017), while external collaborations foster a 'shared language' that supports successful innovation adoption (Hoeber & Hoeber, 2012; Winand & Hoeber, 2017). Svensson et al. (2020) do however note that Hall et al.’s capacity dimensions (e.g., number of volunteers) may not sufficiently capture volunteers’ specialised skillset or dedication often required to successfully innovate, which are dimensions included in for example the Bridgespan Group’s capacity model used in their study.

H2. Human-related capacity determinants are higher for more innovative adopter categories than for less innovative adopter categories.

12.3.1.3 Non-human related organisational capacity determinants

Non-human-related capacity includes financial resources, infrastructure capacity, and planning & development capacity (Doherty & Cuskelly, 2019; Hall et al., 2003). Financial resources like stable revenues, stable expenses, and alternate sources of revenues can enable innovation (Hoeber & Hoeber, 2012; Winand & Hoeber, 2017), yet clubs facing financial constraints may also have more incentives to innovate and develop creative solutions (Winand et al., 2013). Infrastructure captures the availability, use, and quality of day-to-day facilities, where sufficient access and having control through possession may facilitate innovation (e.g., hardware installation) (Hoeber & Hoeber, 2012; Winand & Hoeber, 2017), though others found no

difference for owning versus renting (Corthouts et al., 2020). Planning and development pertain to clubs' strategic planning process and degree of formalisation (Balduck et al., 2015). While formal structures can support strategic planning through R&D or reward systems (Hoeber & Hoeber, 2012; Winand et al., 2013), some bureaucratic regulations imposed by federations may also constrain strategic planning - and hence innovation - in clubs (Winand & Hoeber, 2017). While each aforementioned dimension of non-human-related capacity appears to be required for significant and complex (e.g. infrastructural) innovations, several studies argue that the nature of online technological service innovation like virtual cycling is less resource-intensive (Evsikov et al., 2024; Fouquaert, Tjønndal, De Clerck, et al., 2025), lessening the effect of non-human-related capacity on its adoption (Winand et al., 2016; Wolfe, 1994). Consequently, we assume that non-human related capacity is not a prerequisite for initiating adoption, and hence not different among adopter categories.

H3. Non-human-related capacity determinants between club adopter categories are equal.

12.3.1.4 Environmental Determinants

Factors external to clubs may also impact their innovativeness. First, the engagement of external stakeholders as reflected by the interest and number of stakeholders can stimulate clubs' innovation adoption (Hoeber & Hoeber, 2012). However, this overlaps with the earlier discussed external relationship determinant within human-related capacity (Doherty & Cuskelly, 2019). Second is perceived competition or organisations' competitive position: clubs that face intense regional competition may implement innovations to attract volunteers, members, and resources (Winand et al., 2013; Winand & Hoeber, 2017).

H4. Environmental determinants are higher for more innovative adopter categories than for less innovative adopter categories.

12.3.2 Innovation-Specific Determinants of Organisational Innovativeness

Beyond organisation-specific determinants, several impactful studies have advocated for complementing research on organisational innovativeness with innovation-specific determinants, i.e. the perceived attributes of an innovation living *in* the organisation (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002; Wolfe, 1994). Neglecting these attributes ignores the fact that the same innovation can be perceived differently by each organisation (Corthouts et al., 2020, 2022; Rogers, 2003). For instance, to what extent virtual cycling can be

easily implemented in existing club services, can be compatible with the club, or might offer the club a relative advantage. Combined, they explain clubs' likelihood of adopting an innovation (e.g. adoption-intention towards virtual cycling) and are well established in innovation diffusion frameworks and acceptance models (Davis et al., 1989; Moore & Benbasat, 1991; Rogers, 2003; Venkatesh & Brown, 2001). Despite their high determining power, innovation-specific determinants were not included in many sport innovativeness studies (e.g. Hoeber & Hoeber, 2012; Winand et al., 2013, 2016; Winand & Anagnostopoulos, 2017).

12.3.2.1 Perceived innovation attributes

We rely on Rogers' (2003) perceived innovation attributes (e.g. relative advantage, compatibility, perceived ease of use), which are explained in Table 5. These five attributes are found to be positively related to clubs' innovativeness (Corthouts et al., 2020, 2022). Based on further literature review in a context outside sport (Karahanna et al., 1999; Moore & Benbasat, 1991; Venkatesh & Brown, 2001; Wolfe, 1994), we also included additional attributes repeatedly found as relevant such as image, visibility, result demonstrability, and perceived enjoyment (see explanation of each in Table 5). All of these attributes were consistently found to be positively related to innovation adoption.

H5. Perceived innovation attributes are higher for more innovative adopter categories than for less innovative adopter categories.

12.4 Methodology

12.4.1 Data Collection

Regarding our population and sampling strategy, clubs had to comply with the following criteria to be included in our study: (1) actively offer sports services (e.g., facilitate group training sessions) to their members that (2) are accessible to the public (i.e., everyone can become a member) and (3) be a member of one of the recognised Flemish cycling federations or the Flemish Triathlon Federation. To recruit clubs, the triathlon federation provided a club register that met our target population criteria (as confirmed by the federation), which allowed us to invite all triathlon club representatives in Flanders by email to take our survey. For the cycling clubs, however, the definition of "club" as used in the Flemish cycling federations' registers is very broad. We were made aware by the cycling federations that over half of the register would not meet our selection criteria (e.g., actively offering public sports services),

preventing a guaranteed match. Consequently, we resorted to a self-selection sampling strategy that maximised the chances of cycling clubs being included by spreading the survey as widely as possible. An invitational email to complete our survey was sent out to 301 municipal sport services with the question to forward our email to cycling clubs within their municipality, while additional dissemination was done through the cycling federations' email newsletter (n=3), social media channels (n=6) and at local sport events (n=7). The survey ran for 12 weeks starting from June 2022. Informed consent to participate was obtained for all clubs (see Ethical Statement).

Regarding our unit of analysis, this study considers (innovation-in-)clubs as the unit of analysis. Consequently, we surveyed club representatives (i.e. volunteers with operational oversight like board members or head coaches) rather than general members, whose engagement is typically limited to consuming club services (e.g. training) rather than perform management duties. Surveying key volunteers seemed appropriate since the decision-making process of clubs is “determined by key actors” (Schlesinger et al., 2015, p. 204). We communicated to clubs that only *one* volunteer who is “*highly involved in their club (e.g. board, head coach, highly engaged volunteer)*” may fill in our survey. While we did not explicitly control for their involvement, being able to answer all required questions on capacity, strategy, mission, etc., seems an appropriate proxy.

In total, we collected 100 responses from clubs that have been verified as distinct based on their name given the anonymity at individual level. We checked and cleaned these responses in terms of target population, unreliable or invalid responses (i.e. based on comments, conflicting or all same answers, attention checks), and missing values on variables used in further analyses, after which 70 responses (=n) remained, each response representing a distinct club. We reached around 20% triathlon clubs and 80% cycling clubs in our sample. The majority of clubs (83%) offered public sports services at least weekly. The average number of members was 75 (min = 10, max = 303, median = 50), 93% of clubs had one or more coaches and 20% of clubs had one or more youth coaches. This demographic profile confirms that our sample represents established organisational entities rather than informal groups.

12.4.2 Operationalisation

The survey began with a clear explanation of what virtual cycling is through an exemplary video of Zwift. Following the constructs listed in Table 5, innovation-specific determinants were measured using adapted scales from Moore and Benbasat (1991) and Venkatesh & Brown (2001), while organisation-specific determinants (managerial, organisational, environmental

capacity) were based on Howell et al. (2005), Winand et al. (2013), Balduck et al. (2015), and Doherty and Cuskelly (2019). Where possible, validated scales from literature were used for operationalisation (see section 12.9.1 Appendix). The used scales were subjected to a pilot test ($n = 64$) using a convenience sample of athletes and club volunteers familiar with cycling. Most scales came out reliable ($\alpha > .80$) in the pilot test and those that did not (i.e. often scales with reversed items) had the unreliable items dropped or changed accordingly. In our actual sample, all scales rendered adequate reliability with a Cronbach's Alpha (α) above .80 except for the scale of 'volunteer quality' ($\alpha = .79$). Only for the scale of 'complexity', one item was dropped to increase the reliability from α of .75 to .84 as noted in section 12.9.1. Common method variance was addressed ex ante by measuring all determinants on a 5-point Likert scale while measuring the adopter categorisation variables (i.e., attitude, evaluation and adoption-intention) on a 7-point semantic differential scale.

Individual scales, all thus having adequate reliability, are mean scales that consist of two to six items (see section 12.9.1 Appendix). The classification of determinant constructs (i.e. grouping scales as shown in Table 5) follows established theoretical frameworks (Hall et al., 2003; Rogers, 2003; Winand & Hoeber, 2017; Wolfe, 1994) to allow optimal results comparison to other studies. Each determinant used in our analysis is the mean of their underlying individual scales.

12.4.3 Adopter categorisation

Similar to Corthouts et al. (2022) who clustered their sample of 37 sport federations based on their innovativeness, we categorised clubs by clustering their scores on attitude, evaluation, and adoption-intention using a two-step procedure: Ward's hierarchical clustering (squared Euclidian) of which the optimal solution was used as non-random starting points for iterative k-means clustering. Comparing cluster solutions based on their parsimony, interpretability, Dendrogram and statistical validity, we prioritised a three-cluster solution ($R^2 = .76$) over five clusters ($R^2 = .84$), as the marginal variance gain and slightly lower Duda-Hart index (0.60 vs 0.67) failed to justify the added complexity and loss of stability (smallest cluster $n=6$) in a 5-cluster solution. Also, the Pseudo- T^2 indices indicated superior distinctness for three clusters (15.34) compared to four (7.70) or five (15.17).

While Rogers (2003) traditionally distinguishes five adopter categories, our optimal cluster solution in terms of statistical validation, parsimony and interpretability yields three clusters. Aggregating the theoretical adopter categories from five to three has been performed and validated by previous studies to mitigate the instability of clusters (for an overview of

studies and examples, see Corthouts et al., 2022). Specifically, this has been done by merging the categories in the tails of Rogers' diffusion curve, i.e. Innovators with Early Adopters, and Laggards with Late Majority. This is an approach theoretically grounded in the three distinct diffusion phases of (1) initiation, (2) intermediate mass acceptance, and (3) saturation identified by Martínez and Polo (1996) which maximise the discriminant power of the analysis without compromising the theoretical trajectory of innovation diffusion.

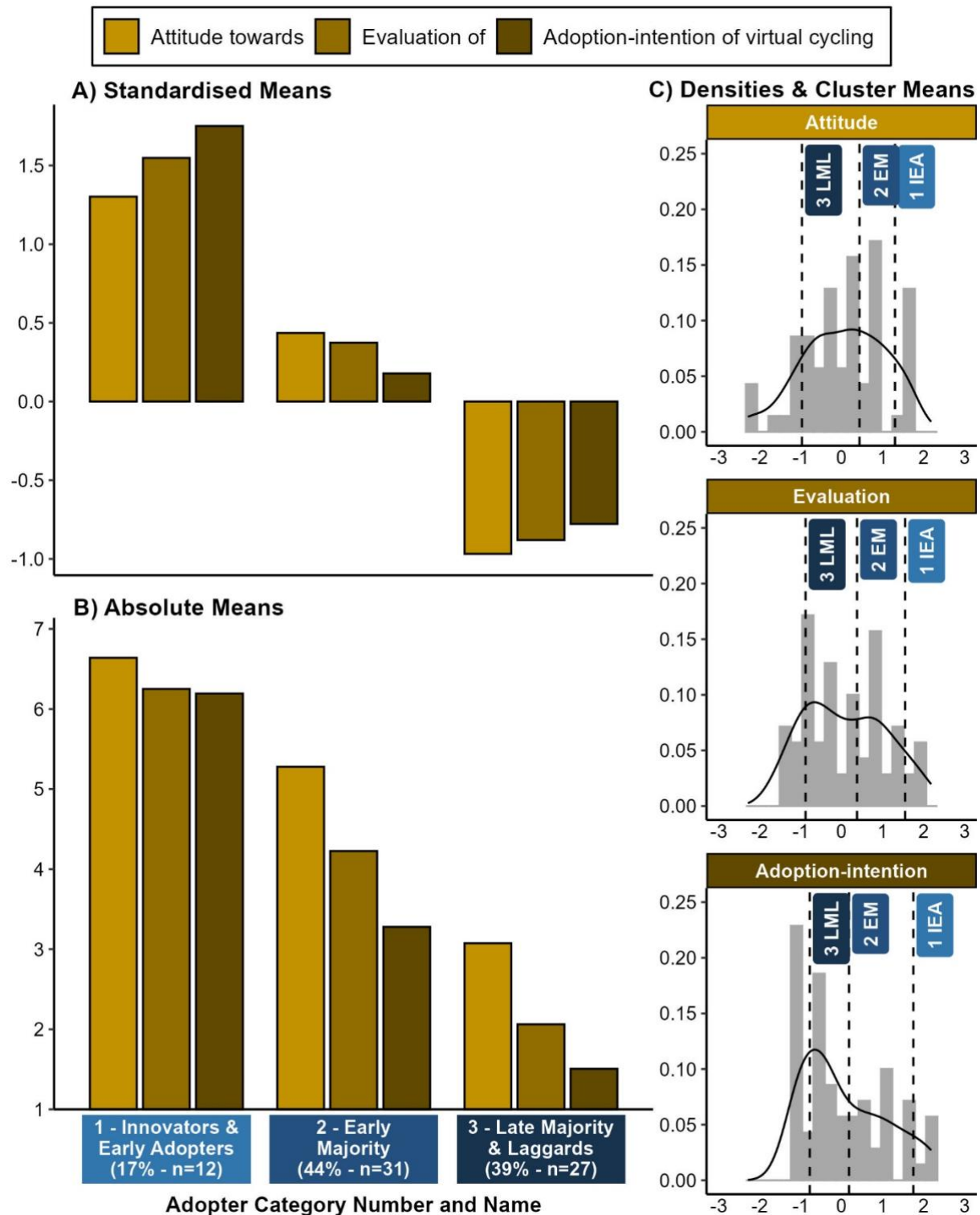
This 3-cluster solution was utilised in a multivariate analysis of variance (MANOVA) to simultaneously compare differences of five innovation determinants between clubs scoring differently on innovativeness (i.e. between clusters). We continued with a linear discriminant analysis (LDA) to test whether the significant different determinants from our MANOVA can sufficiently discriminate among and predict our three adopter categories (clusters). All data met the assumptions for MANOVA and LDA, including multicollinearity ($VIF < 3$), linearity, heteroscedasticity, homogeneity of variance, and normality of residuals. All analyses were conducted in R 4.5.2.

12.5 Results

12.5.1 Adopter categories

Our 3-cluster solution based on clubs' attitude towards, evaluation of, and adoption-intention of virtual cycling, demonstrated the presence of innovative, intermediate and lagging clubs across their three innovativeness measures. Consequently, the clusters were labelled as followed: (1) Innovators & Early Adopters ($n = 12$, 17%), (2) Early Majority ($n = 31$, 44%) and (3) Late Majority & Laggards ($n = 27$, 39%). Our three clusters each showed similar standardised scores for all three of their innovation measures attitude, evaluation and adoption-intention, with a maximum within-cluster difference of 0.45. As presented in Figure 10, the Innovators & Early Adopters cluster consistently displays the highest scores, both standardised (Figure 10 A) and absolute (Figure 10 B) across all measures (attitude $M = 1.30$, $SD = 0.31$; evaluation $M = 1.55$, $SD = 0.37$; adoption-intention $M = 1.75$, $SD = 0.39$). The Early Majority cluster also showed similar scores within but relatively lower, more intermediate, scores between clusters (attitude $M = 0.44$, $SD = 0.50$; evaluation $M = 0.37$, $SD = 0.55$; adoption-intention $M = 0.18$, $SD = 0.65$). The Late Majority & Laggards cluster as well showed similar scores within their cluster but consistently negative standardised scores, which are the relative lowest scores between clusters (attitude $M = -0.97$, $SD = 0.58$; evaluation $M = -0.88$, $SD = 0.43$; adoption-intention $M = -0.78$, $SD = 0.31$).

Figure 10: Clustering of clubs based on innovativeness towards virtual cycling (study 1). Subfigures A and B respectively show standardised and absolute mean scores (Y-axis) for each cluster (X-axis) across three innovativeness measures. Subfigure C shows histogram and density plots of standardised innovativeness measures with the means of each cluster plotted as dashed vertical line. Higher bars in subfigure A and B represent stronger innovativeness toward virtual cycling, while the similar densities and aligned dashed lines in subfigure C illustrate how the clustering variables share similar distributions and are well spread between clusters yet equivalent within clusters.



Looking closer at the density plots in Figure 10 (C), we see that each adopter category is distinctly separated along the spectrum of each measure. The cluster means (dashed vertical lines) align neatly across attitude, evaluation, and adoption-intention, indicating their appropriateness to be used as an index of innovativeness.

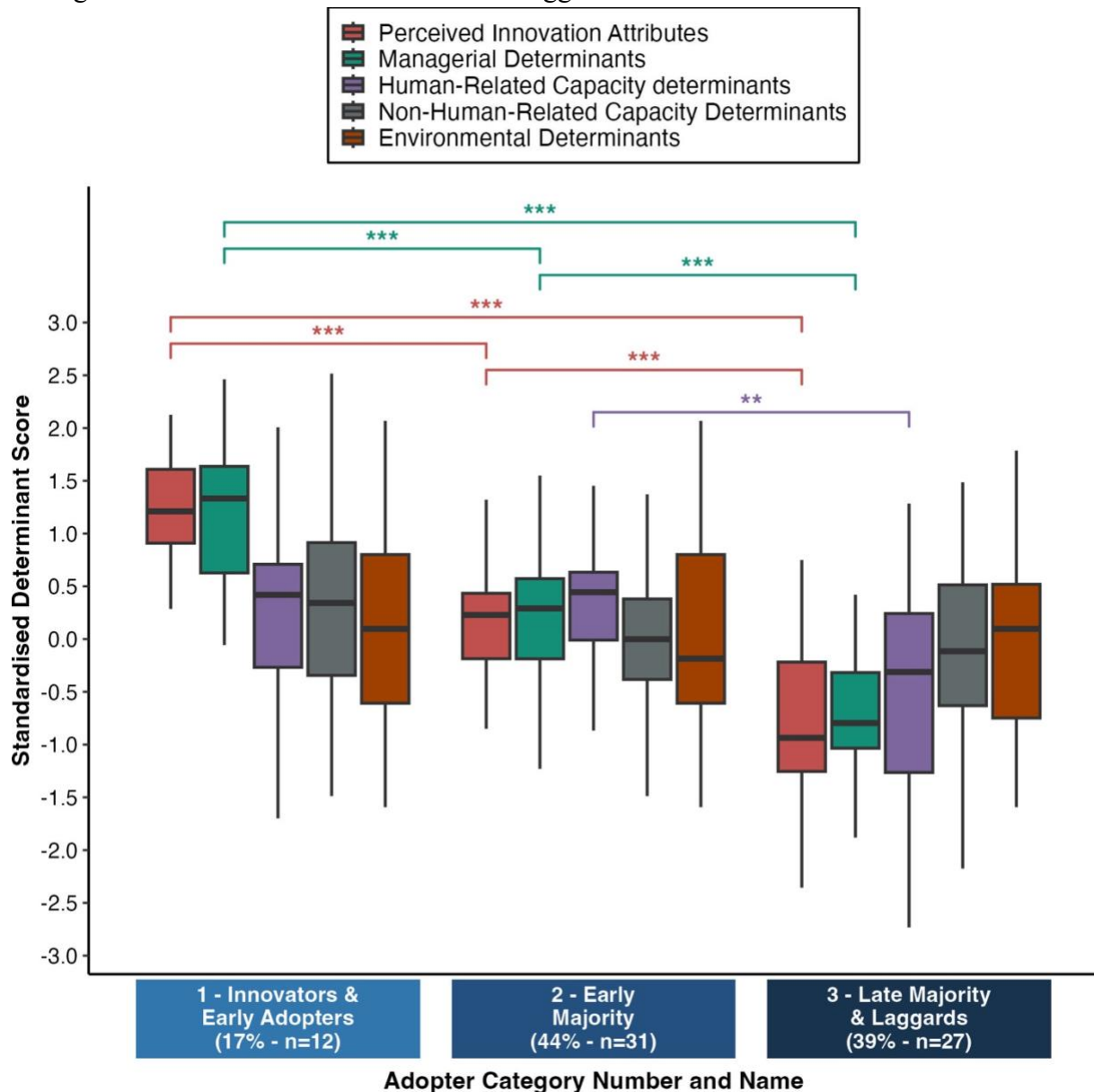
12.5.2 Differences in determinants between adopter categories

Determinants of innovativeness scores across our sample all varied between 2.4 and 3.4 on 5, i.e. perceived innovation attributes ($M = 2.8$, $SD = 0.7$), managerial ($M = 3.2$, $SD = 0.6$), human-related capacity ($M = 3.4$, $SD = 0.7$), non-human-related capacity ($M = 3.4$, $SD = 0.6$), and environmental determinants ($M = 2.4$, $SD = 0.9$).

Figure 11 visually presents a MANOVA type III ($F(2, 126) = 9.95$, $p < .001$, Wilk's $\Lambda = 0.31$, $\eta_p^2 = .39$). To interpret Figure 11, standardised determinant scores are shown in boxplots per cluster, while the brackets at the top highlight the significant differences in determinants between clusters. Post-hoc (Tukey's HSD) we see significant different scores for perceived innovation attributes and managerial determinants based on clubs' adopter category (i.e. cluster). The boxplots in Figure 11 show that both these scores are highest for Innovators and Early Adopter clubs, slightly lower for Early Majority clubs and absolute lowest for Late Majority & Laggard clubs, confirming H5 and H1. Human-related capacity determinants' score is significantly higher for Early Majority clubs compared to Late Majority & Laggard clubs, while there is no significant difference in human-related capacity determinants' score between Innovators and Early Adopter and other adopter categories, which partially confirms H2. No significant differences are observed between adopter categories for non-human-related capacity determinants and environmental determinants, confirming H3 but rejecting H4.

Our MANOVA shows a large effect ($\eta_p^2 = .39$), indicating adopter category variance is substantially explained by both innovation-specific (perceived innovation attributes) and organisation-specific determinants (managerial determinants and human-related capacity). Individual effect sizes are highest for perceived innovation attributes ($\eta^2 = .56$) and managerial determinants ($\eta^2 = .50$), followed by human-related capacity ($\eta^2 = .16$). Non-human-related capacity and environmental determinants have smaller effects ($\eta^2 = .04$ and $\eta^2 = .01$). This highlights the equal importance of perceived innovation attributes and managerial determinants in distinguishing innovative, intermediate, and lagging clubs.

Figure 11: Determinant boxplots of innovativeness across adopter categories (study 1). Boxplots showing standardised determinant scores across adopter categories (clusters), with significant differences indicated between determinants (Tukey's HSD; ** $p < .01$, *** $p < .001$). The horizontal brackets highlight exactly which adopter categories significantly differ from one another, showing a step-down effect in perceived innovation attributes and managerial determinants from innovators to laggards.



12.5.2.1 Differences in determinant subdimensions between adopter categories

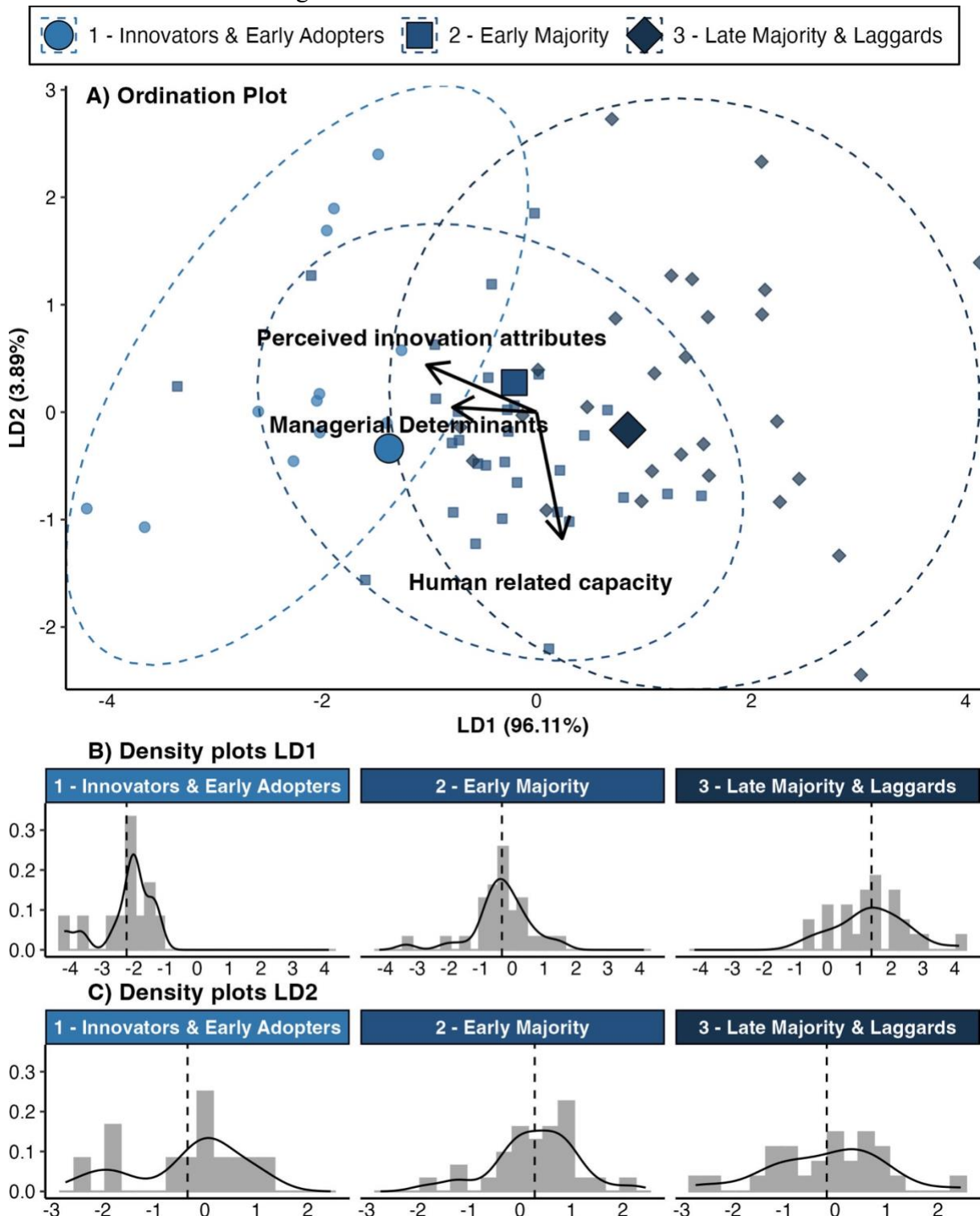
Subsequent ANOVAs with Tukey's HSD on the subdimensions (explained in Table 5) of prior described significantly different determinants show that all subdimensions of perceived innovation attributes differ across all adopter categories ($p < .05$), generally highest for Innovators & Early Adopters and lowest for Late Majority & Laggards. Exceptions are image

(no difference between Early Majority and Late Majority & Laggards) and trialability (no difference between Innovators & Early Adopters and Early Majority). For managerial determinants, innovation champion and subjective knowledge follow the same trend to differ across all adopter categories ($p < .05$), while pro-innovation characteristics only differ between Innovators & Early Adopters and Late Majority & Laggards ($p < .05$). For human-related capacity only two subdimension differ: coaches for adults differs between Innovators & Early Adopters and Late Majority & Laggards ($p < .01$) and between Early Majority and Late Majority & Laggards ($p < .001$); volunteer quality differs between Early Majority and Late Majority & Laggards ($p < .05$).

12.5.3 Determinants discriminating amongst adopter categories: Discriminant analysis

Our LDA showed that perceived innovation attributes, managerial determinants, and human related capacity can significantly discriminate among our three adopter categories based on two functions: LD1 ($F(6, 130) = 15.19$, $p = <.001$, Wilk's $\Lambda = 0.34$) and LD2 ($F(2, 66) = 2.28$, $p = 0.111$, Wilk's $\Lambda = 0.935$). LD1 significantly discriminates between adopter categories and explains 96% of the variance (canonical $R^2 = .63$, Eigenvalue = 1.71), while LD2 did not significantly discriminate between adopter categories and explains 4% of the variance (canonical $R^2 = .06$, Eigenvalue = 0.07). Cross-validation tests show that 79% of clubs were correctly classified within their adopter category and Press' Q test confirmed that the cross-validated classification is significantly better than chance ($p < 0.01$). The coefficients in Table 6 show that perceived innovation attributes and managerial determinants have high discriminant value for LD1, while human related capacity has high discriminant value for LD2. To visualise how these determinants discriminate between and predict a club's adopter category, Figure 12 breaks down the results of our LDA. The ordination plot (Figure 12 A) shows how adopter categories spread out along discriminant functions LD1 and LD2, while the arrows (ordination vectors) show the direction and strength of the determinants. The vectors for perceived innovation attributes and managerial determinants both point strongly to the left (affecting LD1), distinguishing Innovators & Early Adopters clubs (1) away from the Early Majority clubs (2) and Late Majority & Laggards clubs (3). Meanwhile, the vector for human-related capacity points downward (affecting LD2) which distinguishes Early Majority clubs (2) from Late Majority and Lagging clubs (3).

Figure 12: virtual cycling adopter category ordination plot (study 1). Subfigure A shows an ordination plot of adopter categories placed by LD1 and LD2 with group centroids as larger shapes and dashed ellipses indicate confidence values. Subfigures B and C show histogram and density plots of linear discriminant functions LD1 and LD2 respectively across adopter groups with group centroids as vertical dashed line. The direction and length of the arrows in subfigure A indicate which determinants most strongly drive adopter category separation, while a wider distance between group centroids seen in all subfigures demonstrates stronger discrimination between categories.



The density plots for LD1 (Figure 12 B) show group centroids (dashed lines) that are widely spread apart, confirming the strong predictive power of innovation attributes and managerial determinants via LD1. The density plots for LD2 (Figure 12) show much more overlap as group centroids (dashed lines) line up closely, confirming that human-related capacity (via LD2) offers only marginal predictive power.

Table 6: Discriminant functions standardised structure coefficients (study 1), which measure the unique partial contribution of each determinant to the discriminant functions. Variables with the largest weights (in bold) contribute most heavily to the calculation of the function.

Determinant	Function 1 (LD1)	Function 2 (LD2)
Perceived innovation attributes	0.73	0.31
Managerial Determinants	0.55	0.03
Human related capacity	0.22	1.08

12.6 Discussion

This study distinguishes adopter categories of clubs based on a composite measure of their *innovativeness* related to virtual cycling adoption as a complementary technological service (Figure 10). In addition, we demonstrate differences in determinants of club innovativeness among adopter categories (Figure 11), and how these can discriminate between adopter categories (Figure 12). In essence, our results support the use of an integrated innovation model combining “innovation-in-an-organisation”-specific determinants and organisational-specific determinants (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002; Wolfe, 1994, p. 409). We reveal that the adoption of virtual cycling as complementary online technological service is primarily driven by a combination of both types of determinants, and specifically by “soft” determinants (e.g. innovation perception, managerial attitudes, volunteers) rather than “hard” determinants like infrastructure or financial capacity.

12.6.1 Innovative, intermediate and lagging clubs, distinguished by “soft” determinants

To explain *the rate and patterns of innovativeness* (RQ1), we rely on Rogers’ (2003) theoretical underpinned adopter categories. Our resulting adopter categories (Figure 10) distinguish innovative (17%), intermediate (44%) and lagging (39%), which are well in line with the proportions of adopter categories in innovation diffusion theory: Innovators together with Early Adopters (16% vs our 17%), Early Majority (34% vs our 44%) and Late Majority together with Laggards (50% vs our 39%) (Rogers, 2003). Such close resemblance between theoretical and empirical proportions indicates the applicability of innovation diffusion theory to

complementary online technological service innovations in clubs like virtual sports. However, a slight deviation must be acknowledged, confirming prior literature that exact percentages vary depending on the innovation, population and contextual factors (Corthouts et al., 2022; Martínez & Polo, 1996; Park et al., 2024).

To explain *why clubs are innovative* (RQ2), we employ integrated innovation models that have been advocated for over time (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002; Wolfe, 1994) to our adopter categories underpinned by Rogers (2003). Our findings in Figure 11 and III show that “soft” determinants like perceived innovation attributes and managerial determinants are the strongest predictors of clubs’ innovativeness, validating Wolfe’s (1994) critique that the instability in traditional organisational innovativeness research is due to innovation attributes being perceptually based yet often not treated as inherent to the organisation, ultimately requiring a shift from the organisation to “innovation-in-an-organisation” (p. 409). First, our findings indeed confirm that while all clubs are exposed to the same technology to be used as a complementary service in their club (virtual cycling), their categorisation as innovators, intermediates, or laggards is above all predicted by how the club perceives this technology as positive on attributes ($\eta^2 = 0.56$) repeatedly reported in diffusion and innovativeness research (Corthouts et al., 2020, 2022; Karahanna et al., 1999; Moore & Benbasat, 1991; Rogers, 2003; Venkatesh & Brown, 2001; Wolfe, 1994). Viewing virtual cycling as club service innovation and especially the technology behind virtual cycling as significantly more advantageous, compatible with existing services, easy to use, visible to externals, conveying demonstrable results, and fun to use, significantly increases clubs’ innovativeness (i.e. attitude, evaluation and adoption-intention) towards virtual cycling. Second, managerial capacity but also human-related capacity determinants predict clubs’ innovativeness ($\eta^2 = 0.50$ and 0.16 respectively), strengthening the earlier reported importance of possessing knowledge about the innovation, someone to champion the innovation, and the sufficiency in skilled volunteers, coaches and network (Damanpour & Schneider, 2006; Delshab et al., 2022; Hoeber & Hoeber, 2012; Svensson et al., 2020; Winand et al., 2013; Winand & Hoeber, 2017). Third, innovation attributes and managerial capacity discriminate together between all adopter categories (Figure 12), while human-related-capacity allows to further distinguish Innovators and Early Adopter clubs from Early Majority clubs. This means that while each determinant has a significant effect on innovativeness, they affect and predict it differently, which puts forward new evidence of how earlier discovered determinants in a sport club context interact (Hoeber & Hoeber, 2012; Winand et al., 2013)

Effectively, our analyses indicate that innovation-specific and organisational-specific determinants are not only equally important, but they appear to interact in some way. Within a voluntary sport context, we provide a first step to the concern of Wolfe (1994) that “little is known about how the determinants of organisational innovation interact” (p. 409) and Frambach and Schillewaert (2002) concern that “research must also assess the individual *and* organisation culture antecedents” (p. 173). Figure 11 clearly shows how innovation perception and managerial determinants are equally high (innovators) or low (laggards), while Figure 12 (A) shows how they “work together” to discriminate innovators from intermediates from laggards as they show equally high standardised structure coefficients on discriminant function LD1. Damanpour and Schneider (2006) already acknowledged the interaction of both innovation perception and managerial determinants, emphasising how “organisations led by managers with unfavourable attitudes toward innovation (...) do not encourage their members to develop new perceptual frames of innovation” (p. 231). While our study confirms this interaction of innovation perception and managerial capacity within the context of technological service innovations in clubs, further research can be done to uncover the way they interact (e.g., direction of their interrelatedness).

12.6.2 The context of digital service innovations in clubs

Contrary to earlier studies on determinants of technological innovation identifying “the availability of financial resources as important throughout the innovation process” next to “unique organizational resource (i.e. infrastructure ...)” (Hoeber & Hoeber, 2012, p. 222), our findings (Figure 11) align with other studies (e.g. Corthouts et al., 2022; Winand et al., 2013) showing no significant difference in non-human-related capacity (finances, infrastructure and vision). This discrepancy can be explained by the nature of innovation(s) studied; while Hoeber and Hoeber (2012) surveyed a resource-intensive technological innovation (i.e. electronic scoring sheet), Winand et. al. (2013), Corthouts et al. (2022), and our research studied less resource-intensive service innovations. Virtual cycling can be considered a resource-inexpensive online technological “light sport” service innovation in clubs (Corthouts et al., 2023) that for instance (1) relies on members’ existing at-home equipment (e.g. smart trainer, platform subscription), (2) for which infrastructure is relatively accessible (e.g. rent workout studio with smart trainers for a 2 hour not-at-home training session), or (3) that can be planned relatively easily (e.g. use platform features to plan virtual club rides or competitions and communicate these) (Fouquaert, Tjønndal, De Clerck, et al., 2025; Westmattelmann et al., 2021). Another explanation is that technological service innovations like virtual cycling “follow

a bottom-up process as they are mainly initiated at the technical core and implemented at the administrative core” (Damanpour & Schneider, 2006, p. 230), meaning that for such innovations to be *initiated* (i.e. what Innovators & Early Adopters do first), non-human-related capacity is of lesser importance. By building on Corthouts et. al. (2023) “light sport” (here, as a technological club service rather than a sport) and Damanpour and Schneiders’ (2006) “cores” as a lens for online service innovations like virtual sports, we can explain why, under certain conditions, the barriers to adopt shift from resources and structures to attitudes, knowledge and skills (Corthouts et al., 2022; Delshab et al., 2022; Svensson et al., 2020; Winand et al., 2013).

12.6.3 Theoretical contributions

This study advances sport management and marketing literature in three ways. First, we demonstrate how diffusion theory and organisational innovativeness theory can be integrated by considering “*innovation-in-an-organisation*” as the unit of analysis (Rogers, 2003; Wolfe, 1994). Our results show that organisation-specific determinants and innovation-specific determinants (living *in-an-organisation*) interact to shape clubs’ innovativeness, suggesting a need to reassess the to-be-considered innovation determinants in clubs of existing theoretical frameworks (Corthouts et al., 2020; Delshab et al., 2022; Hoeber & Hoeber, 2012; Winand et al., 2013; Winand & Hoeber, 2017). Moreover, it confirms Crossan and Apaydin’s (2010) and Frambach and Schillewaert’s (2002) multi-level framework in a voluntary organisational sport context, suggesting that the club culture and receptiveness of key volunteers towards technological club services are as critical as the technology, organisational resources, and structures themselves.

Our second theoretical contribution is the suggestion to develop a determinant typology for technological service innovation adoption within voluntary sport organisations. We provide an empirical basis that identifies a unique set of determinants simultaneously predicting innovativeness compared to existing literature, which is attributed to our specific context of virtual sports as (1) online “light sport” technological service innovations, (2) completing the technical (vs. administrative) core, (3) examined at the early-stage of technological service adoption, (4) in voluntary clubs. This clearly maps to Wolfe’s (1994) recommendation to specify “the characteristics of the innovation(s) studied, the stage(s) of the innovation process considered, and the type(s) of organizations” (p. 424) and Damanpour and Schneiders’ (2006) recommendation to specify the innovation-affected “cores” (as described above). Consequently, we argue that sport management and marketing theory is in need of a determinant typology for innovation adoption. This would allow to explicitate important

dimensions affecting innovation determinants (e.g. four dimensions aforementioned) and classify the relevant to-be-studied determinants into optimal “configurations” of determinants for its context. Our results could provide one such configuration already (i.e. adoption-initiation of virtual sports as technological service innovations in voluntary sports clubs). Such a typology could be similar or even complementary to the service innovation archetype typology of Gerke and Dickson (2025). For instance, any of their service innovation archetypes (e.g. output-based or process-based) naturally occurs within a given context for which, during each study, the most relevant determinants need to be considered on a theoretical — yet empirically grounded — basis.

Third, we validate the use of a composite innovativeness index (attitude, evaluation and adoption-intention) for early-stage technological service innovations in clubs. For innovations investigated yet scarcely adopted or implemented already, such index appears an effective operationalisation opposed to the often used “composite score based on the number of innovations adopted” (Wolfe, 1994, p. 408) in both diffusion and innovativeness studies (Corthouts et al., 2022; Davis et al., 1989; Moore & Benbasat, 1991; Rogers, 2003; Venkatesh & Brown, 2001; Winand et al., 2013; Wolfe, 1994).

12.6.4 Practical contributions

The findings of this study hold implications for practice in two ways: for policy and for industry. First, for federations, the finding that non-human-related capacity is not a significant discriminator for club innovativeness suggests that, in the case of resource-inexpensive technological service innovations in clubs, federations should shift their support more towards perception and knowledge management. We echo Delshab et al. (2022) that clubs are “able to improve their innovative capacity if they create a network to share and discuss new ideas” (p. 152) and should foster innovation champions, for which it would indeed be fruitful “to recognise and reward knowledgeable staff members.” (p. 153). Federations’ role can be to act as the central node and facilitate this in three ways: (1) through connecting clubs that exchange knowledge, (2) by linking external knowledge like virtual sports providers (e.g. Zwift) to clubs, and (3) by hosting programs themselves that focus on sharing knowledge, changing perceptions, or directly offer skilled volunteers who may temporarily collaborate with the club and act as their innovation champion until enough managerial and human-related capacity is developed (Fouquaert, Tjørndal, De Clerck, et al., 2025).

Second, to explain marketing strategy implications for industry (RQ3), our findings suggest that marketing strategies for virtual sports require a segmented approach. We refer to

the in-depth analysis of barriers and facilitators by Evsikov et al. (2024) for individual adoption of technological services like virtual sports, while adding the following advice for club-level adoption. Aimed at the intermediate and lagging clubs, marketers may address clubs' perceptions and knowledge including: (1) ease-of-use, (2) compatibility with the clubs' operations (e.g. offer club-paid access or licenses, offer club specific features), (3) clearly communicating the relative advantages of virtual cycling as club service over traditional cycling as club services, (4) boosting the status and image of virtual cycling adopters, (5) allowing clubs (physical) access to virtual cycling to try it out free of obligation by providing hardware and software, (6) allowing clubs to showcase the results and impact of adopting virtual cycling (e.g. feature them on social media), and (7) the fun aspect virtual cycling may bring to club workouts (Fouquaert, Tjørndal, De Clerck, et al., 2025; Rogers, 2003; Wolfe, 1994). To change club managers' and key volunteers' perceptions, knowledge and attitudes, marketers may prioritise on (1) developing an accessible way through which club volunteers may become knowledgeable about their product and its specific features, and on (2) communicating club-level features and initiatives to consumers who may act as innovation champions for virtual sports within their clubs (e.g. frequent users) (Delshab et al., 2022; Hoeber & Hoeber, 2012). In addition, marketing surveys can be designed with the goal of determining the clubs' adopter category by measuring only three determinants (perceived innovation attributes, managerial determinants, and human-related capacity) that together yield a high predictive power of 79%. Insights from these surveys can in turn be used to implement targeted marketing strategies based on adopter categories (Martínez & Polo, 1996).

Third and last, our findings offer direct, actionable insights for voluntary sports clubs themselves. As our findings show that financial and infrastructural capacities do not significantly differentiate innovative from lagging clubs, a lack of financial means or dedicated facilities should not be viewed as unconquerable barriers by the club's board to start offering virtual cycling as a service. Instead, clubs should focus their efforts on cultivating "soft" determinants such as managerial attitudes and human-related capacity. This involves identifying and empowering an "innovation champion" as a formal role within their existing members, athletes or volunteers. Active athletes who are already frequent users of virtual cycling platforms could for instance serve this formal role effectively (Delshab et al., 2022; Hoeber & Hoeber, 2012). By sharing their knowledge and enthusiasm, these champions can help demystify the technology for the board, volunteers, and other members, thereby addressing any (mis)perceived complexity while also demonstrating the relative advantages, ease of use, and fun aspects of the service (Evsikov et al., 2024; Wolfe, 1994). Such an approach allows

clubs to shift their overall managerial attitude towards a more innovative stance with regards to resource-inexpensive technological service innovations, while not needing significant financial or structural investments (Winand et al., 2013). Furthermore, clubs could explore virtual cycling as a service to offer online events. These could range from leisure rides for fun and training to engaging fun-competitions and even serious competitive races (Fouquaert, Tjønndal, De Clerck, et al., 2025; Tjønndal, 2022; Westmattelmann et al., 2021).

12.6.5 Future research

Our findings indicate that technological service innovations adoption like virtual sports in clubs is mostly determined by individual-level (*in-an-organisation*) factors, yet studied at the organisational-level. To bridge this gap, future research could adopt Crossan and Apaydin's (2010) recommendation to employ a "Practice-Based View" (p. 1178) (PBV). As they note, only 5% of studies on organisational innovation operate at the individual level, yet our findings suggest that in a sport club context, this is a critical level of analysis. PBV considers the actions at micro-level (e.g. volunteers), their consequences at club-level, and "the feedback loop from contextual and organisational variables back to the actors" (Crossan & Apaydin, 2010, p. 1178). Additionally, research could explore the "tensions" occurring at this micro-level, for instance, the belief that "abundant [digital] information may actually distract from the essence of the sports" (Standaert & Mazurova, 2025, p. 33) and their feedback loop from and to traditional club cultures, which could be the case for technological service innovations like Zwift which are complementary to the core of the traditional sport.

12.6.6 Limitations

This study has several limitations. First, it relies on a relatively small sample size and was therefore unable to account for discriminant validity. In addition and while insightful for future research, we have not measured and therefore could not control for clubs' current positioning in innovation stage (e.g. initiation, implementation). Nonetheless, a pilot test with the federations confirmed that (1) clubs were merely aware of virtual cycling's existence (i.e. first step in initiation) and (2) only a handful of clubs are known to do something with virtual cycling, which was a collective individual use rather than club service. Second, our understanding of determinants is cross-sectional, meaning there is no knowing if and how clubs will decide to adopt or implement virtual cycling. Longitudinal designs are necessary to track the transition from initiation to implementation. Last, while one key volunteer's opinion is critical in the decision-making process of the whole club (Schlesinger et al., 2015), we have not measured or could not account for beliefs or attitudes varying within one club. Future research

could employ a multi-level approach to account for this.

12.7 Conclusion

This study demonstrates that the adoption of virtual cycling in voluntary sports clubs is driven by a distinct hierarchy of determinants, where perceived innovation attributes, managerial capacity and human-related capacity significantly outweigh non-human-related and environmental capacity for clubs to be innovative with virtual cycling (e.g. Zwift). Revisiting Rogers' (2003) adopter categorisation to distinguish clubs in their innovativeness, we observe innovative, intermediate and lagging clubs related to their attitude towards, evaluation of and adoption-decision regarding virtual cycling. By applying an integrated innovation model (Frambach & Schillewaert, 2002) combining insights from diffusion and organisational innovativeness (Rogers, 2003; Wolfe, 1994), we reveal that for “light sports” technological service innovations like virtual cycling (Corthouts et al., 2023), the barrier to adopt is not resources but the human capabilities to perceive and manage technological change. Innovators and Early Adopter clubs show significant more positive perceptions about virtual cycling and significant higher commitment and knowledge regarding virtual cycling. In contrast, Late Majority and Laggard clubs exhibit lower perceptions and engagement, highlighting the need for targeted efforts to turn around prejudiced negative perceptions and increase commitment to and knowledge about virtual cycling.

12.8 References Study 1

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12.9 Appendix

12.9.1 Innovation Determinant and Outcomes Measures (appendix study 1)

Table 7: Study variables, measuring scales and Cronbach's Alpha (appendix study 1).

Measurement Scales	Items	α
Perceived innovation attributes determinants		
Relative advantage ¹	Integrating virtual cycling allows us to achieve recreational club working. (1)	.93
	Integrating virtual cycling improves the quality of our recreational club working. (2)	
	Integrating virtual cycling helps us achieve a recreational club working more easily. (3)	
	Integrating virtual cycling increases the effectiveness our recreational operations. (4)	
	Virtual cycling seems to be a great way to strengthen our recreational club working. (5)	
Compatibility ¹	Integrating virtual cycling is compatible with all aspects of our club. (1)	.91
	Integrating virtual cycling is fully compatible with the current situation of our club. (2)	
	Virtual cycling fits within our club. (3)	
	Virtual cycling is integrable in our club operation. (4)	
Image ¹	Clubs that integrate virtual cycling have more prestige than cycling clubs that do not offer it. (1)	.91
	Clubs that integrate virtual cycling into club offerings are highly regarded. (2)	
	Clubs that integrate virtual cycling have a certain status. (3)	
Ease of use (complexity reversed) ¹	It is clear and understandable how virtual cycling can be integrated into a recreational club working. (1)	.84
	We believe it is easy to grow as a club by integrating virtual cycling. (2)	
	(dropped)	
Trialability ¹	Developing an offer with virtual cycling that strengthens our recreational club working seems easy to us. (3)	.86
	Before deciding whether or not to offer virtual cycling, we were able to try out the games and/or hardware properly. (1)	
	We have been able to use a test setup of virtual cycling games sufficiently to see what we can do with them for our club operations and offerings. (2)	
Visibility ¹	When another club integrates virtual cycling, it is clearly visible. (1)	.81
	The fact that we offer virtual cycling as a club is visible to supporters, athletes or other clubs. (2)	
	As a club, we have good visibility on how other clubs expand or enhance their offerings with virtual cycling. (3)	
Result demonstrability ¹	We would have no difficulty in convincing other clubs of the benefits we gain from offering virtual cycling. (1)	.89
	The opportunity of virtual cycling within our recreational club working is clear. (2)	
	It is easy to explain why integrating virtual cycling is useful or not. (3)	
	We can make it clear to other clubs or members what the outcomes can be of virtual cycling within our club offer. (4)	

<i>Measurement Scales</i>	<i>Items</i>	<i>α</i>
Perceived enjoyment ²	Integrating virtual cycling into our club can help increase the fun aspect of our workouts for our members. (1) Integrating virtual cycling into our club can help make our members feel good about the club offer. (2) Integrating virtual cycling into our club can help motivate our members to participate in training sessions (3) Integrating virtual cycling into our club can help to engage our members to participate in training sessions or club activities. (4)	.88
Managerial determinants		
Innovation champion ³	Within our club, there is at least one person who is eager to and will promote the benefits of virtual cycling. (1) Within our club, there is at least one person who will be able to convince others of the potential of virtual cycling for our club. (2) Within our club, there is at least one person who will remain involved in virtual cycling until it is integrated into our club offer. (3) Within our club, there is at least one person who will involve key decision-makers in the integration of virtual cycling. (4)	.94
Pro-innovation characteristics ⁴	Our club should live up to new expectations of our federation and/or members. (1) We should consider suggestions from federations and/or members. (2) Our club should invest in developing new services. (3) Our club should make more (financial) investments to develop new services for our members. (4) To achieve our club goals, we sometimes need to take risks. (5) Change is usually a good thing for sports clubs. (6)	.80
Subjective knowledge ⁵	Within our club, directors, coaches or volunteers are quite knowledgeable about virtual cycling (1). Within our club, administrators, coaches or volunteers are very knowledgeable about virtual cycling. (2) Within our club, we have at least one “expert” on virtual cycling. (3) Within our club, administrators, coaches or volunteers know more about virtual cycling compared to others. (4)	.91
Human-related capacity determinants		
Coaches (for adults) ⁶	Our sports club has enough competent trainers for an adult operation. (1) Our trainers for adult offerings have the right competences for carrying out their job through relevant experience. (2) Our trainers for adult offerings have the appropriate competences for carrying out their job based on a relevant diploma. (3)	.86
Youth coaches ⁶	Our club has enough competent trainers for youth offerings. (1) Our trainers for youth offerings have the right competences to perform their job through relevant experience. (2) Our trainers for youth offerings possess the right competences for performing their job based on a relevant diploma. (3)	.97
Volunteer sufficiency ⁶	Our club can rely on sufficient volunteers for the canteen (e.g. bar service). (1)	.93

Measurement Scales	Items	α
Volunteer quality ⁷	Our club has sufficient volunteers for support at competitions. (2)	.79
	Our club has sufficient volunteers for day-to-day operations of the club (e.g. laundry of jerseys, maintenance of grounds). (3)	
	Our club supports volunteers, board members and/or trainers in obtaining or updating essential skills, trainings or certificates. (1)	
	People within our club are passionate about developing the sport further. (2)	
	Our club guarantees a smooth transition of key people within the volunteers, board members or trainers. (3)	
Board ⁶	The club has sufficient board members to properly perform all tasks. (1)	.81
	There is a defined division of tasks (each board member has their own tasks). (2)	
	Board members have the appropriate competences for keeping accounts. (3)	
	Board members have appropriate competences for raising sponsorship. (4)	
	Board members have the right competences for organising activities (competitions/tournaments/club activities/events) (5)	
Inter-organisational relationships ⁷	Board members have the right competences for club-related communication (e.g. creating the website /club magazine /sending newsletters). (6)	.92
	Our sports club communicates regularly with partners, stakeholders. (1)	
	Our sports club keeps in close contact with the partners, stakeholders. (2)	
	There is two-way communication between our sports club and partners. (3)	
	Our sports club and its partners share up-to-date knowledge from their field or sports discipline. (4)	
	The relationship with our club's partners is characterised by mutual trust. (5)	
	Our club and its partners share an interest in our sports field. (6)	
Non-human-related capacity determinants		
Financial capacity ⁷	Our club currently has sufficient financial resources to implement its policies. (1)	.87
	Current financial difficulties hinder the operation of our club. (2) (reversed)	
	The financial situation of our club allows it to launch new initiatives. (3)	
	Our club has financial resources set aside for unforeseen circumstances. (4)	
	Our club provides income that covers costs which keeps us at least break-even. (5)	
Infrastructure capacity ^{6,7}	Our club has sufficient sports facilities to accommodate all training for the current number of members (e.g. are there enough sports halls, equipment...). (1)	.86
	Our club has sufficient hours in the sports facility considering the current number of members. (2)	
	With the current amount of sports infrastructure, our sports club can accommodate more members. (3)	
	The sports facility used by our sports club is safe. (4)	
	The sports facility used by our sports club is in good condition. (5)	
Planning and development capacity ⁶	Our club has a written long-term policy plan. (1)	.87
	Our club has developed an operating plan/policy plan for the coming year. (2)	
	Our club has an organisational chart. (3)	
Environmental determinants		

Measurement Scales	Items	α
Competitive environment ⁴	We compete heavily with other clubs for subsidies. (1) Within our sport, there is a high degree of competition between clubs to attract members. (2) Within our sport, there is a high degree of competition between clubs to attract volunteers and/or coaches. (3) Competition for subsidies is high. (4)	.83
Innovativeness variables		
Clubs' attitude towards virtual cycling ⁸	Our attitude towards integrating virtual cycling into our recreational club offerings: Don't like ... like (1) Unfavourable ... favourable (2) Negative ... positive (3)	.91
Evaluation ^{8,9}	To what extent is virtual cycling being considered or discussed within the club (e.g. to be able to respond to the needs of your members or to create new opportunities). Very unlikely ... very likely (1) Definitely not ... definitely (2) Very uncertain ... very certain (3)	.97
Adoption-intention	The probability that we will purchase software or hardware in the near future, start collaborations to obtain these, or, for example, develop an offer for members who have virtual cycling equipment at home is: Very unlikely ... very likely (1) Definitely not ... definitely (2) Very uncertain ... very certain (3)	.96

Note: Items that were dropped to increase Cronbach's Alpha are indicated with 'dropped'. α indicates the Cronbach's Alpha.

¹ adapted from Moore and Benbasat (1991)

² adapted from Venkatesh and Brown (2001)

³ adapted from Howell et al. (2005)

⁴ adapted from Winand et al. (2013), Winand and Anagnostopoulos (2017)

⁵ adapted from Flynn and Goldsmith (1999)

⁶ adapted from Balduck et al. (2015)

⁷ adapted from Doherty and Cuskelly (2019)

⁸ adapted from Uhrich (2022)

⁹ based on the definition of Damanpour and Schneider (2006)

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Ethical statement

Ethical approval was obtained from the ethical commission of the Faculty of Medicine and Health Sciences of Ghent University (code: 2021-106). Permission to conduct the online data collection for the purposes of this research was obtained by all respondents (i.e. clubs), who were fully informed about the purposes of this research and how their responses would be used and stored (i.e. anonymised).

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We have no conflicts of interests to disclose.

Data Availability Statement

Only anonymous data are available from the corresponding author, upon reasonable request.

13 Study 2: Virtual Cycling in Sports Clubs: Innovation Adoption within Institutional Logics

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Abstract

This study investigates voluntary sports clubs' varied beliefs about whether and how virtual sports can be a useful tool to achieve both competitive sports performance goals and sport-for-all goals. For this purpose, this study also explores archetypes' beliefs about virtual sports. We draw on three theoretical concepts: archetypes, institutional logics and innovation adoption categories. Factor analysis, clustering and parametric tests have been used on survey data of 90 voluntary sports clubs in Flanders (Belgium). The results show that clubs believe that there are three overarching goals that may be achieved through virtual cycling: (1) inclusive participation, (2) skill development, and (3) attractiveness of the club's services. Based on the scores across all three goals, five club archetypes are presented: (1) performer clubs, (2) unconvinced clubs, (3) sceptic clubs, (4) hesitant clubs and (5) holistic clubs. These archetypes demonstrate that voluntary sports clubs hold different beliefs about the potential of virtual cycling. The archetypes also show that virtual cycling is believed to hold more potential as a tool to achieve sport performance goals rather than to achieve sport-for-all goals. Our contributions demonstrate how the self-evident use of virtual sports within a result-oriented logic may undermine its potential use within a sport-for-all logic.

Keywords: Virtual Cycling, Sport-for-All, Institutional Logics, Voluntary Sports Clubs, Archetypes, Zwift, Innovation Diffusion

13.1 Introduction

For voluntary sports clubs, virtual sports represent radical innovations that offer new opportunities to grow and diversify their services. Virtual sports are mixed reality sports games that can be offered as sports services by clubs to their members by using interactive peripheral sport equipment, for example online club rides on the virtual cycling platform Zwift. Virtual sports are considered radical innovations because they disrupt the notions of how competitive and leisure sports are practiced and organised. More precisely, virtual sports combine (1) the competitiveness of esports, (2) the physicality of sports and (3) the gamification of exergames (Goebeler et al., 2021). This distinctive combination could enable clubs to attract new members while expanding their services in terms of sporting formats, complementing their core activities. Considered a form of social innovation, virtual sports may also have the potential to serve as a valuable tool for clubs in achieving sport-for-all policy goals. Virtual sports are already recognised for their social benefits, such as fostering identity, community, and connectivity (Reed et al., 2023), while also offering practical advantages for athletes' training and competition (Fouquaert et al., 2025; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021). When integrated into club services, they can for example: (1) enable individuals to develop a sense of identity and community by training together, regardless of ability or disability; (2) enhance club offerings with gamification elements that cater to both leisure and competitive settings; and (3) boost participation across all levels, from beginners to elite athletes.

The potential of virtual sports is reflected in the policies of international sport governing bodies such as the International Olympic Committee (IOC), The Union Cycliste Internationale (UCI) and World Triathlon. For instance, the UCI is investing in Cycling Esports as new discipline while the IOC is investing in virtual sports through events such as The Olympic Virtual Series and The Olympic Esports Week. The IOC also recently announced that the Olympic Esports Games will be taking place in 2027, featuring both esports and virtual sports (International Olympic Committee, 2025).

In large parts of Europe, voluntary sports clubs are publicly funded and considered the basis for developing athletic talent and elite sports (Grix & Carmichael, 2012). At the same time, those clubs are also expected to promote sport-for-all policy goals. In other words, to democratise sport and provide leisure activities in an inclusive way (Mittag & Naul, 2021; Waardenburg, 2016). While elite sports and sport-for-all are not necessarily mutually exclusive endeavours, many researchers have demonstrated how these activities often represent two

conflicting institutional logics (De Bock et al., 2022; Skille, 2009; Stenling & Fahlén, 2009). That is, elite sports, talent development, and sports performance enhancement are often based on a result-oriented logic, while leisure and sport-for-all are closely tied to a welfare logic. Against this backdrop, our research aims are twofold. First, we aim to examine voluntary sports clubs' beliefs about whether and how virtual sports can be a useful tool to achieve result-oriented and sport-for-all goals. This includes identifying the overarching goals for which clubs believe that virtual sports may hold potential. Second, we categorise clubs into archetypes based on their beliefs, thereby understanding the different convictions and hesitations clubs have about virtual sports.

Empirically, we analyse cycling and triathlon clubs' beliefs about virtual cycling because cycling is the most advanced and widely adopted form of virtual sports in 2025 (Botwina et al., 2024; Devine et al., 2024). Virtual cycling's well-developed technology offers extensive opportunities for voluntary sports clubs including structured training, competition formats, and community engagement, enabling online club rides (e.g., via Zwift or MyWhoosh) when weather or other constraints prevent outdoor riding. By transcending practical, physical or capacity limitations within clubs, virtual cycling represents an ideal case for examining the beliefs about the potential a new innovation like virtual sports hold to address result-oriented and sport-for-all goals in a club setting.

13.2 Literature review

There are two research fields that inform our research: (1) virtual cycling, digital tools and innovation in sports clubs, and (2) institutional logics in voluntary sports clubs.

13.2.1 Virtual cycling, digital tools and innovation in sports clubs

Research examining virtual cycling remains limited. This gap may partly be attributed to the fact that virtual cycling is a relatively new phenomenon. However, recent literature increasingly covers the individual and self-organised use of virtual cycling (Botwina et al., 2024; Reed et al., 2023; Westmattmann, Grotenhermen, Sprenger, Rand, et al., 2021). This can be linked to the increase in popularity and availability of virtual cycling platforms during the COVID-19 pandemic, with several studies indicating that both recreational and professional athletes increasingly relied on cycling platforms for online training and competitions during the pandemic (Helsen et al., 2022; Tjønndal, 2022). Many athletes perceived virtual cycling as a positive alternative to mainstream cycling during lockdowns (Westmattmann, Grotenhermen, Sprenger, & Schewe, 2021). Virtual cycling platforms also include potential economic benefits,

as they offer a new online space for various internet-based sponsorship strategies that may be utilised by clubs who organise online events, knowing that online platforms have significantly transformed sponsorship and fan engagement (Koronios et al., 2023; Koronios & Dimitropoulos, 2020).

Sports performance and inclusion aspects about virtual cycling have also been studied in the aftermath of the pandemic lockdowns. Devine and colleagues (2024) found that virtual cycling provides a valuable alternative to in-person elite competitions while also providing an accessible platform for leisure participants to compete against opponents of similar skill levels for the first time. Westmattelmann and colleagues (2021) identified several utilitarian benefits of virtual cycling, including safety, training efficiency and health-consciousness, which all act as motivators for both leisure and elite athletes to engage with these virtual platforms. In addition, hedonic benefits such as gamification and socialising elements have also been found as motivators to engage in virtual cycling (Devine et al., 2024). While these enjoyment-oriented aspects were important motivators for leisure use, competitive cyclists have also been found to benefit from them (Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021). Finally, Reed and colleagues (2023) recognised virtual cycling platforms as a leisure space that fosters social connections by supporting community and connectivity. Platforms do this by enabling users to customise avatars and exercise together with other cyclists, regardless of their physical capabilities. In addition to leisure, Reed et al. (2023) also identifies benefits for professional athletes, including increased visibility and the ability to connect and ride with fans online.

To inform us further about the potential use of virtual sports and virtual cycling in voluntary sports clubs specifically, we can build on studies of digital tools and innovation in clubs. Ehnold and colleagues (2020) found that clubs primarily use digital tools for internal and external communication, administrative tasks, and to pursue goals in competitive sporting success and institutional cooperations. However, they found no impact of these tools in advancing sport-for-all goals, such as conviviality or service orientation. While other studies have focussed on the determinants of innovation implementation in voluntary sport clubs, little attention was given to their relation in achieving organisational goals such as delivering sport-for-all services or excelling in competitive sport (Behnam et al., 2022; Corthouts et al., 2020; Hoeber et al., 2015; Hoeber & Hoeber, 2012). Primarily administrative, incremental and technical innovations have been discussed (e.g. electronic scoring sheet), indicating how such innovations enhance clubs' member engagement and administrative efficiency (Hoeber et al., 2015; Hoeber & Hoeber, 2012). To achieve successful innovation implementation, managerial determinants (i.e. attitude towards and knowledge about change) and innovation perception (i.e.

positive and negative beliefs) were found to be important (Corthouts et al., 2020; Delshab et al., 2022; Hoeber & Hoeber, 2012).

The way an innovation is perceived within an organisation has consistently been found as one of the best-predicting determinants of potential innovation integration (Rogers, 2003, Chapter 10; Wolfe, 1994), also within sport clubs (Corthouts et al., 2020). Perceived innovation attributes may be primary (e.g., size or cost) or secondary (e.g., relative advantage, compatibility). The latter are shaped by subjective beliefs, experiences and norms and can be redefined or reinterpreted within an organisation (Wolfe, 1994).

13.2.2 Conflicting institutional logics in voluntary sports clubs

Over the past two decades, many national governments have intensified pressure on clubs to implement sport-for-all policies (De Bock et al., 2022). The increased focus on clubs as vehicles for the promotion of ‘social goods’ such as health (Skille, 2010) has led sport management scholars to examine clubs as policy implementers through the lens of institutional logics (Skille & Stenling, 2018; Stenling, 2014; Szerovay, 2022). Institutional or ‘dominant’ logics refer to certain rules, values, and norms that are derived from institutions, informing organisations on legitimate ways to arrange their activities (Bettis & Prahalad, 1995; DiMaggio & Powell, 1983). Voluntary sports clubs find themselves involved in a struggle between logics, i.e. between contrasting or even conflicting goals and purposes. For instance, Stenling and Fahlén (2009) identified three competing institutional logics in their study of voluntary sports clubs in Sweden: (1) the sport-for-all logic, (2) the result-oriented logic, and (3) the commercialization and professionalization logic. Stenling and Fahlén (2009) describe the sport-for-all logic as promoting democratic, social, and moral values, while the result-oriented logic prioritises competitive sport and performance. The commercialisation logic, closely tied to the result-oriented logic, focuses on revenue-generating activities which often undermine the implementation of the sport-for-all logic (Stenling & Fahlén, 2009).

Similarly, in his studies of Norwegian voluntary sports clubs, Skille (2010, 2011) found that institutional logics of competitiveness and performance are central to club activities, while health promotion and sport-for-all activities receive less emphasis. Many scholars have argued that the sport-for-all logic derives from social inclusion in and through sports (i.e. mass participation and democratic values) while the result-oriented logic maintains social exclusion (i.e. selection of athletes, skill development) (Spaaij et al., 2014).

The sport-for-all logic is rooted within EU’s broader sport-for-all policy. Today, the European sport-for-all charter outlines thirteen articles that focus on participation, health,

inclusion, but also improving performance and professional sport (Council of Europe, 2001; Mittag & Naul, 2021). Voluntary sports clubs play a key role in achieving these sport-for-all goals (Mittag & Naul, 2021; Waardenburg, 2016). Yet, it is well documented that both national sport organisations and voluntary sports clubs experience difficulties in fulfilling sports-for-all policy goals (May et al., 2013; Skille, 2009). To give an example: Out of 218 sport-for-all projects in the sport organisations in Flanders (Belgium), only 26% were open-for-all (e.g. start-to-run), 12% open-for-disabled, and 10% open-for-elderly (De Bock et al., 2022).

Both logics – result-oriented logic and sport-for-all logic – are however not bound to the European context. For instance, Robertson et al. (2023) discuss how business and social logics compete in Australian professional sport teams delivering community activities, Parent (2018) discusses how CSR initiatives conflict with business logic in North American professional sport leagues (NFL, NBA), while similar competing logics involving Sport-for-Development versus high-performance sports have been discussed within the Global South (e.g. Latin America) (Añorve Añorve, 2021). These competing logics can furthermore affect each other. Scholars advocate that the legacy of elite sports is seen in increased grassroots participation, often referred to as trickle-down effects. However, empirical evidence remains inconclusive as recent reviews (Lion et al., 2023; Storm & Denstadli, 2024) highlight that trickle-down effects are often assumed rather than demonstrated, and where evidence exists, it shows mixed or context-dependent outcomes.

13.3 Theory and conceptual framework

To examine voluntary sports clubs' beliefs about virtual cycling, its potential benefits, and how these beliefs shape different club archetypes, we draw on three theoretical concepts: archetypes, institutional logics and innovation adopter categories.

The rationale to apply the concept of archetypes in this study is rooted in the fact clubs may represent a limited number of archetypes that are both shaped by and shape institutional logics, including the sport-for-all logic and the result-oriented logic (Bettis & Prahalad, 1995; Stenling & Fahlén, 2009). The concept of archetypes is based on the notion that organisations typically function within a finite set of configurations, encompassing their structure, strategy, and environment (Greenwood & Hinings, 1988). Archetypes were originally concerned with the dynamics of organisational change and structure, while later studies focussed more on how archetypes are legitimated by institutional fields (Greenwood & Hinings, 1993; Kikulis et al., 1995b, 1995a; Slack & Hinings, 1994). Due to this development, it has been argued that the

starting point for the identification and classification of archetypes is the interpretive scheme of institutional logics, or as Hinings and Malhotra (2008) have described it: “the search for archetypes starts within an institutional sphere, for it is there that institutional prescriptions for organizing will be found” (Hinings & Malhotra, 2008, p. 109).

Within the field of sport management, many have applied the concept of archetypes to national sport organisations (Amis et al., 2004; Hoye et al., 2020; Kikulis et al., 1995b, 1995a; Slack & Hinings, 1994; Thibault et al., 1993). For instance, Hoye and colleagues (2020) showcase how archetypes have been leveraged to explain dynamics of organisational change in national sport organisations. Archetypes have in fact been acknowledged as ‘a fundamental driver for future research’ (Hoye et al., 2020, p. 577) and ‘important building blocks of organizational theory’ (Hoye et al., 2020, p. 578). Following the development of empirical elaboration of different archetypes in sport (Hoye et al., 2020), our work identifies archetypes aimed at understanding “clusters of ideas, values and beliefs” (Greenwood & Hinings, 1988, p. 295). More specifically, collective beliefs that define and are defined by institutional logics (Greenwood et al., 2002; Lion et al., 2023; Storm & Denstadli, 2024). In particular, we explore the beliefs surrounding the potential use of virtual cycling in voluntary sports clubs drawing on their beliefs of how virtual cycling could fulfil both sport-for-all goals and result-oriented goals.

To further identify how clubs form and maintain these beliefs, we turn to conceptualisations of innovation diffusion, where adopter categories are identified. Adopter categories are ‘ideal types, concepts based on observations of reality that are designed to make comparisons possible’ (Rogers, 2003, p. 282) and therefore parallel to archetypes as an analytical tool to describe the complex phenomena of innovation adoption. Adopter categories are also used to describe organisations’ proposition towards innovation (Rogers, 2003, Chapter 10) and are therefore relevant when aiming to explain clubs’ beliefs of the potential of virtual cycling to fulfil a sport-for-all logic or result-oriented logic. While literature on archetypes does not link to Rogers’ adopter categories, it is appropriate to do so here because our research sits at the intersection of understanding beliefs about an innovation (i.e. the use of virtual cycling in voluntary sports clubs) and how its potential use aligns with broader ‘configurations’ of clubs that encompass their institutional logics of sport-for-all and result-oriented.

The five adopter categories, ranging from 'innovators' to 'laggards', describe a continuum of archetypes based on their level of innovativeness. Each category presents main distinct characteristics of the adopting unit (e.g., venturesome vs. traditional). These categories can therefore be used to detail and characterise various archetypes. Building on the foundational work of Kikulis, Slack and Hinings (1995b) who identified institutionally legitimised and

emerging archetypes in sport organisations, and the work of Rogers (2003) who differentiates organisations along a continuum between innovative and lagging, we combine these concepts to examine voluntary sports clubs' beliefs about the potential use of virtual cycling. This approach allows us to understand how clubs align with their dominant logic or logics. Identifying variations in beliefs about virtual cycling can provide critical insights into the future implementation of any virtual sport or mixed-reality sporting technology, regardless of clubs pursuing a result-oriented logic, a sport-for-all logic, or a combination of both.

13.4 Methods

13.4.1 Study setting and context

Our study takes place in Flanders (Belgium), which is characterised by high levels of participation in sports clubs compared to other European countries (Rask et al., 2024). The average club size of 81 (based on member count) (Sport Vlaanderen, 2024) is rather small compared to other European clubs such as Dutch, Danish, or German clubs, who have an average size between 320 and 410 (Breuer et al., 2017). Sport clubs have a central role in providing sport services within the Flemish organised sports sector and are supported by public funded federations. In total, 17.198 clubs are member of a federation that is recognised and subsidised by the government (cf. 'federation clubs') (Sport Vlaanderen, 2024). Consequently, our target population is defined as clubs who are member of the subsidised Flemish Cycling or Triathlon Federation and who actively offer sports services (e.g. facilitate group trainings) to their members, which must be accessible to the public (i.e. everyone can become a member).

Cycling clubs within the cycling federation are uni-sport clubs and are the second most popular type of club in Flanders, with a total share of around 21% of all federation clubs ($n = 3658$). Triathlon is less popular, only representing 0.6% of all federation clubs ($n = 102$). However, exact numbers on the amount of cycling clubs are hard to interpret and not representative for our target population criteria. We were informed by the cycling federation that only about 800 clubs would meet our target population criteria.

13.4.2 Sample and data collection

Using a cross-sectional study design, we sampled triathlon clubs from the triathlon federation using their club register which was confirmed to meet our target population criteria (subsidised federation clubs that actively offer sports services to their members which are accessible to the public). We invited each triathlon club representative in Flanders by email to take our online survey. To sample cycling clubs, the provided register by the cycling federation could not be

filtered on our target population criteria. Hence, we decided to spread the online survey as widely as possible to clubs that we knew would likely meet our population criteria, which was then controlled for at the beginning of the survey. We sent out invitational emails and enquired Flanders' 301 municipal sport services to forward our invitational email to cycling clubs within their municipality that would fit our criteria. Additional dissemination of our online survey was done through the cycling federations' email newsletter and website, social media channels (n=6) and at local sport events (n=7). The survey ran for 12 weeks starting from June 2022.

Our data was checked to fit the target population and for unreliable or invalid responses (e.g., all same answers, attention checks) after which 90 (=n) responses remained of the initial dataset of 124 clubs, encompassing approximately 10% of cycling clubs and 10% of triathlon clubs in Flanders. The average amount of club members in our sample was 75 (min = 10, max = 303, median = 50, n = 46), which is in line with the average club size in Flanders of 81 member (Sport Vlaanderen, 2024). Informed consent to participate was obtained for all clubs (see Ethical Statement).

13.4.3 Measurement instrument

The survey revolved around the exemplary case of the virtual cycling game Zwift. Considering the novelty of research on Zwift and other virtual sports as tools for sport performance enhancement and achieving sport-for-all goals, we developed our own set of items to measure this potential.

Similar to the development process of the Youth Sport Values Questionnaire by Lee, Whitehead, and Balchin (2000), four consecutive steps were taken to develop our questionnaire on sport policy (i.e. sport-for-all and result-oriented) values and goals. First, we developed an initial set of statements that probed into the extent to which virtual cycling can help to fulfil sport-for-all goals (e.g., To which extent do you think that virtual cycling can help to focus on youth sport services) and result-oriented goals (e.g., ensure that members who want to perform better can do so in an informed and scientific way). Second, we compared our items to the articles in the sport-for-all charter goals (Council of Europe, 2001; Mittag & Naul, 2021) and assessed if each goal was clearly and equally addressed. Based on this, we adjusted the items to a draft set of which we thought it best covered both sport-for-all goals and result-oriented goals while remaining comprehensible for clubs. Third, a panel of researchers and club board members gave feedback on the statements. We subsequently changed, removed, and added items to land on a set of 18 statements (cf. Table 8).

Table 8: Statements and loadings on clubs' beliefs about virtual sports goals (study 2). Items' means are based on five-point Likert scale. The statements were preceded with "To which extent do you think that virtual cycling can help to...". Only factor loadings $\geq .40$ are shown

Statement	Mean (SD)	Factor 1 Inclusive Participation	Factor 2 Skills Development	Factor 3 Attractiveness of Offering
1 make it possible for everyone to be introduced to cycling.	2.68 (1.03)	0.56		
2 involve people who would otherwise not find their way to our club.	2.58 (1.04)	0.55		
3 focusing on youth (< 18y) and a youth programme.	2.79 (1.03)	0.44		
4 focusing on initiatives for disadvantaged groups (e.g. people living in poverty, people with physical or mental disabilities, people from migrant backgrounds, etc.).	2.21 (1.03)	1.01		
5 focus on the elderly and senior citizens (60+) with adapted operations or initiatives.	2.66 (1.07)	0.41		
6 better reach disadvantaged groups (e.g. people living in poverty, people with a physical or mental disability, people with a migration background...)	2.17 (1.00)	1.07		
7 make cycling or triathlon more accessible.	2.62 (0.95)			0.43
8 clearly communicate that cycling is a sport for everyone and all ages.	3.1 (0.99)	0.45		
9 integrate people with disabilities (e.g. physical disability, deaf, blind) into our operation for members without disabilities.	3.02 (1.11)	0.66		
10 teach our members how to practice their sport in a correct, responsible way (e.g. technique, load).	3.08 (0.94)		0.72	
11 ensure that members who want to perform better (at leisure level) can do so in an informed and scientific way.	3.59 (0.98)		0.98	
12 create a safe sports environment.	3.51 (1.03)		0.50	
13 raise the quality of our training sessions.	3.36 (1.07)	-	-	-
14 provide target group-specific activities (e.g. training or (fun) competitions for beginners, people with disabilities...).	3.02 (0.97)	-	-	-
15 pay more attention to social activities besides training or competition.	2.59 (1.05)	-	-	-
16 create solidarity among club members.	3.03 (1.11)			0.87
17 increase the fun aspect of our training sessions for our members.	3.19 (1.09)			0.87
18 make our members enjoy our club offerings.	3.01 (1.11)			0.71
19 make our training sessions more challenging.	3.19 (1.09)			0.65
20 motivate our members to participate in training sessions.	3.09 (1.01)			0.79
21 engage our members to participate in training sessions or club activities.	2.94 (0.98)			0.81

In the fourth and last step, we pilot tested our survey with 65 individuals who were closely involved in voluntary sports clubs, including head coaches, board members, and other volunteers (n=65). The survey also included open-ended questions to gather feedback on clarity and ran for 1 month in May 2022. After having conducted exploratory factor analysis to explore the possible factor structure of the statements, we changed statements that shared too much similarity based on content validity and factor loadings (Hair et al., 2019, Chapter 3), while minor changes were made to improve clarity. Three new statements (nr. 19-21) were added to the survey. All statements were rated on a five-point Likert scale (completely disagree – completely agree). The statements were shown in a randomised order to participants to improve reliability and exclude order effects that may have further influenced the factor analysis. The final set of 21 statements is presented in Table 8.

13.4.4 Data analysis

Data analysis was conducted with R (version 4.3). Before conducting a factor analysis, the data was checked with the Kaiser-Meyer-Olkin (KMO) test (=0.91) and Bartlett's test of sphericity ($p < .001$), showing adequate data to continue. We also checked for values of more than three standard deviations above or below the mean (i.e. outliers), but no outliers were found.

First, descriptive statistics and correlations of the 21 statements were explored. Second, we aimed to explore the factor structure of the statements as to reduce them to similar sport-for-all and result-oriented policy goals. To do so, we ran an exploratory factor analysis which yielded three factors based on the Eigenvalues greater than 1, visual scree plot inspection, and greater than 60 percent of total explained variance (Hair et al., 2019, p. 144). The resulting loadings had one item (14) that did not load sufficiently on any factor (standardised loading $< .40$ following Hair et al. (2019, p. 151)) and two cross-loading items (13, 15) (one or more standardised loadings of $\geq .40$). These statements included aspects of quality of trainings, providing target group-specific activities and social club activities. We removed these three items for a second iteration of the factor analysis, resulting in an overall good factor structure with no cross-loadings nor items that did not load on any factor (Hair et al., 2019, Chapter 3). The final factor structure is presented in Table 8.

Lastly, we used the resulting three factors as a starting point to categorise clubs into archetypes based on their beliefs about virtual cycling's potential. To do so, a two-step procedure using hierarchical and non-hierarchical clustering methods was implemented. Ward's hierarchical method (i.e. squared similar Euclidian distance) was used to identify and compare several possible cluster solutions based on their parsimony, dendrogram and

interpretability of each cluster. Then, the most favourable cluster solution, yielding five clusters, was used as non-random starting points in an iterative non-hierarchical k-means procedure fixed to five clusters (Hair et al., 2019, Chapter 4).

13.5 Results

The results first present how clubs believe virtual cycling has potential to achieve result-oriented and sport-for-all goals. Second, we detail our archetyping of sport clubs and their characteristics based on how these beliefs interrelate.

13.5.1 Three addressable goals using virtual cycling in clubs: a way to pursue sport-for-all and a result-oriented logics

Categorising sport-for-all and result-oriented goals in relation to their achievability through virtual cycling, our factor analysis resulted in two overarching sport-for-all goals (1) *inclusive participation* and (2) *attractiveness of the clubs' offer*, and one result-oriented goal (3) *skill development*. This process exposed a structure with three types of goals encompassing both sport-for-all and result-oriented goals, enabling us to pinpoint specific beliefs about virtual cycling's potential to pursue a sport-for-all logic and result-oriented logic.

The first type of goal that appears from our statements is *inclusive participation*. This factor includes eight statements that highlight virtual cycling's potential to engage diverse target groups and reduce participation barriers in cycling. Statements included 'make it possible for everyone to be introduced to cycling', 'reach and serve disadvantaged groups', 'focus on youth and seniors', or 'clearly communicate that cycling is a sport for everyone and all ages', amongst others (see Table 8 for a complete list). The statements within this factor relate to the sport-for-all policy dimensions of social inclusion, participation at all levels, health, and economic impact (e.g. revalidation) (Council of Europe, 2001; Mittag & Naul, 2021). Compared to the two other factors, inclusive participation was believed to be the least achievable by means of implementing virtual cycling in the club with a mean (scale 1-5) of 2.65 (SD = 0.77, Mdn = 2.62, IQR = 2.12 – 3.12).

The second type of goal covers *attractiveness of the club's offer*. This factor includes seven statements related to the potential of virtual cycling to make sports a more attractive, shared and motivating experience. Statements included 'create solidarity among club members', 'increase the fun aspect of our training sessions for our members' or 'motivate and engage our members to participate' (see Table 8 for a complete list). The statements within this factor relate to the sport-for-all policy dimensions of sociality and fun, sustainability, and urban

planning and infrastructure (Council of Europe, 2001; Mittag & Naul, 2021). Compared to the other factors, attractiveness of the club's offer was believed to be the second most achievable by means of implementing virtual cycling in the club with a mean (scale 1-5) of 3.01 (SD = 0.84, Mdn = 3.00, IQR = 2.29 – 3.57).

The third type of goal covers *skill development*. Skill development is the only sport-for-all goal that can be said to align with a result-oriented logic (Stenling & Fahlén, 2009). This factor includes three statements assessing virtual cycling's capacity to safely and effectively improve cycling skills. Statements included 'teach our members how to practice their sport in a correct, responsible way', 'Ensure sports performance in an informed and scientific way' and 'create a safe sports environment'. Compared to the other factors, skill development was believed to be the most achievable by means of implementing virtual cycling in the club with a mean (scale 1-5) of 3.39 (SD = 0.83, Mdn = 3.50, IQR = 2.67 – 4.00).

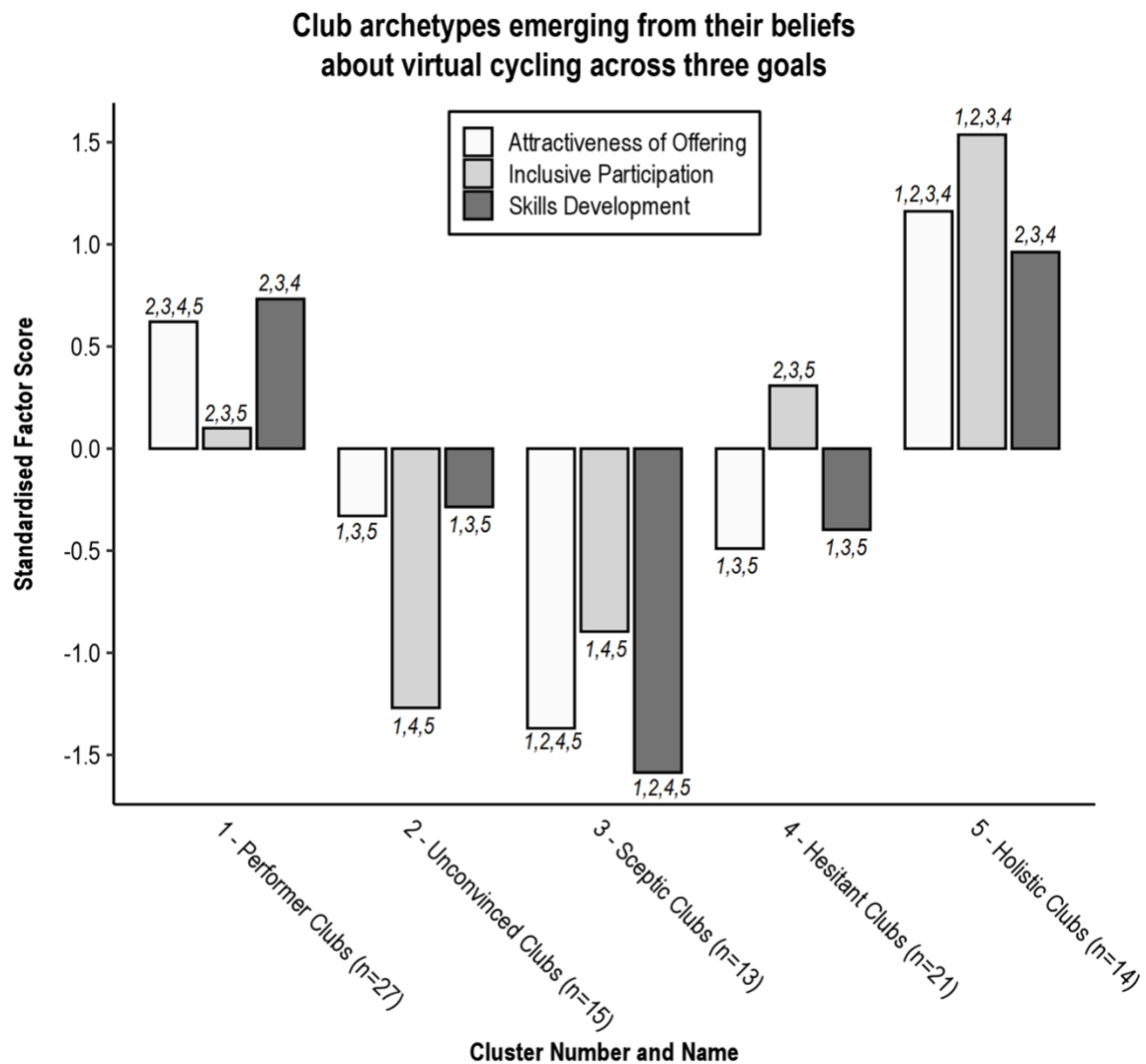
13.5.2 Archotyping clubs based on their beliefs about virtual cycling's potential

Our second research aim is to categorise clubs into archetypes based on their beliefs about whether and how virtual cycling can be a useful tool to achieve sport-for-all and result-oriented goals. We do this by clustering clubs, resulting in five archetypes that showed good parsimony, interpretability, and acceptable cluster sizes. Figure 13 illustrates the archetypes' Z-score on the three sport-for-all goals described above. These factors' Z-scores enable us to assess whether the score of each factor is above (positive Z-score) or below (negative Z-score) the mean score of that same factor per club archetype, in terms of times standard deviation. Each factor's Z-score was compared to the same factor's Z-score in all other clusters and significant differences with other clusters are indicated above or below each bar in Figure 13.

The five archetypes (Figure 13) are, based on their Z-scores, separated in 'innovator clubs' (i.e. 1 and 5) who believe virtual cycling's potential to be more positive, and 'laggard clubs' (i.e. 2, 3 and 4) who believe virtual cycling's potential to be neutral or negative. Club archetypes are labelled as followed: (1) performer clubs, (2) unconvinced clubs, (3) sceptic clubs, (4) hesitant clubs and (5) holistic clubs.

Within the innovator clubs, we find performer clubs (archetype 1) and holistic clubs (archetype 5). Performer clubs are characterised by their positive to neutral beliefs about using virtual cycling to improve both sports performance and achieve sport-for-all goals. Their motives to utilise virtual cycling stem from recognizing its potential to enhance members' skill development ($Z = 0.73$), as well as to augment the appeal of their club services ($Z = 0.62$). Performer clubs are however relatively neutral about the potential of virtual cycling to achieve inclusive participation within their club ($Z = 0.10$) (e.g. attract youth, elderly).

Figure 13: Standardised factor scores of goals for five club archetypes (n=90) (study 2). Each goals' Z-score was compared to the same goals' Z-score across all archetypes and, per Z-score, significantly different archetypes are indicated above or below each Z-score bar with its number in italics.



Holistic clubs are characterised by their overall positive beliefs about the potential of virtual cycling. Their motives to utilise virtual cycling is rooted in all three overarching goals, which all have high Z-scores. The highest potential is in fact seen in inclusive participation ($Z = 1.54$), followed by increasing their club services' attractiveness ($Z = 1.16$), closely followed by skill development ($Z = 0.96$). When comparing between the two positive club archetypes (1 and 5), we see the biggest difference in the perceived potential to use virtual cycling for inclusive participation, which is high for the holistic clubs, whereas the performer clubs remain neutral.

Within the laggard clubs, we find unconvinced clubs (archetype 2), sceptic clubs (archetype 3) and hesitant clubs (archetype 4). Hesitant clubs are characterised by an overall neutral stance towards virtual cycling's potential to improve the clubs' offer (i.e. services) and

achieve any of the three goals of inclusive participation ($Z = 0.31$), skill development ($Z = 0.40$) or attractiveness of their club's offer ($Z = -0.49$). Their motives to utilise virtual cycling are thus not attributed to a specific factor, nor particularly pro or contra.

Unconvinced clubs are characterised by a neutral stance on the potential of virtual cycling, except for their negative beliefs about inclusive participation ($Z = -1.27$) through virtual cycling. These clubs are thus unconvinced (cf. their labelling) that virtual cycling has the potential to improve their club's inclusivity and participation rates, while they think it *might have* potential (i.e. neutral stance) for skill development ($Z = -.029$) and attractiveness of their club's offer ($Z = -0.33$). Inclusive participation is thus undervalued by these clubs.

Sceptic clubs are characterised by an overall negative stance towards the virtual cycling's potential to achieve any of all three goals, i.e. inclusive participation ($Z = -0.90$), skill development ($Z = -1.59$) and attractiveness of their club's offer ($Z = -1.37$). Their overall belief about virtual cycling is negatively affected because they believe that none of the goals could be achieved by using virtual cycling in their club. In other words, virtual cycling does not seem like a fruitful innovation to engage in for the sceptic club archetype to achieve sport-for-all nor result-oriented goals.

13.6 Discussion

The discussion of our results is divided into two sections according to our two research aims. We begin by discussing voluntary sports clubs' beliefs about the potential of virtual cycling to improve sports performance and achieve sport-for-all goals. We continue by considering how institutional logics shape clubs' beliefs about virtual cycling's potential through our archetypes of clubs.

13.6.1 Do sport clubs believe that virtual cycling has the potential to pursue institutional logics?

Our results identified a structure of clubs' beliefs about virtual cycling across a range of potential uses that align with a result-oriented and a sport-for-all logic (Skille, 2009; Stenling & Fahlén, 2009). Our identified structure (based factor analysis in Table 8) revolves around two sport-for-all goals: *inclusive participation* and *attractiveness of the club's offer*, and one result-oriented goal: *skill development*. Out of these three, inclusive participation was believed to be the least achievable by means of implementing virtual cycling. Skill development was seen as most achievable, whereas attractiveness of the club's offer was believed to be the second most achievable goal through the use of virtual cycling. Understanding virtual cycling as a

social (service) innovation in sports (Corthouts et al., 2023), this finding both aligns with and nuances previous research on social innovations in sport. Specifically, studies have illustrated how social inclusion and mass participation are often a main driver for, and achievement resulting from, social innovation processes in sport organisations that do not involve radical technological innovations (Corthouts et al., 2020; De Bock et al., 2023; Nilssen & Tjønndal, 2021; Segrave, 2021). Still, there is limited research regarding technological social innovations in sport. Therefore, our findings indicate that (1) beliefs about the potential of technological social innovations such as virtual cycling may differ from (non-technical) policy-based social innovations, and that (2) clubs might still be ignorant about the fact that virtual cycling can serve as a dual-purpose innovation fostering both inclusivity and economic benefits. For example, inclusive virtual cycling events can equally serve as a strategic tool for sports sponsorship given that internet-based sponsorship becomes increasingly important (Koronios et al., 2023; Koronios & Dimitropoulos, 2020).

Out of the three goals, skill development is the only goal that aligns with a result-oriented logic (Stenling & Fahlén, 2009). Skill development is also perceived as the most readily achievable goal through virtual cycling by clubs, suggesting that clubs may view virtual cycling primarily as a tool for enhancing performances and results in competitive sports rather than achieving sport-for-all objectives. This interpretation aligns with studies showing that clubs often use of other digital tools to support competitive sporting success (Ehnold et al., 2020; Hoeber & Hoeber, 2012; Tjønndal, 2021). Consequently, the potential for technological innovations such as virtual sports to advance sport-for-all goals may remain underdeveloped. In other words, the self-evident use of virtual sports for result-oriented goals may undermine its potential use for sport-for-all goals.

Overall, we find that almost half ($n=41$) of the clubs included in our sample ($n=90$) believes that virtual cycling has potential to enhance sport-for-all goals as described by the sport-for-all Charter of the Council of Europe (2001) based on clubs' Z-scores across all three goals. Related to literature on how clubs engage in innovation, this finding underscores that of previous studies, demonstrating how beliefs of an innovation determine its potential use (Best et al., 2021; Corthouts et al., 2020; Hoeber & Hoeber, 2012).

13.6.2 How do beliefs about virtual cycling differ? Developing five archetypes of sports clubs

Using the concept of archetypes (Greenwood & Hinings, 1988; Hoye et al., 2020) we identify configurations of sports clubs' beliefs about the usefulness of virtual sports for achieving result-

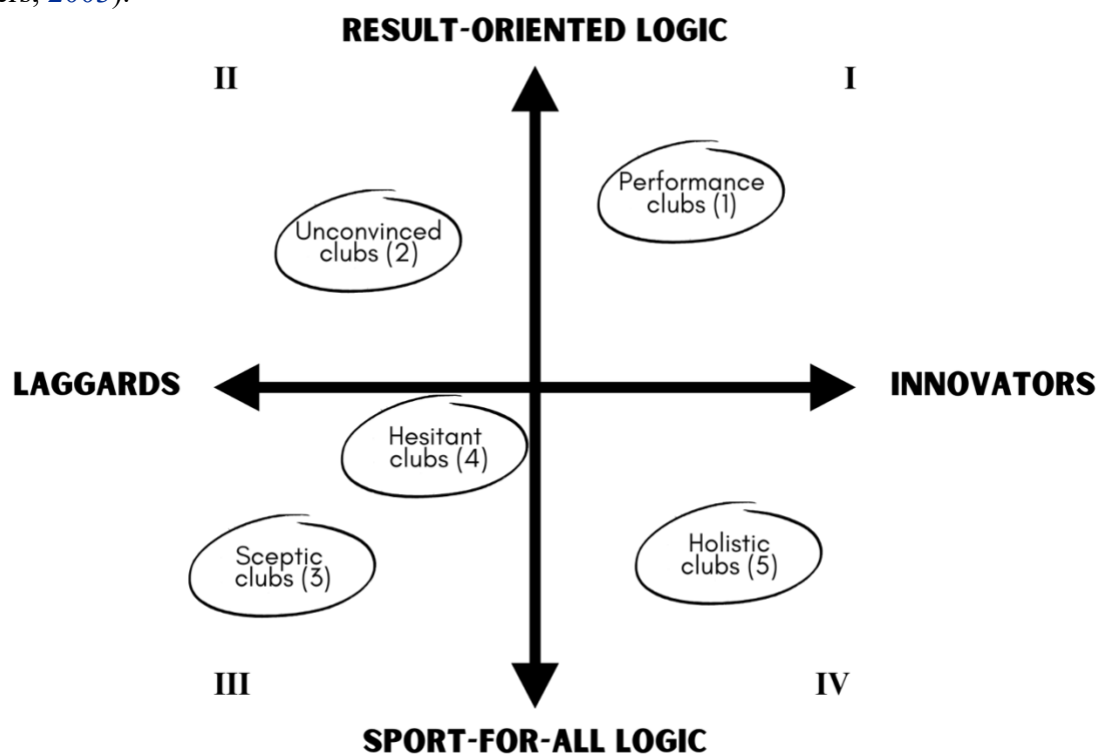
oriented and sport-for-all goals. The five club archetypes developed through this study (Figure 13) demonstrate how some clubs believe that there is high potential in virtual cycling to achieve inclusive participation, skill development, and increase the attractiveness of the club's offer, while others are hesitant or reject the idea that virtual cycling may hold potential to achieve such goals. Our results also show how all three overarching types of goals are meaningful in shaping the beliefs about the potential of using such technology to pursue a result-oriented logic or sport-for-all logic.

The results indicate that clubs' beliefs about virtual cycling for one type of goal do not necessarily hinder its perceived value for another. In other words, a low Z-score for the potential in one goal does not preclude clubs from believing that virtual cycling can be beneficial in other areas. For example, performer clubs (archetype 1) may believe that virtual cycling is not the right tool to strengthen inclusive participation in their clubs, but this does not negatively impact or impedes their belief that it has the potential to enhance skill development and increase the attractiveness of their services. The finding that clubs may not see potential in virtual cycling for one goal (e.g., skill development) without dismissing its value for another (e.g., inclusivity) underscores the significance of both institutional logics and innovation beliefs in understanding how clubs approach new technologies. Research on virtual cycling has shown that such platforms can serve varied purposes, such as enhancing sports performance (Devine et al., 2024; Fouquaert et al., 2025; Westmattelmann, Grotenhermen, Stoffers, & Schewe, 2021) to broadening social and participatory aspects (Botwina et al., 2024; Reed et al., 2023). Meanwhile, studies of digital innovations in sport organizations suggest that perceived utility often depends on whether club leaders (e.g. club presidents) see an alignment with their clubs' core objectives (Corthouts et al., 2020; Ehnold et al., 2020). By drawing on institutional logics (Skille, 2010; Stenling & Fahlén, 2009) and archetypes (Greenwood & Hinings, 1988; Hoye et al., 2020), we can interpret clubs' divergent views on virtual cycling as reflecting deeper configurations shaped by the conflicting logics of sport-for-all versus result-oriented. In parallel, Rogers' (2003) adopter categories provide a lens for examining how different archetypes may vary in their openness to integrate virtual cycling.

Through the development of our five archetypes, we identify that voluntary sports clubs' beliefs about the potential of virtual cycling is not arbitrary or incidental. Rather, voluntary sports clubs' beliefs interrelate with existing – often conflicting - institutional logics (Skille & Stenling, 2018; Stenling & Fahlén, 2009). Since our five archetypes emerge from their beliefs about virtual cycling across three addressable goals (Figure 13), we find it appropriate to discuss the possibility that these archetypes could be further characterised by

connecting them to (1) institutional theory (result-oriented logic vs. sport-for-all logic) in organised sports (Skille, 2010) and (2) the innovation adopter category continuum (Rogers, 2003). In summary, Figure 14 aligns our archetypes of clubs within these institutional logics and within the adopter category continuum, where we theorise that the archetypes could be placed along the axes of the result-oriented versus sport-for-all logic and the innovators versus laggards continuum of innovativeness in terms of using virtual sports.

Figure 14: Archetypes within institutional logics and innovation diffusion (study 2). Axes represent the institutional logics of result-oriented versus sport-for-all (Stenling & Fahlén, 2009) and the innovators versus laggards adopter categorisation continuum of innovativeness (Rogers, 2003).



Within the first quadrant (I), we theorise the archetype of performer clubs (1) to be better understood as innovators within the institutional result-oriented logic as their dominant rationale. It raises further questions if the narrative of technology and performance enhancement monopolises the sports sector and impedes the use of technological innovations for sport-for-all purposes. Further research may therefore investigate if and how policy design and implementation could overcome such biased focus in order to have its full potential unlocked and equally address the sport-for-all logic.

Moving to the second quadrant (II), we theorise that unconvinced clubs (2) are to be better understood as laggards within the institutional result-oriented logic as their dominant rationale. Here, knowledge gaps remain where future research could dive into, that is, how

policy and sport governing bodies could convince clubs to experiment with virtual sports and leverage its potential, and if done so, how clubs could then be convinced to explore the use of virtual sport to also address a sport-for-all logic.

For the third quadrant (III), we theorise that sceptic clubs (3) and hesitant clubs (4) are to be better understood as laggards within the sport-for-all logic as their dominant rationale. For sceptic clubs, future research can focus on which types of sport innovations other than technological innovations (c.f. typology of Tjønnedal, 2017) are more likely to be successfully adopted and therefore worth investing in. In line with Rogers (2003), laggards may in fact never adopt an innovation, meaning that it could not be worth pursuing the adoption of virtual sports for this archetype at all. For hesitant clubs, it could be argued that these clubs need some extra support during the innovation adoption process to successfully leverage the potential of virtual sports. Future research may therefore gain a deeper understanding of key innovation determinants (Corthouts et al., 2020; Hoeber & Hoeber, 2012) for this archetype so that sport governing bodies may act upon these in their support.

Last, within the fourth quadrant (IV), we theorise that holistic clubs (5) are to be better understood as innovators within the institutional sport-for-all logic as their dominant rationale. It could be argued that this archetype represents some sort of ideal type when it comes to leveraging the potential of virtual sports to achieve sport-for-all goals. Future research could expand on the idea of using holistic clubs as a resource of ‘best practice’ or ‘innovation champions’ when implementing virtual sports in other archetypes, as such approach is known to be efficient (Corthouts et al., 2020; Hoeber & Hoeber, 2012; Rogers, 2003).

Naturally, our discussion around Figure 14 is not something we evidence through our data or results. It is a connection we raise for further research, based on coupling literature from sport innovation management literature with both literature on the innovation adopter category continuum (Corthouts et al., 2022; Rogers, 2003) and literature on institutional logics in organised sports (De Bock et al., 2023; Skille, 2010).

13.6.3 Limitations of the study

There are some methodological limitations in this study that are important to recognise. The presented factor structure could have been more fine-grained, encompassing factors and more sport-for-all policy goals of Mittag and Naul (2021) in comparison to the three overarching themes we discovered. This might show that the 21 statements (Table 8) might not have been able to capture all sport-for-all goals thoroughly. It can further be questioned how investigating a different sport having virtual variants (e.g. rowing, climbing, table tennis) would impact the

study's results. Clubs' beliefs about the addressable sport-for-all goals may differ for sports that have greater impact on the environment or on sustainability (e.g. ski jumping, Formula 1), especially where its virtual discipline (e.g. Formula 1 Esports) could cope with aspects like addressing environmental or sustainability issues.

Another limitation of this study concerns the lack of existing and validated measurement instruments capturing most notable policy goals within the sport-for-all logic and the result-oriented logic, requiring us to develop our own instrument based on existing policy charters (Council of Europe, 2001; Mittag & Naul, 2021). Following these policy texts, some items were formulated in an equally broad manner (e.g. item 7 'make cycling or triathlon more accessible') which could, based on its interpretation, theoretically fit within multiple of the resulting factors. Therefore, we strictly followed our empirical model during factor development when factor loadings were statistically valid (standardised loadings > .40) though open for discussion (e.g. item 7 loading on the third factor of attractiveness of offering). Nonetheless, the fact that the statements were asked in a random order and then all but one statement still loaded together with similar adjacent statements further confirms the validity of our factor analysis.

13.6.4 Future Research

As already highlighted in our discussion of Figure 14, future research can build on our findings in several ways. First, although our data capture clubs' immediate beliefs of how virtual cycling might be useful to achieve result-oriented and sport-for-all goals, longitudinal studies could investigate how these beliefs evolve as clubs gain more exposure to and experience with virtual cycling platforms and activities. Operationalisation could be possible through panel surveys tracking the same clubs over multiple years, through ethnographic case studies that follow selected clubs during its adoption and implementation process of virtual cycling, or through the combination in a mixed-method design integrating survey data with in-depth interviews. However, alternative designs to capture both attitudinal changes and behavioural practices over time, are equally encouraged in future studies. Second, examining how different types of virtual sports (e.g. virtual running, virtual rowing, augmented table tennis) could provide a broader understanding of which innovations align best with specific institutional logics. Third, further analyses of how club leaders negotiate possible tensions between sport-for-all values (e.g., inclusive participation) and a result-oriented mindset (e.g., skill development) could reveal the internal decision-making processes that either facilitate or hinder innovation adoption. Finally, exploring governmental and policy-level interventions tailored to each archetype could help identify best practices for supporting clubs' varied goals, ensuring that innovation serves both

competitive excellence and broader social inclusion in sport.

13.7 Conclusion

The aim of our study was twofold. First, to examine voluntary sports clubs' beliefs about whether and how virtual cycling can be a useful tool to achieve result-oriented and sport-for-all goals. Second, we aimed to categorise clubs into archetypes based on their beliefs about virtual cycling. Our results show that almost half of the clubs (41 of 90 clubs) believe that virtual cycling has potential to achieve either or both result-oriented and sport-for-all goals. We have identified this by seeing how our factor structure of three overarching goals differs amongst clubs. The resulting factor solutions ties into the sport-for-all logic and the result-oriented logic, presenting three overarching addressable goals by clubs: (1) inclusive participation, (2) skill development, and (3) the attractiveness of the club's offer (Table 8). Inclusive participation and attractiveness of the club's offer relate to a sport-for-all logic and sport-for-all goals (Council of Europe, 2001; Mittag & Naul, 2021), while skill development relates to a result-oriented logic focusing on enhancing sports performance. Inclusive participation was believed to be the least achievable by means of implementing virtual cycling, while skill development was seen as the most achievable. This suggests that the clubs in our sample believed that virtual cycling currently holds more potential as a tool for sport performance enhancement, catering a result-oriented logic over a sport-for-all logic.

In response to our second research aim we employed a cluster analysis which identified five archetypes of clubs: (1) performer clubs, (2) unconvinced clubs, (3) sceptic clubs, (4) hesitant clubs and (5) holistic clubs. Archetypes 1 and 5 were labelled as 'innovator clubs', while archetypes 2, 3 and 4 were labelled as 'laggard clubs'. The significant differences between club archetypes shows the potential of virtual cycling to address both sport-for-all goals and result-oriented goals (Figure 13). Our club archetypes demonstrate that virtual cycling adoption is not a one-size-fits-all story: clubs have different beliefs about the goals virtual cycling could address. Using club archetypes, we can theorise for each archetype possible goals are important 'reasons for' adoption and which 'reasons against' adoption to avoid, to which governmental programs and support can be tailored.

Last, we theorise that our archetypes align with both institutional theory (result-oriented logic vs. sport-for-all logic) in organised sports (Skille, 2010), and the innovation adopter category continuum (Rogers, 2003) (Figure 14). If policy and sport governing bodies aim to leverage innovative practices within the sport-for-all logic using virtual sports, we recommend

them to consider clubs' beliefs about this – potentially social – innovation using an individualised approach as demonstrated using our archetypes. We believe that virtual sports will continue to gain popularity because of their ability to fulfil a wide array of customers' needs. However, it remains difficult to anticipate on this popularity increase as virtual sports sit at the intersection between public (i.e. sports) and private domains (e.g. commercial virtual sports platforms). It would therefore be a missed opportunity for policy and sport governing bodies not to benefit from its commercial and result-oriented success by implementing it in such a way that it equally caters a sport-for-all logic.

13.8 References Study 2

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Ethical statement

Ethical approval was obtained from the ethical commission of the Faculty of Medicine and Health Sciences of Ghent University (code: 2021-106). Permission to conduct the online data collection for the purposes of this research was obtained by all respondents (i.e. clubs), who were fully informed about the purposes of this research and how their responses would be used and stored (i.e. anonymised).

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We have no conflicts of interests to disclose.

Data Availability Statement

Only anonymous data are available from the corresponding author, upon reasonable request.

14 Study 3: How Virtual Sports Became Sports: Legitimisation Strategies of Virtual Cycling as Disruptive Innovation

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Abstract

This study examines how sport governing bodies engage with the development of virtual cycling as a new sporting discipline. Drawing on theories of innovation diffusion, disruptive innovation theory and institutional theory, we examine how governing bodies in cycling and triathlon legitimise and delegitimise virtual cycling, as well as the different mechanisms they employ in their process. Data were collected through 16 semi-structured interviews with representatives from international and national federations, the International Olympic Committee, athletes, and platform developers. Findings reveal two legitimisation strategies: (1) constructing new evaluation criteria (i.e. attributing normative value to new features like safety, inclusivity, ecological value, or revenue models) and (2) developing new markets by attracting youth, women, para-athletes. Alongside two legitimisation strategies, we also identified one delegitimation strategy covering externalised and passive governance of virtual cycling. Interviewees framed virtual cycling as lacking credibility due to its dependence on commercial digital platforms, and persistent technical underdevelopment. Platform fragmentation, the absence of independent verification systems, and limited interoperability (e.g., cross-platform rankings) illustrate how a technical strategy has been passively neglected relative to the employed institutional strategy actively shaping the new norms of cycling through virtual cycling's unique features. Combined, the imbalance between legitimising and delegitimising mechanisms highlights a case of “decoupling” where sport governing bodies' public endorsement of virtual cycling conflicts with the practical challenges of developing the technical infrastructure, particularly where it intersects with others' intellectual property.

Keywords: Disruptive innovation; Legitimacy; Virtual sports; Institutional strategy; Innovation diffusion

14.1 Introduction

Virtual sports have developed as new sport disciplines where mixed reality games are intertwined with physical sports via interactive peripheral sport equipment (Westmattelmann et al., 2025). For sport governing bodies, these innovations offer opportunities such as attracting new audiences and developing novel spectating formats (Westmattelmann et al., 2021). Yet, whether these new disciplines gain acceptance and recognition as legitimised competition formats depend on how sport governing bodies engage with virtual sports. While several sport governing bodies, such as those for taekwondo, archery and sailing have already developed and implemented virtual competition formats, cycling and triathlon currently offer the most advanced cases of virtual sports. Since its launch in 2014, virtual cycling has evolved immensely, with competition platforms emerging, merging, or exiting the market. As of 2024, two commercial platforms dominate the virtual cycling market: MyWhoosh and Zwift. Operating independently of sport governing bodies, these platforms exemplify how technology providers shape the development of sport while challenging traditional power hierarchies (Fouquaert et al., 2025). The dependence on commercial competition platforms is a trait unique to virtual sports, something that poses novel challenges for sport governing bodies, who typically represent authorities of power in sport.

Considering the skewed stakeholder power of technology providers in virtual sports (Fouquaert et al., 2025), sport governing bodies are likely to react to, and engage with, these innovations in different ways. Some might seek to legitimise the development of virtual sports as new disciplines, event types, or competition formats, while others would seek to delegitimise these innovations. From this starting point, this study aims to examine how sport governing bodies react to, and engage with, the development of virtual cycling as disruptive sport innovations. Specifically, the following research question guides our analysis: How do governing bodies in cycling and triathlon legitimise or delegitimise virtual cycling, and what mechanisms do they employ in these processes?

We examine both cycling and triathlon organisations as they rely on the same technology and stakeholders for their virtual competition formats (virtual cycling and virtual triathlon). While virtual triathlon also includes pool swimming and virtual running using treadmills, both virtual sports share similar virtual cycling platforms and formats. As both sports are thus comparable cases, we refer to them collectively as ‘virtual cycling’. A central premise for our study is the recognition that virtual sports shares characteristics of both sport (e.g., mimicking parent sport with gross motoric physical exertion) and esports (e.g., digital mediation and platform dependence). Therefore, our literature review draws on both studies in

sport and esports legitimisation. Analytically we approach virtual cycling as a disruptive innovation within sport (cycling and triathlon) and not esports, and hence do not aim to expand the debate on whether virtual sports or esports should be considered sports (Borggreffe & Hoffmann, 2024).

14.2 Literature and theoretical framework

14.2.1 Legitimisation of innovations in sports and esports

To understand how sport governing bodies engage with technological innovation, we ground our understanding of legitimacy in neo-institutional theory. This posits that organisations compete for resources or customers, while also being driven by the need to ensure that their organisational practices are perceived as desirable and appropriate within a system of norms that they proactively shape themselves (DiMaggio & Powell, 1983). Hence, legitimacy is a strategic process through which organisations shape cultural and cognitive frames of stakeholders, thereby constructing new evaluation criteria, as highlighted by Das and Van de Ven (2000). Suchman (1995) identifies three primary forms of legitimacy: pragmatic legitimacy, based on the self-interest of stakeholders who believe they derive benefits like support or shared values; moral legitimacy, based on normative judgments of whether an activity is “right” within a value system; and cognitive legitimacy, based on comprehensibility and taken-for-grantedness of institutional norms.

In the context of sport, research indicates that legitimacy is often driven by pragmatic and moral legitimacy, rather than being driven by technical features and adoption (Stauff & Vogan, 2021; Stenling, 2014). For instance, Pizzo et al. (2022) demonstrate how traditional sport organisations integrated esports by framing them as complementary activities to align with existing stakeholder expectations. Similar to virtual cycling as a new race format in cycling, Read (2023) illustrates how legitimacy of new formats like “The Hundred” in cricket was secured using a strategy positioned it as a distinct category of cricket through the use of seven mechanisms (e.g., gender equity, new sponsorship, new broadcasting). Another illustrative study of legitimacy being socially constructed shows how new practices like “drive-in sports” in Sweden gained acceptance through problem-solving narratives (e.g., addressing norm-breaking behaviour) and alignment with sport-for-all principles (Stenling, 2014). Similarly, but outside the context of sport, Ben-Slimane et al. (2020) examined legitimisation strategies for cardboard coffins, reporting a shift from mainstream evaluation criteria (e.g., material quality) to new evaluation criteria such as price, ecological impact and personalisation. These studies highlight

how legitimisation often involves strategic framing and alignment with prevailing cultural value through socially constructing different types of legitimacy (Suchman, 1995). For instance, pragmatic legitimacy is observed in the economic incentives and new market access described in Pizzo et al.'s (2022) study showed that sports integration created the perception of direct benefit with stakeholders. In the studies of Read (2023), Stenling (2014), and Ben-Slimane et al. (2020), moral legitimacy is developed through framing the innovation (i.e. The Hundred, drive-in sports, and cardboard coffins) around normative goals that align with societal values like gender equity, Sports-for-All implementation, and ecological impact. Last, Ben-Slimane et al. (2020) also illustrate cognitive legitimacy under the use of instructional videos and design adaptations (e.g., adding handles) to ensure cardboard coffins became fitting, comprehensible and ultimately taken-for-granted within the existing norms and cognitive schemas of the funeral industry.

There are also studies demonstrating how delegitimisation of technological innovations such as virtual cycling, occurs in the context of sport (Reitman et al., 2020; Summerley, 2020). For instance, digital competition environments in esports have been associated with toxic communication and behaviour (Felczak, 2022), exclusionary tribal and negative cultures (Hayday et al., 2021), and institutional promotion where power imbalances force athletes act as unpaid promoters (Chee & Karhulahti, 2020). In virtual cycling specifically, these risks extend to integrity issues like digital weight manipulation and algorithmic exploitation (Richardson, 2025; Westmattelmann et al., 2025), potential concerns on overly focussing on numbers and the quantified-self (Felczak & Filiciak, 2025), or vulnerabilities regarding data privacy, commodification of athletes, and exclusion of athletes based on gender or origin (Fouquaert et al., 2025; Tjørndal et al., 2025).

Last, pursuing legitimacy can lead to tensions between the structure of organisations and their internal technical operations. Neo-institutional literature describes this as “decoupling”: in seeking legitimacy, organisations adopt policies to satisfy societal norms (e.g., inclusivity or fairness) while their internal technical operations present a conflicting reality to these claimed policies (Boxenbaum & Jonsson, 2008). Washington and Patterson (2011) highlight that, in sport, mechanisms for legitimacy can become decoupled from performance outcomes when institutional pressures regarding rules and norms conflict with technical realities. This suggests that sport governing bodies who embrace the idea of a technological innovation can nonetheless struggle with its technical implementation because they lack the needed structures, resources and organisational capacity to bridge this gap.

14.2.2 Legitimisation of virtual cycling: a disruptive innovation perspective

We examine virtual cycling through a theoretical lens of disruptive innovation, rooted in our neo-institutional understanding of how sport governing bodies seek social acceptance for novel technologies. Disruptive innovations are typically defined as products or services that initially underperform on mainstream criteria but introduce new functionalities and benefits to niche or non-consumers, such as accessibility, affordability, or convenience (Christensen et al., 2015). Virtual cycling can be interpreted to fit this classification: while it may currently lack the robust anti-doping or standardised governance of traditional cycling (e.g., universally recognised rankings and events) (Dyer & Taylor, 2023; Fouquaert et al., 2025; Richardson, 2025), it introduces new features that may meet unserved customer needs, including affordability (one-time equipment costs offset travel), accessibility (indoor cycling regardless of time or weather), gamification and customisation (e.g., Zwift's PowerUps feature), safe participation (avoiding crashes), and individual adaptability of the required efforts (e.g., in group rides, include para-athletes) (Bentvelzen et al., 2022; Evsikov et al., 2024; Reed et al., 2023; Westmattelmann et al., 2021). The introduction of these new features can be linked to neo-institutional theory and the fact that disruptive innovations like virtual cycling inherently suffer from a lack of legitimacy (Ben-Slimane et al., 2020). For virtual cycling to develop legitimacy, it must be assessed against the institutional and societal norms of sports and in particular of cycling. This assessment ultimately requires new performance evaluation criteria because disruptive innovation (i.e. virtual cycling) introduces new functionalities that mainstream performance evaluation criteria (i.e. of cycling) cannot adequately assess (Christensen & Bower, 1996).

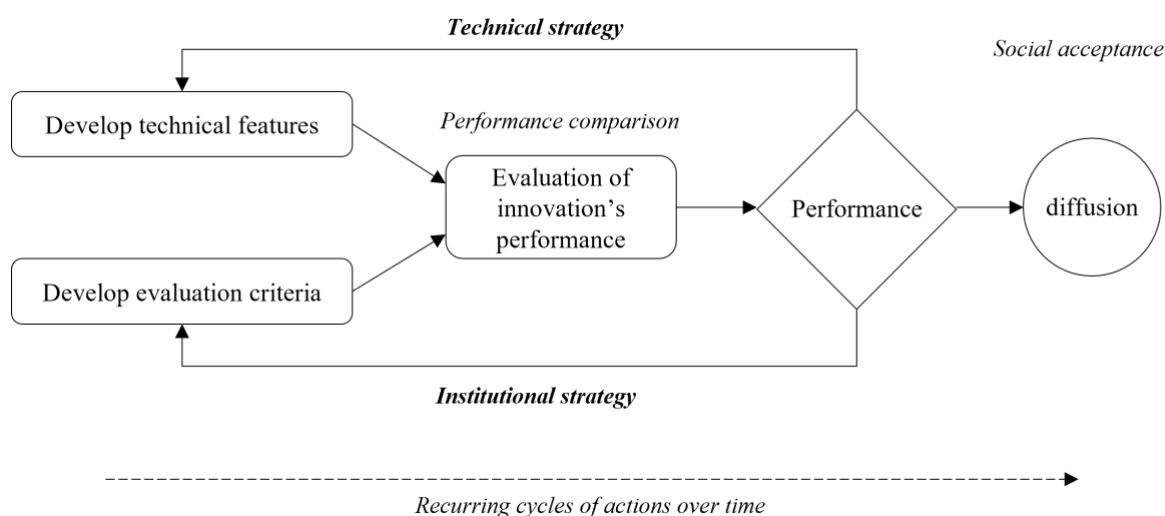
However, applying a disruptive innovation perspective to virtual cycling requires nuance regarding the organisational structures and particularities of the adopting sport governing bodies. Disruptive innovation theory often assumes that the firm seeking to disrupt and gain legitimacy in the market with a new technology, also controls its technological development. Sport governing bodies (SGBs) are regulatory institutions that are constrained in their structure, capacity and resources to develop the hardware (e.g., mechanical doping detection, trainer accuracy) and software (e.g., cross-platform integration, algorithm standardisation) of virtual cycling. Instead, they depend on commercial platforms (e.g., Zwift, MyWhoosh, Suito) to provide the technical infrastructure. This structural reality can be related to a potential decoupling as it challenges sport governing bodies in their legitimisation process, who must seek to validate a new discipline despite lacking direct control over its technical

features. Hence, the organisational constraints sport governing bodies creates a dependency on external stakeholders like commercial platforms that need strategic navigating.

14.2.3 A theoretical model of innovation diffusion strategies to legitimise virtual cycling

To analyse how sport governing bodies navigate their organisational constraints during the development of virtual cycling, we utilise Das and Van de Ven's (2000) process model of innovation diffusion strategies, where diffusion can be interpreted as a form of social acceptance, and hence, legitimisation. This model distinguishes between two strategies: (1) technical strategy, which are actions that manage the changes in technological features, and an (2) institutional strategy, which are actions that manage the changes in criteria used to evaluate the innovation. As illustrated in Figure 15, innovations' 'performance comparison' is thus based on both newly developed technical features and newly developed evaluation criteria. This means that performance (i.e. the value cycling has to offer) can change regardless of actual technical feature development but because of new evaluation criteria to assess performance.

Figure 15: Process model of innovation diffusion strategies (study 3), adapted from Das and Van de Ven (2000).



Das and Van de Ven (2000) propose that the choice of strategy is contingent on the *nature of the technology* and the *market structure* as illustrated in Figure 16. For the nature of technology, evolved technology is incrementally developed while novel technology allows to address new functional needs. Virtual cycling represents a novel technology compared to other, mainstream, cycling disciplines. For the market structure, dispersed markets have many well-informed potential customers, contrary to concentrated markets where few potential customers may negotiate over performance evaluation criteria (Das & Van de Ven, 2000). The market structure of virtual cycling is, however, complex: it involves a concentrated market of institutional

adopters (federations, IOC) who negotiate rules, and a dispersed market of individual end-users. Because we focus on institutional adopters, we assume the market in our case to be concentrated, requiring us to analyse if and how both a technical and institutional strategy are employed by sport governing bodies.

Figure 16: Innovation diffusion strategies based on technology and market (study 3), adapted from Das and Van de Ven (2000).

		<i>Nature of market</i>	
		Dispersed	Concentrated
<i>Nature of technology</i>	Evolved	<i>Technical strategy</i>	<i>Technical & Institutional Strategy</i>
	Novel	<i>Technical & Institutional Strategy</i>	<i>Institutional Strategy</i>

Applying this model to sport governing bodies requires incorporating the organisational nuances discussed above. While the model prescribes a mixed strategy (i.e. technical and institutional), sport governing bodies are potentially constrained by their structure, capacity and resources to actively pursue a technical strategy, given their core expertise lies in policy development and not technological development. Such organisational constraints potentially force a reliance on institutional strategy over technical strategy, while both are needed to legitimise and diffuse new technology (Das & Van de Ven, 2000). Next, we will turn to our methodological approach allowing us to explore this tension.

14.3 Methodology

14.3.1 Sampling, data collection and participants

Recognising innovation legitimisation as a socially constructed process shaped by strategic framing, negotiation, and technological development, we adopted a qualitative research design. This methodological approach facilitates the examination and capture of mechanisms (including normative arguments, narrative constructions, and evolving evaluation criteria) that underpin the legitimisation of virtual cycling.

Data collection relied on a multistage purposeful sampling design (Palinkas et al., 2015). The first stage begins with a broader view emphasising variation, for which we

purposely sampled a variety of stakeholders (e.g., managers within international federations, elite athletes, developers, reporters) likely to possess specific knowledge of the legitimisation process of virtual cycling. The second stage moves to a narrow view emphasising in-depth information on central tendencies, for which we supplemented purposeful sampling with snowball sampling to be chain-referred to experts who are vocal about specific earlier uncovered mechanism that drive virtual cycling legitimisation strategies.

Data was eventually collected from a sample of 13 participants (see Table 9) through 16 qualitative semi-structured interviews. To align with our theoretical model (Das & Van de Ven, 2000), which highlights the duality of achieving legitimacy through both an institutional and technical strategy, participants were sampled to include both strategies in our empirical material (see Table 9). About half our sample consisted of institutional and policy making entities in cycling and triathlon. This includes representatives from UCI, World Triathlon, the IOC, and two national federations. These actors have oversight over and influence on the mechanisms that make up legitimisation strategies potentially aligning with the theoretical notion of an institutional strategy, for instance virtual cycling's market development, Olympic inclusion, rules, calendars, events, or policy. The other half of our sample consisted of technical innovators and practitioners including SuperTri as format and league developer, a platform developer, elite athletes and specialised reporters. These actors have oversight over and drive demand of mechanisms and technical features that make up legitimisation strategies potentially aligning with the theoretical notion of a technical strategy, for instance collaborations, feature development, interoperability.

Table 9: description of interview participants (study 3)

Pseudo-nym	Gender	Theoretical-alignment	Expertise / Role	Follow-up interview
Stan	Male	Institutional	Innovation manager, national sport governing body	-
Robert	Male	Institutional	Management, international sport governing body	Yes
Kim	Male	Technical	Reporter & content creator	-
Floyd	Male	Institutional	Management, international sport governing body	Yes
Claressa	Female	Technical	Elite virtual cycling athlete	-
Manny	Male	Technical	Elite virtual cycling athlete	-
Mike	Male	Technical	CEO and developer virtual cycling platform	-
Elvene	Female	Technical	Elite virtual cycling athlete	-
Ben	Male	Institutional	Management, (inter)national sport governing body	-
Ali	Male	Both	Management, SuperTri, developing virtual league	-
Rika	Female	Technical	Athlete, content creator & reporter	Yes
Tony	Male	Institutional	Management, international sport governing body	-
Lucas	Male	Institutional	Department manager, IOC	-

To ensure anonymity, all participants are referred to using pseudonyms. The study received ethical approval from Nord University, SIKT – The Norwegian Agency for Shared Services in Education and Research. All participants provided informed voluntary consent.

Our interview guide comprises five topics related to the (de)legitimisation of virtual cycling: (1) personal or organisational experience with virtual cycling, (2) mechanisms contributing to the (de)legitimisation of virtual cycling, (3) experience with (organisational) policy and governance of virtual cycling, (4) participation in virtual cycling, and (5) technology and change within interviewee's organisation. One interview was conducted face-to-face, while 15 interviews were conducted using online video calls. Interviews were either conducted by the first author or collectively by the first two authors and lasted between 40 to 80 minutes. They were recorded and transcribed in full. The interview of Stan was not used in further analysis because it was considered out of scope.

14.3.2 Data analysis

We applied Eggebø's (2020) four-step collective qualitative analysis. First, the first two authors each drafted one-page summaries of every transcript and shared them with each other. Second, they independently coded data-driven themes and displayed these on Post-its for collaborative mapping. Third, themes were merged or refined into 20 analytical themes, each annotated on A4 sheets with subthemes, illustrative quotes, and interviewee pseudonyms. Analytical themes were also visually linked to identify patterns, further aiding our understanding of how both legitimising and delegitimising mechanisms were employed. Fourth, we devised an outline for manuscript development based on these categories. This study focuses on three of the twenty analytical categories to inform our research question: (1) legitimisation, (2) delegitimation and (3) stakeholders. While these themes could be considered as primary to our analysis, other themes (e.g., Olympic Esports Games, national federations, hardware) naturally overlapped and secondarily informed our analysis. Going forward with our three primary analytical categories, all transcripts were revisited to extract exemplar quotes.

14.4 Findings and Discussion

The following sections outline our findings that revolve around two main strategies to legitimise virtual cycling: (1) developing new evaluation criteria; (2) developing new markets; and one main strategy to delegitimising virtual cycling: (3) externalised & passive governance. A summary of the strategies with examples from our data are shown in Table 10.

Table 10: Summary of virtual cycling (de)legitimation strategies (study 3). Examples are illustrative for how strategies and mechanisms are employed by sport governing bodies.

Legitimisation strategies	Evaluation criteria & Strategic mechanism	Examples
Constructing new evaluation criteria	Improve safety, reduce risks	Avoiding peloton and crashes Controlled indoor environment
	Reduce costs and logistics	Reducing travel Fewer restrictions and permits
	Develop sport-for-all in cycling	Online talent scouting Independent of infrastructure
	Develop sustainable events	Online mass participation events Online (elite) competitions
	Diversify revenue streams	Streaming by individual athletes Spectating entry fees indoor events
Developing new markets	Customising cycling experience	New training and racing formats Gamification and social features
	Targeted inclusion	Engage in different communities Inclusion of women & para-athletes
Delegitimisation strategies	Strategic mechanisms	Examples
Externalised & passive governance	Sustaining platform reliance	Commercial nature of platforms Lack of platform interoperability
	Lacking organisational policies	Lacking guidelines for federations Inconsistent support of federations

14.4.1 Legitimising virtual cycling through the construction of new evaluation criteria

A first strategy to legitimise virtual cycling entails various mechanisms that can be interpreted as the development of new evaluation criteria, suggesting that sport governing bodies rely on an institutional strategy (Das & Van de Ven, 2000). Within this institutional strategy, sport governing bodies actively shape the perception of what criteria can be considered important and societally relevant to cycling. Virtual cycling is described to outperform mainstream cycling disciplines in relation to these new evaluation criteria. Specifically, we identified five such evaluation criteria in our data (see Table 10). First, *improving safety and reducing risks* is found as an important criterion:

A lot of people are just not ready or feel like there's the barrier to go and actually do a biking race is massive, right? They feel unsafe to ride in a pack, cornering, and so on.

There's an opportunity with virtual cycling to actually bring those communities into sports and in a social environment, right? (Ali)

Two small children depend on me. I can't afford to be ran off the road or you know someone not looking and hitting me. So, I've moved everything indoors [...]. (Claressa)

From the criteria of safety in sport and protecting athletes' wellbeing, virtual cycling can be perceived as positive as there are fewer inherent risks compared to outdoor cycling (crashes or route obstructions) strengthening current policies of prioritising athlete safety in cycling (Reed et al., 2023).

Second, *reducing costs and logistics* is an evaluation criterion introduced by our participants, as they explain the benefits of virtual cycling with regards to (1) participating in events independent of space and time, and (2) organising events without the logistical administrative overhead of outdoor events:

The benefit to virtual cycling, eliminating the need for travel and costly expenses like hotel rooms and flights and things will allow us to have more consistent racing. [...] It's still minimizing the cost of that travel and the effects of that travel, but legitimising the sport. (Robert)

[...] try to close the road up for a cycling event or a triathlon event. I mean, it's impossible almost, right? (Ali)

These explanations can be interpreted as introduction of new evaluation criterion to assess the performance of different cycling disciplines, favouring virtual cycling. While participants point out the advantages of virtual cycling compared to the challenging nature of organising mass-participation outdoor sports events in public spaces, we know from previous research that virtual cycling also holds unique challenges, like (mechanical) doping (Dyer & Taylor, 2023; Richardson, 2025), exclusion (Tjønndal et al., 2025), algorithmic transparency (Westmattelmann et al., 2025), data privacy, performance verification and equal qualification pathways (Fouquaert et al., 2025). On a more critical note, the absence of these challenges in our interviews suggests a selective legitimising narrative. By emphasizing normative advantages while neglecting addressing technical limitations, governing bodies engage in a form of “decoupling”. As recognised earlier, this occurs when institutional ambitions (e.g., claiming to offer a fair, low-cost sport) are projected publicly, despite not being supported by

the technical reality required to enforce them (e.g., independent performance verification, cross-platform qualification system) (Boxenbaum & Jonsson, 2008; Washington & Patterson, 2011). This decoupling is arguably driven by the earlier recognised organisational constraints of sport governing bodies (e.g., the lack of intellectual property rights), which limits their ability to develop the required technical infrastructure.

Third, *developing sport-for-all* in cycling is something sport governing bodies attribute value to, as they mention how virtual cycling allows for online talent detection or to make cycling more accessible:

We're trying to establish virtual cycling as an activity in that position between not riding your bike or riding outdoors. And from [our] perspective, it's a fantastic tool for our national federation partners to use, perhaps in developing countries as a good example, somewhere like Colombia that has a large cycling population, but really bad road infrastructure. Or [where] in summertime it's simply too hot to ride outdoors. (Robert)

[Federations] can find talent in a corner of the country that is not normally known for good riders rather than finding it in the normal traditional ways, through the traditional clubs. (Manny)

From Robert and Manny's explanation, one might perceive virtual cycling to outperform traditional cycling as a social inclusion initiative in talent development and selection. This and other sport-for-all development initiatives continue to hold substantial influence and importance in national and international sport policies (Mittag & Naul, 2021). In this case, this inclusion argument is utilised in support of the legitimisation of virtual cycling. As found in other studies, Sport-for-All development is often used to legitimise new practices in sport (e.g., drive-in sports) (Stenling, 2014). However, the case of virtual cycling might differ from other innovation in sport, as studies have also identified context social exclusion issues in virtual cycling (Tjørndal et al., 2025). Some of these issues relate to the accessibility of virtual cycling (e.g. purchasing equipment, need for physical space for such equipment) (Evsikov et al., 2024).

The development of sport-for-all also ties into the fourth evaluation criterion introduced in our data, *developing sustainable events*. Here, participants describe how virtual cycling represents a new way of organising cycling events that is socially, economically and ecologically more sustainable compared to traditional cycling events:

We don't have to come to each other's cities [to race]; it's minimizing the cost of that travel and the effects of that travel. (Robert)

We might have 2,000 people competing online. So, we are giving more opportunities thanks to this technology ... without increasing the footprint of carbon... . [M]ore people around the world are able to compete ... without traveling. (Lucas)

And so that ease of access, the kind of democratisation of sport, is incredibly valuable in terms of growing the sport. I genuinely think that it's a way of bringing more people into competitive sport who wouldn't be able to go out and compete in the real world for any number of reasons. (Mike)

This new evaluation criterion can be understood against increased policy focus on sport events that are both environmentally and socially sustainable (Mittag & Naul, 2021). In relation to social sustainability, virtual mass participation events like the Cancer Council Tasmania's virtual Women's 5 km have been shown to provide practical and inclusive benefits (Hookway et al., 2025), thus contributing to developing the pragmatic legitimacy of virtual cycling (Suchman, 1995).

Last, *diversifying revenue streams* also appeared in our data as a new evaluation criterion of importance. Participants raised several arguments to why virtual cycling may “outperform” other cycling disciplines when evaluated against traditional revenue structures and athlete-driven monetisation:

virtual cycling competitions are yearlong, it's every week ... nine times out of ten, they [athletes] are streaming live on Twitch. If they're not streaming their races, they're streaming their training. (Rika)

[many athletes] stream consistently from their home. A low-cost way to generate their own revenue. (Robert)

When you look at traditional esports, they have generated significant revenue through things like live streaming, TV rights, as well as sponsorship of teams and athletes. (Robert)

These quotes illustrate how newly introduced evaluation criteria may reinforce each other. For instance, this revenue shift supports the reduced costs and sport-for-all criteria as it allows non-professional riders to earn income via streaming while lowering participation costs. In addition, these quotes also illustrate a mechanism by which pragmatic legitimacy (Suchman, 1995) for virtual cycling is developed as they emphasise virtual cycling's revenue potential through

(indoor) ticket sales, livestreaming, and shifting sponsorship from teams to organisers and athletes. This reflects a key point in the literature where esports are known to rely almost entirely on online viewership, reducing costs tied to TV broadcasting and media rights (Kim & Kim, 2020).

14.4.2 Legitimising virtual cycling through the development of new markets

As outlined by Christensen & Bower (1996), disruptive innovations serve both the unmet needs of existing customers and the needs of new customers. Relatedly, Table 10 shows that sport governing bodies relied on developing new markets to legitimise virtual cycling by using two mechanisms. First, several interviewees voiced *customising the cycling experience* as a way of developing new markets:

Both getting young people to get into sports from their home, but you also want to organize some like e-sports live event for the professional athletes ... by having all these opportunities that technology is giving us. (Lucas)

People would argue that cycling in general is not really a very exciting sport to watch. You watch a six-hour race, how much of it do you actually watch? I think virtual cycling has a huge potential in that. The World Championships was over in an hour. And it was three, you know, action-packed stages. (Kim)

This illustrates how virtual cycling allows the cycling industry to expand in different ways: the audience who spectates virtual cycling and the athlete as consumer who adopts it. Virtual cycling introduces new training and racing formats for cyclists from leisure to elite participation, but also for those watching sports competitions (Fouquaert et al., 2025; Reed et al., 2023; Rojas-Valverde et al., 2022). Our data highlights that virtual cycling platforms do this by offering gamified sports competitions where participants now “have to be tactically smart and clever and be able to use the game and its construct in [their] advantage” (Tony). Simultaneously, virtual cycling attracts competitive athletes who praise the “opportunity to win over someone that might be far superior from an athletic performance perspective” (Tony). This distinction allows governing bodies to shape a legitimising narrative of traditional cycling becoming customisable through for instance gamification in virtual cycling, now also capturing those seeking gamified experiences. This can be understood as part of an institutional strategy (Das & Van de Ven, 2000) as customisation can be considered a new performance evaluation criterion where virtual cycling can outperform mainstream cycling. Mike (platform CEO)

described how platforms are developed to cater to a specific target audience of “different people” who might be attracted to virtual cycling for different reasons (e.g., gamification) compared to outdoor cycling:

[Some platforms] are not competitors - in the same way that Mario Kart is not a competitor to Grand Turismo - they are different products for different people. (Mike)

Mike’s quote can be interpreted as an act of pragmatic legitimisation (Suchman, 1995). This is based on the self-interest of virtual cycling platforms as stakeholders in virtual cycling who are not in competition with each other, but rather cater the preferences of their individual communities (e.g., reality-seeking virtual cyclists vs. gamification-seeking virtual cyclist) (Bentvelzen et al., 2022; Reed et al., 2023).

Second, the mechanism of *targeted inclusion* emerged from our interviews. Similar to the legitimising narratives found in esports (Pizzo et al., 2022) and or new tradition sporting formats (e.g., Cricket) (Read, 2023), sport governing bodies described virtual cycling as a tool to develop the cycling market in terms of age and gender:

I do think it has the opportunity to conquer new markets, right? And conquer new audiences. [...] We were part of the Olympic esports Week in Singapore. There was definitely a younger audience there. (Ali)

Obviously, we all know that youth are a little bit more engaged on the gaming side of things. They're a little bit more technically savvy, so it's an easier hill to climb with them. (Robert)

we want to reach the young audiences where they are. [...] We also want to be where they are. And also, it's important to highlight that the event [Olympic Esports Games] is a new product. (Lucas)

Ali’s statement about a ‘younger audience’ during the Olympic Esports Week can be interpreted as targeting a specific community (e.g., gamers, Gen-Z), working to serve new markets for virtual cycling as disruptive innovation (Christensen et al., 2015) and legitimise it via an institutional strategy that attaches importance to the new evaluation criteria of inclusion for differing ages, genders and capabilities (Das & Van de Ven, 2000). Taken together, the quotes from Ali, Robert and Lucas could also be understood as expressions of cognitive legitimacy (Suchman, 1995) within international sport governing bodies. Engaging with youth is a

longstanding policy goal of sport federations and the Olympic movement (Mittag & Naul, 2021). Hence, if virtual cycling is believed to be able to attract youth both as audiences and participants, it is evident that sport governing bodies support the development of virtual cycling and consider it as a new socially accepted norm, necessary, and inevitable (Suchman, 1995). While arguments for the potential of virtual cycling to attract youth was a recurring theme in our material, Floyd nuanced this sentiment by describing what he felt was the majority of virtual cycling participants:

The data...would suggest that it's an older cohort. It's not 15-year-olds. It's not 18-year-olds or 21-year-olds. It's more the older groups. You start to have a lot of probably uptake at 35, the general paying subscriber. Now, whether that is because there is a financial barrier to entry, you need to buy a device and pay a subscription and all those things. [...] you know, it was 88, 90 percent male, 12, 10 percent female (Floyd)

Floyd's description of the demographics of virtual cycling participants is a known issue in literature that can be a delegitimising mechanism to virtual cycling if sport governing bodies do not work to resolve it (Bork et al., 2015; Fouquaert et al., 2025). Furthermore, Floyd's comment of the gender distribution in virtual cycling is also important, as sport governing bodies often using the argument of attracting more women in their legitimising narrative:

we want to also be part of more female engagement in traditional esports, because now this is going to happen under our umbrella [Olympics]. (Lucas)

You can get more women involved. Perhaps it's a lower barrier to entry. It's less intimidating. (Robert)

The notion that virtual cycling may serve as a tool to enhance gender equity in sport participation represents a form of moral and cognitive legitimacy (Suchman, 1995), aligning with dominant sport policy values of international and national governing bodies (Mittag & Naul, 2021). It is however important to remain critical about the feasibility of this legitimising mechanism as literature on esports suggests that gender equity cannot be taken for granted (Rogstad, 2022), while literature on virtual cycling also demonstrates gender inequality issues (Tjønndal et al., 2025).

Targeted inclusion as legitimising strategy is also observed by interviewees highlighting the potential for virtual cycling to lower barriers for para-athletes in competitive cycling:

We could have different categorization of para-athletes. [...] athletes who perhaps have vision impediments or even hearing impediments, then you can become an organization that can help them compete on their own. [However, we need] commercial entities to help support it (Robert)

We are in front of everyone because we could push the Paralympic button tomorrow and have Paralympians at the Esport Games as well. And for us and for the IOC, that's a huge benefit where we automatically get like an advantage of everybody else. (Tony)

Developing parasports can, next to age and gender, also be understood as a form of moral legitimacy (Suchman, 1995) by sport governing bodies. That is, virtual cycling is framed using a legitimising narrative as a tool for social innovation that promotes inclusivity of diverse ages, women and para-athletes (Tjønndal, 2017). Despite virtual cycling's promise as an inclusive innovative tool, several issues and concerns are however still faced in its legitimisation, to which the following section dives deeper.

14.4.3 Delegitimising virtual cycling through externalised & passive governance

While most interviewees described virtual cycling's potential advantages, which can be considered similar to the advantages described in esports studies (Abanazir, 2022; Felczak & Filiciak, 2025; Reitman et al., 2020; Stauff & Vogan, 2021; Summerley, 2020), some also expressed concerns about virtual cycling. Our analysis recognises one delegitimation strategy centred around externalising governance and passively governing virtual cycling. More specifically, two mechanisms contribute to this strategy (Table 10). First, *sustaining platform reliance* emerged as delegitimising mechanism. Several interviewees mentioned how commercial platform reliance presented a challenge for the governance of virtual cycling:

You need the platforms to create race series and race events ... they have so much power over the event. (Manny)

You've got a few large commercial companies who [satisfy] their shareholders, not build something designed for athletes and credibility. (Mike)

Platforms are commercial entities in competition with each other, making it very difficult to ... progress with an international [platform-independent] ranking system, an international calendar, getting an independent performance verification service... (Kim)

To begin, this lack of coordination complicates the institutional integration of virtual competitions, leading to a fragmented landscape where UCI-sanctioned World Championships and Olympic-branded events operate in isolation from the platform's own commercial leagues that for instance, opposed to some traditional sports, do not grant World Championship qualification (Fouquaert et al., 2025). In addition, the reliance on commercial platforms introduces risks that extend into the daily governance and integrity of cycling. While platform dependence is a well-documented challenge in esports (Abanazir, 2022), in virtual cycling it manifests as issues related power asymmetries which impacts both sport governing bodies and athletes by fostering “institutional promotion” of commercial entities (Chee & Karhulahti, 2020). For governing bodies in cycling, the absence of interoperability between platforms presents new issues compared to traditional outdoor cycling, where interoperability between race organiser could be assured through managing non-technological mechanisms like ranking systems or competition calendars. In virtual cycling, technological feature development is needed to assure and implement such mechanisms (Fouquaert et al., 2025), which falls outside the capacities and resources of governing bodies and consequently presents a barrier to pursue a technical strategy. Such organisational constraints of governing bodies makes that responsibilities become redistributed between commercial and institutional entities: unlike traditional cycling where the federation controls the “rules of the game”, virtual cycling is faced with intellectual property rights of platforms that impedes such control (Fouquaert et al., 2025). From this perspective, Kim's statement can be interpreted as an example of how institutional strategy is dominantly employed over a technical strategy as little efforts appear to be taken to develop interoperability and independent services (e.g., performance verification), potentially contributing to the delegitimisation of virtual cycling and slowing down its diffusion process.

Second, *lacking organisational policies* seems to be a mechanism that works to delegitimise virtual cycling. For instance, Kim elaborates on the challenges associated with being reliant on commercial platforms rather than being able to independently develop technical features:

It's up to the UCI to make that an initiative and follow through with it, because otherwise all the platforms are just going to go in separate directions and the sport will never be unified. [...] One of the ways that the UCI is falling short is that they don't have a set of guidelines. (Kim)

Kim's quotes illustrate how commercial, technological and institutional environments intersect in virtual cycling and creates a decoupling between the technological reality and institutional

ideal, due to governing bodies organisational constraints in independently developing technological features, or even guidelines for those who do implement them (Washington & Patterson, 2011).

Moreover, the support given to virtual cycling appears to be not the same across the different sport governing bodies like national federations. In fact, our data suggests a lack of unified organisational policies. Manny, an elite virtual cyclist, talked about the discrepancies between national federations:

There is a discrepancy in terms of federations that are definitely embracing it and wanting to grow it. But then you can see the other ones that are still apprehensive. Look at British cycling, they're still very apprehensive about it and not really sure about it. So, you can see there's not that much support behind it. (Manny)

One of our interviews with a representative of a national federation supported Manny's experience:

No, it's impossible [to develop virtual cycling]. [...] Where we make the difference is being outdoor sports, so, I prefer to get the money to do outdoor sports than not to do indoor sports because the cost will be really high. And I don't see a potential venue where this can be done. (Ben)

Ben's statement illustrates the lack of programmatic embedding for virtual cycling within sport governing bodies, whose organisational logics remain rooted in non-virtual "analogue" cycling (Fouquaert et al., 2025; Reed et al., 2023). In our interviews with Manny and other athletes, it also became clear how differences in support from the national federation impacts athletes' opportunities and conditions to compete:

I had nothing from my federation. Absolutely nothing. They didn't even email me. They didn't talk to me. (Manny)

We had a lot of great racers drop out because they weren't able to support it. (Elvene)

[another national federation] had coaches there at Worlds [championship virtual cycling 2024]. They had team tactics meetings. They had athlete meetings. Like, they were just so supported with everything they did. I mean, the former world champion [...], he had like a whole team with him, everything was dialled in for him. So, I just think that makes like a huge difference to have the support of a federation. (Claressa)

In comparison to the *verbal* toxicity observed in broader esports (Felczak, 2022; Hayday et al., 2021) or the technical integrity issues found in virtual cycling (Dyer & Taylor, 2023; Fouquaert et al., 2025; Richardson, 2025; Westmattelmann et al., 2025), the lack of federation support is interpretable as a distinct form toxicity, which we define as *institutional* toxicity. This occurs when governing bodies benefit from the visibility of virtual cycling (e.g., World Championships) without fulfilling their duty of care to participating athletes. Such neglect leaves athletes in a precarious position because they act as “institutional promoters” (Chee & Karhulahti, 2020) for the sport while being denied the structural support (e.g., coaching, funding) required to compete on equal terms. The *laissez-faire* approach of the UCI on the inequality in support athletes receive from their federation, as interpreted from our interviews, may in fact be another act of delegitimisation. By failing to mandate minimum support standards across its member federations, the UCI effectively endorses a competition structure where success is partly contingent on arbitrary organisational support. This failure to secure a level playing field contradicts the institutional norms the UCI officially upholds, further deepening the decoupling between their institutional ambitions for the discipline and the precarious reality of the athletes competing in it (Boxenbaum & Jonsson, 2008).

However, as Claressa indicated, the difference in support between national federations is natural and could be expected as virtual cycling is a disruptive innovation, meaning it is unrealistic for all federations to be enthusiastic about adopting this innovation (Christensen & Bower, 1996).

14.5 Conclusion

This study examined how sport governing bodies work to legitimise virtual cycling as a disruptive innovation. Drawing on Das and Van de Ven’s (2000) process model of innovation diffusion, we identified two complementary legitimisation strategies presented in Table 10: (1) constructing new evaluation criteria and (2) developing new markets. The first legitimisation strategy reflects an institutional strategy, where sport governing bodies establish virtual cycling’s superiority by attributing value to five new performance evaluation criteria (safety, reduced costs, inclusivity, sustainability, and diversified revenues), which echo evolving societal values and institutional policies. This legitimisation strategy aligns with disruptive innovation theory as disruptive innovations introduce new features that ultimately require new evaluation criteria to assess (Christensen et al., 2015; Das & Van de Ven, 2000). The second legitimisation strategy also aligns with disruptive innovation theory, showcasing how virtual

cycling targets niche cycling markets. Both strategies illustrate how the legitimisation of disruptive innovations in sport are socially constructed and enable sport governing bodies to legitimise technically inferior innovations like virtual cycling. This is demonstrated through the ways in which the interviewees describe the development of new performance criteria through narratives of what cycling *is*, what it should *be*, and how virtual cycling could support this development of the sport. Meaning is thus given to what is considered important in cycling (i.e. its ‘performance’, see Figure 15) and how these closely align with societal values such as safety, cost-effectiveness, inclusivity, or sustainability.

Next to legitimisation strategies, we also observed several mechanisms to delegitimise virtual cycling. The combination of a protective stance towards traditional disciplines, high reliance on commercial platforms and technical underdevelopment are critical in the interviewees delegitimising arguments of virtual cycling. Issues such as platform fragmentation, lack of independent verification systems, or absence of platform interoperability (e.g., cross-platform rankings) illustrate how technical strategy has been largely neglected compared to institutional strategy in the development of virtual cycling as a new competition format (Das & Van de Ven, 2000). In this way, the delegitimising arguments found in our data demonstrate how the imbalance between the use of an institutional strategy and the use of a technical strategy, creating a decoupling between what governing bodies say (i.e. legitimising narrative) and what they do (i.e. the challenging reality of developing technical features over others’ intellectual property) (Washington & Patterson, 2011). Legitimacy is, in this context, not simply a matter of aligning with evolving societal values and institutional policies, but also something which requires technical features to support consistent governance, athlete and federation trust, and avoid monopolistic markets. As virtual sports continue to evolve, their successful innovation diffusion will hinge on both normative alignment and the capacity to deliver technically accurate, interoperable, and inclusive virtual participation, whether that is for leisure or elite sports.

Practically, our findings offer guidance for sport federations seeking to expand their disciplines with a virtual competition format. Specifically, our findings demonstrate the importance of employing a technical strategy (e.g., develop platform interoperability) as a lack of key technical features is shown to limit legitimacy and increase dependence on commercial stakeholders. Solely employing an institutional strategy based on societal and normative values is thus insufficient to legitimise virtual sports. Hence, sport governing bodies must actively take responsibility in developing technical features (e.g., cross-platform ranking, calendar, doping-control) in close collaboration with technology providers (e.g., game publishers).

Finally, our study illustrates how legitimacy is shaped through normative language and framing. For example, positioning virtual cycling as a tool for youth engagement, gender equity, and para-athlete inclusion reflects moral and cognitive legitimacy-building mechanisms (Suchman, 1995). Yet, such framing is not universally accepted as we also observed resistance from national federations which, together with platform fragmentation, continue to impede broader diffusion. Despite the understanding of virtual cycling (de)legitimation this study develops, the extent to which the physical parent-sport-mimicking nature of virtual sports influences their social acceptance compared to sedentary esports remains unclear, something that future research may investigate more closely. For instance, to which extent does “physical” or “emotional” closeness impacts social acceptance, institutional and technical evaluation criteria, and how does this translate into (de)legitimation strategies and mechanisms? Such potential future research question must also account for Borggreffe and Hoffmann’s (2024) analysis that “sport and esports differ in that the communication of physical performance is an end in itself in sport, whereas in esports it is limited to the operation of an input device and is thus a means to the end of controlling an avatar.” (p. 1204), meaning that operating equipment (e.g. mouse, cycling ergometer) does not communicate *of* or *about* physical performances. With regards to our case of virtual cycling, there *is* communication of physical performance in itself. Still, these physical performances occur on stationary equipment, which render them unable to communicate *of* or *about* the physical performance itself as spectators cannot observe the performance (e.g. who is winning?). In this way, virtual cycling requires a digital translation between what the player does and what happens (Borggreffe & Hoffmann, 2024). However, this rationale does not universally apply to all virtual sports. For example, augmented table tennis using mixed-reality technology can meaningfully communicate *of* or *about* physical performances as the performance itself allows spectators to observe victory or defeat. Consequently, future studies on virtual sports must recognise such differences, connections to sport and esports, and their effect on legitimisation.

Expanding on the abovementioned suggestions for future research, new studies could also focus on further identifying the differences between institutional and technical legitimisation strategies and the key stakeholders impacting these strategies. This is important as the differences between these strategies and the role that stakeholders play are not clear-cut or easily identifiable. In addition, future studies could focus on normative language and practices used, by whom, how and with what effect, but also take a more quantitative innovation diffusion approach. We also promote future research on virtual sports beyond cycling, for example sailing or running, to see how their legitimacy will evolve in the future.

14.6 References Study 3

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Data Availability Statement

For ethical reasons, only pseudonymised data of this study are available from the corresponding author, upon reasonable request.

15 Study 4: “There are a lot of vulnerabilities still left”: Governance Issues in Virtual Sports

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Abstract

International sport governing bodies are investing considerable time and effort into the development of competitive virtual sports, with two prominent examples being virtual cycling and virtual triathlon. As virtual cycling, triathlon and other virtual sports sit between the intersection of sport and esports, these innovative sporting formats create novel governance dilemmas. Against this backdrop, this study examines the governance issues that stakeholders in virtual sports are currently facing. Methodologically, the study is based on qualitative interviews with representatives from national and international sport governing bodies, elite athletes, journalists and content creators in virtual cycling and virtual triathlon. Central to our findings is the notion that governance issues in virtual sports both mirror and nuance those of esports and sport. The analysis identified four types of governance issues related to (1) doping, (2) platform fragmentation, (3) transparency and (4) commitment and decision-making. In particular, governance issues experienced by stakeholders encompass physical-, cognitive-, software-, and hardware-related manipulation, as well as concerns around the intersection of commercial interests, data privacy, and performance verification. Overall, the unanimous call for stronger commitment of international governing bodies point to a critical gap in proactive governance, social responsibility, and athlete support within the emerging discipline of virtual cycling.

Keywords: sport governance, esports governance, good governance, virtual cycling, Olympic esports games, Zwift, MyWhoosh, integrity issues

15.1 Introduction

Virtual Sports are emerging as an established hybrid between sport and esports, integrating physical activity with technology-mediated competition. Defined as interactive sports games that utilise interactive peripheral sports equipment, virtual sports are also classified as esports when organised in competitive formats (Parry & Giesbrecht, 2023). Virtual sports are expanding across disciplines (e.g. virtual rowing, virtual taekwondo), witnessing a significant growth in popularity, amplified up to 30% by the COVID-19 pandemic (Helsen et al., 2022; Westmattmann et al., 2021). Amongst the most advanced cases are virtual cycling and virtual triathlon, which gained formal recognition from the Union Cycliste Internationale (UCI) and World Triathlon as legitimate sports disciplines. Both UCI and World Triathlon have featured yearly officially sanctioned esports World Championships since 2020 and 2021. The rapid professionalisation of virtual cycling has since grown steadily to a competitive ecosystem with total prize money reaching \$3.1 million in 2024 (Schwenker, 2025). Last but not least, the International Olympic Committee (IOC) has also recognised the significance of virtual sports with the inauguration of the Olympic Esports Games in 2027 (International Olympic Committee, 2025).

These developments showcase how virtual sports is becoming increasingly governed at the intersection of both governing bodies within sport and esports. This hybrid nature of virtual sports may however nurture new governance issues. For instance, Peng et al. (2020) demonstrates that the complex dynamics (e.g., power, conflict, and interest) between game publishers and other diverse stakeholders in esports are a root cause for governance issues. These dynamics raise questions about the governance of hybrid sports games, which involve game publishers, international federations and various other actors. Hence, while governance in sport is a well-established research field (Thompson et al., 2023), esports governance adds new complexities that are still under development (Kelly et al., 2022; Peng et al., 2020). To our knowledge, the governance of virtual sports remains unexplored in scholarly literature, despite growing recognition of related issues such as doping (Dyer & Taylor, 2023), integrity breaches (Dyer, 2022; Richardson et al., 2022), software manipulations, and hardware manipulations (Dowd et al., 2024; Fincoeur & Bongiovanni, 2021; Richardson et al., 2023).

Our study examines governance issues in virtual sports, focusing on virtual cycling and virtual triathlon. When referring to virtual cycling, this equally covers virtual triathlon since both industries rely on the same key stakeholders. Against this backdrop, our research question is: How do stakeholders experience current governance issues in virtual cycling? To explore

this, we rely on interviews with stakeholders, including representatives from the UCI, World Triathlon, the IOC, elite virtual cycling athletes, platform publishers, and broadcasting commentators. The article continues with outlining previous research on governance in sport and esports before drawing on Kelly and colleagues' (2022) esports governance framework as theoretical underpinning.

15.2 Literature review: Governance issues in sport, esports and virtual sports

Virtual sports governance blends elements of both sport and esports. International sport federations (e.g. UCI) are obliged to collaborate closely with virtual sport platform publishers (e.g. MyWhoosh) and hardware providers (e.g. Garmin) (Fincoeur & Bongiovanni, 2021). Therefore, building on research of governance issues from both domains is necessary to understand the current governance issues facing virtual sports.

15.2.1 Sport governance

Governance models in sport have been a focal point of research over the last decade. For instance, Thompson et al.'s (2023) literature review of sport governance models identified 258 unique governance principles across 73 records, revealing a set of central pillars including transparency, accountability, democracy, and social responsibility. However, the rate and extent to which these principles are adopted by governing bodies varies greatly. While an increasing amount of sport governing bodies have formalised governance codes, challenges to adopt governance frameworks remain due to entrenched power dynamics and resistance to reform (Thompson et al., 2023).

The adoption of good governance principles in sport (e.g. transparency, accountability, democracy, and social responsibility) is influenced by institutional pressures and cultural contexts (McLeod et al., 2020). This underscores the importance of structural dimensions (e.g. organisational networks and processes) and historical dimensions (e.g. cultural, political, educational, financial) when implementing governance reforms in sporting environments. Adding to the complexity of vastly different determinants of successfully implementing good governance principles, additional complexities arise due to inconsistencies in the practical implementation assessment of governance codes, or implementation requirements (Chappelet & Mrkonjic, 2019; Parent & Hoye, 2018). For instance, transparency could range from the publication of annual reports to board minute meetings, while adherence to good governance principles may range from legislative obligation to a voluntary opt-in choice.

Organisations who govern sport often rely on self-regulation, which is a voluntary commitment to adopt good governance principles in order to create legitimacy and minimise reputation costs (Geeraert, 2019). While self-regulation mechanisms like persuasion and social learning are promoted measures to ensure good governance, the lack of external, independent monitoring and sanctions limits effective compliance (Geeraert, 2019). This suggests that self-regulation alone is insufficient and external accountability such as governmental and societal oversight is necessary to ensure good governance.

15.2.2 Esports and virtual sports governance

Similar to sports, governance in esports has become increasingly prominent as the industry grows, while also facing challenges related to regulation, fairness, and player wellbeing. A fundamental issue in esports governance is the fragmented regulatory landscape, which lacks the centralised governing bodies seen in sport (Chao, 2017; Gupta et al., 2021; Kelly et al., 2022; Peng et al., 2020). Unlike sport, where governing bodies like international federations establish standardised rules and enforcement mechanisms, esports governance is often dictated by a few dominant stakeholders, i.e. game publishers and their leagues due to their control over game intellectual property (Peng et al., 2020). This fragmented governance structure creates inconsistencies in rules, integrity standards and enforcement mechanisms (Peng et al., 2020). Complicating things further, esports governance involves a range of stakeholders, including game publishers, league organisers, contracted teams, contracted players, sponsors, and content distributors (Chao, 2017). The emergence of additional actors with diverse and sometimes competing interests, such as esports federations, government agencies, or self-proclaimed industrial guardian organisations, adds to the fragmentation (Peng et al., 2020; Wong & Meng-Lewis, 2023). Although research on virtual sports governance remains limited (e.g. Postma et al., 2022), the governance challenges observed in esports, such as stakeholder complexity and regulatory fragmentation, are equally relevant to virtual sports, and may be even more pronounced due to the additional involvement of mainstream sport governing bodies.

A similarity between sport, esports and virtual sports governance is the need to mitigate integrity issues related to doping and performance enhancement. Since these issues have pertained both sport and esports (Gupta et al., 2021; Holden et al., 2019; Hurst et al., 2024; Tweedie et al., 2023), virtual sports are expected to be equally plagued. Dyer and Taylor (2023) reported that doping in virtual cycling competitions, for example by manipulating avatar weight, is perceived to be common practice by athletes yet regarded as highly undesirable. Dyer (2022) and Richardson et al. (2022) have both summarised a range of other potential integrity

breaches in virtual cycling, including: (post-race) data manipulation and hacking, hardware manipulation, using banned substances (e.g. EPO), misrepresentation (e.g. gender, avatar), deceptive functional threshold power, or ‘sandbagging’ (i.e. intentionally underperforming or entering races ranked below your ability). Both studies conclude that long-term measures are needed, including creating a ‘digital performance passport’, adapting esports cheating frameworks to address physical asset abuse, collaborating with the Court of Arbitration for Sport (CAS), adopting the full World Anti-Doping Agency (WADA) anti-doping code, establishing an athlete appeal system, and continuing to refine integrity policy and processes.

Focussing on hardware and software manipulation, something extensively covered in esports integrity issues (Schubert et al., 2024), the lack of accuracy must also be considered a governance issue in virtual sports. For instance, recent studies report power differences in power measured on virtual cycling equipment ranging from 4.2% (Bjärehed & Bjärehed, 2022) to 16% (Dowd et al., 2024). While the majority of virtual cyclists want clearer and enforcing rules against aforementioned breaches, something that is addressed by platform-developed anti-doping policies, many of the breaches can be, and still are, exploited (Fincoeur & Bongiovanni, 2021; Richardson et al., 2022, 2023).

There are also nuances and differences in governance issues between sport, esports and virtual sports, rendering various challenges. First, esports rules are subject to the publisher’s platform and the updates they receive (Chao, 2017). Secondly, esports are prone to new, technological and digitally mediated, forms of doping (Schubert et al., 2022; Tjønndal, 2023). The use of performance-enhancing substances, combined with the absence of overseeing and sanctioning organisations like WADA or the International Testing Agency (ITA) in esports, presents complex challenges (Gupta et al., 2021; Richardson, 2024; Schubert et al., 2022). Third, the oligopolistic nature of the publisher market (e.g. for virtual sports platforms, game titles) induces anticompetitive and even discriminant behaviour (Chao, 2017). Players can also be banned without proper ways to appeal, which could suspend their participation in large parts of the sport (Chao, 2017).

15.3 Theory: Esports Governance Framework

The complicated relationships between stakeholders in esports necessitates a stakeholder-driven approach, to which Kelly and colleagues (2022) have proposed an esports governance framework. Based on multi-theoretical analysis incorporating stakeholder, agency, legitimacy, and actor-network theory, the model incorporates Burger and Goslin’s (2005) seven governance

pillars: accountability, responsibility, transparency, social responsibility, independence, fairness, and discipline.

(1) Accountability ensures that governing bodies and its stakeholders are accountable to each other and helps define liability. It can be addressed by implementing requirements and policies to protect players, and especially minors, from online harassment and the risk of health issues. It has however been argued that within the esports industry, its fragmentation, absence of central authority, and the dominant position of game publishers may undermine accountability as publishers rarely face consequences for governance failures (Peng et al., 2020; Wong & Meng-Lewis, 2023). (2) Responsibility is closely related to accountability but ensures actual consequences for who fail governance duties. For governing bodies, responsibility should be ensured by the law in which the body is incorporated, while its members can be appropriately held responsible for (in)action by enforcing codes of conduct such as having the required athlete safeguarding, anti-doping and performance verification measures. While esports and virtual sports introduce new threats such as privacy concerns and data misuse, formal responsibilities among relevant stakeholders and industry-wide standards remain inadequate (Czakó et al., 2023). To guard against these threats, updated responsibilities for protecting athlete well-being are needed within esports and virtual sports governance (Qi et al., 2024). (3) Transparency is crucial in esports given the commercial nature of the publishers and could help to clarify the interest in which some decisions have been taken, which allows to protect athletes. However, publisher-led governance in esports has been argued to lack transparency (Peng et al., 2020). (4) Social responsibility ensures esports potential contribution towards creating inclusive and safe environments, while mitigating potential harms such as online harassment, doping, and exclusion. Governing bodies can ordain regulations coupled with actions as education initiatives to inform stakeholders with the consequences of potential harm while policy guidelines to keep tech-mediated environments inclusive and psychologically safe are advocated for (Qi et al., 2024). (5) Independence is necessary to ensure dispute resolution, fairness and prevent conflicts. Current independent bodies, such as the Arbitration Court for Esports (ACE) and the Esports Integrity Commission (ESIC), should pursue greater unification while also establishing specialised arbitral units tailored to the distinct needs of different esports and virtual sports disciplines. These independent bodies oversee player rights and market dynamics to reduce the risk of stakeholders' interests being undermined. However, esports (and by extension, virtual sports) remain vulnerable to integrity and governance challenges, as the independence of these organisations is often limited. This limitation stems primarily from the control that game publishers exert through their intellectual

property rights, which frequently grants them ultimate authority over governance and dispute resolution (Peng et al., 2020; Wong & Meng-Lewis, 2023). (6) Fairness ensures that the needs of all stakeholders are balanced, which is in particular important in esports considering power of commercial publishers. Governing bodies can achieve fairness by consulting independent experts when developing regulations next to adopting governance standards comparable to those used in traditional sport (e.g. Olympics) (Postma et al., 2022). (7) Discipline is argued to be critical for esports' governance and legitimacy, yet is currently lacking due to no consistent and proportional sanctioning of players who violate integrity norms. To mitigate inappropriate behaviour, governing bodies must develop clear procedural rules and apply them consistently, for example when to ban an athlete and for how long. However, esports and virtual sports remain particularly susceptible to such issues, as their fragmented, publisher-led systems of self-governance often result in reactive rather than proactive responses to integrity breaches (Peng et al., 2020; Wong & Meng-Lewis, 2023).

Drawing on the current literature, we view the esports governance framework proposed by Kelly et al. (2022) as a useful starting point for understanding governance issues in virtual sports. However, emerging studies highlight important differences between esports and virtual sports governance, particularly in terms of stakeholder composition and interdependence. For example, the involvement of sport governing bodies and national Olympic committees in virtual sports. These distinctions challenge the applicability of Kelly and colleagues' (2022) framework to the virtual sports context. We also critically question the need for separate governance frameworks for esports and virtual sports, as the seven good governance principles are broadly applicable across traditional sport, esports, and virtual sports. It is not the principles themselves that are context-specific, but rather the nature and sources of threats to these principles that vary across sporting contexts.

15.4 Methods

15.4.1 Research Setting

Since Zwift's 2014 launch, virtual cycling has greatly evolved. New platforms have emerged while others ceased to exist or merged; and by 2024, Zwift and MyWhoosh dominantly hosted esports racing. MyWhoosh is leading in prize money, offering \$3.75 million annually through 'Sunday Race Club', while Zwift's racing series offered \$128,000 (Schwenker, 2025). While other platforms exist (e.g. TrainingPeaks Virtual, Rouvy), these focus on structured workouts or races without prize money. Next to platforms' self-hosted races, independent leagues exist

that offer limited prize money (e.g. \$10,000) and collaborate with a national cycling federation (e.g. Echelon Racing League). For virtual triathlon, SuperTri in collaboration with World triathlon will offer \$57,000 in prize money for the 2025 World Championship (World Triathlon, 2024). Altogether, these advancements led some athletes to become professional in esports racing (Schwenker, 2025).

UCI has been hosting the virtual cycling world championships since 2020, with the 2024 event first held live in Abu Dhabi. Also SuperTri has been organising virtual triathlon races since 2020, partnering with World Triathlon in 2021 for its first E Triathlon World Championship. Both organisations initially used Zwift but switched to MyWhoosh in 2024 after it won their tender. Beyond cycling and triathlon, the IOC also announced a 12-year deal with Saudi Arabia to host the Olympic Esports Games in Riyadh starting in 2027 which will feature virtual sports (International Olympic Committee, 2025).

15.4.2 Participants

To understand how stakeholders in virtual cycling experience governance issues, we conducted 16 qualitative interviews with 13 participants who collectively represent key stakeholders in virtual cycling and virtual triathlon. Follow-up interviews were conducted when the initial interview did not cover all intended topics, or when researchers identified additional relevant information that the interviewee could provide. We selected participants based on the criteria that our sample should: (1) represent (virtual) cycling and triathlon organisations located at different levels (national and international), (2) represent a diversity of stakeholders in virtual cycling and triathlon and (3) include both men and women participants. We recruited participants by contacting UCI, World Triathlon, SuperTri, IOC, and two national cycling/triathlon federations, of which elected and employed leaders from all of the selected organisations agreed to be interviewed. We also contacted two international journalists in the field, who both agreed to be interviewed and who put us in contact with elite athletes. The characteristics of the participants are summarised in Table 11. All the participants have been given pseudonyms to protect their identities. The study was approved by Nord University, SIKT – The Norwegian Agency for Shared Services in Education and Research. All participants provided active consent and were informed of their right to withdraw from the study at any time.

Table 11: Description of interview participants (study 4)

Pseudonym	Gender	Expertise / Role	Follow-up interview
Stan	Male	Innovation manager, national sport governing body	-
Robert	Male	Management, international sport governing body	Yes
Kim	Male	Journalist & content creator	-
Floyd	Male	Management, international sport governing body	Yes
Claressa	Female	Elite virtual cycling athlete	-
Manny	Male	Elite virtual cycling athlete	-
Mike	Male	Tech developer/virtual cycling platform	-
Elvene	Female	Elite virtual cycling athlete	-
Ben	Male	Management, national & international sport governing body	-
Ali	Male	Management, international sport governing body	-
Rika	Female	Athlete, content creator & journalist	Yes
Tony	Male	Management, international sport governing body	-
Lucas	Male	Department manager, IOC	-

15.4.3 Data collection

We conducted semi-structured interviews and developed the interview guide around five topics related to the development and legitimisation of virtual cycling: (1) the interviewees' background and experience with virtual cycling, (2) legitimisation and delegitimization of virtual cycling, (3) policy and governance, (4) participation, and (5) technology and innovation. Some exemplary questions for each topic were: (1) What are the experiences of you/organisation X in virtual cycling/triathlon? What solutions could virtual cycling bring to your/organisation X's challenges? (2) Why would you/organisation X (avoid) engage in virtual cycling/triathlon? (3) How would you describe organisation X's policy on virtual cycling/triathlon? Can you give some examples of how this policy is evaluated? (4) who do you think is attracted to virtual cycling/triathlon? (5) How do novel technologies impact the development of sport Y?

One interview was conducted face-to-face while the other 15 interviews were conducted using online video calls. Eight interviews were conducted by the first author while the remaining eight interviews were collectively conducted by the first and second author together. The interviews lasted between 40 to 80 minutes. They were recorded and transcribed in full. The interview with Stan (see Table 11) was discarded from further analysis because it was found to be irrelevant to inform the research questions of this paper.

15.4.4 Data analysis

We analysed the data by employing Eggebø's (2020) collective qualitative analysis. This approach is a methodological tool to enhance the rigour of the analytical process and to foster creativity in the development of thematic categories. The approach consists of four steps, which are all done collectively (Eggebø, 2020). The first step is for multiple researchers to read through transcriptions of interview data and write abstracts from each interview. The first two authors each composed one-page summaries of each interview, which were then shared with all authors. Step two consists of mapping data through joint discussions and construction of data-driven themes. The first two authors independently reviewed all interview summaries, identifying predominant themes within each interview. These data-driven themes and their subthemes were mapped on Post-its and displayed on a wall to facilitate collaborative analysis. The transcripts were constantly at hand to aid theme selection based on the transcript summaries, for example when additional details were required. This allowed for step three which is sorting the data in analytical themes (Eggebø, 2020). Closely related data-driven themes were either merged or refined to capture distinct concepts, resulting in a final set of 20 analytical themes. Each analytical theme was written on a A4-paper, which were again all displayed on a wall to facilitate collaborative development and analysis of them. Each analytical theme A4 was further annotated with important subthemes, interview observations and relevant interviewee pseudonyms, again with the transcripts and their summaries at hand. Interrelated analytical themes were visually connected to aid understanding of our data. Following this, we made a disposition and outlined a workplan for the writing process guided by our twenty analytical themes, which is step four (Eggebø, 2020). For the current study, three analytical themes were found to be relevant to inform our research question: (1) governance, (2) integrity issues and (3) commercial platform dynamics. The remaining 17 analytical themes fall outside of our study's scope and will be reported in subsequent studies. To finalise Eggebø's collective qualitative analysis, the first two authors went through the transcriptions again, based on earlier noted relevant pseudonyms per analytical theme, to extract quotes and descriptions fitting each analytical theme.

15.5 Findings and discussion

The interviews yielded insights into four types of governance issues in virtual cycling: (1) doping, (2) platform fragmentation and commercial interests, (3) transparency and open criteria and (4) commitment and decision-making.

15.5.1 *Doping*

We found that participants experience doping, both intentional and unintentional, as a governance issue in virtual cycling. Manny, an elite athlete, described conventional doping methods as an issue in virtual cycling:

There will always be people cheating. The one stereotypical way of the cheating is the doping and the EPO because there is no real way to control that from home (...) One thing all the elite people would like to see is more anti-doping (Manny)

Manny's concerns about doping and gaining an unfair advantage in virtual cycling mimic findings from elite athletes in mainstream sports (Hurst et al., 2024). In esports, doping concerns primarily centre on cognitive enhancing prescription drugs such as Adderall and Ritalin which contain dextroamphetamine and methylphenidate, both banned substances by the Esports Integrity Commission (Holden et al., 2019; Schubert et al., 2024). In contrast, elite virtual cycling athletes are mainly concerned with traditional performance enhancers like EPO (erythropoietin), which provides significant physiological advantages in endurance sports and is prohibited by WADA. This is further complicated by the non-adoption of other WADA standards (e.g. whereabouts and out-of-competition testing) in virtual cycling. While previous studies have recognised positive developments on this front, such as platform-led performance verification like Zwift Accuracy and Data Analysis (ZADA) (Fincoeur & Bongiovanni, 2021; Richardson et al., 2022) or collaborations with WADA or ITA through international federations (Gupta et al., 2021; Tweedie et al., 2023), the need for unified doping governance in virtual cycling still remains.

Similarly to findings from other virtual cycling studies (Dyer & Taylor, 2023; Richardson et al., 2022) and esports research (Schubert et al., 2022), interviewees spoke about new software- and hardware-related ways to dope. For example, Floyd and Tony spoke about how performance data can be manipulated in virtual cycling:

In the virtual world, there is potential to manipulate inputs such as weight and speed. (Floyd)

It's the credibility of the software, being able to make sure that there's no... Just like in doping outside, we have to make sure that the software is able to be checked and is compliant in a way which then creates an equal environment for everybody. (Tony)

Mike, a virtual cycling platform developer, gave insights into how doping is enabled by

platform's design, since many platforms were initially not designed for competitive virtual cycling:

You know, you can't take an existing box, put a padlock on it and call it a safe. [...] You can't just add something to an existing design and say, hey, this is secure now. [...] That first generation of platforms that came out and the rampant cheating that happens on them put a lot of people off. (Mike)

However, Mike continues by nuancing this through the idea that performances can also be unintentionally manipulated by, for example, uncalibrated equipment which results in an “undue advantage”.

I like to think that not everybody's out to cheat. There is a reasonable proportion of people who just have got badly calibrated equipment, for example. So, it's not so much about cheating, although clearly some people will. (Mike)

Despite the unintentional performance enhancement, other interviewees shared insights into intentional cheating. For example, Ali mentions how hardware could be manipulated intentionally by athletes and teams.

You hear all the tricks of mechanical doping, put your trainer in the freezer for an hour and then unpack it from the freezer and then race and you'll be much better (Ali)

Our findings on doping in virtual cycling highlight how virtual sports sit at the intersection of sport and esports, sharing doping-related challenges from each domain. This includes the risks of both (1) physical and (2) cognitive performance enhancing substances, but also (3) mechanical doping through software, hardware and data manipulation. However, the latter may very well occur “unintentional” (e.g. uncalibrated equipment), which is an additional complexity for virtual sports to take into account. These three types of doping demonstrate how the governance pillars of social responsibility and fairness (Burger & Goslin, 2005; Kelly et al., 2022) remain challenged in virtual cycling and require new, collaborative actions to mitigate.

Additionally, performance verification could also be undermined by commercial interests or legal responsibilities when competitions are organised via commercial platforms. The protection of privacy enforced by, for example, regulations such as GDPR can limit the actions that can be undertaken by governing bodies to verify users' data:

A lot of the platforms have signed up their subscribers and told them that they will never share their data with third parties. Then you have the governing body trying to ensure athletic authenticity. They can't get access to the data. And so that becomes a battle. (Floyd)

These market dynamics between integrity, commercial interest and legal obligations tie into both pillars of independence and fairness (Kelly et al., 2022). In this case, stakeholders' needs are unbalanced since athletes' interest (e.g. competing in a verified environment) are undermined by the terms of service of platforms, which are written in favour of attracting paying subscribers. While an independent body could oversee and advocate for players' rights, it could also collaborate with legal experts to develop platform terms of service. These terms would include tailored exemptions, allowing athletes' data to be used for integrity purposes without compromising their privacy.

15.5.2 Platform Fragmentation and Commercial Interests

A common issue raised by the athletes in our sample relates to the commercial interests of both virtual cycling platform providers and sport governing bodies. Here, several of the interviewees spoke about the contract between UCI and MyWhoosh to exemplify how commercial interests often take precedence over athletes' interests:

You've got a few large commercial companies who are by definition trying to do what's best for their shareholders, not build something that is designed for athletes and credibility. (Mike)

There are two major platforms on which most people are racing, which makes it quite limited in terms of what is able to be done. (Manny)

Since UCI signed a contract with MyWhoosh, everyone's trying to do their races on MyWhoosh now [...] I guarantee you everyone would still be on Zwift if UCI didn't sign that contract with MyWhoosh. (Claressa)

I'm afraid that the athletes have little choice. I think this will be a commercially driven discussion. (Ali)

As discussed by Mike, Manny, Claressa and Ali, the commercial interests of platform providers and international sports governing bodies are problematic because these decisions fail to take into consideration the needs of the athletes. Here, the governance pillar of transparency becomes relevant (Kelly et al., 2022) to ensure consistency for the athletes. This is important, because the aims of virtual cycling platforms to sell subscriptions to leisure athletes should not undermine the interests of elite athletes and vice versa.

The choice of virtual platform emerged as a significant concern among athletes, with interviewees highlighting differences in game physics across platforms and the resulting impact on virtual cycling races:

Most of the platforms are measuring performance variables in different ways. (Floyd)

Learning the bike physics of each platform has been very, very challenging. (Elvene)

One of the things you sadly have to accept is just you have to learn the platforms. (Manny)

While governing bodies of sport often rely on self-regulation (Geeraert, 2019), the quotes above illustrate how virtual cycling mirrors the governance issues of esports related to a fragmented regulatory landscape (Chao, 2017; Peng et al., 2020). Several interviewees from international sport governing bodies identified platform differences as a governance issue, noting that virtual cycling races are inherently dependent on the intellectual property rights held by platform publishers. Nuancing studies of esports governance (Czakó et al., 2023; Kelly et al., 2022; Richardson et al., 2022; Wong & Meng-Lewis, 2023), Robert and Floyd spoke about the mitigation of these issues through standardisation:

We need to standardise competitions across platforms. And when I say platforms, I mean the virtual cycling software. They are all fair in their own way. We just need to bring standardisation to it so that athletes can expect the same experience each time. (Robert)

[The UCI should develop ...] some sort of a common ground so that we can progress with an international ranking system, an international calendar, getting an independent performance verification service, getting standardization across all the platforms. I think it's up to the UCI to take that initiative and follow through with it, because otherwise all the platforms are just going to go in separate directions and the sport will never be unified. (Floyd 2)

Mike, however, offered a more critical perspective on standardisation, arguing that platforms should adopt such standards only if their primary focus is competitive virtual cycling over mass participation, since *“focusing on both is really hard”*. From his viewpoint, industry-wide standards are not a necessity for all platforms but rather a mechanism to encourage strategic prioritisation. As he explained, standards *“will force companies to choose and maybe a company will look at those standards and go this is too much effort”*. The standardisation that Floyd, Robert and Mike describe can be related to the governance pillars of responsibility and accountability (Kelly et al., 2022), for instance by ensuring that platforms align in terms of performance measurement and verification, ranking and calendar. In his interview, Robert

continued to describe how there are also vulnerabilities in the virtual platforms that need to be addressed in order to ensure fair competitions:

Data security between the hardware that you're using at home and the internet's cloud and then back to the servers. There are lot of vulnerabilities still left in that process. If you're talking about elite level sports competition, it needs to be secure. If somebody hacks into that data stream and we can't trust the data that we're getting, we don't know if that avatar is being propelled legitimately or not [...]. There's a long way for us to go on the elite side of things. (Robert)

This is an example of how accountability and responsibility is underdeveloped in the governance of virtual cycling, and further, how these issues also intersect with fairness and discipline as good governance pillars (Kelly et al., 2022). The example of Robert can also be connected to findings of integrity breaches such as avatar weight manipulation (Dyer & Taylor, 2023) and data manipulation (Dyer, 2022; Richardson et al., 2022, 2023). Other examples were raised by interviewees relating to the challenges in accountability and responsibility such as intellectual property rights. These seem to complicate the conflict of interests and outline of responsibilities. For instance, Tony explained how this impeded his federation's governance work: *"To be perfectly frank, we don't have our hands on it so much. It's more because that's their IP and they're dealing with it"*. Tony's statement illustrates how governance responsibilities blurs between international sport federations and commercial platform providers in virtual cycling.

15.5.3 Transparency and Open Criteria

As discussed by Kelly and colleagues (2022) transparency is a crucial pillar of good governance in esports, due to the commercial nature of game developers and the complexities of the governance structures. Relatedly, the virtual cycling stakeholders in our study described how transparency about qualification pathways was a significant challenge:

Different federations have different paths to qualify, and we are working with all of them to understand what is the best to be done [...]. In order for us to select the games that are going to be part of the Olympic esports games, we have a set of criteria [...] for all these games to be approved. (Lucas)

The lack of unified qualification pathways as described by Lucas, shares similarities with governance issues identified in esports (Chao, 2017; Czako et al., 2023; Gupta et al., 2021;

Wong & Meng-Lewis, 2023). Just as there are inconsistencies in rules and enforcement mechanism within esports games, there are disparities in the pathways to qualify for major virtual cycling events. The transparency challenges of qualification pathways are related to the governance pillar of social responsibility (Kelly et al., 2022) in terms of ensuring equal opportunities for athletes worldwide to qualify. This was underlined as virtual cycling is in the process of becoming part of the Olympic Esports Games:

No matter what, the IOC will always want to have this pathway where every single person has the opportunity to compete in the Games. There has to be a pathway everywhere. And at the moment, that's not applicable in virtual racing because (..) we don't have enough events. So by 2027 we have let's say one event in every region and we have regional qualifiers, and then we have a situation where we create a qualification pathway where you basically have an opportunity to get in from your regional side, then we'll have 30 or 40 athletes or more, 50 athletes, that will be able to go to the eSport Games. (Tony)

As Tony's quote illustrates, when virtual cycling is included in events governed by the IOC, their values and criteria for social responsibility comes into play, putting increased demands on international federations and publishers to adhere to the integrity frameworks of the IOC. Ensuring equal qualification possibilities for athletes worldwide may be more difficult in virtual cycling compared to many mainstream sports given its digitally mediated features. With increased digitisation of sports, digital inequalities become a challenge for social responsibility in virtual cycling. Tony went on to explain how there were also issues related to a lack in transparency between the international sport governing bodies and the IOC regarding the development of the Olympic Esports Games:

What is the benefit of having three [virtual sports events] versus having one when you don't really know what it's for? You haven't got a clear indication on what the esports games are doing. You haven't got a clear indication of what the performance pathway of qualification is. (Tony)

However, on a more critical note it should be noted that even if there were open criteria, we can simply not expect all federations will honour these. For instance, Kim described a situation where a national federation prevented two athletes from participating in the world championship:

That federation had two athletes that qualified through the open qualification pathway, and they didn't recognise them. So those athletes didn't get supported and weren't able to race. They [the federation] wouldn't provide them with the needed UCI license to race. (Kim)

This situation relates to the governance pillar of responsibility, as this federation did not fear any consequences by not recognising athletes, but also to the pillars of fairness, independence and transparency since a federation took an unexplained decision against the interest of their athletes who were unable to dispute this decision (Kelly et al., 2022). This is in turn exacerbated by lack of well-developed unified specialised arbitral units and independent bodies to oversee and resolve such conflicts in esports and virtual sports (Czakó et al., 2023; Kelly et al., 2022; Wong & Meng-Lewis, 2023). Last, the lack of transparency in qualification pathways is illustrated by Kim's statement that *"Every year [publishers and the UCI] tweak things and change around the selection criteria that the racers have to go through [...]"*, highlighting a congruence in the governance issues of virtual cycling and those observed in esports (Chao, 2017).

15.5.4 Commitment and Decision-making

Lack of commitment from both national and international federations was raised as a governance issue. For instance, some athletes expressed that national federations provided limited support or offered assistance only when it aligned with the federation's own interests:

We had a lot of great racers drop out because they weren't able to support it. [...] [The NF] is going to be following whatever is shiny and new. [...] We got temporary support and visibility from [NF] for virtual cycling when I got my medal last year. (Elvene)

I had nothing from my federation. Absolutely nothing. They didn't even email me. They didn't talk to me. They didn't talk to me afterwards [...] We had to organize amongst ourselves as riders [...] so you can see there's not that much support behind it. (Manny)

They just have no interest in it. (Claressa)

There's a lot of federations who have not caught up and not on the ball and are not treating it as a proper discipline. They're seeing where it goes (Rika)

From the perspective of a national federation, Ben reinforced this perceived lack of interest, explaining that his federation views virtual cycling as practically complex and offering limited potential for enhancing the organisation's credibility within the sport.

[Hosting virtual triathlon/cycling event] is impossible [...], I don't see a potential venue where this can be done. We have credibility because we are a sport that is seen in the streets [...] so I don't see people getting attracted to be inside an arena. (Ben)

In addition, Kim noted the common belief that the UCI is expected to provide guidance in developing virtual cycling: “If not the UCI, then who?”. Yet they appear to fall short in terms of governance structures and providing guidelines:

What the UCI is there to do is provide guidance. And there isn't a commission, there isn't any guidance whatsoever. One of the ways that the UCI is falling short is that they don't have a set of guidelines. (Kim)

While it could be debated which governing body (i.e. esports or sport) is best suited to develop virtual cycling as a discipline, it is evident in our data that the interviewees (regardless of roles and background) see the UCI as a logical choice, since they have already recognised virtual cycling as legitimate discipline and sanctioned world championships in collaboration with platform publishers.

Notwithstanding the consensus in our data regarding the UCI as ‘logical choice’ for governing body in virtual cycling esports, it remains imperative to critically assess the measures needed for governing bodies to more effectively align their practices with good governance principles (Kelly et al., 2022). We urge governing bodies aiming to engage with the fast-evolving field of virtual cycling esports to demonstrate their commitment by promptly developing any organisational prerequisites needed to establish appropriate governance structures and guidelines that address the aforementioned governance issues. Inaction leaves a vacuum that invites governance issues and undermines good governance principles such as fairness, accountability and responsibility (Kelly et al., 2022). Governance efforts should in that sense also aim to establish clear accountability mechanisms for all stakeholders involved (Richardson et al., 2022). Governing bodies should be liable for their responsibilities, for instance through national law or oversight forums such as CAS. At the same time, they must also hold their members (e.g. national federations) and partners (e.g. platform providers) liable for implementing essential governance practices, such as performance verification and athlete licensing.

15.6 Limitations and Future Research

In our analysis we have aimed to give an overview of the governance issues that stakeholders currently experience in virtual cycling. However, virtual cycling incorporates physical performance from sport with the digital infrastructure from esports and introduces novel configurations where current frameworks, when applied independently, may limit our understanding to the threats — and not the pillars — of virtual sports good governance as these are context specific. This makes comparative studies more challenging as well.

In addition, the emerging nature of virtual cycling means that governance structures are still developing and rapidly changing, which may limit the generalisability of our findings in time (e.g. only for a few years) but also for specific platforms (e.g. new platforms with new game physics and rules may emerge). Future research could delve deeper into each of the four governance issues identified in our material. This would allow for nuanced and in-depth analyses of how different stakeholders (e.g. athletes, national federations, international sport governing bodies, platform developers) experience issues of doping, platform fragmentation, lacking transparency and insufficient commitment and responsibility from other stakeholders in virtual sports. Further examinations are also needed into how digital inequalities, such as unequal access to reliable hardware or high-speed internet, shape fairness and inclusivity in competitive virtual cycling, or other virtual sports.

15.7 Conclusion

The aim of this article was to understand how stakeholders experience governance issues in virtual cycling. Central to our findings is the notion that governance issues in virtual cycling both mirror and nuance those of esports and sport. Through interviews, we have identified four key governance issues in virtual cycling: (1) doping, (2) platform fragmentation and commercial interests, (3) transparency and open criteria, and (4) commitment and decision-making. We find that governance issues experienced by stakeholders converge with those of sport and esports, encompassing physical-, cognitive-, software-, and hardware-related manipulation, as well as concerns around the intersection of commercial interests, data privacy, and performance verification.

Stakeholders expressed clear frustrations with current fragmented governance structures where commercial interests overshadow good governance principles of social responsibility, accountability, fairness and independence. Athletes voiced particular concerns regarding the

lack of clear qualification pathways, as well as the absence of robust guidelines and support dedicated to virtual cycling from national and international federations.

In terms of contributions to practice, our findings offer explicit directions for reform. Stakeholders call for unified anti-doping protocols that address both physiological and mechanical manipulations, as well as for platform standardization in performance metrics, rankings, and transparent open qualification processes across all competitive tiers. We also observe the urge for international governing bodies to unify and act swiftly by establishing comprehensive guidelines and foster collaboration with national federations to legitimize virtual cycling.

In terms of theoretical contributions, our study demonstrates the applicability of esports governance models (Kelly et al., 2022) within the context of virtual sports. Good governance pillars appear robust yet the issues pertaining them must be understood to the context-specific dynamics of virtual cycling, highlighting that good governance issues are not directly generalisable between sports, esports, and virtual sports. This nuanced understanding opens avenues for future research to refine governance models that effectively bridge traditional and virtual competitive environments.

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Ethical statement

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Data Availability Statement

For ethical reasons, only pseudonymised data of this study are available from the corresponding author, upon reasonable request.

16 Study 5: When Algorithms Define Sports: Algorithmic Differences and Governance Gaps in Virtual Cycling

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Abstract

Research question: This study investigates how algorithmic differences between virtual cycling platforms shape competition outcomes and governance practices. Specifically, we examine how technical designs of platforms (i.e. algorithms) differ in translating athletes' physical exertion into virtual race outcomes, and what governance issues pertain to such differences. This addresses a gap in sport management literature on the governance of sport digitalisation where technology mediates sports.

Research methods: A sequential mixed-methods design was employed. Quantitatively, we analysed scraped data of 102 virtual cycling races (Zwift: $n=52$; MyWhoosh: $n=50$) to compare potential differences between them. Qualitatively, 15 semi-structured interviews with international federations (e.g., UCI, IOC), platform publishers, and elite athletes explored governance responses to these differences.

Results and findings: Quantitative findings reveal significant differences between platform's explanatory power, that is, the extent to which variance in competition results is explained by physical performance. MyWhoosh exhibits higher explanatory power ($R^2=0.87$) than Zwift ($R^2=0.76$), indicating that algorithmic design influences competitive dynamics and that physical performance significantly better explains race results on MyWhoosh. Interviews show how both elite athletes and representatives of national and international governing bodies are aware of these platform differences. The qualitative data also illustrate how participants tended to underplay the significance and potential integrity issues associated with these differences, resulting in governance gaps in transparency, accountability, and democracy.

Implications: The study advances theory by introducing the concept of dual governance, uniting sport federations and platform publishers. Practically, it calls for certification frameworks, scrutability-without-disclosure models, and formal dual-governance agreements to safeguard athlete rights.

Keywords: Virtual sports; Good Governance; Algorithmic transparency; Sport digitalisation; Mixed-methods

16.1 Introduction

Hybrid virtual sports disciplines sitting at the intersection of sport and esports are rapidly expanding, introducing new opportunities and governance challenges for sport organisations (Fouquaert et al., 2025; Westmattelmann et al., 2022, 2025). The UCI's official virtual cycling championships illustrate this trend. Despite virtual sports becoming increasingly popular, issues like doping, platform incompatibility, or lacking commitment remain critical governance gaps (Fouquaert et al., 2025).

By mimicking the exertion and physicality of the parent sport using interactive technology, virtual sports translate actual sporting performances into the virtual realm (e.g., athletes physically performing on smart trainers connected to the virtual cycling platform Zwift) (Fouquaert et al., 2026; Parry & Giesbrecht, 2023, p. 216). Similar to sports, virtual sports competitions have become institutionalised and rule-governed (Parry & Giesbrecht, 2023). However, its governance is not easily comparable to that of sports, as new stakeholders appear in a unique ecosystem combining sports (e.g., international sport federations) and esports (e.g., virtual platforms and game publishers) (Fouquaert et al., 2025; Haffner et al., 2025). This gives rise to novel governance challenges, one of which is the impact of virtual platforms mediating sporting performances through its algorithms and ultimately influencing competition results. This study focuses on the governance of virtual platforms by sport governing bodies and, in particular, how sport governing bodies cope with algorithmic differences between virtual sports platforms in the translation of physical performance (i.e., input) into (virtual) competition results (i.e., output). We use virtual cycling as exemplary case because it is the most legitimised institutionalised sport in which performance is well-defined and measurable (i.e. W/kg) (Fouquaert et al., 2025, 2026; Parry & Giesbrecht, 2023; Reed et al., 2023; Westmattelmann et al., 2025).

Performance is central to elite sport, yet in virtual cycling, results depend not only on physical performances but also on virtual platform algorithms and tactical choices. In Zwift, for example, variance in competition results is not fully explained by physical performance (i.e. W/kg). Physical performance explains between 77% to 99% of race outcomes (i.e. rank), depending on stage characteristics and gender (Westmattelmann et al., 2022). Previous studies have however quantified this performance-result relationship only for the Zwift platform (Westmattelmann et al., 2022) while focussing heavily on technological and athlete-centric aspects (Westmattelmann et al., 2025). Consequently, there is a critical gap in understanding whether and how this relationship varies across different platforms (e.g. Zwift vs. MyWhoosh),

and what do differences in platforms' algorithms mean for governance in sport management. The way algorithms act as governance objects in virtual sports, requiring institutional oversight by sport governing bodies, remains largely unexplored in contemporary literature. Yet, such a governance perspective is highly relevant because differences in platforms' algorithms means that commercial publishers directly shape competition outcomes. Commercial decisions, like the UCI's short-notice shift from Zwift to MyWhoosh for the 2024 Esports World Championships, consequently raise questions about integrity and governance mechanisms related to democracy, transparency and accountability (Geeraert, 2019; Shilbury et al., 2016; Thompson et al., 2023) in virtual sports. Such instances also raise questions about governance mechanisms needed to safeguard athletes and warrant competition integrity (Fouquaert et al., 2025; Qi et al., 2024), further illustrating the need for virtual sports governance contributions to sport management literature (Funk et al., 2018; Kelly et al., 2022). Recent integrative reviews of esports scholarship have explicitly echoed this call, urging sport management research to move beyond conceptualisation debates and to engage with the platform-specific and technical nuances of digitally-mediated competition rather than treating esports and virtual sports as a homogeneous category (Pizzo et al., 2022).

Against this backdrop, the following research question is examined: how do differences between virtual sports platforms shape virtual sports competitions and their governance? To explore this, we conducted two complementary studies under a mixed-method approach. First, a quantitative study provides insights into the differences in competition results between Zwift and MyWhoosh using scraped race data. Secondly, a qualitative study provides insights into how platform differences are perceived and shape proactive or reactive governance mechanism by sport governing bodies in virtual cycling using interview vignette data from representatives within the UCI, World Triathlon, the IOC, elite virtual cycling athletes, platform publishers and media spokespersons. Thirdly, insights from both studies are combined to discuss meta-inferences. Combined, our study contributes to existing literature in three ways: it presents the first cross-platform comparison of the performance-result relationship while also adopting a governance perspective as we link quantitative algorithmic differences to qualitative insights of sport governing bodies. Although our empirical setting is virtual cycling, insights may additionally inform the broader governance of platform-mediated sports.

We continue by reviewing literature on governance within the broader process of sport digitalisation, focusing on how technological mediation and stakeholder dynamics influence competition outcomes. We then introduce our theoretical perspective, anchored in universal

good governance principles (Geeraert, 2021) and complemented by Kelly et al.'s (2022) esports governance framework.

16.2 Literature review

16.2.1 Sport Digitalisation and Governance

Sport digitalisation refers to integrating digital technologies into sport practices, governance, and consumption (Westmattelmann et al., 2025). It is not about creating new sports through technology (i.e. “technology-driven” sports) but rather about augmenting or mediating existing sports with technology (Thomas & Tierney, 2021). Sport digitalisation has occurred in successive waves. The first wave introduced technology for performance monitoring and officiating (e.g., VAR in football, Hawk-eye in tennis), assuming superior accuracy but later revealing biases or inconsistent outcomes (Fouché, 2017; Skirbekk et al., 2025). The second wave introduces technology that mediates the sport itself, giving rise to hybrid environments and virtual disciplines where physical performance is translated into virtual results (Parry & Giesbrecht, 2023; Westmattelmann et al., 2022). Also in this wave, rules are put in place to protect the ‘purity’ of the sport and ‘unnatural’ technological intrusion (Magdalinski, 2009).

The progression of sport digitalisation illustrates how technology increasingly shapes sports’ ontology. This presents unprecedented challenges and dilemmas that become especially relevant as virtual sports also begin to meet the criteria of competitive sport, involving voluntary competition, physical and strategic skills, and institutionalised governed rules (Loy, 1968; Naraine, 2021; Suits, 2007). Virtual cycling is a prime example, as platforms such as Zwift allow sports to be physically executed yet digitally governed, spectated and consumed (Fouquaert et al., 2025; Tjønndal et al., 2025; Westmattelmann et al., 2025). Governance often becomes relevant in these cases as publishers have an interest in legitimising these new disciplines through establishing rules, standards and governance structures that also help to preserve their commercial interests (Fouquaert et al., 2025; Thomas & Tierney, 2021; Tjønndal et al., 2025). This dynamic mirrors esports governance more broadly, where Heidenreich et al. (2022) demonstrate that newly established esports associations can only achieve partial legitimacy because they must continually navigate the tension between traditional sport governance logics and the property rights held by commercial game publishers. Knowing this, our focus is on governance interventions across sport digitalisation, particularly in the second wave where technology mediates competition.

Based on prior cases across different waves of sport digitalisations (e.g. hydrodynamic swimwear, continuous glucose monitors in cycling), governing bodies are known to respond slowly to technological change, a “phase lag” (Synrem, 2019) explained by new integrity dilemmas and institutional inertia that require lengthy consensus-driven processes (Marchant, 2011; Washington & Patterson, 2011). In virtual cycling, this lag is evident as cross-platform regulations or standards are currently absent (Dyer, 2022; Fincoeur & Bongiovanni, 2021; Fouquaert et al., 2025, 2026; Richardson et al., 2022; Tjønndal et al., 2025), leaving publishers free to undisclosedly define how algorithms transform physical performances into virtual results. Such autonomy creates information asymmetry and a transparency gap (Westmattelmann et al., 2025), which might especially be problematic with short-notice platform changes like UCI’s switch from Zwift to MyWhoosh shortly before the 2024 Cycling Esports World Championships. Fairness and accountability consequently depend on commercial agreements and proprietary systems beyond traditional oversight (Bjärehed & Bjärehed, 2022; Postma et al., 2022).

16.2.2 Translating Real Performances into Virtual Results

There is limited research on virtual cycling platform algorithms that mediate athletes’ performances. Based on two studies of Westmattelmann et al. (2022, 2025), a ‘performance–result gap’ is evident on Zwift. Physical performance (i.e. W/kg) explains 77-99% of the variance in competition results depending on stage characteristics (i.e. route topography) and gender (i.e. men 84–99% and women 77–98%) (Westmattelmann et al., 2022), while a follow-up study confirmed a similar gap of 72-96% of explained performance variance (Westmattelmann et al., 2025). This indicates that, similar to road cycling, other factors besides athletes’ performance impacts their final result. In this case, tactical behaviour, drafting or platforms’ physics engine (Kyle, 2003; Westmattelmann et al., 2021).

However, these other influencing factors remain generally unknown and undisclosed, creating a “transparency gap” (Westmattelmann et al., 2025). This gap is unique because in virtual sports, input (i.e., W/kg) and output (i.e., riding time) are precisely known, unlike in physical competitions (Kyle, 2003). Influencing factors that explain this gap likely include tactical behaviours such as timing in-game efforts around latency or applying drafting strategies. In light of the performance-result and transparency gaps, comparing platforms is critical to determine whether these gaps are isolated or systemic across virtual sports platforms. Yet, no research to our knowledge has investigated such differences.

16.3 Theoretical Framework: Good Governance in Sports and Esports

Virtual cycling occupies a hybrid space between traditional sport and esports (Fouquaert et al., 2026). When focussing on governance within virtual cycling, it becomes clear that traditional sport governance models often struggle to address the distinct power dynamics introduced by commercial platform publishers (Fouquaert et al., 2025; Geeraert, 2021). Because publishers like Zwift or MyWhoosh own and control the platforms mediating the competition, virtual cycling shares fundamental governance characteristics with esports (Wong & Meng-Lewis, 2023). On the contrary, many established stakeholders from traditional sports (e.g. UCI, World Triathlon, National Federations) are involved while virtual cycling also, in part, demonstrates a physical nature (Fouquaert et al., 2026), both highlighting the relevance of traditional sports governance approaches to virtual cycling.

We adopt a sport governance lens anchored in universal good governance principles (Geeraert, 2018, 2021), complemented by Kelly and colleagues' (2022) esports governance framework. This combined approach provides the fundamental good governance principles needed to approach new governance dilemmas at the intersection of sports and esports, making it a particularly relevant lens to virtual sports. While Kelly et al.'s (2022) esports governance framework discusses seven principles based on the sport governance principles of Burger and Goslin (2005), traditional governance frameworks often combine these principles in three widely accepted dimensions that are “mutually reinforcing and mutually constitutive” (Geeraert, 2018, p. 16, 2021). To maintain conceptual clarity and focus on the issues most pertinent to algorithmic mediation, we concentrate on three universal core principles described in both sports and esports governance: transparency, accountability, and democracy.

16.3.1 Transparency

Transparency is defined in traditional sport governance as the “rules and practices that ensure that information is made available about an actor” (Geeraert, 2021, p. 21), allowing other stakeholders to monitor the behaviour and performance of such actor (e.g. governing body) (Meijer, 2014). In the context of esports, transparency extends to clarifying the interests and mechanisms behind commercial and technical decisions of publishers that (co-)govern competitions (Kelly et al., 2022). The hybrid nature of virtual cycling creates unique challenges. For instance, it involves observable physical performances that are mediated through publishers' proprietary algorithms and give way to the earlier described platform-results gap. This also creates a “transparency” gap, as information relating physical input (power) to virtual output (race time) is not transparently available (Westmattelmann et al., 2025). In such a case,

the transparency dimension also relates to the concept of organisational transparency, defined as “the perceived quality of intentionally shared information from a sender” (Schnackenberg & Tomlinson, 2016, p. 1788). More specifically, Schnackenberg & Tomlinson (2016) conceptualise organisational transparency under three subdimensions: disclosure (providing relevant information), clarity (presenting it in an understandable way), and accuracy (ensuring factual correctness). For virtual cycling, publishers (i.e. sender) intentionally share or retain information about their platform, on which sport governing bodies rely to regulate and communicate. This in turn reaches other stakeholders who form their perception for each transparency subdimension based on the information availability such as platform dynamics, scheduled features, the features allowed to use within certain races, or the platform used for races. This ultimately creates two layers affecting transparency: first by publishers and consequently by other stakeholders like sport governing bodies. Such dynamics within virtual cycling inherently increase the complexity and risks associated with fulfilling transparency (Meijer, 2014), potentially also affecting other governance principles such as the perceived *fairness* of the competition or the demarcation in terms of *responsibility* of assuring transparency (Czakó et al., 2023; Geeraert, 2021; Kelly et al., 2022).

16.3.2 Accountability

Accountability in governance refers to “a relationship between an actor and a forum, in which the actor has an obligation to explain and to justify his or her conduct, the forum can pose questions and pass judgement, and the actor may face consequences” (Bovens, 2007, p. 450). Both within traditional sports (Bovens, 2007; Geeraert, 2018, 2021) and esports governance (Kelly et al., 2022), accountability encompasses several related subdimensions such as: *responsibility* (enforcing consequences for governance failures), *discipline* (consistency in sanctioning inappropriate behaviour), and *independence* (unbiased oversight crucial for fair operations). In traditional sports, these mechanisms are usually managed by governing bodies and independent tribunals. In virtual cycling, the commercial ownership of the digital platform challenges this traditional structure. The power of publisher (e.g. monopoly position, technical complexity) can hinder independent oversight and limit the enforcement of industry-wide standards (Czakó et al., 2023; Fouquaert et al., 2025, 2026). Because virtual cycling involves traditional sport governing bodies but relies on publishers’ infrastructure, traditional accountability mechanisms require adaptation. Sport governing bodies must navigate ways to hold platform publishers accountable for things like unannounced algorithmic updates or other technical patches that directly impact athletes' performances and, as such, their procedural

rights. Similar to transparency, accountability in virtual cycling is also double-layered because governing bodies themselves (e.g. UCI) must be consistently held accountable for their actions by unbiased overseers having the right to sanction (Fouquaert et al., 2025). For instance, the empirical case of this study (i.e. UCI's short-notice Zwift to MyWhoosh platform change before the 2024 Cycling Esports World Championships) is an exemplary kind of action related to virtual cycling where traditional sport governing bodies must be held accountable for the impact on the athletes.

16.3.3 Democracy

Democracy within sport governance is “a system of rules that establishes competition (electoral competition between political alternatives), participation (affected actors' influence over collective decisions) and deliberation (fair and open debates)” (Geeraert, 2021, p. 22). Within esports, Kelly et al. (2022) capture these democratic dimensions through the principles of *fairness* and *social responsibility*. This respectively includes balancing needs of new stakeholders like publishers (Postma et al., 2022) and promoting inclusive and safe digital environments while mitigating risks inherent to them (e.g. toxicity, exclusion, privacy, algorithmic bias) (Hayday et al., 2021; Tjønndal et al., 2025). In traditional sports, athletes and national federations typically have formal representation in governance structures. Conversely, esports ecosystems are often governed unilaterally by game publishers (Fouquaert et al., 2025; Qi et al., 2024; Wong & Meng-Lewis, 2023). This undermines the rights of traditional stakeholders (e.g. traditional sport governing bodies, athletes, or national federations) in electing decision-making representatives, participate in decision-making processes and engage in deliberation, increasing the likelihood that inappropriate behaviour remains unnoticed (Warren, 2014).

As virtual cycling connects new esports and established sports stakeholders, it clearly requires adaption and mixing of sports and esports dimensions. For instance, athletes physically perform and are sanctioned by traditional bodies like the UCI, yet they are subjected to the digital realities installed by commercial publishers (Fouquaert et al., 2026). Hence, the democracy dimension of virtual cycling also becomes multi-layered much like transparency and accountability. This requires new structures where the needs and rights of athletes and traditional governing bodies are fairly represented and balanced against the commercial interests of platform publishers. The development of democratic structures can be facilitated by acknowledging new stakeholders like publishers as owners of good governance outcomes, which creates an incentive to strive for mutual benefits and reciprocity in terms of loyalty,

fairness, responsibility, and ethical treatment (Fassin, 2012). In that sense, virtual cycling may need to recognise publishers as ‘stakeowners’. Stakeowners are stakeholders with a legitimate interest that strive for mutual benefits and with whom dual governance is ultimately necessary (Ferkins & Shilbury, 2015).

16.4 Methodology

16.4.1 Research Setting

The use of virtual cycling makes a particularly relevant case to study differences between platforms’ impact on performance and its impact on sport governance. While virtual cycling is organised and structured similar to esports, virtual cycling is uniquely different because (1) the sporting performance (i.e., physiological parameters like power output) is transformed to performance metrics in the virtual platform (i.e., riding time and race results) and (2) it is governed by a sport federation (i.e., UCI) and not by an esports federation (e.g. International eSports Federation; Westmattelmann et al., 2021, 2022). The technology used in platforms such as Zwift and MyWhoosh provides real-time feedback on a rider’s performance, allowing for a detailed examination of how algorithms translate physical effort into competitive results. As cycling federations are key governing bodies in establishing and maintaining competition rules, the involvement of virtual platform providers adds a layer of complexity to the governance structure, as these platforms are responsible for the algorithms that determine competition outcomes.

Zwift and MyWhoosh are leading platforms in virtual cycling, both enabling riders to compete in real-time virtual environments using smart trainers that measure performance metrics like power and cadence. Athletes’ relative power output (W/kg) is the main metric used to translate their real performances into virtual performances using algorithms that simulate terrain, drafting, and other real-world cycling dynamics. Both platforms offer gamified experiences, competitive events, and social features that foster community engagement and support structured training formats, and leisure to competitive events and leagues (e.g., the Virtual Tour de France on Zwift).

Despite their similarities, there are notable differences between the two platforms that may influence race outcomes (for an overview, see Minařík, 2025). Zwift has a broader user base and offers a diverse mix of real-world and fictional routes (e.g., UCI World Championship course in Innsbruck), as well as training and social features. MyWhoosh, by contrast, focuses more narrowly on serious leisure and professional-level racing, with distinct race formats,

unique courses, and larger-scale events and leagues offering substantial prize money every week (e.g. Sunday Race club). In addition, MyWhoosh offers free access, whereas Zwift operates on a subscription-based model of around €20/month, potentially attracting different segments of the virtual cycling market.

These platform-specific differences suggest potentially divergent algorithmic approaches to simulating cycling performance. Such differences also raise critical questions about how performance is mediated and whether identical inputs (e.g., in watts/kg) yield consistent outcomes across platforms. This is especially relevant as governing bodies like the UCI begin to formally engage with platform providers, underscoring the need to understand the role of platform algorithms in shaping fairness and legitimacy in virtual sport.

16.4.2 Mixed-Methods Design

To address our research question, we implemented a sequential mixed-methods design (Tashakkori & Teddlie, 1998; Venkatesh et al., 2013, 2016), which is summarised in Table 12. As Venkatesh et al. explain, this method leverages “elements of both the quantitative and qualitative approaches” (Venkatesh et al., 2016, p. 5). Our design follows the guidelines provided by Venkatesh et al. (2013, 2016) for conducting mixed-methods research, specifically adhering to the purpose of “complementarity”. In this context, the qualitative study was used to “gain additional insights on the findings from a quantitative study” (Venkatesh et al., 2016, p. 26). A sequential rather than a parallel design was appropriate because Study 5.2 was intentionally built on Study 5.1. More specifically, Study 5.1 first established whether and to what extent the input-output relationship differed between Zwift and MyWhoosh, and these findings then informed the vignette used in Study 5.2, which explores how possible differences between virtual sport platforms are perceived and acted upon by sport governing bodies. Mixed-methods guidelines recommend such a sequential design when one strand is intended to inform the next (Venkatesh et al., 2013, 2016). We maintain a constructivist perspective for both the quantitative and qualitative components of the study (Tashakkori & Teddlie, 1998). Such a design is classified as partially mixed-methods, with a sequential and multistrand structure. The mixing of the two methods occurs during data analysis and the inferential stages of the overall study.

Our study began with the collection of quantitative competition data. To ensure consistency and comparability across platforms, we focused exclusively on virtual competitions at the highest performance level. This resulted in a sample of 52 races on Zwift and 50 races on MyWhoosh. This sampling decision was purposive, as our aim was not to represent the full

population of virtual cycling events, but to construct a controlled cross-platform comparison in which differences in the input-output relationship could be examined under relatively similar competitive conditions. Restricting the sample to the highest performance level reduced heterogeneity that would likely arise in lower-level races due to larger variation in rider ability and tactical behaviour. Once the competition data were analysed, we conducted semi-structured interviews with a diverse set of stakeholders in virtual cycling (see Appendix section 16.13.3 for full overview). We discussed the quantitative findings with them through semi-structured interviews including a vignette (see Appendix section 16.13.4) to gain further insights into the governance of virtual sports platforms.

Table 12: Overview of mixed-methods studies, objectives and methodology (study 5)

Study	Research Objective	Data Collection	Sample Size	Analysis
5.1	Comparing objective input-output-relation between virtual cycling platforms	Scraped performance data and competition results	N = 102 virtual races	3 rd order polynomial regressions; t-tests
5.2	Examining sport governing bodies' governance mechanisms related to differences in virtual cycling platforms	Semi-structured interviews using vignettes with representatives from the UCI, World Triathlon, the IOC, elite virtual cycling athletes, platform publishers, and broadcasting commentators	N = 16	Collective Qualitative Analysis (Eggebo, 2020)

16.5 Study 5.1: Quantitatively Comparing Input-Output Between Virtual Sports Platforms

16.5.1 Study 5.1 Research Design

Study 5.1 quantitatively compares the relationship between sporting performance and competition outcomes on the virtual cycling platforms Zwift and MyWhoosh. Using scraped performance data, we analyse how each platform's algorithms mediate race results by examining the input-output relationship between relative power (W/kg) and riding time.

16.5.1.1 Scraping and Curve Fitting Input and Output Variables.

To ensure comparability, we focused exclusively on races from the highest competition category on each platform, scraping data from 52 Zwift and 50 MyWhoosh races. As outlined in the Mixed-Methods Design section, this purposive sampling approach aimed to maximise

internal comparability between the platforms. The resulting sample provided near-symmetrical coverage of the two focal platforms. Consequently, each race served as the unit of analysis for assessing differences in the explanatory power of physical performance. Following (Westmattelmann et al., 2022), we used average relative power output (W/kg) as the proxy for sporting performance (input). Competition outcome (output) was measured by individual riding time in milliseconds, as it provides a more granular measure of performance than finishing rank.

Based on the physics of road cycling (Kyle, 2003) and previous research (Westmattelmann et al., 2022, 2025), we assumed a cubic relationship between relative power output and riding time. We validated this by testing linear, quadratic, and cubic models for each race using the curve fit function in SPSS 28. The cubic regression consistently provided the best fit, confirming it as the most appropriate model for our analysis (see R^2 values in section 16.13.1 and 16.13.2 Appendix).

16.5.1.2 Analysis and Comparison of the Input-Output-Relationship.

We analysed the input-output relationship for each race using third-order polynomial regressions (see Eq. 1; Stimson et al., 1978) using SPSS 28. We verified that all assumptions for multiple regression were met (i.e., no multicollinearity, independence of residuals, homoscedasticity, and normal distribution of residuals) and mean-centred the independent variable to reduce multicollinearity (Aiken et al., 1991).

$$Y = b_0 + b_1 * x + b_2 * x^2 + b_3 * x^3 + e \quad (1)$$

In Equation 1, Y represents the avatar's riding time in milliseconds (dependent variable), while x denotes the mean-centred relative power output in watts per kilogram of body weight (independent variable). The coefficients for the 1st, 2nd, and 3rd order polynomials are represented by b_1 , b_2 , and b_3 , respectively, with b_0 as the intercept and e as the error term (Westmattelmann et al., 2022). By performing these third-order polynomial regressions, we assess how relative power output explains race outcomes, providing a deeper understanding of how physical performance translates into virtual results. Following Stimson et al. (1978), we used the global fit index R^2 to evaluate the model's explanatory power, as individual coefficients in polynomial regressions are not interpretable in isolation.

Once the explanatory power (R^2) for each virtual sports competition was calculated (in total 102 R^2 indices), we proceed to compare the resulting R^2 indices between the two platforms, Zwift and MyWhoosh. To determine statistical differences in R^2 indices, we employed Welch's t-test given their unequal variances based on Levene's test ($F(1,100) = 9.226, p = .003$). Also

for other variables reported, we employed a Welch's t-test given their unequal variances based on Levene's test (mean W/kg winner $F(1,100) = 30.467$, $p = <.001$; participants $F(1,100) = 21.153$, $p = <.001$).

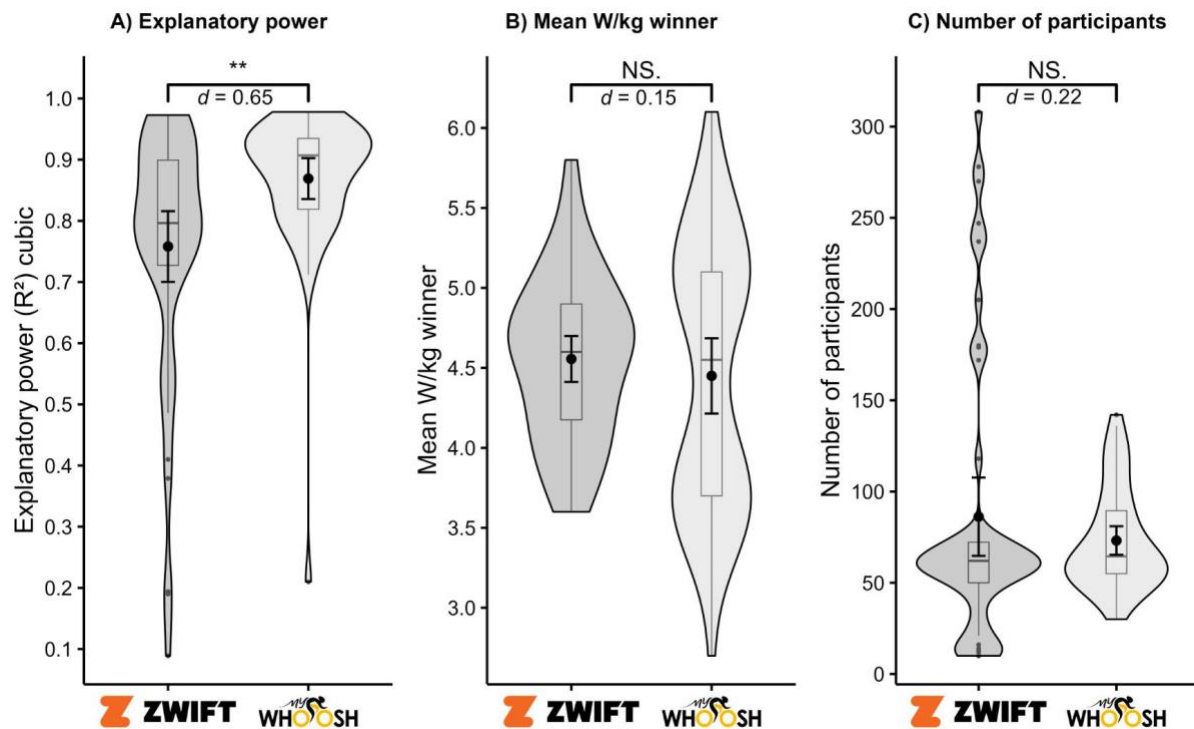
Last, to ensure the stability of our results, we conducted several robustness checks. We systematically removed races deviating more than one standard deviation from their platform's mean on key variables (explanatory power, winner's W/kg, and number of participants), first one at a time and later simultaneously. In each of these subsets, all differences between Zwift and MyWhoosh (i.e. each variable) remained robust, meaning they either remained statistically insignificant or significant.

16.5.2 Results Study 5.1

For Zwift, the R^2 values across the 52 races varied between 0.089 and 0.973 with an average explanatory power of 0.758 (SD = 0.208). For MyWhoosh, the R^2 values ranged from 0.210 to 0.978, with a mean of 0.869 (SD = 0.118).

Results are presented in Figure 17 using violin-boxplots, showing each variable's distribution shape and density next to its mean, median, quartiles, and potential outliers for easier comparison. The comparison of the explanatory power between Zwift and MyWhoosh as shown in Figure 17A reveals a statistically significant difference ($t(81.3) = 3.34$, Cohen's $d = 0.658$, $p = 0.001$). This significant difference remained after performing our aforementioned robustness checks ($t(46.8) = 2.85$, $d = 0.734$, $p = 0.006$). MyWhoosh exhibits higher explanatory power (mean diff. = 11%pt) for the relationship between performance and competition outcomes compared to Zwift ($M_{Zwift} = 0.76$, $M_{MyWhoosh} = 0.87$). Despite the differences in explanatory power, Figure 17B shows that the average sporting performance required to win races (mean W/kg) did not significantly differ between Zwift and MyWhoosh ($t(81.5) = -0.77$, $d = 0.1553$, $p = 0.443$). Similarly, Figure 17C shows that the number of participants in the selected races was also not significantly different between Zwift and MyWhoosh ($t(64.29) = -1.15$, $d = 0.226$, $p = 0.254$).

Figure 17: Violin-boxplots of differences between Zwift and MyWhoosh (study 5). Figures show means (dot) and error bars at 95% confidence-interval for (A) explanatory power of the race, (B) mean W/kg of the race winner and (C) number of participants in the race. Significance levels Welch's t-test: NS = $p \geq 0.05$, * = $p < 0.05$, ** = $p < 0.01$, *** = $p < 0.001$. d = Cohen's d as effect size used to indicate the standardised difference between two means.



16.5.3 Conclusion of Study 5.1

Virtual races on either Zwift or MyWhoosh can be regarded as highly similar since both the average sporting performance (W/kg) required to win the race and the number of participants are not significantly different. However, there is a difference in explanatory power between Zwift and MyWhoosh, where sporting performance (W/kg) explains significantly better the race results (riding times) in MyWhoosh compared to Zwift. In other words, riders' average power per kg bodyweight throughout the race is better able to explain their final result in that race on MyWhoosh compared to Zwift, meaning that factors other than sporting performance (W/kg) such as drafting, race tactics, or equipment choices, have a greater impact in determining the race results in Zwift compared to MyWhoosh. This confirms the performance results gap, as there remains 24%pt of unexplained performance in Zwift compared to 13%pt unexplained performance in MyWhoosh. Where these differences are rooted, remains however, a black box of opaque algorithms.

16.6 Study 5.2: Qualitatively Assessing Governance Mechanisms Towards Algorithmic Differences Between Virtual Sports Platforms

16.6.1 Study 5.2 Research Design

Study 5.2 aims to explain the differences identified in study 5.1 from a governance perspective, which shows how riders' average relative power throughout the race better explains the final results in a race on MyWhoosh compared to Zwift. Study 5.2 examines how this algorithmic difference is perceived and addressed through proactive or reactive governance mechanisms by sport governing bodies in virtual cycling.

16.6.1.1 Participants and Data Collection

We conducted 16 interviews with stakeholders representing national and international sport governing bodies and elite athletes, including managers within UCI, World Triathlon, SuperTri, the IOC, and two national federations. Participants had to meet the following criteria: (1) represent a cycling and triathlon governing body at the national or international level, (2) represent a diversity of stakeholders in virtual cycling, and (3) maintain gender balance in our sample. Recruitment was done by directly mailing the abovementioned organisations and two international broadcast commentators who introduced us to several elite athletes. We refer to Appendix Section 16.13.3 for a full overview of interviewees and their pseudonyms. Ethical approval was granted by Nord University, SIKT – The Norwegian Agency for Shared Services in Education and Research, and all participants gave informed consent, with the right to withdraw at any time.

Semi-structured interviews comprised one face-to-face and fifteen via video call, each lasting 40 to 80 minutes. Eight were conducted by one researcher, while the remaining eight were jointly conducted with two researchers. The interview guide addressed six domains: (1) personal/organisational background; (2) (de)legitimisation of virtual cycling; (3) policy and governance; (4) participation (5) innovation and change in general, and (6) a vignette. The vignette (see Section 16.13.4 Appendix) presented to the participants described UCI's abrupt yet impactful switch from Zwift to MyWhoosh shortly before the 2024 Cycling Esports World Championship. It included the quantitative findings from study 5.1 (i.e. differences between Zwift and MyWhoosh) to gain further insights into participants' awareness of these differences and their opinion related to possible integrity and governance concerns of virtual sports platforms. The vignette approach was selected because it provides a concrete, contextualised scenario that helps elicit participants' perceptions, values, and normative beliefs regarding complex or sensitive situations (Barter & Renold, 1999). As explained in our mixed-methods

design, the use of a vignette aligns well with our sequential approach, as it allowed us to present the abstract quantitative differences within the context of a recent and impactful platform switch, which enables stakeholders to reflect more tangibly on the resulting governance implications. Because the interview with Stan was found irrelevant to our RQ, qualitative analysis was performed on the remaining 15 interviews (see Section 16.13.3 appendix) and mainly focussed on the performance-results gap presented in the vignette, while qualitative data from the other categories are presented in (Fouquaert et al., 2025, 2026; Tjønndal et al., 2025).

16.6.1.2 Data Analysis

We applied Eggebø's (2020) four-step collective qualitative analysis. First, two researchers each drafted one-page summaries of every transcript and shared them with all authors. Second, they independently coded data-driven themes and displayed these on Post-its for collaborative mapping. Third, related themes were merged or refined into 20 analytical themes, each annotated on A4 sheets with subthemes, illustrative quotes, and participant pseudonyms. Analytical themes were also visually linked to reveal patterns. Fourth, we devised an outline for manuscript development based on these themes. Following Venkatesh et al.'s (2013, 2016) mixed-method approach, the current study focuses on the parts of the interview data that provided complementarity and additional insights on the findings of study 5.1. Specifically, we focus on the participants' reflections after being presented with the vignette (Section 16.13.4 Appendix). Three interviews did not include the vignette (Robert, Ben, Ali) because the interview process didn't lend itself to it (e.g. too little time, unknowledgeable about specific platforms and their differences). Two of these (i.e. Robert, Ali) still revealed insights on governance issues around platforms and their algorithms, resulting in data of 14 out of 16 interviews incorporated.

16.6.2 Results Study 5.2

When presented with the vignette, many of the stakeholders we interviewed described experiences of differences in how platforms affected 'riding style', tactics, and drafting mechanisms in virtual cycling competitions. Furthermore, participants described the (lack of) governance related to platform choice. This is illustrated in appendix section , where we present quotes from the vignette data and connect them to the three governance dimensions of transparency, accountability and democracy (Geeraert, 2021). Additionally, results are refined with subdimensions from Kelly et al.'s (2022) esports framework.

Regarding transparency, stakeholders highlight that they lack insights into factors behind platforms' performance-results gap, noting that there is an incomparability between

platforms' different performance-results gaps. Some interviewees link these issues to platform algorithms (Floyd and Manny, Section 16.13.5 Appendix), which can be interpreted as a gap regarding the *transparency* dimension. More specifically, both the disclosure and clarity of algorithms appear insufficient as publishers fail to provide providing relevant and understandable information how performances are handled (Schnackenberg & Tomlinson, 2016). Given the lack of information, stakeholders also remain unable to assess the accuracy about such information (Schnackenberg & Tomlinson, 2016). Participants report differences between Zwift and MyWhoosh based on their own experiences rather than based on available information (Section 16.13.5 Appendix), suggesting an information asymmetry skewed towards publishers of how physical performance translate into virtual results. However, some interviewees, like Kim (Section 16.13.5 Appendix) dispute the magnitude of platform differences, which further indicates a lack of disclosure and clarity on the magnitude of the performance-results gap.

Accountability issues were primarily identified through a lack of procedural protection, independent oversight, and clear responsibilities. We observe a mismatch between the expectations of different stakeholders. Athletes expect clear rules and guidelines regarding unprecedented platform switches (e.g., timely notice, transition support) safeguarding their procedural rights. Having read the vignette, Mike explains the lack of expectation management while Ali and Manny illustrate how commercial decisions may undermine athletes' rights without subsequent liability of both publishers and governing bodies. Furthermore, there are gaps in discipline and independence. While interviewees like Tony, Robert, Claressa and Mike (Section 16.13.5 Appendix) do not perceive the performance-results gap as unfair, they do however indicate unexpected algorithm updates or platform switches as problematic threats to governance, especially liability and sanctioning lacks. As Tony and Ali note (Section 16.13.5 Appendix), power is skewed towards those with algorithmic access, making it difficult for independent bodies to verify virtual performances and enforce standardisation.

Finally, gaps in the democracy dimension are evidenced by significant power imbalances and a lack of social responsibility. Platform differences are experienced to impact the fairness of competitions, as the algorithms induce different drafting dynamics. Nonetheless, many stakeholders appear to tolerate such differences, describing them as intrinsic to 'playing the game'. The acceptance of such differences highlights a democratic deficiency, because stakeholders feel they must unilaterally adapt to commercial decisions rather than participating in fair decision-making and deliberation. Additionally, our data suggests that stakeholders currently disregard some form of social responsibility (Section 16.13.5 Appendix). Mike

mentions the negligence of hardware developers in delivering highly accurate smart trainers due to its trade-off with accessibility and cost. Meanwhile, Claressa reveals how platform switches are driven by commercial reasons rather than athlete wellbeing, illustrating a lack of balanced stakeholder representation.

16.6.3 Conclusion of Study 5.2

Study 2 complements study 1 by illustrating that stakeholders recognise platform-dependent differences in how physical performance translates into results, yet tend to tolerate these differences as imperative or as features of virtual cycling. Section 16.13.5 in the Appendix maps the dynamics behind these algorithmic differences against three universal governance dimensions and several specific subdimensions (Geeraert, 2021; Kelly et al., 2022), revealing how governance gaps pertain to virtual cycling. First, transparency gaps exist due to the non-disclosure of the input-output relationship and a lack of clarity regarding algorithmic differences. Second, accountability is challenged by limited procedural rights safeguarding athletes, no defined consequences for non-compliance, a lack of version-control oversight, and the non-scrutability of platform algorithms by independent bodies. Third, democracy is hindered by power imbalances that undermine perceived fairness and a decoupling of commercial actions from athletes' wellbeing.

16.7 Discussion and Meta-Inferences

This study demonstrates how algorithmic differences between virtual cycling platforms shape competitive outcomes and governance practices. While prior research by Westmattelmann et al. (2022, 2025) quantified the performance-result gap within the Zwift platform, our study 5.1 extend these findings by evidencing that its magnitude varies across different platforms. This suggests that algorithmic design is not a neutral mediator but an active determinant of competitive dynamics, aligning with sport digitalisation literature that highlights how technology increasingly influences the ontology of sport (Cranmer et al., 2021; Fouquaert et al., 2026; Goebeler et al., 2021; Parry & Giesbrecht, 2023). The complementary insights of study 5.2 illustrate how stakeholders acknowledge platform-dependent differences yet largely tolerate them, framing them as intrinsic imperative features of virtual cycling rather than governance risks. Such tolerance may be understood as the “phase lag” phenomenon in sport governance (Marchant, 2011; Synrem, 2019). Fouquaert et al. (2026) argue that technological complexities and commercial interest lie at the root cause of this phenomenon. A complementary explanation for this tolerance can be drawn from Anthony's (2021)

ethnographic study of analytical technologies in “expert organisations”, as we may understand federations similarly in terms of expertise level (Willem et al., 2019). Within expert organisations, knowledge workers routinely 'black box' the algorithmic tools they rely on, trusting outputs without engaging with the underlying calculations. This consequently constrains their own expertise and capacity for oversight. Hence, the tolerance observed among federation representatives may reflect a similar dynamic, in which the technical complexity of platform algorithms is implicitly delegated to publishers as a matter of working routine rather than actively engaged with or scrutinised. Such approaches may however reinforce rather than mitigate the governance gaps identified in this study.

Building on the works of Geeraert (2021), Kelly et al. (2022), and Schnackenberg and Tomlinson (2016), we observe how governance gaps in transparency, accountability, and democracy remain unaddressed. This supports the idea that sport governing bodies perceive no such gaps or lag, as also shown by Fouquaert et al. (2025, 2026). Nonetheless, proprietary undisclosed algorithms in virtual cycling impedes trust development between stakeholders as it creates deficits across all three dimensions of organisational transparency (Schnackenberg & Tomlinson, 2016). Disclosure is compromised because publishers do not intentionally share information about how performances (W/kg) are transformed into virtual performances, leaving stakeholders reliant on unverifiable assumptions. Clarity lacks because any provided information is often complex and lacking specific technical interpretability. Finally, accuracy is compromised as there is no real way to independently verify the factual correctness of utilised hardware (e.g. ergometers) and software (e.g. algorithms). These deficits collectively exacerbate the transparency gap introduced by Westmattmann et al. (2025), challenging trust in governance processes and reinforcing the need for semi-black-boxes, i.e. scrutability-without-disclosure models that balance sport governing bodies' interest, athletes' interests and commercial platform's interest.

Following Venkatesh et al.'s (2013, 2016) recommendation to integrate quantitative and qualitative insights through meta-inferences, Table 13 highlights the governance-critical nature of algorithmic variability. In order to mitigate athletes' vulnerability to abrupt platform switches and asymmetric power relations, sport governing bodies must navigate a dual governance reality: one of esports publishers combined with sports stakeholders. Hutchinson et al. (2025) similarly characterise the commercial dependency in esports as a 'wild west' in which players and other key stakeholders feel incapable in partaking in decision-making. This imbalance reinforces the need for integrating publishers into formal decision-making structures to safeguard athletes' procedural rights and ensure democratic representation (Geeraert, 2021;

Warren, 2014). In line with this, the study of Hutchinson et al. (2025) also proposes an integrated multistakeholder governance model as a rectification to this imbalance.

Last, these findings contribute to debates about fairness within the second wave of sports digitalisation, where technology increasingly mediates sport performances. While interviewees often drew an analogy between platform variability and environmental variability in road cycling, such analogies remain however ignorant of the systemic commercial nature of algorithmic differences. Unlike the external environment (e.g. route, terrain), algorithms are developed and maintained within publishers' commercial and strategic interest. This raises normative questions about legitimacy and competitive integrity within sport digitalisation that sport management literature has yet to fully address (Fouquaert et al., 2026; Geeraert, 2019; Thompson et al., 2023).

Table 13: Meta-inferences based on quantitative and qualitative inferences (study 5)

Research question	How do differences between virtual sports platforms shape virtual sports competitions and their governance?
Quantitative inferences	Algorithmic differences between virtual sports platforms exist, in this case between Zwift and MyWhoosh, and effectively shape the outcomes of virtual sports competitions.
Qualitative inferences	While sport governing bodies are aware of algorithmic differences, they are unaware of their potential integrity issues in terms of transparency (disclosure, clarity, and accuracy), accountability(competition integrity, commercialisation) and democracy (asymmetric power relations, participation, and fairness). This allows governance gaps to exist.
Meta-inferences	Algorithmic differences: Platform-specific algorithms significantly influence competition outcomes, though not perceived to be of significant influence. Impact ignorance: Algorithms' impact is underestimated, impeding the development of governance mechanisms to address fairness and transparency. Need for standards: Establishing standards for virtual physics engines and cross-platform calendars could allow differences to exist yet maintain a fair and transparent competition while preserving platform's identity and commercial interest. Safeguarding athletes: Multi-platform access and clear transition protocols should be provided by federations to prevent financial and competitive disadvantages for athletes when UCI-sanctioned virtual racing would switch platforms or take place on multiple platforms.

16.8 Theoretical Contributions

16.8.1 Extending Sport Digitalisation: From Mediation to Algorithmic Ontology

Our study extends sport digitalisation theory (e.g. Haffner et al., 2025; Parry & Giesbrecht,

2023) by moving beyond technology as a mediator to the role of underlying platform algorithms in shaping competitions. Building on the ‘performance-result gap’ (Westmattelmann et al., 2022, 2025), we provide the first comparative analysis between platforms, revealing significant differences in how performances translate into virtual results. Our analysis offers a novel empirical lens for examining algorithmic environments and allows to argue for an *algorithmic ontology of virtual sports*, referring to how fundamental rules and physics are not fixed but variably defined by proprietary algorithms. As our qualitative data shows, algorithms shape different riding styles, transforming the "playing field" to the algorithmic engine itself. This reframes technology not to be just a mediator but a constitutive foundation of sports’ reality that is moreover commercially underpinned (Fouquaert et al., 2025).

16.8.2 Advancing Sport Governance: A Dual Governance Structure

We advance sport governance theory by challenging the traditional “phase lag” view for technology regulation (Marchant, 2011; Synrem, 2019). Rather than a mere delayed response, sport governing bodies develop tolerance, accepting algorithmic differences as imperative features to the “game”. Consequently, our first contribution is introducing the concept of a dual governance structure where federations govern institutional aspects while the commercial platform publishers govern fundamental physics via undisclosed algorithms. Understanding how such a dual governance structure complicates fundamental and universal governance dimensions of transparency, accountability, and democracy (Geeraert, 2021; Kelly et al., 2022) provides a basis for future theoretical frameworks capturing the governance of technologically-mediated sports.

This dual governance structure links to the dual layer of power observed in our data. While sport governing bodies hold institutional authority, platform publishers hold power over the digital environment. This power asymmetry leaves sport governing bodies and their members (federations, athletes) vulnerable to unilateral technical changes that may for instance occur instantly and unannounced. Such dynamics resonate with Westerbeek and van Schaik's (2025) platform-capitalism analysis of sport governance, which shows that even if digital platforms appear to empower athletes, they remain structurally constrained by corporate-controlled infrastructures (e.g. rules programmed within black-box algorithms) over which they have little insight or control. Hong's (2023) multi-stakeholder analysis of esports reaches a complementary conclusion from the welfare side, demonstrating that responsibility for athlete health and wellbeing must be distributed across the full ecosystem: players, publishers, federations, sponsors and tournament organisers, rather than resting on a single or few actors.

Translated to algorithms in virtual cycling platforms, the integrity consequences of publishers' decisions and federations' regulatory decisions cannot be neatly separated and should be evaluated against their joint impact on athletes, further reinforcing the case for shared rather than siloed governance. Because control over competition mechanics is skewed towards publishers, establishing reciprocal, collaborative, and ethically grounded relationships (Fassin, 2012) becomes a prerequisite for legitimate governance. Future research could for instance explore how recognising publishers as 'stakeowners' (Ferkins & Shilbury, 2015) might facilitate this dual structure.

16.8.3 Deepening Organisational Transparency: Algorithmic Opacity as New Frontier

Our research extends organisational transparency theory (Schnackenberg & Tomlinson, 2016) to the challenge of algorithmic opacity. As earlier discussed, Westmattelmann et al.'s (2025) 'transparency gap' undermines Schnackenberg and Tomlinson's (2016) principles of disclosure, clarity, and accuracy. While a degree of uncertainty is inherent to and even exciting in sport, undisclosed algorithms create black boxes (challenging disclosure), are complex (challenging clarity) and limit the ability to ensure factual correctness (challenging accuracy). Mazurova et al. (2022) have also discussed the paradoxical tensions around 'black-boxed' technologies, which informs the need for algorithmic opacity in virtual cycling. In their study, algorithmic systems like AI-powered are shown to be black-boxes requiring better explainability and interpretability. While such systems are actually designed to increase transparency and objectivity, yet they might paradoxically achieve the opposite as literature suggests that transparent processes yield positive satisfaction and acceptance of competition outcomes (Mazurova et al., 2022). Similarly, virtual cycling may benefit from increased algorithmic opacity to avoid perceptions around unfair competition, algorithmic biases and a lack of interpretability.

Black-boxed systems consequently affect the entire transformation chain from hardware measurement to virtual results. This analysis shifts the theoretical paradigm from mere information sharing to scrutability, i.e. the ability to examine and verify algorithmic logic. Scrutability can be argued critical for managing algorithms in organizations (Berente et al., 2021) and managing trust in organisation-stakeholder relationships (Schnackenberg & Tomlinson, 2016) when a core mechanism is intentionally opaque, offering future research directions on algorithmic transparency in digitised sports.

16.9 *Practical Implications*

Sport governing bodies need to become aware of and proactively manage governance risks in virtual sports. Our findings show how algorithmic differences are tolerated without considering their actual impact. Consequently, sport governing bodies must first acknowledge that platform choice influences the competition as we demonstrated that switching from Zwift to MyWhoosh alters which athletic attributes are rewarded. Before sanctioning events, sport governing bodies should therefore mandate or conduct analyses to understand platforms' performance-result gap differences, allowing adequate management of potential issues and ensuring alignment with competition goals.

To support this alignment, sport governing bodies are urged to develop minimum standards certification framework for core virtual physics (e.g. drafting) featuring scrutability-without-disclosure. Such framework should maintain publishers' innovative and commercial freedom while demonstrating their algorithms' compliance without revealing proprietary code.

Implementing these frameworks effectively requires formal partnerships with platform publishers. Within such partnerships, it is important to acknowledge the commercial nature of publishers. They often prioritise their main revenue drivers such as gamification, user engagement, and monetisation over strict competitive fairness. However, they risk legitimacy if fairness is questioned. This could in turn undermine their own commercial interests. Publishers should therefore proactively collaborate with sport governing bodies to develop the aforementioned minimum standards, as increasing transparency and democracy will build trust, enhance integrity, and secure long-term commercial stability. For example, information can be shared about publishers' physics engine. In addition, these dual governance agreements should also touch upon the accountability and define mutual responsibilities to safeguarding athletes' procedural rights, including a communication protocol for algorithm changes, a minimum notice period for major platform transitions, and a platform transition protocol. To address the strategic risks of transparency (e.g. competitors copying shared insights on physics engines), our proposed "scrutability-without-disclosure" model is highly valuable in avoiding such risks.

Protecting these procedural rights is critical because our research shows that athletes are most vulnerable under current governance. Differences in performance–result gaps directly impact athletes' careers and training strategies. Athlete representative bodies should use our empirical evidence to lobby for abovementioned procedural rights that should at least include required consultation and minimum notice before platform changes in major races, safeguarding them from being passive recipients of commercial decisions.

16.10 Limitations and Future Research

Several limitations of this study should be acknowledged, opening avenues for future research. First, our findings are specific to two virtual cycling platforms, Zwift and MyWhoosh. The performance–result gap and related governance challenges may differ in other virtual sport (e.g., rowing, running) having different physiological inputs and algorithmic models. Future research could conduct similar comparative analyses in other virtual disciplines to build a more comprehensive and generalisable understanding.

Second, our quantitative analysis relied on scraped data and used average W/kg as the primary performance proxy, omitting other factors like tactical behaviours that may explain unexplained variance. Future studies using more granular proprietary data or athlete survey data could enable more robust models.

Third, this study is cross-sectional in nature, capturing one moment in a rapidly evolving technological and commercial landscape. As algorithms and governance structures continuously change, longitudinal research is needed to track these dynamics over time (e.g. effect of software updates, evolution of governance practices).

Finally, qualitative insights are limited by stakeholder diversity. While interviews included federations, publishers, and athletes, perspectives from hardware developers and independent integrity bodies were absent. Future work could broaden stakeholder inclusion and employ triangulation with observational data.

16.11 Conclusion

This study provides initial empirical evidence that algorithmic differences between virtual cycling platforms significantly shape competitive outcomes. Our findings illustrate that the way platforms convert physical inputs into virtual results is not uniform. By combining quantitative and qualitative insights, we demonstrate that Westmattelmann et al.’s (2022, 2025) performance-result gap is not only present within platforms but varies across them, challenging assumptions of interchangeable virtual sport environments. While governing bodies acknowledge these differences, they tend to see them as imperative features of virtual cycling rather than addressing their implications for fairness, transparency, and athlete safeguarding. Tolerating such differences reflects significant governance gaps, as several issues pertain to the fundamental governance dimensions of transparency, accountability, and democracy in both sports (Geeraert, 2021) and esports (Kelly et al., 2022).

Our findings underscore that proprietary algorithms create structural deficits in transparency (Schnackenberg & Tomlinson, 2016), limiting disclosure, clarity, and accuracy. Addressing these deficits requires proactive measures. These include minimum standards for virtual physics engines and developing scrutability-without-disclosure models. Furthermore, sport governing bodies must actively navigate the new dual governance reality. Because a dual layer of power exists, federations must establish reciprocal and formalised partnerships with publishers. This is necessary to manage the publishers' control over competition mechanics (e.g. algorithms) and to safeguard the procedural rights of athletes. Without appropriate governance interventions, virtual cycling risks undermining its legitimacy as it transitions from a niche innovation to an institutionalised sport.

16.12 References

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16.13 Appendix Study 5

16.13.1 Scraped MyWhoosh races (appendix study 5)

Table 14: Scraped MyWhoosh races (appendix study 5). Reported R^2 (linear, quadratic, cubic) is computed from participants' relative power output (W/kg) and riding time (milliseconds)

Race	Gender	distance km	elevation m	participants	mean W/kg winner	mean W/kg	max W/kg	R^2 linear	R^2 quadratic	R^2 cubic
1	men	45	914	136	5.0	3.7	5.1	0.841	0.966	0.974
2	men	70	465	111	5.1	3.5	5.2	0.741	0.741	0.750
3	men	70	815	102	2.7	3.6	5.1	0.041	0.150	0.210
4	men	70	735	99	5.2	3.5	5.2	0.788	0.809	0.849
5	men	70	1320	92	5.5	3.7	5.5	0.870	0.889	0.908
6	men	70	1735	90	4.9	3.7	5.0	0.822	0.825	0.847
7	men	40	185	94	6.1	4.2	6.1	0.921	0.950	0.950
8	men	70	495	88	5.1	3.7	6.2	0.453	0.697	0.828
9	men	45	704	126	4.9	3.7	5.4	0.830	0.930	0.938
10	men	45	666	119	5.1	3.4	5.1	0.878	0.961	0.962
11	men	44.4	550	81	5.6	3.6	5.8	0.809	0.860	0.862
12	men	45	1017	55	4.9	3.4	5.5	0.836	0.881	0.882
13	men	44.8	835	68	5.5	3.6	5.6	0.846	0.921	0.932
14	men	45	1041	142	5.1	3.6	5.3	0.847	0.957	0.966
15	men	45	605	82	5.7	3.5	5.7	0.792	0.831	0.831
16	men	45	900	77	5.1	3.0	5.6	0.686	0.758	0.773
17	men	45	740	56	5.1	3.1	5.9	0.692	0.792	0.795
18	men	45	492	112	5.3	3.6	5.3	0.779	0.903	0.909
19	men	45	1011	57	5.5	3.6	5.5	0.856	0.961	0.978
20	men	45	997	124	5.0	3.6	5.1	0.866	0.957	0.964
21	men	45	1305	67	5.0	3.2	5.6	0.750	0.805	0.814
22	men	45	506	121	4.6	3.4	5.3	0.638	0.638	0.799
23	men	45	814	68	5.1	3.5	5.6	0.851	0.928	0.934
24	men	45	318	36	5.6	3.5	5.9	0.850	0.921	0.926
25	men	45	333	66	5.0	2.9	5.2	0.794	0.860	0.885
26	women	35	704	48	3.6	3.0	4.3	0.714	0.722	0.725
27	women	35	735	58	5.1	3.5	5.1	0.900	0.967	0.967
28	women	35	502	76	3.5	3.1	3.9	0.918	0.933	0.937
29	women	50	335	63	3.6	3.0	4.5	0.672	0.856	0.879
30	women	50	540	61	3.7	3.0	4.1	0.859	0.923	0.923
31	women	50	515	56	3.7	2.8	3.8	0.827	0.871	0.881
32	women	50	900	55	3.7	3.0	3.8	0.837	0.940	0.946
33	women	50	1100	56	4.1	3.0	4.2	0.866	0.901	0.911
34	women	30	145	53	4.5	3.2	4.6	0.878	0.935	0.935
35	women	50	340	52	3.5	2.9	4.0	0.734	0.798	0.802
36	women	35	905	59	3.8	2.8	3.9	0.892	0.935	0.935
37	women	35.8	595	55	4.2	3.3	4.8	0.860	0.917	0.918
38	women	35	274	41	3.3	2.8	3.9	0.770	0.811	0.816
39	women	35	331	47	3.8	3.1	4.1	0.831	0.877	0.878
40	women	35	593	70	4.0	2.8	4.1	0.680	0.922	0.969
41	women	35	633	68	3.8	2.9	3.9	0.862	0.930	0.930
42	women	35	600	61	3.6	2.8	3.9	0.773	0.781	0.781
43	women	35	720	50	3.5	2.9	4.2	0.785	0.801	0.803
44	women	35	645	30	3.5	2.8	4.3	0.677	0.711	0.712
45	women	35	537	54	3.7	2.8	3.8	0.776	0.785	0.868
46	women	35	324	59	3.9	2.8	3.9	0.835	0.927	0.930
47	women	35	935	43	3.5	2.5	3.6	0.743	0.798	0.806
48	women	35.5	440	52	3.9	3.3	4.4	0.858	0.903	0.906
49	women	35	425	75	3.5	2.9	3.8	0.847	0.847	0.915
50	women	35	535	46	3.8	3.1	4.4	0.821	0.914	0.914

16.13.2 Scraped Zwift races (appendix study 5)

Table 15: Scraped Zwift races (appendix study 5). Reported R^2 linear, R^2 quadratic and R^2 cubic similar to Section 16.13.2.

Race	Gender	distance	elevation	participa	mean	mean	max	R ²		
		km	m	nts	W/kg			R ² linear	quadratic	R ² cubic
51	men	46.8	309	308	4.7	3.9	5.2	0.765	0.809	0.809
52	men	35.5	378	278	4.9	4.1	5.7	0.709	0.795	0.795
53	men	54.6	380	237	4.4	4.0	5.2	0.746	0.789	0.789
54	men	39.3	898	180	5.1	4.0	5.4	0.842	0.903	0.903
55	men	37.4	440	172	4.9	4.3	5.5	0.728	0.794	0.797
56	men	27.8	489	64	4.4	4.2	5.1	0.668	0.793	0.796
57	men	30	254	118	4.1	3.7	4.7	0.814	0.876	0.878
58	men	54.9	939	62	4.4	3.8	4.9	0.656	0.766	0.775
59	men	42.6	338	247	4.6	4.3	5.3	0.710	0.748	0.752
60	men	45.3	385	237	4.6	4.3	5.3	0.656	0.719	0.719
61	men	28.3	1161	270	5.8	4.4	5.9	0.907	0.957	0.962
62	men	52.9	165	205	4.7	4.3	5.2	0.691	0.768	0.775
63	men	41.5	732	179	4.6	4.4	5.4	0.783	0.844	0.845
64	men	50.035	483	57	4.7	4.6	5.5	0.529	0.728	0.729
65	men	54.8	941	78	4.5	4.3	5.1	0.213	0.327	0.379
66	men	35.1	313	41	4.8	4.3	5.1	0.925	0.954	0.961
67	men	12.8	119	57	5.3	4.7	5.4	0.690	0.964	0.964
68	men	15.1	118	25	5.0	4.6	5.4	0.691	0.926	0.938
69	men	13	480	12	5.6	5.0	5.6	0.870	0.947	0.968
70	men	29.5	378	13	4.8	4.4	5.2	0.818	0.961	0.961
71	men	24.3	165	68	5.0	4.8	5.7	0.533	0.778	0.778
72	men	38.4	192	63	4.8	4.6	5.3	0.102	0.289	0.486
73	men	39.1	670	70	4.2	4.5	5.1	0.235	0.387	0.410
74	men	24.8	132	72	5.4	5.2	6.4	0.041	0.086	0.089
75	men	33.1	318	73	4.3	4.4	5.1	0.618	0.720	0.739
76	men	27.7	333	71	5.2	4.7	5.3	0.811	0.931	0.934
77	women	46.8	309	60	3.9	3.3	4.5	0.848	0.937	0.937
78	women	35.5	378	65	3.7	3.6	4.7	0.791	0.854	0.854
79	women	54.6	380	53	3.6	3.5	4.4	0.792	0.868	0.869
80	women	39.3	898	53	4.7	3.5	4.7	0.893	0.955	0.956
81	women	37.4	440	47	4.6	3.6	4.6	0.853	0.905	0.920
82	women	27.8	489	10	4.7	4.1	4.7	0.580	0.627	0.722
83	women	30	254	14	4.3	3.8	4.3	0.766	0.792	0.801
84	women	54.9	939	10	4.3	3.8	4.3	0.757	0.758	0.769
85	women	42.6	338	56	5.1	3.6	5.1	0.864	0.890	0.898
86	women	45.3	385	60	3.9	3.5	4.5	0.794	0.813	0.817
87	women	28.3	1161	70	4.7	3.6	4.7	0.926	0.970	0.973
88	women	52.9	165	62	4.0	3.6	4.7	0.690	0.759	0.759
89	women	41.6	732	55	4.9	3.8	5.0	0.812	0.880	0.882
90	women	50.035	483	47	4.4	3.9	4.5	0.509	0.798	0.819
91	women	54.8	941	72	3.7	3.8	4.6	0.664	0.780	0.780
92	women	35.1	313	31	4.0	3.6	4.1	0.613	0.770	0.772
93	women	13.9	245	51	3.9	4.0	4.6	0.097	0.322	0.526
94	women	14.1	112	21	3.6	3.7	4.4	0.460	0.664	0.696
95	women	13	480	12	4.8	4.1	4.8	0.811	0.912	0.923
96	women	29.5	378	16	4.7	3.9	4.7	0.808	0.875	0.875
97	women	24.3	165	61	4.1	4.1	5.0	0.457	0.544	0.570
98	women	38.4	192	56	3.9	4.0	4.8	0.309	0.517	0.536
99	women	29.1	670	62	4.1	3.9	4.6	0.180	0.184	0.190
100	women	24.8	132	62	5.2	4.4	5.3	0.183	0.183	0.194
101	women	33.1	318	62	4.2	4.0	4.7	0.543	0.570	0.571
102	women	27.7	333	58	5.1	4.1	5.1	0.837	0.874	0.879

16.13.3 Overview of interviewees (appendix study 5)

Table 16: description of interview participants (appendix study 5).

Pseudonym	Gender	Expertise / Role	Follow-up interview	Vignette
Stan	Male	Innovation manager, national sport governing body	-	-
Robert	Male	Management, international sport governing body	Yes	-
Kim	Male	Journalist & content creator	-	Yes
Floyd	Male	Management, international sport governing body	Yes	Yes
Claressa	Female	Elite virtual cycling athlete	-	Yes
Manny	Male	Elite virtual cycling athlete	-	Yes
Mike	Male	Tech developer/virtual cycling platform	-	Yes
Elvene	Female	Elite virtual cycling athlete	-	Yes
Ben	Male	Management, national & international sport governing body	-	-
Ali	Male	Management, international sport governing body	-	-
Rika	Female	Athlete, content creator & journalist	Yes	Yes
Tony	Male	Management, international sport governing body	-	Yes
Lucas	Male	Department manager, IOC	-	Yes

16.13.4 Vignette used in interviews (appendix study 5)

Mark, an 18-year-old cyclist with professional aspirations, excels in the one-day races or classics in Belgium. However, racing abroad is costly for Mark, with his team covering most, but not all, expenses. Seeking an alternative to showcase his talent, Mark turns to UCI eSports races, held on Zwift, a virtual cycling platform he's used in his training for years. Virtual cycling races allow Mark to compete in more events and demonstrate his talent and strengths.

Though virtual cycling differs from outdoor racing, Mark understands Zwift's dynamics. He knows how the virtual drafting effects work, how the virtual course's topography affects the racing strategy, and how the number of participants affects the racing strategy. These factors, much like outdoor races, influence strategy which means that virtual cycling is not as easy as 'the rider with the highest average W/kg wins the race'. While the Zwift algorithm isn't transparent, Mark's experience on the platform gives him a competitive edge because he knows 'the rules of the game'.

Mark qualifies for the UCI Virtual Cycling (eSports) World Championships but faces a new challenge when the UCI suddenly switches from Zwift to MyWhoosh, a platform he's unfamiliar with. Worried about the differences, he researches the new platform's dynamics. Scientific studies reveal that on Zwift, average W/kg explains 75.8% of the competition results, whereas on MyWhoosh, it explains 86.9%. This indicates that, physical performance is significantly more important on MyWhoosh, while factors like tactical behaviour and platform-specific knowledge are less relevant on MyWhoosh. This leaves Mark concerned that his chances of securing a professional contract are impacted by the algorithms of the different virtual cycling platforms, and how this impacts fairness of (eSports) Virtual Cycling races.

16.13.5 Quotes from study 5.2 supporting governance issues (appendix study 5)

Table 17: Quotes illustrating platform-results gap governance issue (appendix study 5). Governance issues are mapped to the governance dimensions of Geeraert (2021) and subdimensions of complementary literature (Kelly et al., 2022; Schnackenberg & Tomlinson, 2016).

Principle	Governance issue in vignette data	Quotes
Governance Dimension 1: Transparency		
Disclosure, Clarity	Lacking insights between and in factors behind platform's performance-results gap.	There's different drafting dynamics for each of the platforms. They have a different algorithm, okay? (Floyd) MyWhoosh is trying to make their platform a lot more realistic [...], definitely 100% or way more geared towards physical performance than actual tactics [compared to Zwift]. (Claressa) Different platforms have different algorithms and different biases towards styles of riding [...]. MyWhoosh is known more as a pure watts than watts per kilo [...], where Zwift is more, the draft is slightly bigger and more efficient [...]. (Manny) I tried that same strategy [I used on Zwift] on MyWhoosh and couldn't do it because the physics were different. (Elvene) If there was such a huge discrepancy, then you would have two totally different pools of athletes that were winning on both platforms. And that's not happening. (Kim) In the case of MyWhoosh, I also felt it myself that it's more realistic in comparison to what I do on the streets. (Lucas)
Governance Dimension 2: Accountability		
Liability	Lacking protection of athletes' rights.	I think the problem in that scenario is about expectation management and it's about treating athletes with respect and not suddenly changing the rules with no notice (Mike) Athletes have little choice [in] a commercially driven discussion. (Ali) So it's a case that he [Mark in vignette] needs to accept that there are different platforms and he needs to learn different platforms. (Manny)
Responsibility	Algorithm standardisation and decision-making power.	We won't prescribe to software platforms, how their drafting algorithm should work. We'll allow them to continue to do what they feel is in their best commercial interests, but we'll have a standardization in terms of how data is recorded and reported, how that watt per kilogram algorithm is managed and things like that. (Robert) The ability for an organizer or the IOC or UCI to say, hey, you [a publisher] have to change your code based and your algorithms that you work on because that's the standard and we want to have the same thing as Zwift. They [publishers] might just say, well, I'm out now. (Ali)
Independence	Independent bodies need to have insight	It's the credibility... [...] we have to make sure that the software is able to be checked and is compliant in a way which

	in platform algorithms.	then creates an equal environment for everybody. And at the moment, there aren't enough checks to be able to 100% say that that's the case. (Tony)
Discipline	Lack of consistency in sanctioning integrity breaches	One of the things that makes esports so engaging is the ability to manipulate the environment [...]. It creates competitive advantages for different types of competitors, different types of athletes in our case. And it keeps the engagement a little bit more fresh or a little bit higher. (Robert) MyWhoosh is changing every update. [...] it felt like every two weeks there was, like, a new update, a new change. It was really stressful [...], we were just all trying to figure it out. (Claressa) I hesitate to say it's a fault of the UCI, but it's a fault of the process because you just wouldn't do that in the road scenario. (Mike)
Governance Dimension 3: Democracy		
Fairness	Power imbalances amongst stakeholders	All of a sudden, you know, you learn that this is the platform we're using. So, everyone's rushing over to the platform trying to learn it. (Claressa) It's not that it's unfair [the algorithmic differences]. Or at least it's no more unfair than road cycling is sometimes flat and it's sometimes hilly. (Mike) You go to the Tour de France one year, there's mountains. You go to the Tour de France the next year, not so many mountains [...]. So, in this sense, you have to be able to adapt. And 99% of the time, the best rider would still win. And I don't think that then the platform is essentially controlling the ultimate result. (Tony) But at the end, it is what it is [the differences between Zwift and MyWhoosh]. So, I think that virtual sports are another category of cycling. So, you have to be used to the system in order for you to achieve your best. (Lucas)
Social Responsibility	Actions needed to protect players and create inclusive and accessible virtual environments.	I guarantee you everyone would still be on Zwift if UCI didn't sign that contract with MyWhoosh. (Claressa) most hardware has been designed just for [leisure] training. (Mike)

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Disclosure statement

We have no conflicts of interests to disclose.

Ethical statement

Ethical approval was obtained from Nord University, SIKT – The Norwegian Agency for Shared Services in Education and Research. All participants provided active consent and were informed of their right to withdraw from the study at any time

Data Availability Statement

The scraped raw data is available in appendix Section 16.13.1 and 16.13.2. For ethical reasons, only pseudonymised qualitative data of this study are available from the corresponding author, upon reasonable request.

Part V

Discussion of the Findings

17 Introduction to the Findings and General Discussion

Part V presents the overall discussion of the empirical results derived from the five studies comprising this doctoral thesis. While virtual cycling serves as the primary empirical context for these studies, the following discussion elevates these findings to contextualise and understand the broader phenomenon of virtual sports. In addition, we seek to synthesise our focused insights into a more interconnected theoretical understanding of managing virtual sports, diffusion, and governance.

To achieve this, Part V is structured as follows. Section 17.1 provides a summary of the main findings mapped against our RQs (see Table 2). Section 17.2 introduces the structure guiding this discussion, identifying three intersecting domains: (1) digitisation, (2) institutional dynamics, and (3) innovation strategies. In the sections that follow, micro, meso, macro, and combined levels of analysis are discussed. This results in twelve different perspectives of analysis (see Table 19). Within each perspective and (sub)section, we reflect upon our results, the theories used, how these are relevant or fall short, and how future research is recommended to improve upon them.

Section 18 explores the impact of sport digitisation (i.e. technological features) in our understanding of managing virtual sports. Section 19 examines the (re)institutionalisation of virtual sports and the impact of its competing logics. Section 20 analyses strategic innovation diffusion and explains this as a balance between technological and institutional forces. Next, section 21 outlines the practical implications and recommendations for key stakeholders while section 22 summarises and deepens our future research recommendations that have been discussed throughout previous sections. Lastly, section 23 highlights the limitations and strengths of this dissertation.

17.1 Summary and Mapping of Empirical Findings and Research Questions

Before delving into the integrated thematic discussion, we consolidate the outcomes of the empirical work. Table 18 provides a summary of the main empirical findings derived in each study. These findings are mapped directly to their corresponding RQs (see Table 2) to offer a clear overview of how each study contributes to the overarching aims of this thesis. Alongside the studies presented in sections 11 to 16, this summary serves as a point of reference for the multi-domain discussion presented in sections 18 to 20.

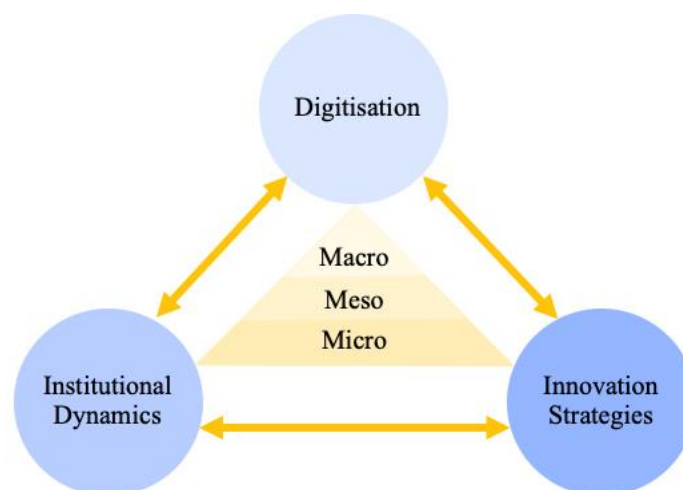
Table 18: Summary of empirical results mapped to RQs (in full in section 4).

Data	Study #	Main findings per RQ (* $p < 0.05$, ** $p < .01$, *** $p < .001$)
 Data collection #1 	Study 1	RQ 2: Clubs cluster in Innovators & Early Adopters (17%), Early Majority (44%), and Late Majority & Laggards (39%) based on attitude towards, evaluation of and adoption-intention of virtual cycling as-a-club-service. More innovative clubs have significant positively higher innovation attribute perceptions*** ($\eta^2 = 0.56$), managerial*** ($\eta^2 = 0.50$), and human-related capacity** ($\eta^2 = 0.16$). Perceived innovation attributes and managerial determinants together predict clusters by 76%. The above configuration indicates that virtual cycling adoption is people-driven rather than infrastructural resource-driven, requiring tailored segmented strategies.
	Study 2	RQ 3: Clubs identify three goals for virtual cycling as-a-club-service: inclusive participation, attractiveness of services, and skill development. Skill development aligns with the result-oriented logic and is perceived most potentially valuable, while inclusive participation and attractiveness of services align with the sport-for-all logic and is perceived less valuable. Clubs cluster in five archetypes based on the goals they value: Performer (30%), Unconvinced (17%), Sceptic (14%), Hesitant (23%), and Holistic (16%). Given their different beliefs and dominant logic, adoption of virtual cycling is not one-size-fits-all and support processes need tailoring.
 Data collection #2	Study 3	RQ 4: Sport governing bodies employ two legitimisation strategies: (1) constructing new evaluation criteria around safety, reduced costs, inclusivity, sustainability, and diversified revenues, and (2) develop new markets through customised experiences and the targeted inclusion of youth, women, and para-athletes. Conversely, delegitimising behaviour was also observed under overreliance on commercial platforms and lacking organisational policies. Overall, an institutional strategy (normative framing) is predominantly employed while a technical strategy (developing features) is lacking, despite their equal importance for successful diffusion.
	Study 4	RQ 5: Four types of governance issues pertain virtual cycling. (1) Doping, which extends beyond traditional substances to mechanical doping via hardware and software manipulation. (2) Platform fragmentation, where commercial interests and IP rights create a fragmented market and marginalise athlete needs. (3) Lacking transparency, in terms of equal qualification pathways, Olympic development, and national federation compliance. (4) Insufficient commitment and support, from both UCI and national federations. While (e)sport governance frameworks are useful, many issues remain specific to virtual sports, like platform standardisation.
 Data set #3	Study 5	RQ 6: Virtual platform algorithms actively determine competition outcomes rather than acting as neutral mediators. Physical performance explains significantly more variance in race results on MyWhoosh ($R^2=0.87$) versus Zwift ($R^2=0.76$)***. While stakeholders recognise platform differences, they frame them as features rather than issues. Such tolerance masks governance issues, including algorithmic opacity and asymmetric power relations. Dual governance and scrutability-without-disclosure models are needed, where commercial platforms act as stakeowners who share regulatory control with federations to ensure competition integrity.

17.2 The Domains affected by Virtual Sports

Based on our literature review in Part I, our ontological pre-study in Part III, and our empirical studies in Part IV, it becomes apparent how different domains help to construct, and are in turn constructed by, virtual sports. As shown in Figure 18, these domains include (1) Digitisation, (2) Institutional Dynamics, and (3) Innovation Strategies. Throughout section 2, we have discussed the aspects of digitisation (i.e. the “technological” what) versus digitalisation in sport (i.e. the how and why). The aspects encompassing digitisation can be considered as closely intertwined with the digitalisation of sport, and consequently have a profound effect on how virtual sports take shape. The dimensions described in section 9.5 as the technical foundation of virtual sports (e.g. virtuality with its representational fidelity, movement compression, physical-digital interreality through algorithms) will, for example, affect how individuals and athletes use and cope with virtual cycling platforms (studies 4 & 5). Furthermore, these dimensions also affect how clubs consider it to be a valuable tool to complement their services (studies 1 & 2), or how institutions approach and communicate about virtual cycling (studies 2 & 3).

Figure 18: Intertwined domains and analysis levels shaping virtual sports. Source: Own conception and illustration.



Both in our section on defining virtual sports (section 9), our theoretical outline (section 3), and our empirical studies (Part IV), we have covered literature in depth that shows how digitalisation of sport affects the shaping of virtual sports, in particular through institutional dynamics and innovation diffusion strategies. For instance, institutional dynamics like competing institutional logics are connected to different beliefs about and innovations adoption strategies of virtual cycling in clubs (studies 1 & 2). Additionally, other latent institutional

norms and pressures help to shape the most effective diffusion strategies for virtual sports at the (inter)national federation level (study 3). Meanwhile, the design choices of software and hardware that open up integrity issues (i.e. digitisation) are in turn again entangled with the institutional legitimacy and the actions needed within certain innovation adoption strategies (studies 3, 4 & 5).

From both our literature review in Part I and Part III, and our empirical work in Part IV, it also becomes clear how the three different domains shaping virtual sports manifest themselves at three different levels: the micro-, meso- and macro-level. While these domains and levels occurred naturally from the aggregation of the different Parts within this dissertation, it is not an uncommon approach to structure and integrate different theories, perspectives and results. For instance, Meng-Lewis et al. (2025) applied a similar structure in what they call unified multilayered esports framework as a way to provide a holistic discussion of different esports studies. Next, we shortly provide some examples of each domain at different levels.

The digitisation of virtual cycling, for example, will have an effect on individual perceptions (studies 1 & 2), affect organisational culture, capacities, and intended use cases (studies 1 & 2), and impact broader societal phenomena. These phenomena include datafication and the quantified self (Lupton, 2016), the delegitimising integrity issues pertaining to virtual cycling as a new sporting discipline, and the good governance of it (studies 3, 4 & 5). Institutional dynamics in virtual sports are also manifested at three different levels. For instance, the (slow) development of governance structures in light of legitimacy development has a significant effect on competing athletes, who need to navigate digital doping, equipment calibration, or algorithmic fairness of competitions (studies 4 & 5). Additionally, individual participants in a non-competitive context are affected by governance structures such as the “main” institutional logic pursued for virtual cycling (i.e. Sport-for-All versus Result-oriented), as this effectively shapes the “main” market served by platforms and feature development for individual consumers (studies 2 & 3). At the meso-level, institutional logics will guide the behaviour of organisations such as clubs (study 2), sport governing bodies, and platform developers (studies 3 & 5). Conversely, at the macro-level, the shaping of virtual sports within two conflicting logics (studies 2 & 3) can be interpreted as different trajectories of a deinstitutionalisation-reinstitutionalisation institutional change process (Borgers et al., 2018). Lastly, innovation adoption and diffusion strategies in virtual sports are also manifested at three different levels. While studies 1 and 2 show how both individual-level determinants (i.e. micro) and organisational-level determinants (i.e. meso) affect the organisational adoption attitudes and intentions towards virtual cycling, study 3 shows how diffusion strategies of virtual cycling

at the international level (macro) can be divided into in both a technical and an institutional diffusion strategy.

Rather than discussing our research aims and RQs (see section 4) in silos, for example one aim or one RQ per section, we seek an integrated discussion of our aims and RQs. To do so, we intersect the three domains of (1) Digitisation, (2) Institutional Dynamics, and (3) Innovation Strategies with the overall-, micro-, meso- and macro-levels of analysis. This results in a 3×4 matrix that will guide our discussion and help us to understand how answers to different RQs are contextually interrelated. We outline this matrix in Table 19. The downside of this “silo-avoiding” approach, however, is that it naturally results in slight repetition of certain results, concepts or recommendations as we discuss them from a different domain or level.

Table 19: Discussion matrix: domain $\times$ level, to be used as guide throughout our overall discussion section.

<i>Level of analysis</i>	Sport Digitalisation		
	Sport Digitisation	Institutional Dynamics	Innovation Strategies
Cross-level	Section 18 Aim 1 RQ 1	Section 19 Aim 1 RQ 1	Section 20 Aim 1 RQ1
Micro	Section 18.1 Aim 3 RQs 5 & 6	Section 19.1 Aim 2 & 3 RQs 3, 5 & 6	Section 20.1 Aim 2 RQs 2 & 3
Meso	Section 18.2 Aim 1 & 3 RQs 4, 5 & 6	Section 19.2 Aim 1 & 2 RQs 3 & 4	Section 20.2 Aim 2 RQs 2 & 3
Macro	Section 18.3 Aim 1 & 3 RQs 4, 5 & 6	Section 19.3 Aim 1 RQ 1	Section 20.3 Aim 1 RQ 4

18 Good Sport Digitisation as a Prerequisite of Successful Virtual Sports Diffusion

The technological foundation of virtual sports, which is possible due to the increasing digitisation of sports (Goebeler et al., 2024; Miah, 2017; Qi et al., 2024), may be interpreted as a complex constellation of dimensions. These dimensions are digitised differently than before (e.g. physicality), and some are even entirely new digitised dimensions (i.e. reality). We address **aim 1** and **RQ 1** (i.e. understanding virtual sports, cf. section 4) through our discussion of the positioning of virtual sports within sport digitisation and its implications for sport management.

The definition of virtual sports presented in section 9.6 provides several elements of sport digitisation that are different compared to sports and esports. These include the premise that they “operate within a distinct physical-digital interreality”, that they have a “high movement decompression mimicking a traditional parent sport”, and that they are technically possible because of virtuality (see section 2.3) yet “unbound to specific configurations of remoteness, (a)synchrony or representational fidelity”. These differing elements allow us to interpret virtual sports as *distinct ontological category* between sports and esports, rather than mere simulations *in* sport (see section 9.2.2). While this new ontological category is not easily compared with sports or esports (see Figure 5), virtual sports are still often mistakenly and normatively labelled as sports or esports (Naraine, 2021; Parry, 2019; Parry & Giesbrecht, 2023). We advise future research and practitioners to avoid such “mislabelling” to create clarity around the innovation or digitalisation at hand.

Within the literature concerning the sports, esports, and virtual sports debate, scholars have frequently analysed the role of physical exertion (S. E. Jenny et al., 2016; S. E. Jenny, 2024; S. E. Jenny et al., 2024; Parry & Giesbrecht, 2023; Postma et al., 2022; Thiel & John, 2018). A novel contribution of our definition is that it transcends the “gross motor skills” versus “fine motor skills” debate by viewing the physicality dimension through movement (de)compression (cf. Figure 8). Because Hemmingsen’s (2024) concept of movement decompressions represents a non-static continuum, the relationship between the physical reality of the athlete and the digital environment becomes naturally flexible or dynamic. The amount of movement decompression within one specific (virtual) sport must also be considered as temporary and often redefined in function of advancements in sport digitisation and the industry standards of technical hardware and software designs. Consequently, the amount of movement (de)compression will be different for any virtual discipline. In this way, digitisation directly shapes virtual disciplines.

In addition, by recognising the distinct physical-digital interreality of virtual sports, we are able to reflect on its *algorithmic ontology* and what this means for virtual sports as a sporting discipline. Reflecting on our findings from study 5 (i.e. platform-results gaps), it becomes clear that each platform's black box algorithm shapes sporting performances differently through its effect on the dimensions of sporting physicality and cognitivity (e.g. behaviours, tactics). In other words, they are not neutral mediators. Section 9.5.3 discusses esports' "metagame" as a similar mediation (S. E. Jenny, 2024). Therefore, incorporating the dimension of reality appears as ultimately required to distinguish sports, esports, simulations *in* sport, and virtual sports. Without the reality dimension, it appears impossible to recognise simulations *in* sports as a specific configuration of reality (i.e. 100% transparent translation of performances, resulting in a physical reality), resulting in normatively mislabelling virtual sports as simulations *in* sports and esports (Parry & Giesbrecht, 2023). Therefore, our definition of virtual sports may redirect Parry and Giesbrecht's (2023) questionable reasoning to classify all technology-mediated sporting activities as simulations *in* sport. It opens up a new way of thinking about different applications that may appear a simulation *in* sports (e.g. Zwift), but are evidenced differently (e.g., results of study 5; Westmattelmann et al., 2025) and hold implications for sport management regarding the integrity and governance of (digitised) disciplines of sports (for details, see studies 4 and 5).

Lastly, the fact that our definition explicitly adds the notion of virtual sports being unbound to remoteness, (a)synchrony or representational fidelity, allows researchers and practitioners to approach virtual sports with a liberated view. It prevents researchers and practitioners from being restricted to seeing virtual sports as activities that must always happen in a "metaverse" (Westmattelmann et al., 2025) or on an online remote gaming platform like Zwift (Bentvelzen et al., 2022; Botwina et al., 2025; Dyer, 2020; Evsikov et al., 2024; Parry & Giesbrecht, 2023). While these are indeed most prevalent to date, we like to critically note that there is no "best" specific configuration for a virtual sport, and we encourage their development in different configurations on their remoteness, (a)synchrony or representational fidelity. A good example of a relatively new virtual sport configuration (i.e. compared to Zwift) is the indoor TGL SoFi golf competition⁷ (see also Figure 8), augmented and interactive climbing⁷, or augmented and interactive table tennis⁷.

⁷ For a detailed explanations and videos on these virtual sports examples, and an overview of other examples, please see Thibault's blog post "Virtual Sports: an overview of what exists and what is possible" on the Virtual Sports Research Network at <https://www.virtualsports.be/2025/03/24/virtual-sports-an-overview-of-what-exists-and-what-is-possible/>

Our theoretically future-proof ontology and conceptual broadening hold implications for researchers and practitioners. It prevents new literature from becoming siloed or contemporary literature from becoming obsolete. As new technologies emerge, such as location-based augmented reality or hybrid physical-digital installations (Goebeler et al., 2024), the spatial, temporal and visual boundaries of play are altered. As such, scholars must expand their scope to investigate currently less common virtual sporting phenomena (e.g. asynchronous or co-located), rather than focusing predominantly on interconnected remote ecosystems like Zwift. For practitioners, this expanded ontological view unlocks new pathways for service delivery and event design. For instance, grassroots clubs and event organisers must recognise that they are not restricted to hosting remote and synchronous multiplayer events requiring online simultaneous participation. Instead, they can innovate through co-located or asynchronous competitions. Examples include time-shifting events where athletes compete locally against ‘ghost avatars’ or leveraging low-fidelity but highly gamified platforms that reduce the barrier to entry (Tu et al., 2019). To conclude, both scholars and practitioners must become aware of the different aspects of virtuality (i.e. reality, remoteness, (a)synchrony, and representational fidelity), and how different configurations in these may exist and affect the different dimensions of virtual sports.

To summarise our answer to **aim 1** and **RQ 1** (i.e. understanding virtual sports), we may posit that virtual sports can have different and complex configuration in virtuality aspects, which translates into its nine dimensions. This makes it difficult to directly compare them to sports or esports, and even inaccurate to label it as either. It must be clear that in order to properly understand virtual sports, a specific *virtual sports ontology framework*, rather than an academic debate, is needed. Only then can proper implications for sport management be drawn. Our work in section 9.5 may well serve as the basis for such a framework and be considered as a unique and important theoretical contribution that captures virtual sports as a *distinct ontological sporting category*. No other frameworks of sufficient depth could be relied on and we consequently had to expand and combine existing frameworks and concepts from multiple fields like sports (Loy, 1968; Suits, 2007) and esports (Hemmingsen, 2024; S. E. Jenny, 2024; Westmattmann et al., 2025).

18.1 Design Choices in Sports Digitisation: Implications for Individual Use

This section addresses the micro-level implications of sport digitisation, providing insights into research **aim 3** contextualising **RQs 5 & 6**. From this perspective, our results provide insights

into how individual athletes navigate and adapt their sporting behaviour in response to technical designs, but also to the current governance structures partly shaped by digitisation. More specifically, findings from study 4 (section 15) and study 5 (section 16) provide critical insights into these dynamics. Study 4 highlights the array of vulnerabilities athletes face, including physical-, cognitive-, software-, and hardware-related manipulation, but also concerns about the lack of unified qualification pathways, inconsistent athlete support, data privacy and athlete performance verification. For instance, our interview data revealed that competitive integrity is challenged by individual behaviours, such as using erythropoietin (EPO) since there “is no way to control that from home”, or putting a trainer “in the freezer” to manipulate hardware metrics. This shows how the aspect of remoteness (cf. Figure 2) and hardware design enables individuals to engage in unfair play. Another example from study 4 involves the structural hurdles such as GDPR limiting what platforms are allowed to share in terms of athlete data, which conversely means that athletes may have their privacy compromised for external performance verification. Furthermore, study 5 provides quantitative and qualitative evidence of algorithmic differences between platforms. It demonstrates that the extent to which physical performance explains race results varies significantly between platforms, with a higher explanatory power observed on MyWhoosh ($R^2=0.87$) compared to Zwift ($R^2=0.76$) (i.e. athletes’ result in MyWhoosh races is for 11%-points more explained by their physical performance, or conversely remains 11%-points more unexplained in Zwift). Consequently, these algorithmic differences actively alter individuals’ sporting behaviour. For example, interviews with athletes in study 5 reveal how the specific physics engine of a platform dictates their required tactics and riding styles: “I tried that same strategy [I used on Zwift] on MyWhoosh and couldn't do it because the physics were different”.

These findings illustrate how the core digitisation elements discussed at the beginning of section 18 indirectly shape individual user behaviour and experiences. This effect is argued to be indirect, since the direct and mediating factor comes down to a lack of governance in virtual cycling, allowing integrity issues such as doping or algorithmic differences to proliferate. Given that virtual sports operate within an *algorithmic ontology* (cf. section 16.8.1, study 5), approaching RQ 5 and 6 from a micro-digitisation perspective teaches us that algorithms act as an active determinant of athletic behaviour and integrity issues.

Reflecting upon micro-level experiences from a digitisation standpoint also offers complementary insights into the broader strategic and institutional challenges of virtual sports. For instance, individual vulnerability to digital doping informs the macro-level institutional challenges explored under aim 1. These vulnerabilities may work as a delegitimising factor and

cannot be solved by merely pursuing an institutional strategy (i.e. developing new evaluation criteria for cycling, see section 14.4.1 study 3). Instead, they must be addressed through a technical strategy (i.e. implementing features and structures). Similarly, the struggle of individual athletes with hardware calibration and algorithmic transparency provides context for the adoption barriers observed at the organisational level (e.g. clubs, governing bodies), which relates to aim 2. The lack of effective governance on such digitisation elements may be argued to negatively impact significant adoption determinants like perceived innovation attributes (e.g. image, reliability) and managerial determinants (e.g. pro-innovation attitude) (cf. study 1; Evsikov et al., 2024; Wolfe, 1994). Both these sets of determinants have been evidenced in study 1 as significant and strong predicting factors (nearing 79% predictive power, $\eta^2 = 0.56$ and 0.50 respectively) of an organisations' willingness to adopt virtual cycling.

To summarise our reflection upon **aim 3** and contextualisation of **RQs 5 & 6** (i.e. governance issues and platform differences), we posit that micro-level behaviour and experiences are mediated by the physical-digital interreality and integrity issues which are partly rooted in the design choices of sports digitisation such as accuracy or transparency (Richardson, 2025; Richardson et al., 2022; Tweedie et al., 2023). To safeguard virtual sports' integrity, relying on existing good governance frameworks developed for traditional sports (Burger & Goslin, 2005; Thompson et al., 2023) or esports (Kelly et al., 2022) remains insufficient. These models fail to capture the algorithmic ontology or physical-digital interreality that is deeply rooted in virtual sports. Furthermore, these models are not able to capture the blurred governance responsibilities between international sport federations and commercial platform providers (cf. study 4). Consequently, developing new theoretical governance frameworks that capture such virtual sports-specific and complex elements is ultimately required. Our theoretical contribution lies in offering detailed and evidence-based insights required to develop such virtual sports-specific governance frameworks. However, since the development of such framework is beyond the scope of this dissertation, we present this as promising avenue for future research.

18.2 Hardware and Software Discrepancies as Managed by Organisations

This section addresses our research **aims 1 & 3** through contextualising **RQs 4, 5 & 6** (i.e. integrity issues, platform discrepancies, and their effect on legitimisation strategies). The focus here lies on how organisations, specifically sport governing bodies, manage discrepancies in digitisation and conflicting institutional logics within the algorithmic ontology described earlier

(cf. section 16.8.1 study 5).

At the meso level, stakeholders must navigate discrepancies not only in algorithmic design but also in software integration and standardisation. As discussed earlier in section 18.1 and study 5, differences exist in the degree by which physical performance explains racing outcomes between MyWhoosh ($R^2=0.87$) and Zwift ($R^2=0.76$). This results in a different reality and practice of cycling on each platform. Related to organisations at meso-level, study 5 reveals that this divergence is compounded by a lack of regulations on standardised hardware (i.e. smart trainers) and software (i.e. platform algorithms). This lack of standardisation regulation in turn prevents uniformity in race results (e.g. rankings), uniformity in the accuracy in physical data capture and equipment calibration, and multi-platform use within a single virtual discipline. For instance, multi-platform use would mean that UCI Cycling Esports qualifications or ranking points could be obtained across *multiple* platforms, rather than exclusively on MyWhoosh, as was the case for the UCI 2025 Cycling Esports World Championship (UCI, 2025). Stakeholders like sport governing bodies are thus forced to deal with these black-box proprietary algorithms, over which they do not possess intellectual property (IP) rights, through a complex “dual governance structure” (cf. study 5, section 16.8.2). When operating under such conditions, platform publishers are argued to operate as “stakeowners” (Ferkins & Shilbury, 2015) who possess the IP of and control the algorithmic ontology. Because governing bodies lack access to certain digitisation artifacts like source code or data logs, we argue in section 16.9 that they should instead rely on “scrutability-without-disclosure” models. This shifts the theoretical paradigm from information sharing and organisational transparency to managing trust in organisation-stakeholder relationships, collaboration, and verification (Berente et al., 2021; Schnackenberg & Tomlinson, 2016). Such new dynamics address research gap 4 outlined in Table 1, highlighting that sport management literature lacks an understanding of the unique issues and challenges such as the ones described above. Consequently, existing frameworks are unable to develop well-integrated sports and esports governance structures capable of encompassing a dual governance structure where both commercial platforms and traditional sport governing bodies have equal weight (Kelly et al., 2022; Thompson et al., 2023).

However, rather than acknowledging or resolving these technical discrepancies, study 3 shows how organisations often downplay their existence or importance through strategic and normative framing. Acknowledging these technical limitations could directly undermine virtual cycling’s legitimacy within the institutional result-oriented logic required for elite sports, as presented in Study 2 and underpinned by adjacent literature (Skille, 2009; Spaaij et al., 2014; Stenling & Fahlén, 2009). In this way, technical compliance may be interpreted as an

isomorphic pressure, more specifically mimetic and normative pressures (DiMaggio & Powell, 1983). That is because other (successfully perceived) elite sporting disciplines do not have such technical discrepancies. Furthermore, there is also the shared notion that these discrepancies may cause integrity issues such as different types of exclusion (Tjønndal et al., 2025). While we can interpret study 3's results as isomorphic pressures, our results demonstrate how governing bodies react to a limited extent to such pressures and instead engage in institutional decoupling (Boxenbaum & Jonsson, 2008; Washington & Patterson, 2011). That is, they publicly endorse virtual sports using societal values rooted in a Sport-for-All logic, advocating concepts such as inclusivity, safety, and accessibility (i.e. "institutional strategy"), while they do not directly address technical discrepancies through a "technical innovation strategy" (cf. section 14.4.3 in study 3). These insights address research gap 1 outlined in Table 1, uncovering that an unbalanced or even decoupled use of legitimisation strategies leaves governing bodies open to isomorphic pressures and delegitimisation.

To summarise our contextualisation of **aims 1 & 3** through their respective **RQs 4, 5 & 6** from a meso-digitisation perspective, we posit that organisations govern the digitisation of virtual sports *not* by resolving fundamental technical discrepancies, but by complying with normative societal expectations rooted within the Sport-for-All logic (Stenling, 2014) through advocating virtual sports' potential in terms of inclusivity, safety, and accessibility (cf. study 3). Contemporary institutional and governance theories in esports and virtual sports fall short in this domain because they traditionally assume that sport governing bodies hold ultimate authority, which is not the case for esports and virtual sports. Instead, our studies show that a dual governance structure is needed here. Consequently, developing a governance framework that integrates these meso-level power dynamics of platform stakeholders with institutional legitimisation strategies appears necessary in the future. The reflection upon our accumulated results, providing evidence-based insights from multiple points of view, is thereby put forward as a meaningful contribution to theory and practice. Since developing such an integrative governance framework itself is beyond the scope of this dissertation, we advise future research to explore this pathway based on our insights and recommendations.

18.3 Datafication and the Macro-Level Governance of Virtual Sports

This section addresses **aims 1 & 3** and **RQs 4, 5 & 6** since our insights from study 3, 4 and 5 fit best to discuss the macro-level implications of the broader societal and technological shifts that shape sport digitisation. Based on the legitimisation strategies defined in study 3 (section

14), the rise of virtual sports can be interpreted as a next step within a broader societal shift towards datafication and the “quantified self” (Goebeler et al., 2024; Lupton, 2016; Miah, 2017; Miah et al., 2020). The institutionalisation of virtual cycling can be considered to aid development of social acceptance for datafication of cycling as a new norm within cycling, building the cognitive legitimacy development explained in study 3 (Suchman, 1995). In addition, study 5’s main finding of the algorithmic ontology and “platform-result gap” also allows us to reflect on the fact that the quantified-self will not necessarily remain limited to “self-knowledge by numbers” (Lupton, 2016, p. 122) but will arguably evolve to some kind of *co-existing with numbers*. This is because athletic performance and its data are deeply intertwined in virtual sports (cf. physical-digital reality, section 9.5.3). In addition, the *co-existing with numbers* can also be argued as driven by broader developments and expansion of technologies in layer 1 and layer 2 of sport digitalisation (discussed in section 2). For virtual sports, these broader advancements are mainly situated in hardware and software capabilities (Seshadri et al., 2019) that, for example, allow movement compression to increasingly become decompressed as a result of increased accuracy, sensitivity, realistic representation (cf. section 9.5.2).

Still, the macro-level legitimacy of this datafication of sport can be threatened by integrity issues such as a systemic lack of industry-wide hardware standardisation and software transparency. Unlike traditional sports, where legitimacy and integrity issues frequently stem from organisational unresponsiveness or the moral failings of individual athletes engaging in physical doping (Read et al., 2020; Richardson, 2025; Sitko et al., 2023), virtual sports face more structural-technological vulnerabilities (e.g. sector-wide standards or dual-governance models, see sections 18.1 and 18.2). If these vulnerabilities remain unsolved, they actively delegitimise the entire development of new virtual sporting disciplines. Similarly, systemic software exploits and digital doping vulnerabilities have been widely reported in the broader esports literature as undermining sector-wide credibility (Richardson, 2025; Tweedie et al., 2023). Naturally, these vulnerabilities can be considered as a direct result of their lower-level actions, i.e. the misbehaviours or missing collaborations at the micro- or meso-level discussed in sections 18.1 and 18.2.

To secure the future legitimacy of virtual sports, governing bodies are advised to transition from a traditional top-down federation model to a sector-wide regulatory network that operates under a multi- or dual-governance structure (cf. study 5). This macro-level transition seems inevitable for virtual sports because, as Lupton (2016) observes:

It has therefore been contended that algorithms, through their power to intervene in decision-making about the life chances of individuals, have contributed to the soft power of contemporary neoliberal governance. In the face of the rapidly growing influence of this algorithmic authority, it is difficult to challenge it, given that the coding that structures algorithms is generally ‘black-boxed’ and unavailable for direct interrogation. (p. 59)

To overcome such “neoliberal governance” and algorithmic authority, there is a critical need to establish industry-wide technological standards. This involves elevating the concept of “scrutability-without-disclosure” discussed in study 5 and section 18.2 from a meso-level organisational strategy (e.g. specific collaboration between two organisations) to a mandatory macro-level industry standard for independent algorithmic auditing (Berente et al., 2021).

To conclude our evaluation of **aims 1 & 3** and **RQs 4, 5 & 6**, we posit that current theories or concepts applied to the macro-level effects of sport digitisation fall short. While previous works on esports legitimacy demonstrated that macro-level governance and societal legitimisation are often achieved by shifting moral responsibility onto the individual consumer (He et al., 2025), our insights show that such approach will not work for virtual sports because integrity issues are not just moral failings of the user but structural technological vulnerabilities of the proprietary software and hardware (RQs 5 & 6). These vulnerabilities require a sector-wide governance shift (e.g. standardisation and algorithmic auditing) as a required part of virtual sports’ legitimising process (RQ 4) rather than just making the athlete a “self-responsible subject” (He et al., 2025, p. 16; Kelly et al., 2022). While the sports sector can be seen moving in this direction, for instance with IEEE’s (2025) “*Standard for the Quality Assessment of Human and Object Tracking Technologies Used in Field- or Court-based Sports*”, sector-wide accepted standards for virtual sports software or hardware still needs to be initiated and developed. In this sense, it can be argued that traditional sport and esports governance frameworks are not explicit enough about the macro-governance pillar of a sector-wide regulatory network mandating algorithmic auditing and hardware standardisation to safeguard the legitimacy of virtual sports. As a contribution to the theory, we provide evidence-based insights and recommendations on which future governance frameworks on esports or virtual sports can build.

19 Institutionalising Virtual Sports: Logics, Governance, and Use Cases

In the following section, we reflect on our understanding of virtual sports, addressing **aim 1** and **RQ 1**, by positioning virtual sports within institutional logics and exploring their

implications for sport management. Unlike the top-down transitions (e.g. state-led) sometimes seen in eSports (He et al., 2025), the adoption of virtual cycling may initially be considered primarily bottom-up and driven by grassroots hedonic and utilitarian motivations, in particular with the “help” of Covid-19 (Botwina et al., 2024; Evsikov et al., 2024; Reed et al., 2023; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021; Westmattelmann, Grotenhermen, Stoffers, & Schewe, 2021). Not much later, bottom-up competition within virtual cycling started to arise (Berger et al., 2025; Devine et al., 2024; Dyer & Richardson, 2026; Westmattelmann, Grotenhermen, Sprenger, & Schewe, 2021). This made virtual cycling a fractured development pulled by conflicting institutional logics: (1) *the sport-for-all logic* and (2) *the result-oriented logic* discussed in the results of study 2 (see Figure 14). Within study 2, we discuss managers' beliefs about the potential use cases of virtual cycling at the grassroots level. Here, we see how these are distinctly divided between these competing logics. This is evidenced by the “performer club” and “unconvinced club” archetypes, who prioritise the performance-related use of virtual cycling, versus the “holistic club” or “sceptic club” archetypes, who see more merit in non-performance-related use (e.g., use it for adding a fun-factor to training). The development into two logics is also demonstrated in study 3, but then from a strategic legitimisation perspective. At the level of governing bodies (study 3), we discuss the use of normative framing rooted in a sport-for-all logic (e.g., advocating inclusivity and health). This framing occurs despite most of the actions (e.g. organising races) taking place within the result-oriented logic, which can be interpreted as a form of institutional decoupling (Boxenbaum & Jonsson, 2008, p. 2011; Washington & Patterson, 2011).

To pursue either institutional logic as a development trajectory, digitisation and sport policy development can be regarded as interconnected vessels. A result-oriented trajectory requires resolving technological discrepancies. It demands strict algorithmic transparency, uniform hardware standardisation, and appropriate dual governance structures to address the platform-result gaps and vulnerabilities discussed in section 18.2, in studies 3, 4, and 5, and by the literature (Richardson et al., 2022; Westmattelmann et al., 2025). Conversely, a Sport-for-All trajectory is able to tolerate higher movement compression (i.e. lower accuracy and sensitivity) (Hemmingsen, 2024) or different representational fidelity. This was anecdotally described by one of our interviewees in study 3: “[...] in the same way that Mario Kart is not a competitor to Grand Turismo”. In addition, the virtuality aspects of remoteness and (a)synchrony will become increasingly important for virtual sports practised in a sport-for-all logic, given that they enable important “social” and “community” facets of virtual cycling such as online simultaneous from-home participation (Reed et al., 2023).

To summarise our reflection upon **aim 1** and **RQ 1** regarding how to understand virtual sports — here particularly through institutional dynamics —, we posit that micro-level individual uses, meso-level club implementations (e.g., Skille, 2008), and macro-level legitimisation narratives are entangled. Traditional sport management and esports frameworks often only partially capture these interdependencies (Pizzo et al., 2022; Stauff & Vogan, 2021; Tjønndal & Skauge, 2021). Such critique also applies to our own initial framework in study 2. On a self-reflective critical note, we consider study 2 limited as it overlooks the impact of technical discrepancies and their governance on institutional dynamics, such as competing logics, institutional decoupling, and isomorphic pressures (Das & Van de Ven, 2000; Stenling, 2014). Understanding virtual sports requires integrating these technicalities directly into institutional logics and governance analyses, which we therefore advise future research on virtual sports to consider.

19.1 Institutional Dynamics and Lacking Structures as Experienced by Individuals

This section addresses the micro-level implications of institutional dynamics in virtual sports, providing insights into research **aims 2 & 3** through contextualising answers to **RQs 3, 5 & 6**. At the micro-level, athletes face operational challenges due to a lack of governance structures regarding support from their national federation, as highlighted in study 3, where one athlete noted: *“I had nothing from my federation. Absolutely nothing”*. In addition, study 4 highlights how international federations like the UCI currently lack an established and formal commission dedicated to enforcing regulations in cycling esports. This gives way to the absence of guidelines for their members and responsibilities demanded from their members (i.e. national federations). Such regulatory void leaves athletes vulnerable to structural inequities, such as refused support or limited access to qualify for and participate in elite competitions (Tjønndal et al., 2025). Similarly, study 5 describes a lack of formal structures related to the platform-results gap (cf. the algorithmic ontology, section 16.8.1). For example, one sport governing body representative noted that *“at the moment, there aren't enough checks to be able to 100% say that that's the case”*, when talking about structures within governing bodies to check software and hardware compliance with certain standards. The lack of structures on all different fronts (e.g. national federation support, industry-wide integrity standards) ultimately leaves individual athletes vulnerable when engaging in virtual sports. Furthermore, as discussed at the beginning of section 19, the conflicting institutional logics also impact the individual beliefs of managers in sport organisations (e.g. clubs, federations). The results of study 2 show how

individual beliefs often align with either a sport-for-all logic (e.g. virtual cycling as attractive and inclusive club-services) or result-oriented logic (e.g. virtual cycling as training and competition tool). Such individual beliefs can be expected to fulfil the club's "dominant" logic (Szerovay, 2022), but remain difficult to exactly measure.

To conclude our reflection on institutional dynamics at the micro-level related to **aims 2 & 3** and **RQs 3, 5 & 6**, we posit that the institutional theory combined with innovation diffusion theory used in study 2, and with esports good governance frameworks in studies 4 and 5, provide good starting points to capture how institutional dynamics in virtual sports are intertwined with potential individualistic issues and beliefs that are often manifested at a higher (e.g. organisational) level. In this way, we agree with Geeraert (2018), who advises future research to account for "federation's resistance to cultural or structural pressures to implement good governance" (p. 45). This resistance can originate in "pressures from stakeholders that prevent the implementation of good governance" (p. 46), which aligns with virtual sports' institutional dynamics presented in our work. Based on our insights at micro-level, we urge future studies not to neglect individual beliefs, issues or perceptions when investigating resistance to good governance (e.g. composition of commission, managers' beliefs). In addition, we advise future research to also incorporate formal structures, checks, and mechanisms as part of integrity and governance analysis in virtual sports, and in particular how their potential absence affects individual athletes. When critically reflecting on the state of the current literature, the notion of assessing structures, checks, and mechanisms is largely lacking in contemporary esports governance frameworks such as those of Kelly et al. (2022) or Peng et al. (2020). As a result, our own work also lacks to make this notion explicit in relation to virtual sports. Still, accounting for such structures has been long advised in others works such as Chappelet and Mrkonjic's (2013) "Control mechanisms" dimension in their *Basic Indicators for Better Governance in International Sport* framework. While our studies discovered a lack of checks, structures, and mechanisms within the UCI and its stakeholders, as highlighted by our interviewees, we acknowledge that our studies remain limited in encompassing these elements in a more structurally embedded way.

19.2 Institutional Dynamics and Constraints in a Commercially-led Market

Here, we address the meso-level institutional dynamics of virtual cycling diffusion, providing additional insights into **aims 1 & 2** and contextualise the empirically answered **RQs 3 & 4**. First, it is important to address the shared ownership of virtual sports. While federations are

traditionally in full control of the sport they regulate, that is not always the case for virtual sports since these are dependent upon commercial stakeholders (cf. studies 3, 4, and 5). Study 5 draws on Ferkins and Shilbury (2015) to describe in detail how commercial stakeholders become “stakeowners” because they possess the IP of the platforms that provide the sporting environment. The introduction of *stakeowners* rather than *stakeholders* creates technical consequences, such as having to regulate others’ algorithmic discrepancies. As ownership becomes shared, new debates may arise about who exactly should regulate certain aspects of virtual sports. It also creates institutional consequences through the loss of regulatory authority and control over the sport, such as altering what it means to be a sport governing body.

These alterations in authority are moreover amplified by the organisational constraints of governing bodies, for example their lack of technical capacities and resources to manage virtual sports. Sport governing bodies traditionally manage regulatory oversight rather than hardware or software development (e.g., independent virtual performance verification, cross-platform qualification systems). Their technical constraints shift power even more towards platforms. However, as discussed in study 3, this power dynamic should not entirely be viewed as one-sided. Platforms still require the moral legitimacy of governing bodies to host prestigious, officially sanctioned events like the UCI Cycling Esports World Championships. Combining these insights with our earlier discussed insights on the lack of formal structures or commissions (section 19.1), it becomes evident how these dynamics serve as the foundation for the institutional decoupling (Washington & Patterson, 2011) between governing bodies’ legitimising narrative and what they actually are able to do. This is for example seen in the challenge of developing technical systems over others’ IP.

Second, it is important to acknowledge that clubs, federations and governing bodies do not adopt neutral tools. Instead, they adopt a specific platform embedded with a distinct commercial logic that is in turn tied to other institutional logics (Stenling & Fahlén, 2009). Analysing an overview of contemporary (i.e. March 2026) virtual cycling platforms reveals a market that appears to be divided into a result-oriented and sport-for-all logic (Cole, 2026; K. Jenny & Nehr, 2026). Most platforms categorise primarily under a result-oriented performance logic with features such as adaptive training plans, threshold tests, or racecourse recons. (e.g. TrainerRoad, TrainingPeaks Virtual, Rouvy, Velocity, FulGaz, Zwift, MyWhoosh...). Only some of these also implement social and gamification features (e.g. MyWhoosh, Zwift), indirectly serving a sport-for-all logic as they can be considered an unorganised “sport light” alternative to (serious-)leisure cycling (Borgers et al., 2018). Remarkably, across all providers, only one smaller platform (i.e. icTrainer) currently offers subscriptions dedicated to clubs. Such

a landscape may suggest that commercial platforms do not yet consider clubs as potential subscribers. Furthermore, they may also not consider virtual cycling as a (club-)organised activity that may serve a sport-for-all logic in the first place. This reinforces the results of studies 2 and 3 that organisations' beliefs about and implementation of virtual cycling are predominantly performance- and result-oriented. This orientation appears to align with the "dominant logic" (Bettis & Prahalad, 1995; Szerovay, 2022) being a result-oriented logic in the commercial platform landscape. While this dissertation did not analyse "commercialisation logics" in depth (Stenling & Fahlén, 2009), it appears as a significant aspect of virtual cycling diffusion that is fruitful for future research to incorporate.

Concluding our contextualisation of **aims 1 & 2** alongside **RQs 3 & 4**, a required theoretical shift could be argued regarding virtual sports diffusion. Traditional innovation theories (e.g. diffusion, disruptive innovation) generally assume adopting organisations (here, governing bodies) to possess the agency to shape new technologies or choose from a "neutral" market (Christensen et al., 2015; Wolfe, 1994). However, the literature and empirical results show how commercial stakeholders in virtual sports may restrict such agency. This redirects virtual sports development towards a "dominant logic" (Bettis & Prahalad, 1995; Stenling & Fahlén, 2009). In the case of virtual cycling, that appears to be a result-oriented dominant logic (cf. studies 2 and 3). The fact that governing bodies' organisational constraints make them highly dependent on platform developers, makes it particularly difficult for those governing bodies to equally address different institutional logics. Our theoretical contribution lies in highlighting the dependency between institutional and commercial dynamics, and how this translates to traditional sport governing bodies and new commercial stakeholders. Future research could develop integrated frameworks to examine how the commercialisation logic of stakeholders further limits the agency of highly institutionalised governing bodies. Furthermore, it would be interesting to examine how this trickles down to virtual sports beliefs and adoption behaviours within organised sports settings such as voluntary sports clubs. Such a framework could therefore, for example, develop a better understanding of how sport-for-all policy is indirectly influenced by the digitalisation of sports and its commercial software and hardware developers. It could also explore what is needed to address multiple logics with virtual sports, or what is needed to develop virtual sports as a (club-)organised activity serving the sport-for-all logic.

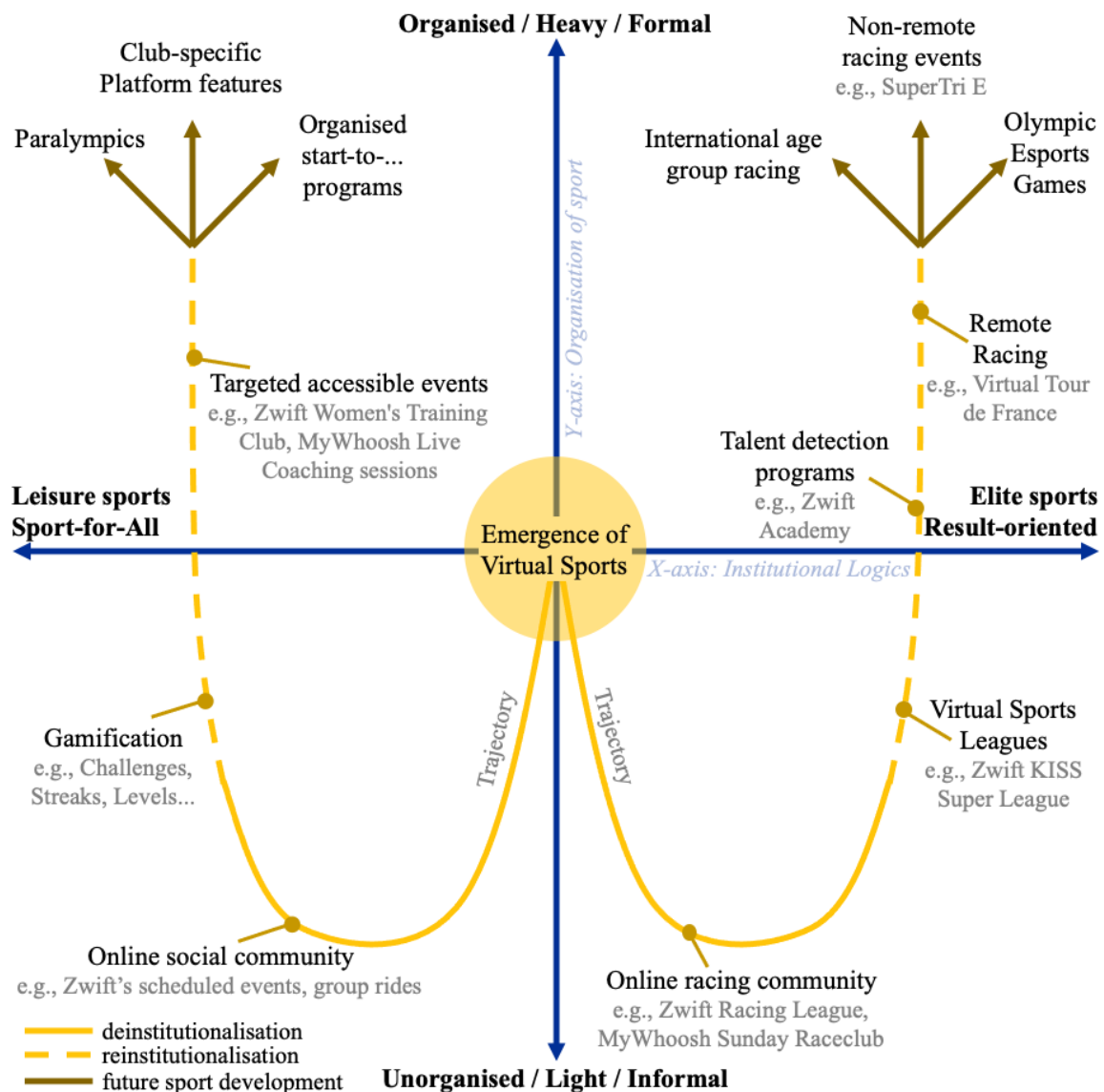
19.3 *Institutional Changes: from deinstitutionalisation to reinstitutionalisation*

Section 18 and 18.2 has already described virtual sports' fractured development into two competing logics (i.e. the result-oriented and sport-for-all logic). This section zooms in on this development, providing insights into **aim 1** and **RQ 1**. Applying the lens of institutional change (cf. section 3.2.3) introduces macro-level insights that links these two logics to distinct trajectories. Each trajectory describes how virtual sports disrupt formal organised sports and force the strategic adaptation of governing bodies who start to experience isomorphic pressures (cf. section 3.2.1). This theoretical understanding is derived from the accumulated findings of our five empirical studies, enriched by examples of significant events in virtual cycling over the past decade.

The emergence of virtual sports can be argued to follow a trajectory of institutional change encompassing both deinstitutionalisation and reinstitutionalisation (Greenwood et al., 2002) within the two competing logics of performance versus sport-for-all (Stenling & Fahlén, 2009). As illustrated in Figure 19, this trajectory is mapped across two axes. The vertical axis represents the participation in and organisation of sport, ranging from unorganised and informal “light” settings to organised and formal “heavy” structures discussed in section 3.2.3. The horizontal axis represents institutional logics, split between the two competing logics of sport-for-all and the result-oriented logic, as discussed in section 3.2.2.

Based on the case of virtual cycling, virtual sports can initially be viewed as a technological disruption that disturbs the established norms of sport (cf. study 3) (Christensen et al., 2015; Washington & Patterson, 2011). Commercial platforms suddenly offer a new way to participate in sports that provides the benefits of organised sports (e.g. social aspects, group rides) while not adhering to the strict schedules and formal structures often found in organised sports (e.g., club membership obligations, availability of specific facilities). In this way, organised sports via “heavily structured sports organisations” are bypassed during this initial stage (Borgers et al., 2018). This stage of the disruptive process is what we argue represents *the deinstitutionalisation of virtual sports* (Figure 19, solid yellow line). Through the emergence of virtual sports platforms (in our case mainly Zwift), participation in sports shifts into an unorganised light sport setting characterised by individual online participation.

Figure 19: Illustration of deinstitutionalisation and reinstitutionalisation of virtual sports.
Source: Own conception



Within this deinstitutionalisation process, we theorise that virtual sports can be considered to diverge along two distinct and “competing” institutional logics (Stenling & Fahlén, 2009; Thornton et al., 2012). The trajectory within the sport-for-all logic includes the initial unorganised light sport phase, which manifests itself primarily through online social communities (for instance seen in Zwift's scheduled events and group rides⁸), gamification features like challenges and missions⁹, or targeted events designed to be more inclusive towards specific audiences like “Women only rides”⁸. Over time, a process of *virtual sports reinstitutionalisation* begins (Figure 19, dashed yellow line) (Stenling, 2014). Based on our

⁸ For instance, see the Zwift event list at <https://www.zwift.com/eu/events>

⁹ For instance, see the MyWhoosh Missions at <https://mywhoosh.com/mywhoosh-missions/>

findings in studies 2 and 3 in particular, interest begins to spark in “heavier” organised sports organisations to engage with these types of virtual sporting activities (here, cycling). For instance, in study 2, clubs see potential in the gamification features of Zwift to make their club services (i.e. training) more attractive, or they see merit in relying on virtual cycling to attract new or different audiences who often face barriers to engage in traditional sports, such as beginners, elder people, or para-athletes. In study 3, our findings show legitimisation strategies in which sport governing bodies show interest in applying organised “heavy” sport-for-all practices to virtual cycling, such as formalising pathways to the sport to increase accessibility (e.g., “that ease of access, the kind of democratisation of sport, is incredibly valuable”, section 14.4.1), targeted inclusivity (e.g. “we want to reach the young audiences where they are”, section 14.4.2), or removing barriers to sport participation (e.g., “We could have different categorization of para-athletes”, section 14.4.2). Moving forward, future sport development within the sport-for-all logic is hypothesised to lead to more formalised practices using virtual sports (i.e. dark yellow arrows in Figure 19). These could include organised start-to-sport programmes, the development of club-specific platform features, integration into the Paralympics, or even serve as a way to bridge the Paralympics to the Olympics by, for example, redefining para-athlete categorisation for virtual sports. In some instances, para-athletes’ disability could become “irrelevant” thanks to the use of technology and allow for a fair and more inclusive integration of para-athletes into regular mainstream competitions usually reserved for non-para-athletes.

Conversely, the trajectory within the result-oriented performance logic can be argued to follow a distinct process of deinstitutionalisation to reinstitutionalisation. As indicated in Figure 19 on the bottom right, this starts with informal “light” (i.e. institutionally unorganised) online racing communities, such as the Zwift Racing League¹⁰ or MyWhoosh Sunday Raceclub¹⁰. In what follows, the environment within this results-oriented logic progressively formalises into heavier structures. Organisations or platforms themselves start to establish virtual sports leagues like the Zwift KISS Super League¹¹, or talent detection programmes in collaboration with UCI Pro and Continental teams such as the Zwift Academy¹². However, during this transition period of deinstitutionalisation to reinstitutionalisation, these first formal leagues and programmes are still not regulated or governed by official sport governing bodies (e.g., UCI). As this trajectory develops, light online racing communities are joined by formal and

¹⁰ More info at <https://www.wtrl.racing/zwift-racing-league/> and <https://mywhooshinfo.com/statistics/src-top>

¹¹ More info at <https://www.zwift.com/eu/news/12122-kiss-super-league>

¹² More info at <https://www.zwift.com/eu/zwift-academy-pro-contender-requirements>

established organisations hosting online remote racing, for instance ASO's Virtual Tour de France during the Covid-19 pandemic¹³. Eventually, formal and official (traditional) sport governing bodies also engage in virtual cycling within this result-oriented logic by facilitating and overseeing officially sanctioned World Championships, which is now already done in triathlon by World Triathlon¹⁴ and in Cycling by the UCI¹⁵. Consequently, we can argue that a significant outcome of this reinstitutionalisation process is the transitioning of virtual sports (in our case, virtual cycling) back into “heavy” organised sporting structures (Borgers et al., 2018). Findings from study 2 show that clubs predominantly perceive virtual sports as useful to address the results-oriented logic (e.g. training or racing on Zwift). In addition, our findings in studies 3, 4 and 5 also showed immense development within the elite and result-oriented trajectory, forecasting that “30 or 40 athletes or more, 50 athletes, that will be able to go to the eSport Games” (section 15.5.3), though interviewees also critically admit the still remaining vulnerabilities and the “long way for us to go on the elite side of things” (section 15.5.2). Nonetheless, we believe that virtual sports will continue to develop in this trajectory (i.e. dark yellow arrows in Figure 19). This could, for example, include international age group racing, elite non-remote racing in other sports, and the inclusion of virtual sports in the Olympic Games or some sort of Olympic Virtual Sports Games, given the IOC's recent communication that they “will develop a new approach to the Olympic Esports Games” (International Olympic Committee, 2025b).

Of course, the future developments of both trajectories (i.e. dark yellow arrows in Figure 19) is something we hypothesise based on what seems technically feasible and how they align with each institutional logic and with the formal, organised, way of implementing sport (Borgers et al., 2018; Stenling, 2014). Our discussion on virtual sports' deinstitutionalisation and reinstitutionalisation helps to explain *why* these new increasingly institutionalised events are being developed (e.g. isomorphic pressures, institutional change), or why events like the Olympic Esports Games are in fact postponed (e.g., increasing institutional complexities and dynamics) (International Olympic Committee, 2025b). On a more critical note, the above explanation does not naturally explain *if* and *why* institutionalisation of virtual sports is actually needed or sensible.

On one hand, the drive towards (re)institutionalisation appears necessary to safeguard the fundamental integrity of the sport and the welfare of its participants (cf. studies 4 and 5).

¹³ More info at <https://www.letour.fr/en/virtual-tour-de-france>

¹⁴ More info at <https://events.triathlon.org/2025-supertri-e-world-triathlon-championships-london>

¹⁵ More info at <https://www.uci.org/discipline/cycling-esports/5awP766tEGOobRlwSdHBDS?discipline=CES>

Leaving virtual sports entirely to the free market pushes the development of a public domain of sports towards a private commercial one. As discussed in sections 18 and 19, it risks exploits in digital doping, algorithmic biases, and the marginalisation of non-profitable user groups. For instance, one participant in study 3 admitted how virtual cycling is mainly used by “older cohorts [starting at age 35 of] 88, 90 percent male”, illustrating how platforms will naturally cater to the most profitable demographics and structurally undermine sport-for-all accessibility. Formal governance structures are required to enforce sector-wide (co-)governance and standardisation regulation but also have commercial stakeholders pursue pathways that develop moral legitimacy (i.e. the right thing to do) like accessibility over profits (cf. section 14.4, study 3). Furthermore, formal institutionalisation (e.g., IOC recognition, national federation integration) unlocks public funding which commercial game developers themselves cannot obtain, though caution remains needed that such funds are spent considerately (Postma et al., 2022).

On the other hand, forcefully redirecting virtual sports back into traditional, “heavy” sporting structures may prove counterproductive. The initial appeal of virtual cycling was precisely its deinstitutionalised nature, offering flexibility and freedom from rigid club obligations (Evsikov et al., 2024; Reed et al., 2023). Re-imposing legacy structures risks alienating this core grassroots and “unbounded” community, a tension that has been evidenced before within the sport digitalisation literature (Standaert & Mazurova, 2025). Furthermore, since commercial stakeholders retain absolute IP over the platforms, traditional federations are currently often poorly equipped to govern this space effectively at the pace that is required. Attempting to apply slow-moving legacy sports governance to a fast-developing technological ecosystem could stifle future software and hardware development, especially if governance regulations are not adequately and timely adapted to new trends and features (Thomas & Tierney, 2021). In addition, reinstitutionalisation may also create a “phase lag” (Marchant, 2011; Synrem, 2019) due to redundant bureaucratic layers rather than solving the underlying technical discrepancies. This means that (re)institutionalisation should not be regarded as an endpoint in itself, but rather be considered where opportunities outweigh risks, and where risks can be mitigated. Consequently, the future of virtual sports may not require a complete redirection to “traditional” institutionalisation, but instead, demands novel regulatory approaches that balances technological agility with traditional management and governance of sport (Thompson et al., 2023). This complex symbiosis is something future research may further look into, and make explicit specific practical recommendations or roadmaps for the industry.

By reflecting on our studies and the development of the virtual cycling landscape from a macro-institutional perspective, we provide an answer to research **aim 1** and **RQ 1** through the lens of institutional change theory. It shows how the concepts of deinstitutionalisation and reinstitutionalisation offer a fruitful framework to analyse the development of virtual sports, particularly when mapped against the competing institutional logics (i.e. performance-oriented versus sport-for-all) (Stenling & Fahlén, 2009; Thornton et al., 2012) and the competing ways of participation in and organisation of sport (i.e. light versus heavy) (Borgers et al., 2018). As visualised in Figure 19, the trajectories of virtual sports are not statically positioned within the sport policy of sport participation and organisation, but instead undergo a continuous process of institutional change where new and informal practices are eventually structured, formalised, and legitimised back into the broader sports ecosystem. However, and on a more critical note, the traditional lens of institutional change might also be limited in truly capturing these change processes and trajectories, given the unique nature of virtual sports compared to traditional sports. As discussed in section 18.2 and 19.2, reinstitutionalisation in virtual sports is heavily impacted by commercial stakeholders who own and develop the platforms, and who ultimately co-create these trajectories by deciding which one receives the most technical support as a way of building pragmatic legitimacy and moral legitimacy (Ferkins & Shilbury, 2015; Suchman, 1995).

20 Strategic Virtual Sports Diffusion: a Technological-Institutional Balance

As illustrated in Table 19, in this final section we reflect on our understanding of virtual sports, addressing **aim 1** and **RQ 1** by positioning virtual sports within different innovation strategies across different levels of analysis. To start from our definition of virtual sports (cf. 9.6), the micro-level of individual user adoption has been substantially studied in sport management and marketing literature (Bentvelzen et al., 2022; Berger et al., 2025; Botwina et al., 2024, 2025; Edwards & Hettinga, 2023; Evsikov et al., 2024; Felczak & Filiciak, 2025; Reed et al., 2023; Rojas-Valverde et al., 2022; Westmattmann, Grotenhermen, Sprenger, Rand, et al., 2021; Westmattmann, Grotenhermen, Sprenger, & Schewe, 2021). These studies highlight an array of reasons for and against adoption, which have been explained in our pre-study (Part III, section 9) and our empirical studies (Part IV, sections 12 - 16). In sum, all dimensions in our definition of virtual sports add to individuals' reasons for or against virtual cycling adoption in different ways. For instance, the physical and cognitive skills, high movement decompression, and distinct physical-digital interreality in our definition are argued to enhance the “immersive

experience”, which is considered a reasons for adoption by cyclists (Devine et al., 2024). Similarly, high representational fidelity such as realistic and non-oppressive graphics is also a reason for adoption (Devine et al., 2024; Reed et al., 2023). Meanwhile, remoteness, (a)synchrony, and the distinct physical-digital interreality can be related to both the social and connected features that recur in many studies as reasons for adoption (Berger et al., 2025; Devine et al., 2024; Evsikov et al., 2024; Reed et al., 2023; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021). In addition, these dimensions can also be related to utilitarian reasons for adoption such as training and time effectiveness (Berger et al., 2025; Westmattelmann, Grotenhermen, Sprenger, Rand, et al., 2021). The dimensions of institutionalised and rule-governed contests in our definition also relate to adoption behaviour, since fraud and doping are reasons against adoption for many cyclists (Dyer, 2022; Dyer & Taylor, 2023; Richardson et al., 2023).

However, since individual innovation adoption is not the scope of this dissertation, innovation adoption strategies have been discussed to either understand virtual sports in itself or understand virtual sports in light of organisational innovation adoption. Looking closer at organisational innovation strategies of virtual sports at the meso and macro levels, our studies show that these strategies diverge significantly. Study 2 shows that voluntary clubs approach virtual sports mainly as a complementary performance-related service (i.e. training) because they can capitalise on its remoteness and (a)synchrony. This allows their members to efficiently and effectively train collectively from home. At the same time, study 2 shows that clubs also consider other innovation strategies (e.g. addressing social goals), though they remain more hesitant towards them. Conversely, sport governing bodies capitalise on multiple dimensions of virtual sports that, when combined, allow them to pursue an innovation strategy that legitimises virtual sports as a new discipline within their sporting context. To do so, studies 3, 4, and 5 show how governing bodies leverage the advancements in the dimensions of increased regulation, high movement decompression, improved representational fidelity, and the distinct physical-digital interreality that is argued as a “feature” rather than a technical governance challenge (cf. study 5). As both organisational levels rely on different innovation strategies, they also experience distinct barriers. While clubs face human-related constraints and a lack of club-specific platform features (cf. study 1), governing bodies face broader governance and integrity issues such as doping and algorithmic opacity (cf. studies 3, 4, and 5).

Ultimately, understanding virtual sports and their implications for sport management from an innovation strategy perspective, which provides insights into **aim 1** and **RQ 1**, requires a comprehensive multi-level view encompassing micro-, meso-, and macro-levels, since they

are fundamentally intertwined. The primary contribution to the literature and implication for sport management is that innovation strategies cannot be treated in isolation. As reflected in the meso- and macro-level institutional dynamics discussed in Section 19.2 and 19.3, the use and development of virtual sports within unorganised sport, grassroots clubs, governing bodies, and the commercial market are all interdependent and steer towards a “dominant” logic. Such effects can be argued to impact the reasons for and against adoption at multiple levels, resulting in vastly different innovation adoption and implementation strategies for virtual sports. This depends on the aims of the adopting organisation, and how well such an aim fits with the dominant logic and dominant dimensions of virtual sports discussed earlier in this discussion and in our pre-study on virtual sports (cf. section 9).

20.1 The Importance of Individual Beliefs in Organisational Innovation Strategies

This section addresses the micro-level inferences of innovation strategies for virtual sports, providing insights into research **aim 2** through contextualising answers to **RQs 2 & 3**. Within the context of clubs, the adoption of virtual sports is also driven by the beliefs and behaviours of micro-level actors such as “key volunteers” (Schlesinger et al., 2015). The results of study 1 reveal that a successful innovation strategy for virtual cycling as-a-club-service must rely on shaping positive perceptions around virtual cycling and acquiring pro-innovation managerial and human-resources capacities (see summary Table 18 and section 12.6.1 in study 1). In other words, “soft” determinants rather than infrastructural or financial resources are found to determine and predict clubs’ innovativeness in relation to virtual cycling. Incorporating individualistic elements in organisational innovation strategies is in fact not uncommon and even advised. For example, one of Wolfe’s (1994) recommendations for organisational innovativeness research reads: “*switching from the organization to the ‘innovation-in-an-organization’ as the unit of analysis, thus treating organization specific innovation attributes as such rather than as innovation inherent attributes*” (p. 409). Hence, by treating “innovation-in-an-organisation” rather than the organisation as the unit of analysis, which is demonstrated in study 1, researchers can account for the beliefs and attitudes about an innovation of the people within the organisation. Such an approach is important given that subjective and individualistic determinants explain a significant amount of variance in organisations’ innovativeness (cf. findings study 1; Frambach & Schillewaert, 2002; Rogers, 2003; Wolfe, 1994). As an illustration, study 1 reports key volunteers’ perceptions about and attitudes towards innovation (in our case, virtual cycling) as the highest predictors of clubs’ innovativeness (see Table 6 and

Figure 12). This means individual-related “soft” determinants are key determinants to consider when crafting an innovation adoption strategy for clubs.

These individualistic perceptions and attitudes can also be argued to be shaped by the digitisation elements of virtual sports (see discussion section 18) and the institutional dynamics of virtual sports (see discussion section 19). Based on innovation diffusion theory, aspects like movement (de)compression or representational fidelity can be expected to be correlated to innovation attributes such as image, perceived complexity, or relative advantage (Evsikov et al., 2024; Park et al., 2024). Consequently, they influence the attitudes and intentions towards the adoption of virtual sports (Kaur Kapoor et al., 2014; Park et al., 2024; Wolfe, 1994). Furthermore, these perceptions are filtered through the overarching normative and dominant logic in which the club operates, as discussed in studies 2 and 3. For instance, a volunteer in a performance-oriented club will evaluate a virtual sport based on its competitive features, while volunteers operating under a sport-for-all logic will seek platforms that prioritise inclusivity and gamification. Hence, adoption determinants and innovation strategies must also be contextualised by broader institutional realities (Stenling, 2014).

Reflecting on these insights and the current theoretical frameworks used in studies 2 and 3, we gain a better contextualised understanding of research **aim 2** and **RQs 2 & 3**. Traditional innovation diffusion models often isolate micro-individual determinants from their meso-organisational and macro-institutional contexts (Lundblad, 2003; Wolfe, 1994). Conversely, organisational innovativeness theory tends to overlook micro-individual or macro-institutional effects (Crossan & Apaydin, 2010; Frambach & Schillewaert, 2002). Our theoretical contribution lies in demonstrating that micro-level adoption drivers living-*in-an-organisation* cannot — and should not — be fully understood without accounting for external logics, such as dominant institutional or commercial logics (cf. discussion section 19). While study 1 relies on an integrative model that includes both micro-individual and meso-organisational determinants, it still falls short in accounting for possible digitisation and macro-institutional determinants. Future research is therefore advised to integrate these different perspectives to improve existing theories, particularly in highly institutionalised and commercialised environments like virtual sports. For instance, scholars could apply the Technology-Organisation-Environment framework (Baker, 2012; Tornatzky et al., 1990) or the multi-level perspective on socio-technical transitions (Geels, 2011). Although rarely, such models have been applied in sport management to better account for digitisation and the environmental context (Kerwin, 2022). In our case, scholars could, for instance, investigate

how specific platform designs or features that are rooted within a dominant institutional logic, actively alter the beliefs of and reasons for or against adoption among key volunteers in clubs.

20.2 Organisational Determinants and External Effects in Innovation Strategies

This section addresses the innovation strategies and their determinants inherent to organisations such as clubs, rather than individual volunteers engaging with *innovation-in-an-organisation* as done in the previous section 20.1. This contextualises **aim 2** and **RQs 2 & 3** from a meso-level view. Drawing on study 1, it becomes clear that a club's innovation strategy mainly relies on its organisational management posture towards new technologies, given that the only significant club-level variables to predict virtual sports services *as-a-club-service* (here, cycling) are managerial determinants and human-related capacity determinants. While managerial determinants predict innovativeness to a large extent, the predictive value of human-related capacity determinants is significant but low (< 5%, see section 12.5.3). This suggests that human resources are important for successful virtual sports adoption, and particularly in the form of positive managerial attitudes and willingness towards implementing virtual sports *as-a-club-service*.

When analysing organisational innovation strategies and their determinants as done in studies 1 and 2, Rogers' (2003) adopter categories have proven methodologically useful to cluster clubs based on their innovativeness, ranging from innovators to laggards. This clustering allows for practical implications to be drawn (cf. section 12.6.4 study 1 and section 13.6.2 study 2). While analytically useful, a more critical reflection on these studies requires acknowledging that organisational innovativeness is a continuum rather than distinct categories. We have accounted for this to a limited extent towards the end of study 2 (see Figure 14). Future research could address this by aiming for larger sample sizes and refined measurement methodologies (e.g. adoption rates or usage times rather than self-reported surveys) that allow researchers to analyse data on a continuous rather than ordinal scale (e.g. regression analysis). This would derive more fine-grained insights to tailor strategies to the club's specific positioning on this continuum (Damanpour, 1991). In addition, researchers are also advised to measure and incorporate individual-rooted determinants (see section 20.1) and digitisation-related determinants (see section 18). While incorporating institutional logics would also be insightful for future research, latent concepts like logics remain difficult to quantify, and any attempt to do so remains a complex enquiry including questionnaires, policy documents, annual reports,

(social) media analysis, and — above all — an examination of the actual execution and scale of projects (De Bock et al., 2022).

Reflecting on these insights and the theoretical frameworks used in studies 1 and 2, we gain a better contextualised understanding of research **aim 2** and **RQs 2 & 3**. While our applied innovation models successfully identified rather straightforward and observable managerial and human-related capacity determinants, they inherently struggle to capture latent macro-institutional and sport-digitisation forces within a single empirical variance model. To advance this field, future research is advised to integrate methods capable of operationalising these latent dynamics within quantitative innovation models. For instance, scholars could employ fuzzy-set Qualitative Comparative Analysis (Ragin et al., 2008) to systematically examine how different configurations of specific managerial resources and dominant institutional logics interact in driving adoption behaviour. Furthermore, anchoring empirical designs in the institutional logics perspective (Skille & Stenling, 2018; Thornton et al., 2012) and utilising structural equation modelling would allow researchers to treat these complex normative beliefs as measurable latent variables. By triangulating survey data with media and policy document analysis (De Bock et al., 2022), future models can also provide a more holistic and integrated evaluation of organisational innovation strategies for virtual sports.

20.3 Innovation Strategies for Sector-wide Diffusion

This section addresses macro-level innovation strategies by reflecting upon sector-wide diffusion and market disruption of virtual sports, thereby contextualising **aim 1** and **RQ 4**. While micro- and meso-level strategies concern individual adoption by consumers or organisational adoption, macro-level strategies focus on the broader social acceptance and strategic spread of a new sporting discipline across the sports ecosystem (Lundblad, 2003; Newell & Swan, 1995). Reflecting on our approach to study virtual sports as an organisational innovation, evaluating the macro-level *innovativeness* (i.e. adoption of a virtual discipline amongst all sports governing bodies) through quantitative diffusion models (Rogers, 2003) seems unrealistic given the prematurity and newness of virtual sports, as explained in section 7.2. Consequently, shifting our research to a study of *innovating* (cf. section 3.1 and 7) can be considered most appropriate and, at this time, most insightful (Wolfe, 1994). These qualitative process-oriented approaches explore how an innovation (e.g. virtual cycling) becomes socially accepted and structurally integrated at a broader level (e.g. within the cycling community, the sports ecosystem, and eventually society), rather than merely measuring and predicting how

many consumers or organisations have adopted and will adopt it.

At this broader sector-wide scale, study 3 proposes the idea that virtual sports act as a disruptive innovation challenging traditional sports (Christensen et al., 2015; Das & Van de Ven, 2000). Governing bodies must consequently manage this disruption and, if they decide to adopt a virtual discipline, achieve broad social acceptance by crafting effective diffusion strategies. As demonstrated in study 3, these strategies revolve heavily around institutional legitimisation, responding to isomorphic pressures, and developing pragmatic, moral and cognitive legitimacy (Suchman, 1995; Washington & Patterson, 2011). In particular, governing bodies are observed to actively construct new evaluation criteria for their sport as a way to make the virtual discipline “performant” on the criteria by which the sport is normatively evaluated by its stakeholders and the public. Our research shows that this is done using narratives around the *ontology of sports*. In our case, this involves the social construction of what cycling *is*, what it should *be*, and how a virtual discipline (i.e. virtual cycling) could support this development of the sport. In doing so, governing bodies for instance emphasise sustainability, inclusivity, or other aspects that normatively align with broader sport policy goals or the IOC agenda (International Olympic Committee, 2021b; Mittag & Naul, 2021) to justify and gain acceptance for a new discipline. Relying on the same normative mechanisms, they strategically develop new markets by aiming to target, for instance, women or para-athletes with virtual sports. These actions can be regarded as institutional work in reaction to both isomorphic pressures (Boxenbaum & Jonsson, 2008; Slack & Hinings, 1994) and commercial pressures (Christensen et al., 2015).

Reflecting on these insights and the theoretical frameworks used in study 3, we gain a better contextualised understanding of research **aim 1** and **RQ 4**. While the process-oriented *innovating* (rather than *innovativeness*) nature of study 3 successfully captures the institutional narratives driving early diffusion, a critical limitation is that qualitative legitimisation frameworks like disruptive innovation theory using a (qualitative) institutional innovation diffusion model (cf. Figure 15) only tend to capture the *intended* strategies of governing bodies (i.e. what they say) rather than capturing their effective translation into *actual* actions, decisions, and policy implementations (i.e. what they do) (cf. section 14.5). In addition, such models fail to capture the actual market disruption in terms of its current position on the adoption-diffusion curve and the rate at which the innovation is adopted (Wolfe, 1994). As such, study 3 is limited by its obscurity on the size and details of institutional decoupling happening within governing bodies (Washington & Patterson, 2011) and on the actual adoption rates as influenced by changing market dynamics and the commercial realities of platforms (see

section 19.2). As the virtual sports market matures, future research is advised to bridge this gap by transitioning towards both longitudinal and quantitative diffusion models to track actual adoption rates (Rogers, 2003). Additionally, future research would also benefit from a closer analysis of institutional decoupling (Boxenbaum & Jonsson, 2008; Washington & Patterson, 2011) and from integrating theories on platform capitalism (Srnicek, 2016). These concepts are major threats to the legitimacy of virtual sports, and developing a better understanding of them could improve the pace and direction of sector-wide virtual sports diffusion to ensure it is in harmony with digitisation developments (see section 18) and commercial-institutional developments (see section 19).

21 Practical Implications and Recommendations

21.1 Contextualising the Practical Scope

Based on our discussion as outlined in Table 19, it is clear that virtual sports represent an emerging and rapidly changing domain within the broader sporting landscape. Given the novelty of this phenomenon, the primary aim of this doctoral thesis is to develop a foundational theoretical understanding of virtual cycling, validated by empirical data. Within our overarching discussion, we draw upon our specific empirical studies to map this new frontier and reflect on the applicability of different theories and concepts commonly used within sport management. This was done using theoretically inspired domains of digitisation, institutional dynamics, and innovation strategies across micro, meso, and macro levels of analysis. At the beginning of this doctoral process, the topic of this dissertation remained unclear for a long time (e.g. seeking links between esports and clubs) to only land on “virtual sports” after several years. This resulted in the fact that the studies in this dissertation were, in such an initial phase, designed to understand the phenomenon of virtual sports and its prevalence. Naturally, our work does therefore not focus on testing specific practical interventions or generating experimental tools for immediate application in sport organisations.

Nonetheless, we offer several insightful practical implications and recommendations in this section. These recommendations are extrapolated from both study-specific and overarching empirical findings, along with theoretical models (e.g., section 19.3) outlined earlier. Despite their cross-sectional or theoretical nature, we believe that they offer robust and evidence-backed strategic guidelines that may assist a diverse range of stakeholders. These range from grassroots voluntary sports clubs to national federations, international governing bodies, policymakers,

and commercial platform developers as they navigate the complexities of adopting and governing virtual sports.

21.2 Empowering Grassroots Adoption through Tailored, Human-Centric Support

For voluntary sports clubs, a perceived lack of financial resources or physical facilities is often viewed as the main barrier to technological adoption. However, empirical findings from study 1 demonstrate that perceptions, managerial attitudes, and human capabilities are stronger predictors of a club's innovativeness than infrastructural or financial capacity. Clubs should therefore focus on cultivating these pro-innovation perceptions, attitudes, and human-related capacities. A practical step is the formal appointment of innovation champions (see section 12.6.4 in study 1), and awarding them with the appropriate recognition. These champions, who are often active members with platform-specific knowledge, can demystify the technology and demonstrate the value of virtual sports to other members and the board.

Furthermore, sport governing bodies and commercial providers should avoid uniform strategies when promoting virtual sports. Study 2 reveals that clubs operate under different dominant institutional logics, prioritising either competitive skill development or inclusive participation. As discussed in detail in section 13.7, marketing efforts and policy support must be segmented (e.g. using personas (LeRouge et al., 2013) or archetypes (Hoye et al., 2020)) to align with the different logics, cultures, and *modi operandi* of clubs. As such, communication strategies from these organisations need to address the specific goals of the target club. For instance, highlighting workout-planning tools appeals to result-oriented clubs, while emphasising social features and accessibility resonates with clubs focused on broad participation.

Additionally, federations can optimise resources by implementing archetype-specific interventions. Holistic clubs demonstrate an ability to balance multiple institutional logics and can serve as role models to guide hesitant clubs through implementation (see section 13.7, study 2). Conversely, policymakers should recognise that sceptic clubs may resist technological innovations, making it more efficient to direct support towards clubs open to experimenting with virtual sports.

21.3 Reconciling Institutional Narratives with Technical Realities

Sport governing bodies face a challenge when promoting virtual sports. Drawing on our insights from study 3, they struggle to rely solely on an institutional diffusion strategy, as this would

ignore technical discrepancies that undermine the discipline's legitimacy. International sport governing bodies should actively pursue a technical strategy alongside their institutional strategy and sport-for-all goals. This dual approach involves developing cross-platform calendars and establishing a framework to address digital doping (Richardson, 2025). Also at the national level, federations should be aware that they play an important role in adhering to these technical standards at local competitive tiers.

Furthermore, national sport governing bodies are advised to transition from passive overseers to active participants in developing the virtual sports landscape. They should formulate a strategy, vision, and mission tailored to the virtual discipline of their specific sport, particularly within both competing logics of sport-for-all and the results-oriented logic (see reinstitutionalisation trajectory section 19.3). Alongside the normative framing conducted within the institutional diffusion strategy, both national and international federations should commit tangible resources, including financial investment and sufficient dedicated staff, to develop the discipline. For instance, in the case of virtual cycling, it is an officially recognised discipline by the UCI where national federations act as its domestic implementers. Assuming other sports will follow this model, national federations share the responsibility for overcoming virtual sports specific governance issues such as a lack of transparency, lacking support to athletes, or dishonouring licences to participate in international virtual sport events at the highest level (see section 15.5.4, study 4). Therefore, national federations must adhere to the seven good governance principles and, above all, consider themselves responsible and accountable for their actions regarding athlete participation, organising national (e.g. qualification) virtual competitions, and building legitimacy for the discipline.

Commercial platform providers also share in these good governance principles. Their design choices in terms of sport digitisation influence the social acceptance of the sport (cf. section 18.2). Platform publishers are advised to proactively collaborate with each other *and* with governing bodies to build interoperable features and unified standard protocols (e.g., anti-doping, physics engine algorithms), rather than operating in isolated commercial silos (see section 15.5.2 study 4). Finally, governing bodies should acknowledge that platform algorithms actively determine competition outcomes and alter athletic tactics, and that improper management of such cases can lead to integrity issues (e.g., too short notice when changing the platform used in competition, cf. study 5). To maintain sporting integrity, governing bodies are encouraged to inform themselves and conduct independent analyses of the (potential) significant differences between platform-result gaps before sanctioning events. However, the

best way to analyse and consequently address this remains open for future research (see section 22).

21.4 Dual Governance and Algorithmic Scrutability

Traditional sport governing bodies are advised to recognise commercial platform providers as “stakeowners” rather than mere stakeholders (Ferkins & Shilbury, 2015). These commercial entities hold IP rights and frequently exercise ultimate control over the digital sporting environment (Kelly et al., 2022; Peng et al., 2020). To ensure competition integrity, a dual governance structure is recommended (see section 16.8.2, study 5). Establishing formal contractual arrangements with these platform publishers appears a logical first step towards operationalising such a dual governance structure. These arrangements preferably specify the requirements and implementation details around technical interventions such as algorithm audits, cross-platform ranking systems, digital anti-doping procedures, or legal procedures. This would provide sport governing bodies with leverage rather than keeping them reliant on the goodwill of publishers. Simultaneously, we advise all stakeholders within a specific sporting landscape to set up and collaborate within a regulatory network for that sport’s virtual discipline. This should distribute shared control among international federations, national federations, and commercial platforms. In addition, we see national federations as a crucial bridge in this network because they are ultimately responsible for translating international good governance guidelines into actionable rules for athletes and local events (see section 15.5.4, study 4).

Furthermore, such collaboration and contractual arrangements between platform publishers and sport governing bodies should optimally lead to solutions that address the main governance issues observed in studies 4 and 5. These include solutions such as minimum standards for physics engines, platform interoperability, and independent performance verification. Such collaboration helps secure long-term commercial stability and builds moral legitimacy and trust within the general public, demonstrating that virtual sports can be regarded as legitimate. Governing bodies should not underestimate the risks of platform fragmentation (i.e. multiple platforms all having their own standards and rules). This has been showcased in esports literature to cause “confusion to people both inside and outside the esports industry” (Peng et al., 2020, p. 12), which makes it challenging to regulate and gain social acceptance as a (virtual) sport.

Lastly, a specific practical recommendation is the mandate of a “scrutability-without-disclosure” model as an industry standard (cf. section 16.9, study 5). As algorithms hold the power to significantly define sporting outcomes, independent verification of their impact becomes a necessity. A scrutability-without-disclosure model allows independent regulatory bodies to verify platform accuracy and algorithmic fairness while protecting the valuable commercial assets (e.g., physics engines) of platform developers. We recommend such model, as it helps reconcile the ongoing tension between public sporting integrity and private commercial interests.

22 Future Research

As the preceding sections have demonstrated, the integration of virtual sports into the traditional sporting landscape is a complex multi-domain phenomenon. Alongside the recommendations for future research in each empirical study, domain-specific recommendations for future research have also been detailed throughout the discussion. Specifically, Section 18 outlined necessary avenues for exploring sport digitisation and theoretical governance frameworks. Section 19 proposed research into the institutional dynamics, competing logics, and commercial trickle-down effects shaping adoption. Finally, Section 20 highlighted the need to further employ large scale innovation diffusion models, examine platform-specific effects, and rely on frameworks that integrate micro, meso, and macro perspectives of each domain, like Technology-Organisation-Environment framework or fuzzy-set Qualitative Comparative Analysis.

Rather than reiterating these specific inquiries, this section sets out a macro-level agenda for the broader field of sport management. While initial research examined virtual sports as a temporary pandemic-induced trend (Tjønndal, 2022; Westmattelmann, Grotenhermen, Sprenger, & Schewe, 2021), contemporary studies do acknowledge a more permanent technological shift in sporting delivery, consumption, and governance (Goebeler et al., 2024; Reed et al., 2023; Tjønndal et al., 2025; Westmattelmann et al., 2025). We align our recommendations for future research to support this trend by recommending (1) the employment of new methodological approaches, (2) expanding on ontological boundaries, and (3) embracing cross-disciplinary and interdisciplinary collaborations required to capture the physical-digital reality of virtual sports.

22.1 *Methodological Recommendations for Sport Digitalisation Research*

The empirical investigations within this dissertation primarily relied on cross-sectional data to map the initial adoption diffusion phases and current governance gaps of virtual cycling. As the market matures, a methodological shift is recommended. Future research should transition towards longitudinal research designs (Wolfe, 1994). Tracking organisations such as clubs or federations over several years will allow scholars to observe how they transition from early adoption to either failed or successful implementation. This approach will provide a more dynamic understanding of the diffusion process and reveal how institutional beliefs, both within the trajectories of a sport-for-all and result-oriented logic, as explained in section 19.3, evolve as users and managers gain sustained exposure to virtual sports hardware and software (Washington & Patterson, 2011).

Furthermore, both the digitisation and digitalisation of sports introduce unique methodological challenges, such as the governance issues and platform-results gap discussed in study 4 (section 15) and study 5 (section 16). Traditional qualitative interviews and self-reported surveys might become insufficient to capture the issues surrounding such a physical-digital interreality. Therefore, sport management scholars are advised to embrace data-driven methodologies, such as algorithmic auditing (Westmattmann et al., 2022, 2025). This includes setting up collaborations with platforms to use primary and confidential algorithm data, relying on data scraping, and employing experimental designs to test specific algorithmic variations under controlled conditions. Such approaches will provide objective insights into the platform-result gaps that currently challenge sporting integrity. However, we remain true to our earlier noted recommendations (cf. sections 19.2 and 20.1) in that researchers should analyse these differences within frameworks like institutional change or the Technology-Organisation-Environment framework, which allows researchers to acknowledge subjective parameters such as institutional (commercialisation) logics or isomorphic pressures perceived by governing bodies or platform developers.

Following up on our practical recommendations in section 21.4, future research is needed to make empirically-grounded actionable recommendations regarding the prioritisation of sport governing bodies' technical interventions aimed at resolving governance issues. To so, future studies are advised to employ a Practice-Based View (PBV) as explained in Section 12.6.5 (study 1). By investigating the micro-level actions and internal resource constraints of key staff and decision-makers within international federations, researchers can develop validated, context-specific roadmaps for implementing virtual sports governance.

Finally, the reinstitutionalisation and adoption trajectories of virtual sports within both the performance and the sport-for-all logic (cf. section 19.3 and Figure 19) would benefit from large-scale quantitative and longitudinal data collection. This research is limited by the fact that it mainly relies on assessing the organisational intent of sport-for-all implementations in virtual sports. To move beyond that limitation and accurately measure the actual impact, researchers are encouraged to capture longitudinal individual-level data. This would allow researchers to evaluate true inclusivity among underrepresented groups, track health metrics, or track psychological indicators of social connectedness. In addition, longitudinal adoption-diffusion modelling would also benefit the empirical underpinning of the reinstitutionalisation and adoption trajectories as explained in sections 19.3 and 20.3. This would render an overview of the current (macro-)adoption of virtual sports and the rate of its diffusion for each of the two trajectories. However, since the reinstitutionalisation of virtual sports is considered a complex interaction between a lack of commitment from national federation support (cf. study 3), varying determinants within adopter categories (cf. study 1), and varying club archetypes within competing institutional logics (cf. study 2), traditional quantitative models such as Rogers' (2003) innovation diffusion model might lack the required in-depth nuances of implicit and unspoken dynamics often captured by qualitative research. Therefore, we recommend that future research complements traditional quantitative diffusion models with more recent and advanced quantitative models capable of operationalising latent dynamics, for instance through computational text analysis and Large Language Models (i.e. LLMs). These advanced natural language processing tools allow researchers to extract and quantify latent institutional logics and narratives from policy documents, media discourse, or user forums (Hannigan et al., 2019; Törnberg, 2025). Large language models like GPT-4 have in fact been shown to interpret implicit or unspoken references with higher accuracy and inter-coder reliability compared to both supervised models and human coders (Törnberg, 2025). By triangulating these computational models with longitudinal tracking and algorithmic auditing, researchers can construct a robust understanding of how virtual sports diffuse and function across all levels of the sporting ecosystem.

22.2 Expanding the Ontological Boundaries of Virtual Sports

The conceptual framework developed in this dissertation defines virtual sports independently of remoteness, (a)synchrony, representational fidelity, and, in a certain way, physicality (cf. section 9.6). This broadened ontological approach provides a foundation for future research to

explore sporting formats that diverge from the current dominance of interconnected, remote platforms, given that the main body of virtual sports research today concerns virtual cycling and platforms such as Zwift or MyWhoosh (see section 2.3 and 9.5). Future studies are encouraged to test our definition and ontological framework against newly emerging formats with different configurations of virtuality (cf. Figure 2), for example, virtual golf as described in section 9.4.2. Exploring these diverse configurations will help ensure the academic literature remains relevant as technologies evolve beyond the current paradigm.

Furthermore, the governance issues and institutional dynamics identified in this thesis are primarily grounded in the empirical context of virtual cycling. To build a more comprehensive understanding of virtual sports, scholars are encouraged to conduct comparative studies across different virtual disciplines, as has for instance been done by Botwina et al. (2025) in their study on Virtual Regatta. Because it remains still debatable if Virtual Regatta could be considered a virtual sport given its very high movement compression (i.e. limited movement decompression) where the game is played sedentary on a mobile device¹⁶, researchers are in particular encouraged to study virtual sporting games that incorporate both parent sports-alike cognitive elements (e.g. strategy) and *high* movement decompression as to increase generalisability (e.g., Virtual Golf, Augmented Wall Climbing, Virtual Biathlon, Virtual Rowing, Virtual Triathlon...¹⁷). Comparative analyses of virtual cycling with activities that possess distinct physiological inputs and algorithmic models, such as virtual rowing or virtual golf, are particularly insightful to help determine whether the governance gaps and legitimisation strategies identified here are prevalent across all virtual sports or specific to the cycling context. Such inquiries are necessary to refine the proposed regulatory networks and ensure they are adaptable to the broader virtual sporting landscape (Peng et al., 2020).

Last, future research may expand our deinstitutionalisation-reinstitutionalisation model of virtual sports (cf. Figure 19) by including a third commercialisation/professionalisation logic as an additional axis. However, researchers are encouraged not to simply reduce the commercialisation logic to solely the financial interests of digital platforms, but also include the exchange value of (virtual) sports and physical infrastructures such as arena companies and hospitality services, as originally set out by Stenling and Fahlén (2009). As virtual sports happen both co-located and online (in fact, mainly online), future researchers must also adapt

¹⁶ More info at https://en.wikipedia.org/wiki/Virtual_Regatta

¹⁷ More examples and info at <https://www.virtualsports.be/2025/03/24/virtual-sports-an-overview-of-what-exists-and-what-is-possible/>

the framework of Stenling and Fahlén (2009) to cover the commercial logic within such hybrid environments.

22.3 *Interdisciplinary Development of Virtual Sports*

As virtual sports mature, the governance of digital platforms demands expertise that extends beyond traditional sport management. Study 5 in this dissertation provides the first empirical evidence to suggest that platform algorithms actively influence sporting outcomes. Such a physical-digital reality (see section 9.5.3) requires an interdisciplinary approach to adequately ensure competitive integrity (see study 4 & 5). Collaboration is needed between sport management, computer science and sports engineering (e.g., Dowd et al., 2024) disciplines to operationalise regulatory frameworks, such as the “scrutability-without-disclosure” model proposed in this dissertation. Cooperative research efforts are also needed to develop secure auditing mechanisms which allow independent regulators to verify the fairness and accuracy of virtual physics engines without compromises for any stakeholder, whether they are commercial or regulatory in nature.

Furthermore, the transition of commercial platform providers into powerful “stakeowners” introduces significant legal and regulatory challenges. Future research should integrate legal research disciplines to address the shifting power dynamics between traditional non-profit federations and private technology companies (Kelly et al., 2022; Peng et al., 2020). Legal scholars are needed to investigate the creation of specialised terms of service and robust data protection frameworks tailored to digitised sports that rely on hardware and software outside of official governing bodies. Future research is also recommended to consider the fact that athletes in virtual sports generate highly sensitive physiological and biological data (Lupton, 2016; Miah, 2017). Interdisciplinary research must explore how to protect athletes’ data under international privacy laws (e.g., GDPR) while simultaneously allowing governing bodies to use it for digital anti-doping and performance verification (see section 15.5.1, study 3). As Downie et al.’s (2025) recent organisational research in an elite sport setting highlights, leveraging biometric data is not merely a technical challenge but a highly relational one because biometric data is inherently decontextualized. Therefore, future studies focussing on the use of biometric data from a virtual sports context are advised to read up on the many pitfalls of using such data and the need for a “shared control of the body” (Downie et al., 2025), which could for instance translate to the need for collaborative interpretation between athletes, coaches and governing bodies in a virtual sporting context. Bridging sport management, computer science,

sports engineering and legal studies will ultimately provide the comprehensive regulatory network required to effectively govern a new era of sports characterised by a physical-digital interreality.

Next to the disciplines mentioned above, future virtual sports research would also benefit from domains adject to sport management. First, we suggest exploring the dimension of fan engagement and virtual spectatorship, something this dissertation has not touched upon despite it being a promising and prospective aspect of virtual sports. The rise of digital fandom introduces new theoretical and practical questions regarding how audiences interact with mixed-reality sporting events that deserve future investigation. Scholars could for instance draw on recent work such as Schmidt (2026) to understand how virtual spectatorship interacts with the commercial and social dimensions of emerging virtual disciplines. Second, we suggest exploring virtual cycling from a sports sciences perspective, as such insights may also indirectly impact sport management. For instance, experimental research could investigate how the physiological and biomechanical differences between indoor cycling and outdoor cycling, such as reduced cycling economy, reduced gross efficiency, higher cadence, reduced critical power (Bertucci et al., 2012; Lipski et al., 2022), affect athletes' performances when indoor cycling is practised regularly (in competition) as a sport. In addition, research may explore how athletes cope with the heightened mental demands of static environments (Irvine et al., 2022) or specific cooling and nutritional practices (Peeters et al., 2024). Such insights may consequently inform how clubs and federations practically support digital athletes, for example by adapting selection programs, coaching, or race support.

23 Limitations and Strengths

23.1 Limitations

Naturally, there are several reasons that our research is constrained. While we have already listed specific theoretical limitations in our discussion (cf. section 18 to 20), this section will discuss more structural and methodological limitations.

23.1.1 Structural Limitations

First, this research is limited by its empirical concentration on virtual cycling. Focussing exclusively on a single discipline might restrict the immediate generalisability of these findings to other emerging virtual sports. Other virtual disciplines, for instance outside of endurance sports like cycling, possess distinct physical and technological requirements that are not

captured here. While core conceptual findings regarding the physical-digital interreality and dual governance models are highly transferable, other empirical insights are inherently tied to cycling's specific culture and individual endurance-based nature. For example, the manifestations of digital doping and seasonal adoption drivers may differ significantly in skill-based disciplines, such as virtual golf, often indoor practiced sports such as climbing, or team sports require a different virtuality. Nonetheless, as outlined in section 7.2, the case selection of virtual cycling was a deliberate methodological *and* theoretically underpinned choice. Virtual cycling serves as a mature proxy for studying the overarching emergence and institutionalisation of virtual sports. Its advanced technological infrastructure and established competitive ecosystems are not found in other virtual disciplines, providing a rich environment for analysis.

Second, another structural limitation stems from researching proprietary platforms that are effectively a “black box” through which “authority is increasingly expressed algorithmically” (Pasquale, 2015, p. 8). Consequently, this research was constrained by a lack of direct access to platforms' undisclosed data and source code, such as virtual physics engines. Alternatively, our analysis had to rely on scraping public output data (cf. study 5), which can be considered a pragmatic workaround for gathering necessary metrics. Data scraping nonetheless remains an inherent limitation regarding data completeness. As indicated before, future research in this domain would benefit from increased transparency by platforms, which may provide access to non-publicly available data for research purposes.

23.1.2 Methodological and Empirical limitations

Third, the rapid evolution of both technology and policy presents a challenge related to capturing a domain that is in continuous flux. The cross-sectional nature of this study means we are, however, limited in capturing such a domain. Sudden software updates, shifting algorithms, and the release of new hardware can quickly alter the empirical reality experienced by users, as even some interviewees noted in study 4. Hence, the issues and dynamics presented in our work might in fact change significantly, even shortly after data collection. Furthermore, this volatility is mirrored by rapid institutional evolutions, such as the announcement of the 2027 Olympic Esports Games at the beginning of 2025 by the IOC (International Olympic Committee, 2025a), which later that year already announced its cancellation (International Olympic Committee, 2025b). Such decisions may frequently shift the policy and actions of many stakeholders involved, making research on the topic difficult to interpret or compare. Therefore, the findings must be interpreted as an analysis of the landscape at the time of study,

which is empirically situated from 2022 to 2025.

Fourth, the geographic scope of the sample and the associated digital inequalities present another important limitation. Our first data collection relies heavily on Flemish grassroots sports clubs. This Flemish context provides valuable insights, yet it requires reflection on which findings are context-specific and which are transferable to other contexts (e.g. geographical). Flemish voluntary sports clubs possess specific organisational structures and a distinct relationship with regional federations. For example, their initiatives often depend on specific volunteers and their capabilities (cf. study 1; Corthouts et al., 2023). The cultural embeddedness of cycling in Flanders (or Western Europe in general), combined with a varied yet complex subsidy structures (e.g. structural, project-based, local...), might shape the adoption of virtual cycling differently than in regions with different structures or commercialised college sports models (e.g. North America). Still, literature has pointed out that, across Europe, voluntary sports clubs share common structural characteristics, such as an inherent reliance on volunteer management and similar financial constraints (Breuer et al., 2017). Therefore, the identified mechanisms such as innovation champions, managerial attitudes, or institutional logics, can be argued to be transferable to other European grassroots settings to at least a certain degree. Beyond the club level, our second data collection focuses predominantly on Western-centric international federations (e.g. UCI, World Triathlon) for its broader institutional analysis. Therefore, our findings may not capture the sporting reality of regions or socio-economic demographics facing digital divides. Sport management literature increasingly highlights these structural disparities. As the literature has pointed to inequalities in participation for both sports (Tjønndal, 2019; Van Tuyckom et al., 2010) and esports (Reitman et al., 2020; Rogstad, 2022; Rogstad & Skauge, 2023), research examining virtual sports adoption should also account for these inequalities and how they impact results. In addition, literature has also discussed the fact that virtual cycling requires a notable financial investment in a smart trainer, equipment (e.g. bike, clothes, device), a platform subscription, and an internet connection (Berger et al., 2025). Combining these barriers, it could be argued that the patterns observed in this research are situated within a privileged digital context. Hence, they should not be universally extrapolated to developing regions without critically considering these barriers to entry.

Fifth and last, this research acknowledges the omission of individuals within clubs such as members or the coach as a distinct unit or level of analysis. Both members and coaches play an important role within the sport club: for members there might be potential misalignment between board-level enthusiasm and member-level engagement while coaches may act as a

potential gatekeeper of technology adoption within their club (Skrubbeltrang et al., 2022). The accounts of both members and coaches may for instance influence the dominant logic behind the implementation of adopted technology (see study 2 and section 19.3). Here, we would like to refer to the fact that we aimed to address this hiatus towards the end of the author's doctoral research process, for which we collected multi-level data that includes responses from 52 sports clubs and 280 coaches regarding the adoption of various MR-technologies in six different sports (i.e. virtual cycling and rowing platforms, interactive football and hockey LED pods, interactive basketball hoop, augmented table tennis). However, a deliberate choice was made to exclude these data and their analysis from this dissertation since the response rate for virtual cycling in particular was too low to meaningfully extract insights from them. Conversely, incorporating the whole dataset would dilute the scope and coherent narrative of this dissertation as we would then have to compare results on virtual cycling with other MR-technologies like LED pods that are not considered virtual sports (see section 9.2.3).

23.2 *Strengths*

Because specific theoretical strengths have already been reflected upon in our discussion (cf. section 18 to 20), this section will discuss more structural and methodological strengths.

23.2.1 *Structural Strengths*

A first strength is the comprehensive theoretical synthesis achieved throughout the research. As shown in Table 1, this dissertation employs an integrative multi-theoretical approach that positions institutional theory as its overarching conceptual foundation. Under this umbrella, it weaves together complementary theories and frameworks, including innovation diffusion theory, organisational innovativeness theory, disruptive innovation theory, and (e)sports good governance frameworks. While we acknowledge the fact that integrating such a diverse array of theoretical lenses carries the inherent risk of conceptual dilution, we believe that this comprehensive approach was nonetheless essential in holistically mapping the unprecedented phenomenon called virtual sports. A single theoretical perspective would in our case be insufficient to capture the nuanced interactions between institutional, technical, commercial, managerial, and regulatory dynamics.

Second, the timing and contemporary nature of this research is regarded as a particular strength. This dissertation addresses the emergence of new sporting disciplines, alongside their issues, as they unfold, rather than retrospectively analysing practices that have already unfolded and achieved broad (re)institutionalisation. This timeliness ensures that our findings are

contemporary and immediately relevant to policymakers and sports managers currently navigating sports digitalisation. However, we must also nuance this contemporary nature with the inherent risks that a rapidly shifting landscape can alter swiftly in unpredictable directions, both positive and negative, as described towards the end of the limitations in section 23.1.

23.2.2 Methodological and Empirical Strengths

Third, a significant strength of this dissertation lies in its balanced methodological design (see section 7, Table 3 and Table 4). We employed two quantitative studies, two qualitative studies, and one mixed-methods study. Such balance facilitates robust methodological triangulation (Skinner et al., 2024) and embraces epistemological pluralism since we use different methodological paradigms to acquire a fuller description of the subject, something we discussed in section 6.1.

Fourth and finally, building upon this methodological foundation, this dissertation benefits from a broad multi-domain and multi-level analytical scope. By relying on both empirical and theoretical reasoning, we are able to capture the phenomenon of virtual sports from a (1) digitisation, (2) institutional, and (3) organisational innovation perspective at the micro-level of individual volunteers and athletes, the meso-level of federations and club archetypes' capacity, and the macro-level of sport policy and institutional international federations and commercial publishers (see Table 19). Such an approach prevents a siloed understanding of virtual sports within the aforementioned subdomains of sport management and demonstrates new linkages between different domains and levels.

24 References Part V

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Part VI

Societal Outreach

25 Societal Outreach: Virtual Sports Research Network



In January 2024, Thibault initiated the Virtual Sports Research Network (VSRN) (www.virtualsports.be) to address two distinct challenges encountered during his doctoral research process. First, there was notable difficulty locating peers studying virtual sports, as researchers were often scattered across different academic disciplines. Second, there was a general confusion among both academics and practitioners regarding the fundamental concept of virtual sports. It was frequently unclear whether these activities should be categorised as traditional sports, exergames, esports, or something else.

To overcome these challenges, Thibault (self-)developed a dedicated platform to translate academic knowledge into accessible formats. The foundational content, featured on the home (www.virtualsports.be) and about page (www.virtualsports.be/about), synthesises our current understanding of the topic. To grow this platform with others' knowledge, Thibault proactively contacted international scholars working on specific different aspects of virtual sports (e.g. cheating, doping, algorithms, social issues), inviting them to be listed on the research network page (www.virtualsports.be/members) and contribute accessible blog posts about their research (www.virtualsports.be/blog). By featuring these scholars and their work, the website evolved into a public directory about virtual sports, backed by evidence-based insights and experts in this field. Over the past two years, this initiative has grown into an active network comprising over ten academic experts (see Figure 20) and featuring more than fifteen dedicated blog posts authored by different network members (see Figure 21). Featuring such network and expertise allows practitioners and researchers to easily uncover new experts and connect with them.

Looking ahead, the vision for the network is to make the complex phenomenon of virtual sports understandable for the broader public, while simultaneously providing robust, evidence-based insights to key stakeholders such as federation employees, policymakers, and academics. Furthermore, Thibault envisions the platform acting as a catalyst for long-term interdisciplinary collaboration. This includes facilitating joint funding applications for network projects and producing co-authored publications. A concrete immediate step in this direction could for instance involve translating the theoretical pre-study of this dissertation (Part III, section 9) into a collaborative publication co-authored by the network members. The ultimate

objective is to sustainably grow the network, expose experts in this field, and disseminate knowledge that effectively bridges the gap between digital sports theory and practice.

Figure 20: Virtual Sports Research Network member page (www.virtualsports.be/members)

VIRTUAL SPORTS

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Virtual Sports Research Network

Connecting expertise from leading academics

We connect both academics and experts closely related with academia who have expertise in virtual sports. We aim to foster future collaboration and provide an accessible overview of academics and experts in the domain of virtual sports. Below, you find an overview (in chronological joining order 🗓️) of all members associated with the Virtual Sports Research Network and their direct contact information. Are you interested in joining the network? [Get in touch!](#)

Looking For Experts

Interested in joining the Virtual Sports Research Network? We're currently still looking for experts on the topics around virtual sports 🗓️!

Thibault Fouquaert
PhD Researcher & Founder of the Virtual Sports Research Network
Sportmanagement, Department of Movement and Sport Sciences, Ghent University

Thibault's PhD focus on the intertwining of technology and sports from a managerial perspective in sport innovation.

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Anne Tjørndal
Professor in Sociology of Sport & Head of Department for Organisation, Leadership and working life.
Response, Faculty of Social Sciences, Nord University

Anne's work focuses on social inclusion in and through sports. To this end, she takes various approaches of which virtual sports is one.

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Andrew Richardson
Postdoctoral Research Associate in Gambling Harms
Population Health Sciences Institute, Faculty of Medical Sciences, Newcastle University

Andrew is Doctor of Social Policy and covers topics on anti-doping, image and performance enhancing drugs, body image, transgender sport, eSports, physical activity and, last but not least, virtual sports.

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Charlotte Van Tuyckom
Lecturer and Researcher
Sports Innovation Campus, Department of Sport and movement sciences, Howest University College

Charlotte is Doctor in sport sociology with interest in the sociological and policy components of sport. Her focus is on the impact of digital innovation on the field of sport and Generation Z.

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Grzegorz Botwina
Assistant Professor of Management, President at Institute for Sport Governance
Faculty of Management, University of Warsaw

Greg's work focuses on good governance and innovation in sport. He has experience in business, government and NGO sectors.

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Daniel Westmattmann
Associate Professor of Digitalization, Sports, and Health Management
Chair of Organization, Human Resource Management and Innovation, Center for Management, University of Münster

Daniel's work focuses on the dynamic interplay between

Willem Standaert
Associate Professor of Marketing
Strategic Marketing Innovation Department, HEC Liège Management School - University of Liège

Willem's research explores the intersection of technology and

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The Virtual Sports science communication blog and Virtual Sports Research Network is an initiative that is originally rooted in the research of the Sport Management Research Group at Ghent University (Belgium).

Figure 21: Virtual Sports Research Network blog post page (www.virtualsports.be/blog)

